libbfd

The Binary File Descriptor Library

First Edition—BFD version < 3.0 % Since no product is stable before version 3.0 :-)
Original Document Created: April 1991

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BFD, 1.5
TEXinfo 2018-01-09.11

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1 Introduction

BFD is a package which allows applications to use the same routines to operate on object files whatever the object file format. A new object file format can be supported simply by creating a new BFD back end and adding it to the library.

BFD is split into two parts: the front end, and the back ends (one for each object file format).

- The front end of BFD provides the interface to the user. It manages memory and various canonical data structures. The front end also decides which back end to use and when to call back end routines.
- The back ends provide BFD its view of the real world. Each back end provides a set of calls which the BFD front end can use to maintain its canonical form. The back ends also may keep around information for their own use, for greater efficiency.

1.1 History

One spur behind BFD was the desire, on the part of the GNU 960 team at Intel Oregon, for interoperability of applications on their COFF and b.out file formats. Cygnus was providing GNU support for the team, and was contracted to provide the required functionality.

The name came from a conversation David Wallace was having with Richard Stallman about the library: RMS said that it would be quite hard—David said "BFD". Stallman was right, but the name stuck.

At the same time, Ready Systems wanted much the same thing, but for different object file formats: IEEE-695, Oasys, Srecords, a.out and 68k coff.

BFD was first implemented by members of Cygnus Support; Steve Chamberlain (`sac@cygnus.com`), John Gilmore (`gnu@cygnus.com`), K. Richard Pixley (`rich@cygnus.com`) and David Henkel-Wallace (`gumby@cygnus.com`).

1.2 How To Use BFD

To use the library, include `bfd.h` and link with `libbfd.a`.

BFD provides a common interface to the parts of an object file for a calling application.

When an application successfully opens a target file (object, archive, or whatever), a pointer to an internal structure is returned. This pointer points to a structure called `bfd`, described in `bfd.h`. Our convention is to call this pointer a BFD, and instances of it within code `abfd`. All operations on the target object file are applied as methods to the BFD. The mapping is defined within `bfd.h` in a set of macros, all beginning with '`bfd_`' to reduce namespace pollution.

For example, this sequence does what you would probably expect: return the number of sections in an object file attached to a BFD `abfd`.

```
#include "bfd.h"

unsigned int number_of_sections (abfd)
bfd *abfd;
{
```

```
    return bfd_count_sections (abfd);
  }
```

The abstraction used within BFD is that an object file has:

- a header,
- a number of sections containing raw data (see Section 2.6 [Sections], page 29),
- a set of relocations (see Section 2.10 [Relocations], page 60), and
- some symbol information (see Section 2.7 [Symbols], page 49).

Also, BFDs opened for archives have the additional attribute of an index and contain subordinate BFDs. This approach is fine for a.out and coff, but loses efficiency when applied to formats such as S-records and IEEE-695.

1.3 What BFD Version 2 Can Do

When an object file is opened, BFD subroutines automatically determine the format of the input object file. They then build a descriptor in memory with pointers to routines that will be used to access elements of the object file's data structures.

As different information from the object files is required, BFD reads from different sections of the file and processes them. For example, a very common operation for the linker is processing symbol tables. Each BFD back end provides a routine for converting between the object file's representation of symbols and an internal canonical format. When the linker asks for the symbol table of an object file, it calls through a memory pointer to the routine from the relevant BFD back end which reads and converts the table into a canonical form. The linker then operates upon the canonical form. When the link is finished and the linker writes the output file's symbol table, another BFD back end routine is called to take the newly created symbol table and convert it into the chosen output format.

1.3.1 Information Loss

Information can be lost during output. The output formats supported by BFD do not provide identical facilities, and information which can be described in one form has nowhere to go in another format. One example of this is alignment information in `b.out`. There is nowhere in an `a.out` format file to store alignment information on the contained data, so when a file is linked from `b.out` and an `a.out` image is produced, alignment information will not propagate to the output file. (The linker will still use the alignment information internally, so the link is performed correctly).

Another example is COFF section names. COFF files may contain an unlimited number of sections, each one with a textual section name. If the target of the link is a format which does not have many sections (e.g., `a.out`) or has sections without names (e.g., the Oasys format), the link cannot be done simply. You can circumvent this problem by describing the desired input-to-output section mapping with the linker command language.

Information can be lost during canonicalization. The BFD internal canonical form of the external formats is not exhaustive; there are structures in input formats for which there is no direct representation internally. This means that the BFD back ends cannot maintain all possible data richness through the transformation between external to internal and back to external formats.

This limitation is only a problem when an application reads one format and writes another. Each BFD back end is responsible for maintaining as much data as possible, and the internal BFD canonical form has structures which are opaque to the BFD core, and exported only to the back ends. When a file is read in one format, the canonical form is generated for BFD and the application. At the same time, the back end saves away any information which may otherwise be lost. If the data is then written back in the same format, the back end routine will be able to use the canonical form provided by the BFD core as well as the information it prepared earlier. Since there is a great deal of commonality between back ends, there is no information lost when linking or copying big endian COFF to little endian COFF, or `a.out` to `b.out`. When a mixture of formats is linked, the information is only lost from the files whose format differs from the destination.

1.3.2 The BFD canonical object-file format

The greatest potential for loss of information occurs when there is the least overlap between the information provided by the source format, that stored by the canonical format, and that needed by the destination format. A brief description of the canonical form may help you understand which kinds of data you can count on preserving across conversions.

files Information stored on a per-file basis includes target machine architecture, particular implementation format type, a demand pageable bit, and a write protected bit. Information like Unix magic numbers is not stored here—only the magic numbers' meaning, so a `ZMAGIC` file would have both the demand pageable bit and the write protected text bit set. The byte order of the target is stored on a per-file basis, so that big- and little-endian object files may be used with one another.

sections Each section in the input file contains the name of the section, the section's original address in the object file, size and alignment information, various flags, and pointers into other BFD data structures.

symbols Each symbol contains a pointer to the information for the object file which originally defined it, its name, its value, and various flag bits. When a BFD back end reads in a symbol table, it relocates all symbols to make them relative to the base of the section where they were defined. Doing this ensures that each symbol points to its containing section. Each symbol also has a varying amount of hidden private data for the BFD back end. Since the symbol points to the original file, the private data format for that symbol is accessible. `ld` can operate on a collection of symbols of wildly different formats without problems.

Normal global and simple local symbols are maintained on output, so an output file (no matter its format) will retain symbols pointing to functions and to global, static, and common variables. Some symbol information is not worth retaining; in `a.out`, type information is stored in the symbol table as long symbol names. This information would be useless to most COFF debuggers; the linker has command-line switches to allow users to throw it away.

There is one word of type information within the symbol, so if the format supports symbol type information within symbols (for example, COFF, Oasys) and the type is simple enough to fit within one word (nearly everything but aggregates), the information will be preserved.

relocation level

Each canonical BFD relocation record contains a pointer to the symbol to relocate to, the offset of the data to relocate, the section the data is in, and a pointer to a relocation type descriptor. Relocation is performed by passing messages through the relocation type descriptor and the symbol pointer. Therefore, relocations can be performed on output data using a relocation method that is only available in one of the input formats. For instance, Oasys provides a byte relocation format. A relocation record requesting this relocation type would point indirectly to a routine to perform this, so the relocation may be performed on a byte being written to a 68k COFF file, even though 68k COFF has no such relocation type.

line numbers

Object formats can contain, for debugging purposes, some form of mapping between symbols, source line numbers, and addresses in the output file. These addresses have to be relocated along with the symbol information. Each symbol with an associated list of line number records points to the first record of the list. The head of a line number list consists of a pointer to the symbol, which allows finding out the address of the function whose line number is being described. The rest of the list is made up of pairs: offsets into the section and line numbers. Any format which can simply derive this information can pass it successfully between formats.

2 BFD Front End

2.1 typedef bfd

A BFD has type `bfd`; objects of this type are the cornerstone of any application using BFD. Using BFD consists of making references though the BFD and to data in the BFD.

Here is the structure that defines the type `bfd`. It contains the major data about the file and pointers to the rest of the data.

```
enum bfd_direction
  {
    no_direction = 0,
    read_direction = 1,
    write_direction = 2,
    both_direction = 3
  };

enum bfd_plugin_format
  {
    bfd_plugin_unknown = 0,
    bfd_plugin_yes = 1,
    bfd_plugin_no = 2
  };

struct bfd_build_id
  {
    bfd_size_type size;
    bfd_byte data[1];
  };

struct bfd
{
  /* The filename the application opened the BFD with.  */
  const char *filename;

  /* A pointer to the target jump table.  */
  const struct bfd_target *xvec;

  /* The IOSTREAM, and corresponding IO vector that provide access
     to the file backing the BFD.  */
  void *iostream;
  const struct bfd_iovec *iovec;

  /* The caching routines use these to maintain a
     least-recently-used list of BFDs.  */
  struct bfd *lru_prev, *lru_next;
```

```
  /* Track current file position (or current buffer offset for
     in-memory BFDs).  When a file is closed by the caching routines,
     BFD retains state information on the file here.  */
  ufile_ptr where;

  /* File modified time, if mtime_set is TRUE.  */
  long mtime;

  /* A unique identifier of the BFD  */
  unsigned int id;

  /* Format_specific flags.  */
  flagword flags;

  /* Values that may appear in the flags field of a BFD.  These also
     appear in the object_flags field of the bfd_target structure, where
     they indicate the set of flags used by that backend (not all flags
     are meaningful for all object file formats) (FIXME: at the moment,
     the object_flags values have mostly just been copied from backend
     to another, and are not necessarily correct).  */

#define BFD_NO_FLAGS                0x0

  /* BFD contains relocation entries.  */
#define HAS_RELOC                   0x1

  /* BFD is directly executable.  */
#define EXEC_P                      0x2

  /* BFD has line number information (basically used for F_LNNO in a
     COFF header).  */
#define HAS_LINENO                  0x4

  /* BFD has debugging information.  */
#define HAS_DEBUG                  0x08

  /* BFD has symbols.  */
#define HAS_SYMS                   0x10

  /* BFD has local symbols (basically used for F_LSYMS in a COFF
     header).  */
#define HAS_LOCALS                 0x20

  /* BFD is a dynamic object.  */
#define DYNAMIC                    0x40
```

```
  /* Text section is write protected (if D_PAGED is not set, this is
     like an a.out NMAGIC file) (the linker sets this by default, but
     clears it for -r or -N).  */
#define WP_TEXT                    0x80

  /* BFD is dynamically paged (this is like an a.out ZMAGIC file) (the
     linker sets this by default, but clears it for -r or -n or -N).  */
#define D_PAGED                   0x100

  /* BFD is relaxable (this means that bfd_relax_section may be able to
     do something) (sometimes bfd_relax_section can do something even if
     this is not set).  */
#define BFD_IS_RELAXABLE          0x200

  /* This may be set before writing out a BFD to request using a
     traditional format.  For example, this is used to request that when
     writing out an a.out object the symbols not be hashed to eliminate
     duplicates.  */
#define BFD_TRADITIONAL_FORMAT    0x400

  /* This flag indicates that the BFD contents are actually cached
     in memory.  If this is set, iostream points to a bfd_in_memory
     struct.  */
#define BFD_IN_MEMORY             0x800

  /* This BFD has been created by the linker and doesn't correspond
     to any input file.  */
#define BFD_LINKER_CREATED       0x1000

  /* This may be set before writing out a BFD to request that it
     be written using values for UIDs, GIDs, timestamps, etc. that
     will be consistent from run to run.  */
#define BFD_DETERMINISTIC_OUTPUT 0x2000

  /* Compress sections in this BFD.  */
#define BFD_COMPRESS             0x4000

  /* Decompress sections in this BFD.  */
#define BFD_DECOMPRESS           0x8000

  /* BFD is a dummy, for plugins.  */
#define BFD_PLUGIN              0x10000

  /* Compress sections in this BFD with SHF_COMPRESSED from gABI.  */
#define BFD_COMPRESS_GABI       0x20000

  /* Convert ELF common symbol type to STT_COMMON or STT_OBJECT in this
```

```
     BFD.  */
#define BFD_CONVERT_ELF_COMMON  0x40000

  /* Use the ELF STT_COMMON type in this BFD.  */
#define BFD_USE_ELF_STT_COMMON  0x80000

  /* Put pathnames into archives (non-POSIX).  */
#define BFD_ARCHIVE_FULL_PATH  0x100000

  /* Flags bits to be saved in bfd_preserve_save.  */
#define BFD_FLAGS_SAVED \
  (BFD_IN_MEMORY | BFD_COMPRESS | BFD_DECOMPRESS | BFD_LINKER_CREATED \
   | BFD_PLUGIN | BFD_COMPRESS_GABI | BFD_CONVERT_ELF_COMMON \
   | BFD_USE_ELF_STT_COMMON)

  /* Flags bits which are for BFD use only.  */
#define BFD_FLAGS_FOR_BFD_USE_MASK \
  (BFD_IN_MEMORY | BFD_COMPRESS | BFD_DECOMPRESS | BFD_LINKER_CREATED \
   | BFD_PLUGIN | BFD_TRADITIONAL_FORMAT | BFD_DETERMINISTIC_OUTPUT \
   | BFD_COMPRESS_GABI | BFD_CONVERT_ELF_COMMON | BFD_USE_ELF_STT_COMMON)

  /* The format which belongs to the BFD. (object, core, etc.)  */
  ENUM_BITFIELD (bfd_format) format : 3;

  /* The direction with which the BFD was opened.  */
  ENUM_BITFIELD (bfd_direction) direction : 2;

  /* Is the file descriptor being cached?  That is, can it be closed as
     needed, and re-opened when accessed later?  */
  unsigned int cacheable : 1;

  /* Marks whether there was a default target specified when the
     BFD was opened. This is used to select which matching algorithm
     to use to choose the back end.  */
  unsigned int target_defaulted : 1;

  /* ... and here: (``once'' means at least once).  */
  unsigned int opened_once : 1;

  /* Set if we have a locally maintained mtime value, rather than
     getting it from the file each time.  */
  unsigned int mtime_set : 1;

  /* Flag set if symbols from this BFD should not be exported.  */
  unsigned int no_export : 1;

  /* Remember when output has begun, to stop strange things
```

```
   from happening.  */
unsigned int output_has_begun : 1;

/* Have archive map.  */
unsigned int has_armap : 1;

/* Set if this is a thin archive.  */
unsigned int is_thin_archive : 1;

/* Set if this archive should not cache element positions.  */
unsigned int no_element_cache : 1;

/* Set if only required symbols should be added in the link hash table for
   this object.  Used by VMS linkers.  */
unsigned int selective_search : 1;

/* Set if this is the linker output BFD.  */
unsigned int is_linker_output : 1;

/* Set if this is the linker input BFD.  */
unsigned int is_linker_input : 1;

/* If this is an input for a compiler plug-in library.  */
ENUM_BITFIELD (bfd_plugin_format) plugin_format : 2;

/* Set if this is a plugin output file.  */
unsigned int lto_output : 1;

/* Set if this is a slim LTO object not loaded with a compiler plugin.  */
unsigned int lto_slim_object : 1;

/* Do not attempt to modify this file.  Set when detecting errors
   that BFD is not prepared to handle for objcopy/strip.  */
unsigned int read_only : 1;

/* Set to dummy BFD created when claimed by a compiler plug-in
   library.  */
bfd *plugin_dummy_bfd;

/* The offset of this bfd in the file, typically 0 if it is not
   contained in an archive.  */
ufile_ptr origin;

/* The origin in the archive of the proxy entry.  This will
   normally be the same as origin, except for thin archives,
   when it will contain the current offset of the proxy in the
   thin archive rather than the offset of the bfd in its actual
```

```
   container.  */
ufile_ptr proxy_origin;

/* A hash table for section names.  */
struct bfd_hash_table section_htab;

/* Pointer to linked list of sections.  */
struct bfd_section *sections;

/* The last section on the section list.  */
struct bfd_section *section_last;

/* The number of sections.  */
unsigned int section_count;

/* The archive plugin file descriptor.  */
int archive_plugin_fd;

/* The number of opens on the archive plugin file descriptor.  */
unsigned int archive_plugin_fd_open_count;

/* A field used by _bfd_generic_link_add_archive_symbols.  This will
   be used only for archive elements.  */
int archive_pass;

/* The total size of memory from bfd_alloc.  */
bfd_size_type alloc_size;

/* Stuff only useful for object files:
   The start address.  */
bfd_vma start_address;

/* Symbol table for output BFD (with symcount entries).
   Also used by the linker to cache input BFD symbols.  */
struct bfd_symbol **outsymbols;

/* Used for input and output.  */
unsigned int symcount;

/* Used for slurped dynamic symbol tables.  */
unsigned int dynsymcount;

/* Pointer to structure which contains architecture information.  */
const struct bfd_arch_info *arch_info;

/* Cached length of file for bfd_get_size.  0 until bfd_get_size is
   called, 1 if stat returns an error or the file size is too large to
```

```
   return in ufile_ptr.  Both 0 and 1 should be treated as "unknown".  */
ufile_ptr size;

/* Stuff only useful for archives.  */
void *arelt_data;
struct bfd *my_archive;      /* The containing archive BFD.  */
struct bfd *archive_next;    /* The next BFD in the archive.  */
struct bfd *archive_head;    /* The first BFD in the archive.  */
struct bfd *nested_archives; /* List of nested archive in a flattened
                                thin archive.  */

union {
  /* For input BFDs, a chain of BFDs involved in a link.  */
  struct bfd *next;
  /* For output BFD, the linker hash table.  */
  struct bfd_link_hash_table *hash;
} link;

/* Used by the back end to hold private data.  */
union
  {
    struct aout_data_struct *aout_data;
    struct artdata *aout_ar_data;
    struct coff_tdata *coff_obj_data;
    struct pe_tdata *pe_obj_data;
    struct xcoff_tdata *xcoff_obj_data;
    struct ecoff_tdata *ecoff_obj_data;
    struct srec_data_struct *srec_data;
    struct verilog_data_struct *verilog_data;
    struct ihex_data_struct *ihex_data;
    struct tekhex_data_struct *tekhex_data;
    struct elf_obj_tdata *elf_obj_data;
    struct mmo_data_struct *mmo_data;
    struct sun_core_struct *sun_core_data;
    struct sco5_core_struct *sco5_core_data;
    struct trad_core_struct *trad_core_data;
    struct som_data_struct *som_data;
    struct hpux_core_struct *hpux_core_data;
    struct hppabsd_core_struct *hppabsd_core_data;
    struct sgi_core_struct *sgi_core_data;
    struct lynx_core_struct *lynx_core_data;
    struct osf_core_struct *osf_core_data;
    struct cisco_core_struct *cisco_core_data;
    struct versados_data_struct *versados_data;
    struct netbsd_core_struct *netbsd_core_data;
    struct mach_o_data_struct *mach_o_data;
    struct mach_o_fat_data_struct *mach_o_fat_data;
```

```
      struct plugin_data_struct *plugin_data;
      struct bfd_pef_data_struct *pef_data;
      struct bfd_pef_xlib_data_struct *pef_xlib_data;
      struct bfd_sym_data_struct *sym_data;
      void *any;
    }
  tdata;

  /* Used by the application to hold private data.  */
  void *usrdata;

  /* Where all the allocated stuff under this BFD goes.  This is a
     struct objalloc *, but we use void * to avoid requiring the inclusion
     of objalloc.h.  */
  void *memory;

  /* For input BFDs, the build ID, if the object has one. */
  const struct bfd_build_id *build_id;
};

static inline const char *
bfd_get_filename (const bfd *abfd)
{
  return abfd->filename;
}

static inline bool
bfd_get_cacheable (const bfd *abfd)
{
  return abfd->cacheable;
}

static inline enum bfd_format
bfd_get_format (const bfd *abfd)
{
  return abfd->format;
}

static inline flagword
bfd_get_file_flags (const bfd *abfd)
{
  return abfd->flags;
}

static inline bfd_vma
bfd_get_start_address (const bfd *abfd)
{
```

```
  return abfd->start_address;
}

static inline unsigned int
bfd_get_symcount (const bfd *abfd)
{
  return abfd->symcount;
}

static inline unsigned int
bfd_get_dynamic_symcount (const bfd *abfd)
{
  return abfd->dynsymcount;
}

static inline struct bfd_symbol **
bfd_get_outsymbols (const bfd *abfd)
{
  return abfd->outsymbols;
}

static inline unsigned int
bfd_count_sections (const bfd *abfd)
{
  return abfd->section_count;
}

static inline bool
bfd_has_map (const bfd *abfd)
{
  return abfd->has_armap;
}

static inline bool
bfd_is_thin_archive (const bfd *abfd)
{
  return abfd->is_thin_archive;
}

static inline void *
bfd_usrdata (const bfd *abfd)
{
  return abfd->usrdata;
}

/* See note beside bfd_set_section_userdata.  */
static inline bool
```

```
bfd_set_cacheable (bfd * abfd, bool val)
{
  abfd->cacheable = val;
  return true;
}

static inline void
bfd_set_thin_archive (bfd *abfd, bool val)
{
  abfd->is_thin_archive = val;
}

static inline void
bfd_set_usrdata (bfd *abfd, void *val)
{
  abfd->usrdata = val;
}

static inline asection *
bfd_asymbol_section (const asymbol *sy)
{
  return sy->section;
}

static inline bfd_vma
bfd_asymbol_value (const asymbol *sy)
{
  return sy->section->vma + sy->value;
}

static inline const char *
bfd_asymbol_name (const asymbol *sy)
{
  return sy->name;
}

static inline struct bfd *
bfd_asymbol_bfd (const asymbol *sy)
{
  return sy->the_bfd;
}

static inline void
bfd_set_asymbol_name (asymbol *sy, const char *name)
{
  sy->name = name;
}
```

```
static inline bfd_size_type
bfd_get_section_limit_octets (const bfd *abfd, const asection *sec)
{
  if (abfd->direction != write_direction && sec->rawsize != 0)
    return sec->rawsize;
  return sec->size;
}

/* Find the address one past the end of SEC.  */
static inline bfd_size_type
bfd_get_section_limit (const bfd *abfd, const asection *sec)
{
  return (bfd_get_section_limit_octets (abfd, sec)
          / bfd_octets_per_byte (abfd, sec));
}

/* Functions to handle insertion and deletion of a bfd's sections.  These
   only handle the list pointers, ie. do not adjust section_count,
   target_index etc.  */
static inline void
bfd_section_list_remove (bfd *abfd, asection *s)
{
  asection *next = s->next;
  asection *prev = s->prev;
  if (prev)
    prev->next = next;
  else
    abfd->sections = next;
  if (next)
    next->prev = prev;
  else
    abfd->section_last = prev;
}

static inline void
bfd_section_list_append (bfd *abfd, asection *s)
{
  s->next = 0;
  if (abfd->section_last)
    {
      s->prev = abfd->section_last;
      abfd->section_last->next = s;
    }
  else
    {
      s->prev = 0;
```

```
      abfd->sections = s;
    }
  abfd->section_last = s;
}

static inline void
bfd_section_list_prepend (bfd *abfd, asection *s)
{
  s->prev = 0;
  if (abfd->sections)
    {
      s->next = abfd->sections;
      abfd->sections->prev = s;
    }
  else
    {
      s->next = 0;
      abfd->section_last = s;
    }
  abfd->sections = s;
}

static inline void
bfd_section_list_insert_after (bfd *abfd, asection *a, asection *s)
{
  asection *next = a->next;
  s->next = next;
  s->prev = a;
  a->next = s;
  if (next)
    next->prev = s;
  else
    abfd->section_last = s;
}

static inline void
bfd_section_list_insert_before (bfd *abfd, asection *b, asection *s)
{
  asection *prev = b->prev;
  s->prev = prev;
  s->next = b;
  b->prev = s;
  if (prev)
    prev->next = s;
  else
    abfd->sections = s;
}
```

```
static inline bool
bfd_section_removed_from_list (const bfd *abfd, const asection *s)
{
  return s->next ? s->next->prev != s : abfd->section_last != s;
}
```

2.2 Error reporting

Most BFD functions return nonzero on success (check their individual documentation for precise semantics). On an error, they call `bfd_set_error` to set an error condition that callers can check by calling `bfd_get_error`. If that returns `bfd_error_system_call`, then check `errno`.

The easiest way to report a BFD error to the user is to use `bfd_perror`.

2.2.1 Type `bfd_error_type`

The values returned by `bfd_get_error` are defined by the enumerated type `bfd_error_type`.

```
typedef enum bfd_error
{
  bfd_error_no_error = 0,
  bfd_error_system_call,
  bfd_error_invalid_target,
  bfd_error_wrong_format,
  bfd_error_wrong_object_format,
  bfd_error_invalid_operation,
  bfd_error_no_memory,
  bfd_error_no_symbols,
  bfd_error_no_armap,
  bfd_error_no_more_archived_files,
  bfd_error_malformed_archive,
  bfd_error_missing_dso,
  bfd_error_file_not_recognized,
  bfd_error_file_ambiguously_recognized,
  bfd_error_no_contents,
  bfd_error_nonrepresentable_section,
  bfd_error_no_debug_section,
  bfd_error_bad_value,
  bfd_error_file_truncated,
  bfd_error_file_too_big,
  bfd_error_sorry,
  bfd_error_on_input,
  bfd_error_invalid_error_code
}
bfd_error_type;
```

2.2.1.1 bfd_get_error

Synopsis

```
bfd_error_type bfd_get_error (void);
```

Description

Return the current BFD error condition.

2.2.1.2 bfd_set_error

Synopsis

```
void bfd_set_error (bfd_error_type error_tag);
```

Description

Set the BFD error condition to be *error_tag*.

error_tag must not be bfd_error_on_input. Use bfd_set_input_error for input errors instead.

2.2.1.3 bfd_set_input_error

Synopsis

```
void bfd_set_input_error (bfd *input, bfd_error_type error_tag);
```

Description

Set the BFD error condition to be bfd_error_on_input. *input* is the input bfd where the error occurred, and *error_tag* the bfd_error_type error.

2.2.1.4 bfd_errmsg

Synopsis

```
const char *bfd_errmsg (bfd_error_type error_tag);
```

Description

Return a string describing the error *error_tag*, or the system error if *error_tag* is `bfd_error_system_call`.

2.2.1.5 bfd_perror

Synopsis

```
void bfd_perror (const char *message);
```

Description

Print to the standard error stream a string describing the last BFD error that occurred, or the last system error if the last BFD error was a system call failure. If *message* is non-NULL and non-empty, the error string printed is preceded by *message*, a colon, and a space. It is followed by a newline.

2.2.2 BFD error handler

Some BFD functions want to print messages describing the problem. They call a BFD error handler function. This function may be overridden by the program.

The BFD error handler acts like vprintf.

```
typedef void (*bfd_error_handler_type) (const char *, va_list);
```

2.2.2.1 _bfd_error_handler

Synopsis

```
void _bfd_error_handler (const char *fmt, ...) ATTRIBUTE_PRINTF_1;
```

Description

This is the default routine to handle BFD error messages. Like fprintf (stderr, ...), but also handles some extra format specifiers.

%pA section name from section. For group components, prints group name too. %pB file name from bfd. For archive components, prints archive too.

Beware: Only supports a maximum of 9 format arguments.

2.2.2.2 bfd_set_error_handler

Synopsis

```
bfd_error_handler_type bfd_set_error_handler (bfd_error_handler_type);
```

Description

Set the BFD error handler function. Returns the previous function.

2.2.2.3 bfd_set_error_program_name

Synopsis

```
void bfd_set_error_program_name (const char *);
```

Description

Set the program name to use when printing a BFD error. This is printed before the error message followed by a colon and space. The string must not be changed after it is passed to this function.

2.2.3 BFD assert handler

If BFD finds an internal inconsistency, the bfd assert handler is called with information on the BFD version, BFD source file and line. If this happens, most programs linked against BFD are expected to want to exit with an error, or mark the current BFD operation as failed, so it is recommended to override the default handler, which just calls _bfd_error_handler and continues.

```
typedef void (*bfd_assert_handler_type) (const char *bfd_formatmsg,
                                         const char *bfd_version,
                                         const char *bfd_file,
                                         int bfd_line);
```

2.2.3.1 bfd_set_assert_handler

Synopsis

```
bfd_assert_handler_type bfd_set_assert_handler (bfd_assert_handler_type);
```

Description

Set the BFD assert handler function. Returns the previous function.

2.3 Miscellaneous

2.3.1 Miscellaneous functions

2.3.1.1 bfd_get_reloc_upper_bound

Synopsis

```
long bfd_get_reloc_upper_bound (bfd *abfd, asection *sect);
```

Description

Return the number of bytes required to store the relocation information associated with section *sect* attached to bfd *abfd*. If an error occurs, return -1.

2.3.1.2 bfd_canonicalize_reloc

Synopsis

```
long bfd_canonicalize_reloc
   (bfd *abfd, asection *sec, arelent **loc, asymbol **syms);
```

Description

Call the back end associated with the open BFD *abfd* and translate the external form of the relocation information attached to *sec* into the internal canonical form. Place the table into memory at *loc*, which has been preallocated, usually by a call to `bfd_get_reloc_upper_bound`. Returns the number of relocs, or -1 on error.

The *syms* table is also needed for horrible internal magic reasons.

2.3.1.3 bfd_set_reloc

Synopsis

```
void bfd_set_reloc
   (bfd *abfd, asection *sec, arelent **rel, unsigned int count);
```

Description

Set the relocation pointer and count within section *sec* to the values *rel* and *count*. The argument *abfd* is ignored.

```
#define bfd_set_reloc(abfd, asect, location, count) \
       BFD_SEND (abfd, _bfd_set_reloc, (abfd, asect, location, count))
```

2.3.1.4 bfd_set_file_flags

Synopsis

```
bool bfd_set_file_flags (bfd *abfd, flagword flags);
```

Description

Set the flag word in the BFD *abfd* to the value *flags*.

Possible errors are:

- `bfd_error_wrong_format` - The target bfd was not of object format.
- `bfd_error_invalid_operation` - The target bfd was open for reading.
- `bfd_error_invalid_operation` - The flag word contained a bit which was not applicable to the type of file. E.g., an attempt was made to set the `D_PAGED` bit on a BFD format which does not support demand paging.

2.3.1.5 bfd_get_arch_size

Synopsis

```
int bfd_get_arch_size (bfd *abfd);
```

Description

Returns the normalized architecture address size, in bits, as determined by the object file's format. By normalized, we mean either 32 or 64. For ELF, this information is included in the header. Use bfd_arch_bits_per_address for number of bits in the architecture address.

Returns

Returns the arch size in bits if known, `-1` otherwise.

2.3.1.6 bfd_get_sign_extend_vma

Synopsis

```
int bfd_get_sign_extend_vma (bfd *abfd);
```

Description

Indicates if the target architecture "naturally" sign extends an address. Some architectures implicitly sign extend address values when they are converted to types larger than the size of an address. For instance, bfd_get_start_address() will return an address sign extended to fill a bfd_vma when this is the case.

Returns

Returns `1` if the target architecture is known to sign extend addresses, `0` if the target architecture is known to not sign extend addresses, and `-1` otherwise.

2.3.1.7 bfd_set_start_address

Synopsis

```
bool bfd_set_start_address (bfd *abfd, bfd_vma vma);
```

Description

Make *vma* the entry point of output BFD *abfd*.

Returns

Returns `TRUE` on success, `FALSE` otherwise.

2.3.1.8 bfd_get_gp_size

Synopsis

```
unsigned int bfd_get_gp_size (bfd *abfd);
```

Description

Return the maximum size of objects to be optimized using the GP register under MIPS ECOFF. This is typically set by the `-G` argument to the compiler, assembler or linker.

2.3.1.9 bfd_set_gp_size

Synopsis

```
void bfd_set_gp_size (bfd *abfd, unsigned int i);
```

Description

Set the maximum size of objects to be optimized using the GP register under ECOFF or MIPS ELF. This is typically set by the `-G` argument to the compiler, assembler or linker.

2.3.1.10 bfd_set_gp_value

Synopsis

```
void bfd_set_gp_value (bfd *abfd, bfd_vma v);
```

Description
Allow external access to the fucntion to set the GP value. This is specifically added for gdb-compile support.

2.3.1.11 bfd_scan_vma

Synopsis

```
bfd_vma bfd_scan_vma (const char *string, const char **end, int base);
```

Description
Convert, like `strtoul`, a numerical expression *string* into a `bfd_vma` integer, and return that integer. (Though without as many bells and whistles as `strtoul`.) The expression is assumed to be unsigned (i.e., positive). If given a *base*, it is used as the base for conversion. A base of 0 causes the function to interpret the string in hex if a leading "0x" or "0X" is found, otherwise in octal if a leading zero is found, otherwise in decimal.

If the value would overflow, the maximum `bfd_vma` value is returned.

2.3.1.12 bfd_copy_private_header_data

Synopsis

```
bool bfd_copy_private_header_data (bfd *ibfd, bfd *obfd);
```

Description
Copy private BFD header information from the BFD *ibfd* to the the BFD *obfd*. This copies information that may require sections to exist, but does not require symbol tables. Return `true` on success, `false` on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *obfd*.

```
#define bfd_copy_private_header_data(ibfd, obfd) \
       BFD_SEND (obfd, _bfd_copy_private_header_data, \
                 (ibfd, obfd))
```

2.3.1.13 bfd_copy_private_bfd_data

Synopsis

```
bool bfd_copy_private_bfd_data (bfd *ibfd, bfd *obfd);
```

Description
Copy private BFD information from the BFD *ibfd* to the the BFD *obfd*. Return `TRUE` on success, `FALSE` on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *obfd*.

```
#define bfd_copy_private_bfd_data(ibfd, obfd) \
       BFD_SEND (obfd, _bfd_copy_private_bfd_data, \
                 (ibfd, obfd))
```

2.3.1.14 bfd_set_private_flags

Synopsis

```
bool bfd_set_private_flags (bfd *abfd, flagword flags);
```

Description

Set private BFD flag information in the BFD *abfd*. Return `TRUE` on success, `FALSE` on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *obfd*.

```
#define bfd_set_private_flags(abfd, flags) \
       BFD_SEND (abfd, _bfd_set_private_flags, (abfd, flags))
```

2.3.1.15 Other functions

Description

The following functions exist but have not yet been documented.

```
#define bfd_sizeof_headers(abfd, info) \
       BFD_SEND (abfd, _bfd_sizeof_headers, (abfd, info))

#define bfd_find_nearest_line(abfd, sec, syms, off, file, func, line) \
       BFD_SEND (abfd, _bfd_find_nearest_line, \
                 (abfd, syms, sec, off, file, func, line, NULL))

#define bfd_find_nearest_line_discriminator(abfd, sec, syms, off, file, func, \
                                            line, disc) \
       BFD_SEND (abfd, _bfd_find_nearest_line, \
                 (abfd, syms, sec, off, file, func, line, disc))

#define bfd_find_line(abfd, syms, sym, file, line) \
       BFD_SEND (abfd, _bfd_find_line, \
                 (abfd, syms, sym, file, line))

#define bfd_find_inliner_info(abfd, file, func, line) \
       BFD_SEND (abfd, _bfd_find_inliner_info, \
                 (abfd, file, func, line))

#define bfd_debug_info_start(abfd) \
       BFD_SEND (abfd, _bfd_debug_info_start, (abfd))

#define bfd_debug_info_end(abfd) \
       BFD_SEND (abfd, _bfd_debug_info_end, (abfd))

#define bfd_debug_info_accumulate(abfd, section) \
       BFD_SEND (abfd, _bfd_debug_info_accumulate, (abfd, section))

#define bfd_stat_arch_elt(abfd, stat) \
       BFD_SEND (abfd->my_archive ? abfd->my_archive : abfd, \
                 _bfd_stat_arch_elt, (abfd, stat))
```

```
#define bfd_update_armap_timestamp(abfd) \
       BFD_SEND (abfd, _bfd_update_armap_timestamp, (abfd))

#define bfd_set_arch_mach(abfd, arch, mach)\
       BFD_SEND ( abfd, _bfd_set_arch_mach, (abfd, arch, mach))

#define bfd_relax_section(abfd, section, link_info, again) \
       BFD_SEND (abfd, _bfd_relax_section, (abfd, section, link_info, again))

#define bfd_gc_sections(abfd, link_info) \
       BFD_SEND (abfd, _bfd_gc_sections, (abfd, link_info))

#define bfd_lookup_section_flags(link_info, flag_info, section) \
       BFD_SEND (abfd, _bfd_lookup_section_flags, (link_info, flag_info, section))

#define bfd_merge_sections(abfd, link_info) \
       BFD_SEND (abfd, _bfd_merge_sections, (abfd, link_info))

#define bfd_is_group_section(abfd, sec) \
       BFD_SEND (abfd, _bfd_is_group_section, (abfd, sec))

#define bfd_group_name(abfd, sec) \
       BFD_SEND (abfd, _bfd_group_name, (abfd, sec))

#define bfd_discard_group(abfd, sec) \
       BFD_SEND (abfd, _bfd_discard_group, (abfd, sec))

#define bfd_link_hash_table_create(abfd) \
       BFD_SEND (abfd, _bfd_link_hash_table_create, (abfd))

#define bfd_link_add_symbols(abfd, info) \
       BFD_SEND (abfd, _bfd_link_add_symbols, (abfd, info))

#define bfd_link_just_syms(abfd, sec, info) \
       BFD_SEND (abfd, _bfd_link_just_syms, (sec, info))

#define bfd_final_link(abfd, info) \
       BFD_SEND (abfd, _bfd_final_link, (abfd, info))

#define bfd_free_cached_info(abfd) \
       BFD_SEND (abfd, _bfd_free_cached_info, (abfd))

#define bfd_get_dynamic_symtab_upper_bound(abfd) \
       BFD_SEND (abfd, _bfd_get_dynamic_symtab_upper_bound, (abfd))

#define bfd_print_private_bfd_data(abfd, file)\
```

```
        BFD_SEND (abfd, _bfd_print_private_bfd_data, (abfd, file))

#define bfd_canonicalize_dynamic_symtab(abfd, asymbols) \
        BFD_SEND (abfd, _bfd_canonicalize_dynamic_symtab, (abfd, asymbols))

#define bfd_get_synthetic_symtab(abfd, count, syms, dyncount, dynsyms, ret) \
        BFD_SEND (abfd, _bfd_get_synthetic_symtab, (abfd, count, syms, \
                                                    dyncount, dynsyms, ret))

#define bfd_get_dynamic_reloc_upper_bound(abfd) \
        BFD_SEND (abfd, _bfd_get_dynamic_reloc_upper_bound, (abfd))

#define bfd_canonicalize_dynamic_reloc(abfd, arels, asyms) \
        BFD_SEND (abfd, _bfd_canonicalize_dynamic_reloc, (abfd, arels, asyms))

extern bfd_byte *bfd_get_relocated_section_contents
  (bfd *, struct bfd_link_info *, struct bfd_link_order *, bfd_byte *,
   bool, asymbol **);
```

2.3.1.16 bfd_alt_mach_code

Synopsis

```
bool bfd_alt_mach_code (bfd *abfd, int alternative);
```

Description

When more than one machine code number is available for the same machine type, this function can be used to switch between the preferred one (alternative == 0) and any others. Currently, only ELF supports this feature, with up to two alternate machine codes.

2.3.1.17 bfd_emul_get_maxpagesize

Synopsis

```
bfd_vma bfd_emul_get_maxpagesize (const char *);
```

Description

Returns the maximum page size, in bytes, as determined by emulation.

Returns

Returns the maximum page size in bytes for ELF, 0 otherwise.

2.3.1.18 bfd_emul_get_commonpagesize

Synopsis

```
bfd_vma bfd_emul_get_commonpagesize (const char *, bool);
```

Description

Returns the common page size, in bytes, as determined by emulation.

Returns

Returns the common page size in bytes for ELF, 0 otherwise.

2.3.1.19 bfd_demangle

Synopsis

```
char *bfd_demangle (bfd *, const char *, int);
```

Description
Wrapper around cplus_demangle. Strips leading underscores and other such chars that would otherwise confuse the demangler. If passed a g++ v3 ABI mangled name, returns a buffer allocated with malloc holding the demangled name. Returns NULL otherwise and on memory alloc failure.

2.3.1.20 bfd_update_compression_header

Synopsis

```
void bfd_update_compression_header
   (bfd *abfd, bfd_byte *contents, asection *sec);
```

Description
Set the compression header at CONTENTS of SEC in ABFD and update elf_section_flags for compression.

2.3.1.21 bfd_check_compression_header

Synopsis

```
bool bfd_check_compression_header
   (bfd *abfd, bfd_byte *contents, asection *sec,
    bfd_size_type *uncompressed_size,
    unsigned int *uncompressed_alignment_power);
```

Description
Check the compression header at CONTENTS of SEC in ABFD and store the uncompressed size in UNCOMPRESSED_SIZE and the uncompressed data alignment in UNCOMPRESSED_ALIGNMENT_POWER if the compression header is valid.

Returns
Return TRUE if the compression header is valid.

2.3.1.22 bfd_get_compression_header_size

Synopsis

```
int bfd_get_compression_header_size (bfd *abfd, asection *sec);
```

Description
Return the size of the compression header of SEC in ABFD.

Returns
Return the size of the compression header in bytes.

2.3.1.23 bfd_convert_section_size

Synopsis

```
bfd_size_type bfd_convert_section_size
   (bfd *ibfd, asection *isec, bfd *obfd, bfd_size_type size);
```

Description
Convert the size *size* of the section *isec* in input BFD *ibfd* to the section size in output BFD *obfd*.

2.3.1.24 bfd_convert_section_contents

Synopsis

```
bool bfd_convert_section_contents
   (bfd *ibfd, asection *isec, bfd *obfd,
    bfd_byte **ptr, bfd_size_type *ptr_size);
```

Description
Convert the contents, stored in **ptr*, of the section *isec* in input BFD *ibfd* to output BFD *obfd* if needed. The original buffer pointed to by **ptr* may be freed and **ptr* is returned with memory malloc'd by this function, and the new size written to *ptr_size*.

2.3.1.25 struct bfd_iovec

Description
The `struct bfd_iovec` contains the internal file I/O class. Each `BFD` has an instance of this class and all file I/O is routed through it (it is assumed that the instance implements all methods listed below).

```
struct bfd_iovec
{
  /* To avoid problems with macros, a "b" rather than "f"
     prefix is prepended to each method name.  */
  /* Attempt to read/write NBYTES on ABFD's IOSTREAM storing/fetching
     bytes starting at PTR.  Return the number of bytes actually
     transfered (a read past end-of-file returns less than NBYTES),
     or -1 (setting bfd_error) if an error occurs.  */
  file_ptr (*bread) (struct bfd *abfd, void *ptr, file_ptr nbytes);
  file_ptr (*bwrite) (struct bfd *abfd, const void *ptr,
                      file_ptr nbytes);
  /* Return the current IOSTREAM file offset, or -1 (setting bfd_error
     if an error occurs.  */
  file_ptr (*btell) (struct bfd *abfd);
  /* For the following, on successful completion a value of 0 is returned.
     Otherwise, a value of -1 is returned (and bfd_error is set).  */
  int (*bseek) (struct bfd *abfd, file_ptr offset, int whence);
  int (*bclose) (struct bfd *abfd);
  int (*bflush) (struct bfd *abfd);
  int (*bstat) (struct bfd *abfd, struct stat *sb);
  /* Mmap a part of the files. ADDR, LEN, PROT, FLAGS and OFFSET are the usual
     mmap parameter, except that LEN and OFFSET do not need to be page
     aligned.  Returns (void *)-1 on failure, mmapped address on success.
     Also write in MAP_ADDR the address of the page aligned buffer and in
     MAP_LEN the size mapped (a page multiple).  Use unmap with MAP_ADDR and
     MAP_LEN to unmap.  */
  void *(*bmmap) (struct bfd *abfd, void *addr, bfd_size_type len,
```

```
                    int prot, int flags, file_ptr offset,
                    void **map_addr, bfd_size_type *map_len);
};
extern const struct bfd_iovec _bfd_memory_iovec;
```

2.3.1.26 bfd_get_mtime

Synopsis

```
long bfd_get_mtime (bfd *abfd);
```

Description

Return the file modification time (as read from the file system, or from the archive header for archive members).

2.3.1.27 bfd_get_size

Synopsis

```
ufile_ptr bfd_get_size (bfd *abfd);
```

Description

Return the file size (as read from file system) for the file associated with BFD *abfd*.

The initial motivation for, and use of, this routine is not so we can get the exact size of the object the BFD applies to, since that might not be generally possible (archive members for example). It would be ideal if someone could eventually modify it so that such results were guaranteed.

Instead, we want to ask questions like "is this NNN byte sized object I'm about to try read from file offset YYY reasonable?" As as example of where we might do this, some object formats use string tables for which the first `sizeof (long)` bytes of the table contain the size of the table itself, including the size bytes. If an application tries to read what it thinks is one of these string tables, without some way to validate the size, and for some reason the size is wrong (byte swapping error, wrong location for the string table, etc.), the only clue is likely to be a read error when it tries to read the table, or a "virtual memory exhausted" error when it tries to allocate 15 bazillon bytes of space for the 15 bazillon byte table it is about to read. This function at least allows us to answer the question, "is the size reasonable?".

A return value of zero indicates the file size is unknown.

2.3.1.28 bfd_get_file_size

Synopsis

```
ufile_ptr bfd_get_file_size (bfd *abfd);
```

Description

Return the file size (as read from file system) for the file associated with BFD *abfd*. It supports both normal files and archive elements.

2.3.1.29 bfd_mmap

Synopsis

```
void *bfd_mmap (bfd *abfd, void *addr, bfd_size_type len,
    int prot, int flags, file_ptr offset,
```

```
void **map_addr, bfd_size_type *map_len);
```

Description
Return mmap()ed region of the file, if possible and implemented. LEN and OFFSET do not need to be page aligned. The page aligned address and length are written to MAP_ADDR and MAP_LEN.

2.4 Memory Usage

BFD keeps all of its internal structures in obstacks. There is one obstack per open BFD file, into which the current state is stored. When a BFD is closed, the obstack is deleted, and so everything which has been allocated by BFD for the closing file is thrown away.

BFD does not free anything created by an application, but pointers into `bfd` structures become invalid on a `bfd_close`; for example, after a `bfd_close` the vector passed to `bfd_canonicalize_symtab` is still around, since it has been allocated by the application, but the data that it pointed to are lost.

The general rule is to not close a BFD until all operations dependent upon data from the BFD have been completed, or all the data from within the file has been copied. To help with the management of memory, there is a function (`bfd_alloc_size`) which returns the number of bytes in obstacks associated with the supplied BFD. This could be used to select the greediest open BFD, close it to reclaim the memory, perform some operation and reopen the BFD again, to get a fresh copy of the data structures.

2.5 Initialization

2.5.1 Initialization functions

These are the functions that handle initializing a BFD.

2.5.1.1 `bfd_init`

Synopsis

```
unsigned int bfd_init (void);
```

Description
This routine must be called before any other BFD function to initialize magical internal data structures. Returns a magic number, which may be used to check that the bfd library is configured as expected by users.

```
/* Value returned by bfd_init.  */

#define BFD_INIT_MAGIC (sizeof (struct bfd_section))
```

2.6 Sections

The raw data contained within a BFD is maintained through the section abstraction. A single BFD may have any number of sections. It keeps hold of them by pointing to the first; each one points to the next in the list.

Sections are supported in BFD in `section.c`.

2.6.1 Section input

When a BFD is opened for reading, the section structures are created and attached to the BFD.

Each section has a name which describes the section in the outside world—for example, `a.out` would contain at least three sections, called `.text`, `.data` and `.bss`.

Names need not be unique; for example a COFF file may have several sections named `.data`.

Sometimes a BFD will contain more than the "natural" number of sections. A back end may attach other sections containing constructor data, or an application may add a section (using `bfd_make_section`) to the sections attached to an already open BFD. For example, the linker creates an extra section `COMMON` for each input file's BFD to hold information about common storage.

The raw data is not necessarily read in when the section descriptor is created. Some targets may leave the data in place until a `bfd_get_section_contents` call is made. Other back ends may read in all the data at once. For example, an S-record file has to be read once to determine the size of the data.

2.6.2 Section output

To write a new object style BFD, the various sections to be written have to be created. They are attached to the BFD in the same way as input sections; data is written to the sections using `bfd_set_section_contents`.

Any program that creates or combines sections (e.g., the assembler and linker) must use the `asection` fields `output_section` and `output_offset` to indicate the file sections to which each section must be written. (If the section is being created from scratch, `output_section` should probably point to the section itself and `output_offset` should probably be zero.)

The data to be written comes from input sections attached (via `output_section` pointers) to the output sections. The output section structure can be considered a filter for the input section: the output section determines the vma of the output data and the name, but the input section determines the offset into the output section of the data to be written.

E.g., to create a section `"O"`, starting at 0x100, 0x123 long, containing two subsections, `"A"` at offset 0x0 (i.e., at vma 0x100) and `"B"` at offset 0x20 (i.e., at vma 0x120) the `asection` structures would look like:

```
section name          "A"
  output_offset   0x00
  size            0x20
  output_section -----------> section name    "O"
                           |  vma             0x100
section name          "B" |   size            0x123
  output_offset   0x20    |
  size            0x103   |
  output_section  --------|
```

2.6.3 Link orders

The data within a section is stored in a *link_order*. These are much like the fixups in `gas`. The link_order abstraction allows a section to grow and shrink within itself.

A link_order knows how big it is, and which is the next link_order and where the raw data for it is; it also points to a list of relocations which apply to it.

The link_order is used by the linker to perform relaxing on final code. The compiler creates code which is as big as necessary to make it work without relaxing, and the user can select whether to relax. Sometimes relaxing takes a lot of time. The linker runs around the relocations to see if any are attached to data which can be shrunk, if so it does it on a link_order by link_order basis.

2.6.4 typedef asection

Here is the section structure:

```
typedef struct bfd_section
{
  /* The name of the section; the name isn't a copy, the pointer is
     the same as that passed to bfd_make_section.  */
  const char *name;

  /* A unique sequence number.  */
  unsigned int id;

  /* A unique section number which can be used by assembler to
     distinguish different sections with the same section name.  */
  unsigned int section_id;

  /* Which section in the bfd; 0..n-1 as sections are created in a bfd.  */
  unsigned int index;

  /* The next section in the list belonging to the BFD, or NULL.  */
  struct bfd_section *next;

  /* The previous section in the list belonging to the BFD, or NULL.  */
  struct bfd_section *prev;

  /* The field flags contains attributes of the section. Some
     flags are read in from the object file, and some are
     synthesized from other information.  */
  flagword flags;

#define SEC_NO_FLAGS                    0x0

  /* Tells the OS to allocate space for this section when loading.
     This is clear for a section containing debug information only.  */
#define SEC_ALLOC                       0x1

  /* Tells the OS to load the section from the file when loading.
     This is clear for a .bss section.  */
```

```
#define SEC_LOAD                          0x2

  /* The section contains data still to be relocated, so there is
     some relocation information too.  */
#define SEC_RELOC                         0x4

  /* A signal to the OS that the section contains read only data.  */
#define SEC_READONLY                      0x8

  /* The section contains code only.  */
#define SEC_CODE                         0x10

  /* The section contains data only.  */
#define SEC_DATA                         0x20

  /* The section will reside in ROM.  */
#define SEC_ROM                          0x40

  /* The section contains constructor information. This section
     type is used by the linker to create lists of constructors and
     destructors used by g++. When a back end sees a symbol
     which should be used in a constructor list, it creates a new
     section for the type of name (e.g., __CTOR_LIST__), attaches
     the symbol to it, and builds a relocation. To build the lists
     of constructors, all the linker has to do is catenate all the
     sections called __CTOR_LIST__ and relocate the data
     contained within - exactly the operations it would peform on
     standard data.  */
#define SEC_CONSTRUCTOR                  0x80

  /* The section has contents - a data section could be
     SEC_ALLOC | SEC_HAS_CONTENTS; a debug section could be
     SEC_HAS_CONTENTS  */
#define SEC_HAS_CONTENTS                0x100

  /* An instruction to the linker to not output the section
     even if it has information which would normally be written.  */
#define SEC_NEVER_LOAD                  0x200

  /* The section contains thread local data.  */
#define SEC_THREAD_LOCAL                0x400

  /* The section's size is fixed.  Generic linker code will not
     recalculate it and it is up to whoever has set this flag to
     get the size right.  */
#define SEC_FIXED_SIZE                  0x800
```

```
  /* The section contains common symbols (symbols may be defined
     multiple times, the value of a symbol is the amount of
     space it requires, and the largest symbol value is the one
     used).  Most targets have exactly one of these (which we
     translate to bfd_com_section_ptr), but ECOFF has two.  */
#define SEC_IS_COMMON                   0x1000

  /* The section contains only debugging information.  For
     example, this is set for ELF .debug and .stab sections.
     strip tests this flag to see if a section can be
     discarded.  */
#define SEC_DEBUGGING                   0x2000

  /* The contents of this section are held in memory pointed to
     by the contents field.  This is checked by bfd_get_section_contents,
     and the data is retrieved from memory if appropriate.  */
#define SEC_IN_MEMORY                   0x4000

  /* The contents of this section are to be excluded by the
     linker for executable and shared objects unless those
     objects are to be further relocated.  */
#define SEC_EXCLUDE                     0x8000

  /* The contents of this section are to be sorted based on the sum of
     the symbol and addend values specified by the associated relocation
     entries.  Entries without associated relocation entries will be
     appended to the end of the section in an unspecified order.  */
#define SEC_SORT_ENTRIES               0x10000

  /* When linking, duplicate sections of the same name should be
     discarded, rather than being combined into a single section as
     is usually done.  This is similar to how common symbols are
     handled.  See SEC_LINK_DUPLICATES below.  */
#define SEC_LINK_ONCE                  0x20000

  /* If SEC_LINK_ONCE is set, this bitfield describes how the linker
     should handle duplicate sections.  */
#define SEC_LINK_DUPLICATES            0xc0000

  /* This value for SEC_LINK_DUPLICATES means that duplicate
     sections with the same name should simply be discarded.  */
#define SEC_LINK_DUPLICATES_DISCARD       0x0

  /* This value for SEC_LINK_DUPLICATES means that the linker
     should warn if there are any duplicate sections, although
     it should still only link one copy.  */
#define SEC_LINK_DUPLICATES_ONE_ONLY  0x40000
```

```
  /* This value for SEC_LINK_DUPLICATES means that the linker
     should warn if any duplicate sections are a different size.  */
#define SEC_LINK_DUPLICATES_SAME_SIZE 0x80000

  /* This value for SEC_LINK_DUPLICATES means that the linker
     should warn if any duplicate sections contain different
     contents.  */
#define SEC_LINK_DUPLICATES_SAME_CONTENTS \
  (SEC_LINK_DUPLICATES_ONE_ONLY | SEC_LINK_DUPLICATES_SAME_SIZE)

  /* This section was created by the linker as part of dynamic
     relocation or other arcane processing.  It is skipped when
     going through the first-pass output, trusting that someone
     else up the line will take care of it later.  */
#define SEC_LINKER_CREATED          0x100000

  /* This section contains a section ID to distinguish different
     sections with the same section name.  */
#define SEC_ASSEMBLER_SECTION_ID    0x100000

  /* This section should not be subject to garbage collection.
     Also set to inform the linker that this section should not be
     listed in the link map as discarded.  */
#define SEC_KEEP                    0x200000

  /* This section contains "short" data, and should be placed
     "near" the GP.  */
#define SEC_SMALL_DATA              0x400000

  /* Attempt to merge identical entities in the section.
     Entity size is given in the entsize field.  */
#define SEC_MERGE                   0x800000

  /* If given with SEC_MERGE, entities to merge are zero terminated
     strings where entsize specifies character size instead of fixed
     size entries.  */
#define SEC_STRINGS                0x1000000

  /* This section contains data about section groups.  */
#define SEC_GROUP                  0x2000000

  /* The section is a COFF shared library section.  This flag is
     only for the linker.  If this type of section appears in
     the input file, the linker must copy it to the output file
     without changing the vma or size.  FIXME: Although this
     was originally intended to be general, it really is COFF
```

```
     specific (and the flag was renamed to indicate this).  It
     might be cleaner to have some more general mechanism to
     allow the back end to control what the linker does with
     sections.  */
#define SEC_COFF_SHARED_LIBRARY     0x4000000

  /* This input section should be copied to output in reverse order
     as an array of pointers.  This is for ELF linker internal use
     only.  */
#define SEC_ELF_REVERSE_COPY        0x4000000

  /* This section contains data which may be shared with other
     executables or shared objects. This is for COFF only.  */
#define SEC_COFF_SHARED             0x8000000

  /* This section should be compressed.  This is for ELF linker
     internal use only.  */
#define SEC_ELF_COMPRESS            0x8000000

  /* When a section with this flag is being linked, then if the size of
     the input section is less than a page, it should not cross a page
     boundary.  If the size of the input section is one page or more,
     it should be aligned on a page boundary.  This is for TI
     TMS320C54X only.  */
#define SEC_TIC54X_BLOCK           0x10000000

  /* This section should be renamed.  This is for ELF linker
     internal use only.  */
#define SEC_ELF_RENAME             0x10000000

  /* Conditionally link this section; do not link if there are no
     references found to any symbol in the section.  This is for TI
     TMS320C54X only.  */
#define SEC_TIC54X_CLINK           0x20000000

  /* This section contains vliw code.  This is for Toshiba MeP only.  */
#define SEC_MEP_VLIW               0x20000000

  /* All symbols, sizes and relocations in this section are octets
     instead of bytes.  Required for DWARF debug sections as DWARF
     information is organized in octets, not bytes.  */
#define SEC_ELF_OCTETS             0x40000000

  /* Indicate that section has the no read flag set. This happens
     when memory read flag isn't set. */
#define SEC_COFF_NOREAD            0x40000000
```

```
  /* Indicate that section has the purecode flag set.  */
#define SEC_ELF_PURECODE           0x80000000

  /*  End of section flags.  */

  /* Some internal packed boolean fields.  */

  /* See the vma field.  */
  unsigned int user_set_vma : 1;

  /* A mark flag used by some of the linker backends.  */
  unsigned int linker_mark : 1;

  /* Another mark flag used by some of the linker backends.  Set for
     output sections that have an input section.  */
  unsigned int linker_has_input : 1;

  /* Mark flag used by some linker backends for garbage collection.  */
  unsigned int gc_mark : 1;

  /* Section compression status.  */
  unsigned int compress_status : 2;
#define COMPRESS_SECTION_NONE    0
#define COMPRESS_SECTION_DONE    1
#define DECOMPRESS_SECTION_SIZED 2

  /* The following flags are used by the ELF linker. */

  /* Mark sections which have been allocated to segments.  */
  unsigned int segment_mark : 1;

  /* Type of sec_info information.  */
  unsigned int sec_info_type:3;
#define SEC_INFO_TYPE_NONE      0
#define SEC_INFO_TYPE_STABS     1
#define SEC_INFO_TYPE_MERGE     2
#define SEC_INFO_TYPE_EH_FRAME  3
#define SEC_INFO_TYPE_JUST_SYMS 4
#define SEC_INFO_TYPE_TARGET    5
#define SEC_INFO_TYPE_EH_FRAME_ENTRY 6

  /* Nonzero if this section uses RELA relocations, rather than REL.  */
  unsigned int use_rela_p:1;

  /* Bits used by various backends.  The generic code doesn't touch
     these fields.  */
```

```
unsigned int sec_flg0:1;
unsigned int sec_flg1:1;
unsigned int sec_flg2:1;
unsigned int sec_flg3:1;
unsigned int sec_flg4:1;
unsigned int sec_flg5:1;

/* End of internal packed boolean fields.  */

/*  The virtual memory address of the section - where it will be
    at run time.  The symbols are relocated against this.  The
    user_set_vma flag is maintained by bfd; if it's not set, the
    backend can assign addresses (for example, in a.out, where
    the default address for .data is dependent on the specific
    target and various flags).  */
bfd_vma vma;

/*  The load address of the section - where it would be in a
    rom image; really only used for writing section header
    information.  */
bfd_vma lma;

/* The size of the section in *octets*, as it will be output.
   Contains a value even if the section has no contents (e.g., the
   size of .bss).  */
bfd_size_type size;

/* For input sections, the original size on disk of the section, in
   octets.  This field should be set for any section whose size is
   changed by linker relaxation.  It is required for sections where
   the linker relaxation scheme doesn't cache altered section and
   reloc contents (stabs, eh_frame, SEC_MERGE, some coff relaxing
   targets), and thus the original size needs to be kept to read the
   section multiple times.  For output sections, rawsize holds the
   section size calculated on a previous linker relaxation pass.  */
bfd_size_type rawsize;

/* The compressed size of the section in octets.  */
bfd_size_type compressed_size;

/* Relaxation table. */
struct relax_table *relax;

/* Count of used relaxation table entries. */
int relax_count;
```

```
/* If this section is going to be output, then this value is the
   offset in *bytes* into the output section of the first byte in the
   input section (byte ==> smallest addressable unit on the
   target).  In most cases, if this was going to start at the
   100th octet (8-bit quantity) in the output section, this value
   would be 100.  However, if the target byte size is 16 bits
   (bfd_octets_per_byte is "2"), this value would be 50.  */
bfd_vma output_offset;

/* The output section through which to map on output.  */
struct bfd_section *output_section;

/* The alignment requirement of the section, as an exponent of 2 -
   e.g., 3 aligns to 2^3 (or 8).  */
unsigned int alignment_power;

/* If an input section, a pointer to a vector of relocation
   records for the data in this section.  */
struct reloc_cache_entry *relocation;

/* If an output section, a pointer to a vector of pointers to
   relocation records for the data in this section.  */
struct reloc_cache_entry **orelocation;

/* The number of relocation records in one of the above.  */
unsigned reloc_count;

/* Information below is back end specific - and not always used
   or updated.  */

/* File position of section data.  */
file_ptr filepos;

/* File position of relocation info.  */
file_ptr rel_filepos;

/* File position of line data.  */
file_ptr line_filepos;

/* Pointer to data for applications.  */
void *userdata;

/* If the SEC_IN_MEMORY flag is set, this points to the actual
   contents.  */
unsigned char *contents;

/* Attached line number information.  */
```

```
  alent *lineno;

  /* Number of line number records.  */
  unsigned int lineno_count;

  /* Entity size for merging purposes.  */
  unsigned int entsize;

  /* Points to the kept section if this section is a link-once section,
     and is discarded.  */
  struct bfd_section *kept_section;

  /* When a section is being output, this value changes as more
     linenumbers are written out.  */
  file_ptr moving_line_filepos;

  /* What the section number is in the target world.  */
  int target_index;

  void *used_by_bfd;

  /* If this is a constructor section then here is a list of the
     relocations created to relocate items within it.  */
  struct relent_chain *constructor_chain;

  /* The BFD which owns the section.  */
  bfd *owner;

  /* A symbol which points at this section only.  */
  struct bfd_symbol *symbol;
  struct bfd_symbol **symbol_ptr_ptr;

  /* Early in the link process, map_head and map_tail are used to build
     a list of input sections attached to an output section.  Later,
     output sections use these fields for a list of bfd_link_order
     structs.  The linked_to_symbol_name field is for ELF assembler
     internal use.  */
  union {
    struct bfd_link_order *link_order;
    struct bfd_section *s;
    const char *linked_to_symbol_name;
  } map_head, map_tail;
 /* Points to the output section this section is already assigned to, if any.
    This is used when support for non-contiguous memory regions is enabled.  */
 struct bfd_section *already_assigned;

} asection;
```

```
/* Relax table contains information about instructions which can
   be removed by relaxation -- replacing a long address with a
   short address.  */
struct relax_table {
  /* Address where bytes may be deleted. */
  bfd_vma addr;

  /* Number of bytes to be deleted.  */
  int size;
};

static inline const char *
bfd_section_name (const asection *sec)
{
  return sec->name;
}

static inline bfd_size_type
bfd_section_size (const asection *sec)
{
  return sec->size;
}

static inline bfd_vma
bfd_section_vma (const asection *sec)
{
  return sec->vma;
}

static inline bfd_vma
bfd_section_lma (const asection *sec)
{
  return sec->lma;
}

static inline unsigned int
bfd_section_alignment (const asection *sec)
{
  return sec->alignment_power;
}

static inline flagword
bfd_section_flags (const asection *sec)
{
  return sec->flags;
}
```

```
static inline void *
bfd_section_userdata (const asection *sec)
{
  return sec->userdata;
}
static inline bool
bfd_is_com_section (const asection *sec)
{
  return (sec->flags & SEC_IS_COMMON) != 0;
}

/* Note: the following are provided as inline functions rather than macros
   because not all callers use the return value.  A macro implementation
   would use a comma expression, eg: "((ptr)->foo = val, TRUE)" and some
   compilers will complain about comma expressions that have no effect.  */
static inline bool
bfd_set_section_userdata (asection *sec, void *val)
{
  sec->userdata = val;
  return true;
}

static inline bool
bfd_set_section_vma (asection *sec, bfd_vma val)
{
  sec->vma = sec->lma = val;
  sec->user_set_vma = true;
  return true;
}

static inline bool
bfd_set_section_lma (asection *sec, bfd_vma val)
{
  sec->lma = val;
  return true;
}

static inline bool
bfd_set_section_alignment (asection *sec, unsigned int val)
{
  sec->alignment_power = val;
  return true;
}

/* These sections are global, and are managed by BFD.  The application
   and target back end are not permitted to change the values in
```

```
   these sections.  */
extern asection _bfd_std_section[4];

#define BFD_ABS_SECTION_NAME "*ABS*"
#define BFD_UND_SECTION_NAME "*UND*"
#define BFD_COM_SECTION_NAME "*COM*"
#define BFD_IND_SECTION_NAME "*IND*"

/* Pointer to the common section.  */
#define bfd_com_section_ptr (&_bfd_std_section[0])
/* Pointer to the undefined section.  */
#define bfd_und_section_ptr (&_bfd_std_section[1])
/* Pointer to the absolute section.  */
#define bfd_abs_section_ptr (&_bfd_std_section[2])
/* Pointer to the indirect section.  */
#define bfd_ind_section_ptr (&_bfd_std_section[3])

static inline bool
bfd_is_und_section (const asection *sec)
{
  return sec == bfd_und_section_ptr;
}

static inline bool
bfd_is_abs_section (const asection *sec)
{
  return sec == bfd_abs_section_ptr;
}

static inline bool
bfd_is_ind_section (const asection *sec)
{
  return sec == bfd_ind_section_ptr;
}

static inline bool
bfd_is_const_section (const asection *sec)
{
  return (sec >= _bfd_std_section
          && sec < _bfd_std_section + (sizeof (_bfd_std_section)
                                       / sizeof (_bfd_std_section[0])));
}

/* Return TRUE if input section SEC has been discarded.  */
static inline bool
discarded_section (const asection *sec)
{
```

```
  return (!bfd_is_abs_section (sec)
          && bfd_is_abs_section (sec->output_section)
          && sec->sec_info_type != SEC_INFO_TYPE_MERGE
          && sec->sec_info_type != SEC_INFO_TYPE_JUST_SYMS);
}

#define BFD_FAKE_SECTION(SEC, SYM, NAME, IDX, FLAGS)                    \
  /* name, id,  section_id, index, next, prev, flags, user_set_vma, */ \
  {  NAME, IDX, 0,          0,     NULL, NULL, FLAGS, 0,               \
                                                                       \
  /* linker_mark, linker_has_input, gc_mark, decompress_status,     */ \
     0,           0,                1,       0,                        \
                                                                       \
  /* segment_mark, sec_info_type, use_rela_p,                       */ \
     0,            0,             0,                                   \
                                                                       \
  /* sec_flg0, sec_flg1, sec_flg2, sec_flg3, sec_flg4, sec_flg5,    */ \
     0,        0,        0,        0,        0,        0,              \
                                                                       \
  /* vma, lma, size, rawsize, compressed_size, relax, relax_count,  */ \
     0,   0,   0,    0,       0,               0,     0,               \
                                                                       \
  /* output_offset, output_section, alignment_power,                */ \
     0,             &SEC,           0,                                \
                                                                       \
  /* relocation, orelocation, reloc_count, filepos, rel_filepos,    */ \
     NULL,       NULL,        0,           0,       0,                \
                                                                       \
  /* line_filepos, userdata, contents, lineno, lineno_count,        */ \
     0,            NULL,     NULL,     NULL,   0,                      \
                                                                       \
  /* entsize, kept_section, moving_line_filepos,                    */ \
     0,       NULL,         0,                                        \
                                                                       \
  /* target_index, used_by_bfd, constructor_chain, owner,           */ \
     0,            NULL,        NULL,              NULL,               \
                                                                       \
  /* symbol,                    symbol_ptr_ptr,                     */ \
     (struct bfd_symbol *) SYM, &SEC.symbol,                           \
                                                                       \
  /* map_head, map_tail, already_assigned                           */ \
     { NULL }, { NULL }, NULL                                          \
                                                                       \
    }

/* We use a macro to initialize the static asymbol structures because
   traditional C does not permit us to initialize a union member while
```

```
   gcc warns if we don't initialize it.
   the_bfd, name, value, attr, section [, udata]  */
#ifdef __STDC__
#define GLOBAL_SYM_INIT(NAME, SECTION) \
  { 0, NAME, 0, BSF_SECTION_SYM, SECTION, { 0 }}
#else
#define GLOBAL_SYM_INIT(NAME, SECTION) \
  { 0, NAME, 0, BSF_SECTION_SYM, SECTION }
#endif
```

2.6.5 Section prototypes

These are the functions exported by the section handling part of BFD.

2.6.5.1 bfd_section_list_clear

Synopsis

```
void bfd_section_list_clear (bfd *);
```

Description
Clears the section list, and also resets the section count and hash table entries.

2.6.5.2 bfd_get_section_by_name

Synopsis

```
asection *bfd_get_section_by_name (bfd *abfd, const char *name);
```

Description
Return the most recently created section attached to *abfd* named *name*. Return NULL if no such section exists.

2.6.5.3 bfd_get_next_section_by_name

Synopsis

```
asection *bfd_get_next_section_by_name (bfd *ibfd, asection *sec);
```

Description
Given *sec* is a section returned by `bfd_get_section_by_name`, return the next most recently created section attached to the same BFD with the same name, or if no such section exists in the same BFD and IBFD is non-NULL, the next section with the same name in any input BFD following IBFD. Return NULL on finding no section.

2.6.5.4 bfd_get_linker_section

Synopsis

```
asection *bfd_get_linker_section (bfd *abfd, const char *name);
```

Description
Return the linker created section attached to *abfd* named *name*. Return NULL if no such section exists.

2.6.5.5 bfd_get_section_by_name_if

Synopsis

```
asection *bfd_get_section_by_name_if
   (bfd *abfd,
    const char *name,
    bool (*func) (bfd *abfd, asection *sect, void *obj),
    void *obj);
```

Description

Call the provided function *func* for each section attached to the BFD *abfd* whose name matches *name*, passing *obj* as an argument. The function will be called as if by

```
func (abfd, the_section, obj);
```

It returns the first section for which *func* returns true, otherwise `NULL`.

2.6.5.6 bfd_get_unique_section_name

Synopsis

```
char *bfd_get_unique_section_name
   (bfd *abfd, const char *templat, int *count);
```

Description

Invent a section name that is unique in *abfd* by tacking a dot and a digit suffix onto the original *templat*. If *count* is non-NULL, then it specifies the first number tried as a suffix to generate a unique name. The value pointed to by *count* will be incremented in this case.

2.6.5.7 bfd_make_section_old_way

Synopsis

```
asection *bfd_make_section_old_way (bfd *abfd, const char *name);
```

Description

Create a new empty section called *name* and attach it to the end of the chain of sections for the BFD *abfd*. An attempt to create a section with a name which is already in use returns its pointer without changing the section chain.

It has the funny name since this is the way it used to be before it was rewritten....

Possible errors are:

- `bfd_error_invalid_operation` - If output has already started for this BFD.
- `bfd_error_no_memory` - If memory allocation fails.

2.6.5.8 bfd_make_section_anyway_with_flags

Synopsis

```
asection *bfd_make_section_anyway_with_flags
   (bfd *abfd, const char *name, flagword flags);
```

Description

Create a new empty section called *name* and attach it to the end of the chain of sections for *abfd*. Create a new section even if there is already a section with that name. Also set the attributes of the new section to the value *flags*.

Return NULL and set bfd_error on error; possible errors are:

- bfd_error_invalid_operation - If output has already started for *abfd*.
- bfd_error_no_memory - If memory allocation fails.

2.6.5.9 bfd_make_section_anyway

Synopsis

```
asection *bfd_make_section_anyway (bfd *abfd, const char *name);
```

Description
Create a new empty section called *name* and attach it to the end of the chain of sections for *abfd*. Create a new section even if there is already a section with that name.

Return NULL and set bfd_error on error; possible errors are:

- bfd_error_invalid_operation - If output has already started for *abfd*.
- bfd_error_no_memory - If memory allocation fails.

2.6.5.10 bfd_make_section_with_flags

Synopsis

```
asection *bfd_make_section_with_flags
   (bfd *, const char *name, flagword flags);
```

Description
Like bfd_make_section_anyway, but return NULL (without calling bfd_set_error ()) without changing the section chain if there is already a section named *name*. Also set the attributes of the new section to the value *flags*. If there is an error, return NULL and set bfd_error.

2.6.5.11 bfd_make_section

Synopsis

```
asection *bfd_make_section (bfd *, const char *name);
```

Description
Like bfd_make_section_anyway, but return NULL (without calling bfd_set_error ()) without changing the section chain if there is already a section named *name*. If there is an error, return NULL and set bfd_error.

2.6.5.12 bfd_set_section_flags

Synopsis

```
bool bfd_set_section_flags (asection *sec, flagword flags);
```

Description
Set the attributes of the section *sec* to the value *flags*. Return TRUE on success, FALSE on error. Possible error returns are:

- bfd_error_invalid_operation - The section cannot have one or more of the attributes requested. For example, a .bss section in a.out may not have the SEC_HAS_CONTENTS field set.

2.6.5.13 bfd_rename_section

Synopsis

```
void bfd_rename_section
   (asection *sec, const char *newname);
```

Description
Rename section *sec* to *newname*.

2.6.5.14 bfd_map_over_sections

Synopsis

```
void bfd_map_over_sections
   (bfd *abfd,
    void (*func) (bfd *abfd, asection *sect, void *obj),
    void *obj);
```

Description
Call the provided function *func* for each section attached to the BFD *abfd*, passing *obj* as an argument. The function will be called as if by

```
func (abfd, the_section, obj);
```

This is the preferred method for iterating over sections; an alternative would be to use a loop:

```
asection *p;
for (p = abfd->sections; p != NULL; p = p->next)
   func (abfd, p, ...)
```

2.6.5.15 bfd_sections_find_if

Synopsis

```
asection *bfd_sections_find_if
   (bfd *abfd,
    bool (*operation) (bfd *abfd, asection *sect, void *obj),
    void *obj);
```

Description
Call the provided function *operation* for each section attached to the BFD *abfd*, passing *obj* as an argument. The function will be called as if by

```
operation (abfd, the_section, obj);
```

It returns the first section for which *operation* returns true.

2.6.5.16 bfd_set_section_size

Synopsis

```
bool bfd_set_section_size (asection *sec, bfd_size_type val);
```

Description
Set *sec* to the size *val*. If the operation is ok, then `TRUE` is returned, else `FALSE`.

Possible error returns:

- `bfd_error_invalid_operation` - Writing has started to the BFD, so setting the size is invalid.

2.6.5.17 bfd_set_section_contents

Synopsis

```
bool bfd_set_section_contents
   (bfd *abfd, asection *section, const void *data,
    file_ptr offset, bfd_size_type count);
```

Description

Sets the contents of the section *section* in BFD *abfd* to the data starting in memory at *location*. The data is written to the output section starting at offset *offset* for *count* octets.

Normally `TRUE` is returned, but `FALSE` is returned if there was an error. Possible error returns are:

- `bfd_error_no_contents` - The output section does not have the `SEC_HAS_CONTENTS` attribute, so nothing can be written to it.
- `bfd_error_bad_value` - The section is unable to contain all of the data.
- `bfd_error_invalid_operation` - The BFD is not writeable.
- and some more too.

This routine is front end to the back end function `_bfd_set_section_contents`.

2.6.5.18 bfd_get_section_contents

Synopsis

```
bool bfd_get_section_contents
   (bfd *abfd, asection *section, void *location, file_ptr offset,
    bfd_size_type count);
```

Description

Read data from *section* in BFD *abfd* into memory starting at *location*. The data is read at an offset of *offset* from the start of the input section, and is read for *count* bytes.

If the contents of a constructor with the `SEC_CONSTRUCTOR` flag set are requested or if the section does not have the `SEC_HAS_CONTENTS` flag set, then the *location* is filled with zeroes. If no errors occur, `TRUE` is returned, else `FALSE`.

2.6.5.19 bfd_malloc_and_get_section

Synopsis

```
bool bfd_malloc_and_get_section
   (bfd *abfd, asection *section, bfd_byte **buf);
```

Description

Read all data from *section* in BFD *abfd* into a buffer, **buf*, malloc'd by this function.

2.6.5.20 bfd_copy_private_section_data

Synopsis

```
bool bfd_copy_private_section_data
   (bfd *ibfd, asection *isec, bfd *obfd, asection *osec);
```

Description

Copy private section information from *isec* in the BFD *ibfd* to the section *osec* in the BFD *obfd*. Return `TRUE` on success, `FALSE` on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *osec*.

```
#define bfd_copy_private_section_data(ibfd, isection, obfd, osection) \
       BFD_SEND (obfd, _bfd_copy_private_section_data, \
                 (ibfd, isection, obfd, osection))
```

2.6.5.21 bfd_generic_is_group_section

Synopsis

```
bool bfd_generic_is_group_section (bfd *, const asection *sec);
```

Description
Returns TRUE if *sec* is a member of a group.

2.6.5.22 bfd_generic_group_name

Synopsis

```
const char *bfd_generic_group_name (bfd *, const asection *sec);
```

Description
Returns group name if *sec* is a member of a group.

2.6.5.23 bfd_generic_discard_group

Synopsis

```
bool bfd_generic_discard_group (bfd *abfd, asection *group);
```

Description
Remove all members of *group* from the output.

2.7 Symbols

BFD tries to maintain as much symbol information as it can when it moves information from file to file. BFD passes information to applications though the `asymbol` structure. When the application requests the symbol table, BFD reads the table in the native form and translates parts of it into the internal format. To maintain more than the information passed to applications, some targets keep some information "behind the scenes" in a structure only the particular back end knows about. For example, the coff back end keeps the original symbol table structure as well as the canonical structure when a BFD is read in. On output, the coff back end can reconstruct the output symbol table so that no information is lost, even information unique to coff which BFD doesn't know or understand. If a coff symbol table were read, but were written through an a.out back end, all the coff specific information would be lost. The symbol table of a BFD is not necessarily read in until a canonicalize request is made. Then the BFD back end fills in a table provided by the application with pointers to the canonical information. To output symbols, the application provides BFD with a table of pointers to pointers to `asymbol`s. This allows applications like the linker to output a symbol as it was read, since the "behind the scenes" information will be still available.

2.7.1 Reading symbols

There are two stages to reading a symbol table from a BFD: allocating storage, and the actual reading process. This is an excerpt from an application which reads the symbol table:

```
long storage_needed;
```

```
asymbol **symbol_table;
long number_of_symbols;
long i;

storage_needed = bfd_get_symtab_upper_bound (abfd);

if (storage_needed < 0)
  FAIL

if (storage_needed == 0)
  return;

symbol_table = xmalloc (storage_needed);
  ...
number_of_symbols =
   bfd_canonicalize_symtab (abfd, symbol_table);

if (number_of_symbols < 0)
  FAIL

for (i = 0; i < number_of_symbols; i++)
  process_symbol (symbol_table[i]);
```

All storage for the symbols themselves is in an objalloc connected to the BFD; it is freed when the BFD is closed.

2.7.2 Writing symbols

Writing of a symbol table is automatic when a BFD open for writing is closed. The application attaches a vector of pointers to pointers to symbols to the BFD being written, and fills in the symbol count. The close and cleanup code reads through the table provided and performs all the necessary operations. The BFD output code must always be provided with an "owned" symbol: one which has come from another BFD, or one which has been created using `bfd_make_empty_symbol`. Here is an example showing the creation of a symbol table with only one element:

```
#include "sysdep.h"
#include "bfd.h"
int main (void)
{
  bfd *abfd;
  asymbol *ptrs[2];
  asymbol *new;

  abfd = bfd_openw ("foo","a.out-sunos-big");
  bfd_set_format (abfd, bfd_object);
  new = bfd_make_empty_symbol (abfd);
  new->name = "dummy_symbol";
  new->section = bfd_make_section_old_way (abfd, ".text");
```

```
  new->flags = BSF_GLOBAL;
  new->value = 0x12345;

  ptrs[0] = new;
  ptrs[1] = 0;

  bfd_set_symtab (abfd, ptrs, 1);
  bfd_close (abfd);
  return 0;
}

./makesym
nm foo
00012345 A dummy_symbol
```

Many formats cannot represent arbitrary symbol information; for instance, the `a.out` object format does not allow an arbitrary number of sections. A symbol pointing to a section which is not one of `.text`, `.data` or `.bss` cannot be described.

2.7.3 Mini Symbols

Mini symbols provide read-only access to the symbol table. They use less memory space, but require more time to access. They can be useful for tools like nm or objdump, which may have to handle symbol tables of extremely large executables.

The `bfd_read_minisymbols` function will read the symbols into memory in an internal form. It will return a `void *` pointer to a block of memory, a symbol count, and the size of each symbol. The pointer is allocated using `malloc`, and should be freed by the caller when it is no longer needed.

The function `bfd_minisymbol_to_symbol` will take a pointer to a minisymbol, and a pointer to a structure returned by `bfd_make_empty_symbol`, and return a `asymbol` structure. The return value may or may not be the same as the value from `bfd_make_empty_symbol` which was passed in.

2.7.4 typedef asymbol

An `asymbol` has the form:

```
typedef struct bfd_symbol
{
  /* A pointer to the BFD which owns the symbol. This information
     is necessary so that a back end can work out what additional
     information (invisible to the application writer) is carried
     with the symbol.

     This field is *almost* redundant, since you can use section->owner
     instead, except that some symbols point to the global sections
     bfd_{abs,com,und}_section.  This could be fixed by making
     these globals be per-bfd (or per-target-flavor).  FIXME.  */
  struct bfd *the_bfd; /* Use bfd_asymbol_bfd(sym) to access this field.  */
```

```
  /* The text of the symbol. The name is left alone, and not copied; the
     application may not alter it.  */
  const char *name;

  /* The value of the symbol.  This really should be a union of a
     numeric value with a pointer, since some flags indicate that
     a pointer to another symbol is stored here.  */
  symvalue value;

  /* Attributes of a symbol.  */
#define BSF_NO_FLAGS            0

  /* The symbol has local scope; static in C. The value
     is the offset into the section of the data.  */
#define BSF_LOCAL               (1 << 0)

  /* The symbol has global scope; initialized data in C. The
     value is the offset into the section of the data.  */
#define BSF_GLOBAL              (1 << 1)

  /* The symbol has global scope and is exported. The value is
     the offset into the section of the data.  */
#define BSF_EXPORT              BSF_GLOBAL /* No real difference.  */

  /* A normal C symbol would be one of:
     BSF_LOCAL, BSF_UNDEFINED or BSF_GLOBAL.  */

  /* The symbol is a debugging record. The value has an arbitrary
     meaning, unless BSF_DEBUGGING_RELOC is also set.  */
#define BSF_DEBUGGING           (1 << 2)

  /* The symbol denotes a function entry point.  Used in ELF,
     perhaps others someday.  */
#define BSF_FUNCTION            (1 << 3)

  /* Used by the linker.  */
#define BSF_KEEP                (1 << 5)

  /* An ELF common symbol.  */
#define BSF_ELF_COMMON          (1 << 6)

  /* A weak global symbol, overridable without warnings by
     a regular global symbol of the same name.  */
#define BSF_WEAK                (1 << 7)

  /* This symbol was created to point to a section, e.g. ELF's
```

```
     STT_SECTION symbols.  */
#define BSF_SECTION_SYM         (1 << 8)

  /* The symbol used to be a common symbol, but now it is
     allocated.  */
#define BSF_OLD_COMMON          (1 << 9)

  /* In some files the type of a symbol sometimes alters its
     location in an output file - ie in coff a ISFCN symbol
     which is also C_EXT symbol appears where it was
     declared and not at the end of a section.  This bit is set
     by the target BFD part to convey this information.  */
#define BSF_NOT_AT_END          (1 << 10)

  /* Signal that the symbol is the label of constructor section.  */
#define BSF_CONSTRUCTOR         (1 << 11)

  /* Signal that the symbol is a warning symbol.  The name is a
     warning.  The name of the next symbol is the one to warn about;
     if a reference is made to a symbol with the same name as the next
     symbol, a warning is issued by the linker.  */
#define BSF_WARNING             (1 << 12)

  /* Signal that the symbol is indirect.  This symbol is an indirect
     pointer to the symbol with the same name as the next symbol.  */
#define BSF_INDIRECT            (1 << 13)

  /* BSF_FILE marks symbols that contain a file name.  This is used
     for ELF STT_FILE symbols.  */
#define BSF_FILE                (1 << 14)

  /* Symbol is from dynamic linking information.  */
#define BSF_DYNAMIC             (1 << 15)

  /* The symbol denotes a data object.  Used in ELF, and perhaps
     others someday.  */
#define BSF_OBJECT              (1 << 16)

  /* This symbol is a debugging symbol.  The value is the offset
     into the section of the data.  BSF_DEBUGGING should be set
     as well.  */
#define BSF_DEBUGGING_RELOC     (1 << 17)

  /* This symbol is thread local.  Used in ELF.  */
#define BSF_THREAD_LOCAL        (1 << 18)

  /* This symbol represents a complex relocation expression,
```

```
     with the expression tree serialized in the symbol name.  */
#define BSF_RELC                (1 << 19)

  /* This symbol represents a signed complex relocation expression,
     with the expression tree serialized in the symbol name.  */
#define BSF_SRELC               (1 << 20)

  /* This symbol was created by bfd_get_synthetic_symtab.  */
#define BSF_SYNTHETIC           (1 << 21)

  /* This symbol is an indirect code object.  Unrelated to BSF_INDIRECT.
     The dynamic linker will compute the value of this symbol by
     calling the function that it points to.  BSF_FUNCTION must
     also be also set.  */
#define BSF_GNU_INDIRECT_FUNCTION (1 << 22)
  /* This symbol is a globally unique data object.  The dynamic linker
     will make sure that in the entire process there is just one symbol
     with this name and type in use.  BSF_OBJECT must also be set.  */
#define BSF_GNU_UNIQUE          (1 << 23)

  /* This section symbol should be included in the symbol table.  */
#define BSF_SECTION_SYM_USED    (1 << 24)

  flagword flags;

  /* A pointer to the section to which this symbol is
     relative.  This will always be non NULL, there are special
     sections for undefined and absolute symbols.  */
  struct bfd_section *section;

  /* Back end special data.  */
  union
    {
      void *p;
      bfd_vma i;
    }
  udata;
}
asymbol;
```

2.7.5 Symbol handling functions

2.7.5.1 `bfd_get_symtab_upper_bound`

Description

Return the number of bytes required to store a vector of pointers to `asymbols` for all the

symbols in the BFD *abfd*, including a terminal NULL pointer. If there are no symbols in the BFD, then return 0. If an error occurs, return -1.

```
#define bfd_get_symtab_upper_bound(abfd) \
       BFD_SEND (abfd, _bfd_get_symtab_upper_bound, (abfd))
```

2.7.5.2 bfd_is_local_label

Synopsis

```
bool bfd_is_local_label (bfd *abfd, asymbol *sym);
```

Description
Return TRUE if the given symbol *sym* in the BFD *abfd* is a compiler generated local label, else return FALSE.

2.7.5.3 bfd_is_local_label_name

Synopsis

```
bool bfd_is_local_label_name (bfd *abfd, const char *name);
```

Description
Return TRUE if a symbol with the name *name* in the BFD *abfd* is a compiler generated local label, else return FALSE. This just checks whether the name has the form of a local label.

```
#define bfd_is_local_label_name(abfd, name) \
       BFD_SEND (abfd, _bfd_is_local_label_name, (abfd, name))
```

2.7.5.4 bfd_is_target_special_symbol

Synopsis

```
bool bfd_is_target_special_symbol (bfd *abfd, asymbol *sym);
```

Description
Return TRUE iff a symbol *sym* in the BFD *abfd* is something special to the particular target represented by the BFD. Such symbols should normally not be mentioned to the user.

```
#define bfd_is_target_special_symbol(abfd, sym) \
       BFD_SEND (abfd, _bfd_is_target_special_symbol, (abfd, sym))
```

2.7.5.5 bfd_canonicalize_symtab

Description
Read the symbols from the BFD *abfd*, and fills in the vector *location* with pointers to the symbols and a trailing NULL. Return the actual number of symbol pointers, not including the NULL.

```
#define bfd_canonicalize_symtab(abfd, location) \
       BFD_SEND (abfd, _bfd_canonicalize_symtab, (abfd, location))
```

2.7.5.6 bfd_set_symtab

Synopsis

```
bool bfd_set_symtab
   (bfd *abfd, asymbol **location, unsigned int count);
```

Description

Arrange that when the output BFD *abfd* is closed, the table *location* of *count* pointers to symbols will be written.

2.7.5.7 bfd_print_symbol_vandf

Synopsis

```
void bfd_print_symbol_vandf (bfd *abfd, void *file, asymbol *symbol);
```

Description

Print the value and flags of the *symbol* supplied to the stream *file*.

2.7.5.8 bfd_make_empty_symbol

Description

Create a new `asymbol` structure for the BFD *abfd* and return a pointer to it.

This routine is necessary because each back end has private information surrounding the `asymbol`. Building your own `asymbol` and pointing to it will not create the private information, and will cause problems later on.

```
#define bfd_make_empty_symbol(abfd) \
       BFD_SEND (abfd, _bfd_make_empty_symbol, (abfd))
```

2.7.5.9 _bfd_generic_make_empty_symbol

Synopsis

```
asymbol *_bfd_generic_make_empty_symbol (bfd *);
```

Description

Create a new `asymbol` structure for the BFD *abfd* and return a pointer to it. Used by core file routines, binary back-end and anywhere else where no private info is needed.

2.7.5.10 bfd_make_debug_symbol

Description

Create a new `asymbol` structure for the BFD *abfd*, to be used as a debugging symbol. Further details of its use have yet to be worked out.

```
#define bfd_make_debug_symbol(abfd,ptr,size) \
       BFD_SEND (abfd, _bfd_make_debug_symbol, (abfd, ptr, size))
```

2.7.5.11 bfd_decode_symclass

Description

Return a character corresponding to the symbol class of *symbol*, or '?' for an unknown class.

Synopsis

```
int bfd_decode_symclass (asymbol *symbol);
```

2.7.5.12 bfd_is_undefined_symclass

Description

Returns non-zero if the class symbol returned by bfd_decode_symclass represents an undefined symbol. Returns zero otherwise.

Synopsis

```
bool bfd_is_undefined_symclass (int symclass);
```

2.7.5.13 bfd_symbol_info

Description

Fill in the basic info about symbol that nm needs. Additional info may be added by the back-ends after calling this function.

Synopsis

```
void bfd_symbol_info (asymbol *symbol, symbol_info *ret);
```

2.7.5.14 bfd_copy_private_symbol_data

Synopsis

```
bool bfd_copy_private_symbol_data
   (bfd *ibfd, asymbol *isym, bfd *obfd, asymbol *osym);
```

Description

Copy private symbol information from *isym* in the BFD *ibfd* to the symbol *osym* in the BFD *obfd*. Return `TRUE` on success, `FALSE` on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *osec*.

```
#define bfd_copy_private_symbol_data(ibfd, isymbol, obfd, osymbol) \
       BFD_SEND (obfd, _bfd_copy_private_symbol_data, \
                 (ibfd, isymbol, obfd, osymbol))
```

2.8 Archives

Description

An archive (or library) is just another BFD. It has a symbol table, although there's not much a user program will do with it.

The big difference between an archive BFD and an ordinary BFD is that the archive doesn't have sections. Instead it has a chain of BFDs that are considered its contents. These BFDs can be manipulated like any other. The BFDs contained in an archive opened for reading will all be opened for reading. You may put either input or output BFDs into an archive opened for output; they will be handled correctly when the archive is closed.

Use `bfd_openr_next_archived_file` to step through the contents of an archive opened for input. You don't have to read the entire archive if you don't want to! Read it until you find what you want.

A BFD returned by `bfd_openr_next_archived_file` can be closed manually with `bfd_close`. If you do not close it, then a second iteration through the members of an archive

may return the same BFD. If you close the archive BFD, then all the member BFDs will automatically be closed as well.

Archive contents of output BFDs are chained through the `archive_next` pointer in a BFD. The first one is findable through the `archive_head` slot of the archive. Set it with `bfd_set_archive_head` (q.v.). A given BFD may be in only one open output archive at a time.

As expected, the BFD archive code is more general than the archive code of any given environment. BFD archives may contain files of different formats (e.g., a.out and coff) and even different architectures. You may even place archives recursively into archives!

This can cause unexpected confusion, since some archive formats are more expressive than others. For instance, Intel COFF archives can preserve long filenames; SunOS a.out archives cannot. If you move a file from the first to the second format and back again, the filename may be truncated. Likewise, different a.out environments have different conventions as to how they truncate filenames, whether they preserve directory names in filenames, etc. When interoperating with native tools, be sure your files are homogeneous.

Beware: most of these formats do not react well to the presence of spaces in filenames. We do the best we can, but can't always handle this case due to restrictions in the format of archives. Many Unix utilities are braindead in regards to spaces and such in filenames anyway, so this shouldn't be much of a restriction.

Archives are supported in BFD in `archive.c`.

2.8.1 Archive functions

2.8.1.1 `bfd_get_next_mapent`

Synopsis

```
symindex bfd_get_next_mapent
   (bfd *abfd, symindex previous, carsym **sym);
```

Description

Step through archive *abfd*'s symbol table (if it has one). Successively update *sym* with the next symbol's information, returning that symbol's (internal) index into the symbol table.

Supply `BFD_NO_MORE_SYMBOLS` as the *previous* entry to get the first one; returns `BFD_NO_MORE_SYMBOLS` when you've already got the last one.

A `carsym` is a canonical archive symbol. The only user-visible element is its name, a null-terminated string.

2.8.1.2 `bfd_set_archive_head`

Synopsis

```
bool bfd_set_archive_head (bfd *output, bfd *new_head);
```

Description

Set the head of the chain of BFDs contained in the archive *output* to *new_head*.

2.8.1.3 `bfd_openr_next_archived_file`

Synopsis

```
bfd *bfd_openr_next_archived_file (bfd *archive, bfd *previous);
```

Description
Provided a BFD, *archive*, containing an archive and NULL, open an input BFD on the first contained element and returns that. Subsequent calls should pass the archive and the previous return value to return a created BFD to the next contained element. NULL is returned when there are no more. Note - if you want to process the bfd returned by this call be sure to call bfd_check_format() on it first.

2.9 File formats

A format is a BFD concept of high level file contents type. The formats supported by BFD are:

- `bfd_object`

The BFD may contain data, symbols, relocations and debug info.

- `bfd_archive`

The BFD contains other BFDs and an optional index.

- `bfd_core`

The BFD contains the result of an executable core dump.

2.9.1 File format functions

2.9.1.1 `bfd_check_format`

Synopsis

```
bool bfd_check_format (bfd *abfd, bfd_format format);
```

Description
Verify if the file attached to the BFD *abfd* is compatible with the format *format* (i.e., one of `bfd_object`, `bfd_archive` or `bfd_core`).

If the BFD has been set to a specific target before the call, only the named target and format combination is checked. If the target has not been set, or has been set to `default`, then all the known target backends is interrogated to determine a match. If the default target matches, it is used. If not, exactly one target must recognize the file, or an error results.

The function returns `TRUE` on success, otherwise `FALSE` with one of the following error codes:

- `bfd_error_invalid_operation` - if `format` is not one of `bfd_object`, `bfd_archive` or `bfd_core`.
- `bfd_error_system_call` - if an error occured during a read - even some file mismatches can cause bfd_error_system_calls.
- `file_not_recognised` - none of the backends recognised the file format.
- `bfd_error_file_ambiguously_recognized` - more than one backend recognised the file format.

2.9.1.2 `bfd_check_format_matches`

Synopsis

```
bool bfd_check_format_matches
```

```
(bfd *abfd, bfd_format format, char ***matching);
```

Description

Like `bfd_check_format`, except when it returns FALSE with `bfd_errno` set to `bfd_error_file_ambiguously_recognized`. In that case, if *matching* is not NULL, it will be filled in with a NULL-terminated list of the names of the formats that matched, allocated with `malloc`. Then the user may choose a format and try again.

When done with the list that *matching* points to, the caller should free it.

2.9.1.3 bfd_set_format

Synopsis

```
bool bfd_set_format (bfd *abfd, bfd_format format);
```

Description

This function sets the file format of the BFD *abfd* to the format *format*. If the target set in the BFD does not support the format requested, the format is invalid, or the BFD is not open for writing, then an error occurs.

2.9.1.4 bfd_format_string

Synopsis

```
const char *bfd_format_string (bfd_format format);
```

Description

Return a pointer to a const string `invalid`, `object`, `archive`, `core`, or `unknown`, depending upon the value of *format*.

2.10 Relocations

BFD maintains relocations in much the same way it maintains symbols: they are left alone until required, then read in en-masse and translated into an internal form. A common routine `bfd_perform_relocation` acts upon the canonical form to do the fixup.

Relocations are maintained on a per section basis, while symbols are maintained on a per BFD basis.

All that a back end has to do to fit the BFD interface is to create a `struct reloc_cache_entry` for each relocation in a particular section, and fill in the right bits of the structures.

2.10.1 typedef arelent

This is the structure of a relocation entry:

```
typedef enum bfd_reloc_status
{
  /* No errors detected.  Note - the value 2 is used so that it
     will not be mistaken for the boolean TRUE or FALSE values.  */
  bfd_reloc_ok = 2,

  /* The relocation was performed, but there was an overflow.  */
  bfd_reloc_overflow,
```

```
  /* The address to relocate was not within the section supplied.  */
  bfd_reloc_outofrange,

  /* Used by special functions.  */
  bfd_reloc_continue,

  /* Unsupported relocation size requested.  */
  bfd_reloc_notsupported,

  /* Unused.  */
  bfd_reloc_other,

  /* The symbol to relocate against was undefined.  */
  bfd_reloc_undefined,

  /* The relocation was performed, but may not be ok.  If this type is
     returned, the error_message argument to bfd_perform_relocation
     will be set.  */
  bfd_reloc_dangerous
 }
 bfd_reloc_status_type;

typedef const struct reloc_howto_struct reloc_howto_type;

typedef struct reloc_cache_entry
{
  /* A pointer into the canonical table of pointers.  */
  struct bfd_symbol **sym_ptr_ptr;

  /* offset in section.  */
  bfd_size_type address;

  /* addend for relocation value.  */
  bfd_vma addend;

  /* Pointer to how to perform the required relocation.  */
  reloc_howto_type *howto;

}
arelent;
```

Description

Here is a description of each of the fields within an `arelent`:

- `sym_ptr_ptr`

The symbol table pointer points to a pointer to the symbol associated with the relocation request. It is the pointer into the table returned by the back end's `canonicalize_symtab`

action. See Section 2.7 [Symbols], page 49. The symbol is referenced through a pointer to a pointer so that tools like the linker can fix up all the symbols of the same name by modifying only one pointer. The relocation routine looks in the symbol and uses the base of the section the symbol is attached to and the value of the symbol as the initial relocation offset. If the symbol pointer is zero, then the section provided is looked up.

- `address`

The `address` field gives the offset in bytes from the base of the section data which owns the relocation record to the first byte of relocatable information. The actual data relocated will be relative to this point; for example, a relocation type which modifies the bottom two bytes of a four byte word would not touch the first byte pointed to in a big endian world.

- `addend`

The `addend` is a value provided by the back end to be added (!) to the relocation offset. Its interpretation is dependent upon the howto. For example, on the 68k the code:

```
char foo[];
main()
        {
        return foo[0x12345678];
        }
```

Could be compiled into:

```
linkw fp,#-4
moveb @#12345678,d0
extbl d0
unlk fp
rts
```

This could create a reloc pointing to `foo`, but leave the offset in the data, something like:

```
RELOCATION RECORDS FOR [.text]:
offset   type      value
00000006 32        _foo

00000000 4e56 fffc          ; linkw fp,#-4
00000004 1039 1234 5678     ; moveb @#12345678,d0
0000000a 49c0               ; extbl d0
0000000c 4e5e               ; unlk fp
0000000e 4e75               ; rts
```

Using coff and an 88k, some instructions don't have enough space in them to represent the full address range, and pointers have to be loaded in two parts. So you'd get something like:

```
or.u     r13,r0,hi16(_foo+0x12345678)
ld.b     r2,r13,lo16(_foo+0x12345678)
jmp      r1
```

This should create two relocs, both pointing to `_foo`, and with 0x12340000 in their addend field. The data would consist of:

```
RELOCATION RECORDS FOR [.text]:
```

```
offset   type      value
00000002 HVRT16    _foo+0x12340000
00000006 LVRT16    _foo+0x12340000

00000000 5da05678           ; or.u r13,r0,0x5678
00000004 1c4d5678           ; ld.b r2,r13,0x5678
00000008 f400c001           ; jmp r1
```

The relocation routine digs out the value from the data, adds it to the addend to get the original offset, and then adds the value of `_foo`. Note that all 32 bits have to be kept around somewhere, to cope with carry from bit 15 to bit 16.

One further example is the sparc and the a.out format. The sparc has a similar problem to the 88k, in that some instructions don't have room for an entire offset, but on the sparc the parts are created in odd sized lumps. The designers of the a.out format chose to not use the data within the section for storing part of the offset; all the offset is kept within the reloc. Anything in the data should be ignored.

```
save %sp,-112,%sp
sethi %hi(_foo+0x12345678),%g2
ldsb [%g2+%lo(_foo+0x12345678)],%i0
ret
restore
```

Both relocs contain a pointer to `foo`, and the offsets contain junk.

```
RELOCATION RECORDS FOR [.text]:
offset   type      value
00000004 HI22      _foo+0x12345678
00000008 LO10      _foo+0x12345678

00000000 9de3bf90     ; save %sp,-112,%sp
00000004 05000000     ; sethi %hi(_foo+0),%g2
00000008 f048a000     ; ldsb [%g2+%lo(_foo+0)],%i0
0000000c 81c7e008     ; ret
00000010 81e80000     ; restore
```

- `howto`

The `howto` field can be imagined as a relocation instruction. It is a pointer to a structure which contains information on what to do with all of the other information in the reloc record and data section. A back end would normally have a relocation instruction set and turn relocations into pointers to the correct structure on input - but it would be possible to create each howto field on demand.

2.10.1.1 `enum complain_overflow`

Indicates what sort of overflow checking should be done when performing a relocation.

```
enum complain_overflow
{
  /* Do not complain on overflow.  */
  complain_overflow_dont,
```

```
  /* Complain if the value overflows when considered as a signed
     number one bit larger than the field.  ie. A bitfield of N bits
     is allowed to represent -2**n to 2**n-1.  */
  complain_overflow_bitfield,

  /* Complain if the value overflows when considered as a signed
     number.  */
  complain_overflow_signed,

  /* Complain if the value overflows when considered as an
     unsigned number.  */
  complain_overflow_unsigned
};
```

2.10.1.2 reloc_howto_type

The `reloc_howto_type` is a structure which contains all the information that libbfd needs to know to tie up a back end's data.

```
struct reloc_howto_struct
{
  /* The type field has mainly a documentary use - the back end can
     do what it wants with it, though normally the back end's idea of
     an external reloc number is stored in this field.  */
  unsigned int type;

  /* The encoded size of the item to be relocated.  This is *not* a
     power-of-two measure.  Use bfd_get_reloc_size to find the size
     of the item in bytes.  */
  unsigned int size:3;

  /* The number of bits in the field to be relocated.  This is used
     when doing overflow checking.  */
  unsigned int bitsize:7;

  /* The value the final relocation is shifted right by.  This drops
     unwanted data from the relocation.  */
  unsigned int rightshift:6;

  /* The bit position of the reloc value in the destination.
     The relocated value is left shifted by this amount.  */
  unsigned int bitpos:6;

  /* What type of overflow error should be checked for when
     relocating.  */
  ENUM_BITFIELD (complain_overflow) complain_on_overflow:2;
```

```
/* The relocation value should be negated before applying.  */
unsigned int negate:1;

/* The relocation is relative to the item being relocated.  */
unsigned int pc_relative:1;

/* Some formats record a relocation addend in the section contents
   rather than with the relocation.  For ELF formats this is the
   distinction between USE_REL and USE_RELA (though the code checks
   for USE_REL == 1/0).  The value of this field is TRUE if the
   addend is recorded with the section contents; when performing a
   partial link (ld -r) the section contents (the data) will be
   modified.  The value of this field is FALSE if addends are
   recorded with the relocation (in arelent.addend); when performing
   a partial link the relocation will be modified.
   All relocations for all ELF USE_RELA targets should set this field
   to FALSE (values of TRUE should be looked on with suspicion).
   However, the converse is not true: not all relocations of all ELF
   USE_REL targets set this field to TRUE.  Why this is so is peculiar
   to each particular target.  For relocs that aren't used in partial
   links (e.g. GOT stuff) it doesn't matter what this is set to.  */
unsigned int partial_inplace:1;

/* When some formats create PC relative instructions, they leave
   the value of the pc of the place being relocated in the offset
   slot of the instruction, so that a PC relative relocation can
   be made just by adding in an ordinary offset (e.g., sun3 a.out).
   Some formats leave the displacement part of an instruction
   empty (e.g., ELF); this flag signals the fact.  */
unsigned int pcrel_offset:1;

/* src_mask selects the part of the instruction (or data) to be used
   in the relocation sum.  If the target relocations don't have an
   addend in the reloc, eg. ELF USE_REL, src_mask will normally equal
   dst_mask to extract the addend from the section contents.  If
   relocations do have an addend in the reloc, eg. ELF USE_RELA, this
   field should normally be zero.  Non-zero values for ELF USE_RELA
   targets should be viewed with suspicion as normally the value in
   the dst_mask part of the section contents should be ignored.  */
bfd_vma src_mask;

/* dst_mask selects which parts of the instruction (or data) are
   replaced with a relocated value.  */
bfd_vma dst_mask;

/* If this field is non null, then the supplied function is
   called rather than the normal function.  This allows really
```

```
     strange relocation methods to be accommodated.  */
  bfd_reloc_status_type (*special_function)
    (bfd *, arelent *, struct bfd_symbol *, void *, asection *,
     bfd *, char **);

  /* The textual name of the relocation type.  */
  const char *name;
};
```

2.10.1.3 The HOWTO Macro

Description
The HOWTO macro fills in a reloc_howto_type (a typedef for const struct reloc_howto_struct).

```
#define HOWTO(type, right, size, bits, pcrel, left, ovf, func, name,   \
              inplace, src_mask, dst_mask, pcrel_off)                  \
  { (unsigned) type, size < 0 ? -size : size, bits, right, left, ovf,  \
    size < 0, pcrel, inplace, pcrel_off, src_mask, dst_mask, func, name }
```

Description
This is used to fill in an empty howto entry in an array.

```
#define EMPTY_HOWTO(C) \
  HOWTO ((C), 0, 0, 0, false, 0, complain_overflow_dont, NULL, \
         NULL, false, 0, 0, false)
```

2.10.1.4 bfd_get_reloc_size

Synopsis

```
unsigned int bfd_get_reloc_size (reloc_howto_type *);
```

Description
For a reloc_howto_type that operates on a fixed number of bytes, this returns the number of bytes operated on.

2.10.1.5 arelent_chain

Description
How relocs are tied together in an `asection`:

```
typedef struct relent_chain
{
  arelent relent;
  struct relent_chain *next;
}
arelent_chain;
```

2.10.1.6 bfd_check_overflow

Synopsis

```
bfd_reloc_status_type bfd_check_overflow
   (enum complain_overflow how,
    unsigned int bitsize,
    unsigned int rightshift,
    unsigned int addrsize,
    bfd_vma relocation);
```

Description
Perform overflow checking on *relocation* which has *bitsize* significant bits and will be shifted right by *rightshift* bits, on a machine with addresses containing *addrsize* significant bits. The result is either of `bfd_reloc_ok` or `bfd_reloc_overflow`.

2.10.1.7 `bfd_reloc_offset_in_range`

Synopsis

```
bool bfd_reloc_offset_in_range
   (reloc_howto_type *howto,
    bfd *abfd,
    asection *section,
    bfd_size_type offset);
```

Description
Returns TRUE if the reloc described by *HOWTO* can be applied at *OFFSET* octets in *SECTION*.

2.10.1.8 `bfd_perform_relocation`

Synopsis

```
bfd_reloc_status_type bfd_perform_relocation
   (bfd *abfd,
    arelent *reloc_entry,
    void *data,
    asection *input_section,
    bfd *output_bfd,
    char **error_message);
```

Description
If *output_bfd* is supplied to this function, the generated image will be relocatable; the relocations are copied to the output file after they have been changed to reflect the new state of the world. There are two ways of reflecting the results of partial linkage in an output file: by modifying the output data in place, and by modifying the relocation record. Some native formats (e.g., basic a.out and basic coff) have no way of specifying an addend in the relocation type, so the addend has to go in the output data. This is no big deal since in these formats the output data slot will always be big enough for the addend. Complex reloc types with addends were invented to solve just this problem. The *error_message* argument is set to an error message if this return `bfd_reloc_dangerous`.

2.10.1.9 `bfd_install_relocation`

Synopsis

```
bfd_reloc_status_type bfd_install_relocation
```

```
(bfd *abfd,
 arelent *reloc_entry,
 void *data, bfd_vma data_start,
 asection *input_section,
 char **error_message);
```

Description
This looks remarkably like `bfd_perform_relocation`, except it does not expect that the section contents have been filled in. I.e., it's suitable for use when creating, rather than applying a relocation.

For now, this function should be considered reserved for the assembler.

2.10.2 The howto manager

When an application wants to create a relocation, but doesn't know what the target machine might call it, it can find out by using this bit of code.

2.10.2.1 `bfd_reloc_code_type`

Description
The insides of a reloc code. The idea is that, eventually, there will be one enumerator for every type of relocation we ever do. Pass one of these values to `bfd_reloc_type_lookup`, and it'll return a howto pointer.

This does mean that the application must determine the correct enumerator value; you can't get a howto pointer from a random set of attributes.

Here are the possible values for `enum bfd_reloc_code_real`:

`BFD_RELOC_64`
`BFD_RELOC_32`
`BFD_RELOC_26`
`BFD_RELOC_24`
`BFD_RELOC_16`
`BFD_RELOC_14`
`BFD_RELOC_8`
: Basic absolute relocations of N bits.

`BFD_RELOC_64_PCREL`
`BFD_RELOC_32_PCREL`
`BFD_RELOC_24_PCREL`
`BFD_RELOC_16_PCREL`
`BFD_RELOC_12_PCREL`
`BFD_RELOC_8_PCREL`
: PC-relative relocations. Sometimes these are relative to the address of the relocation itself; sometimes they are relative to the start of the section containing the relocation. It depends on the specific target.

`BFD_RELOC_32_SECREL`
: Section relative relocations. Some targets need this for DWARF2.

BFD_RELOC_32_GOT_PCREL
BFD_RELOC_16_GOT_PCREL
BFD_RELOC_8_GOT_PCREL
BFD_RELOC_32_GOTOFF
BFD_RELOC_16_GOTOFF
BFD_RELOC_LO16_GOTOFF
BFD_RELOC_HI16_GOTOFF
BFD_RELOC_HI16_S_GOTOFF
BFD_RELOC_8_GOTOFF
BFD_RELOC_64_PLT_PCREL
BFD_RELOC_32_PLT_PCREL
BFD_RELOC_24_PLT_PCREL
BFD_RELOC_16_PLT_PCREL
BFD_RELOC_8_PLT_PCREL
BFD_RELOC_64_PLTOFF
BFD_RELOC_32_PLTOFF
BFD_RELOC_16_PLTOFF
BFD_RELOC_LO16_PLTOFF
BFD_RELOC_HI16_PLTOFF
BFD_RELOC_HI16_S_PLTOFF
BFD_RELOC_8_PLTOFF
For ELF.

BFD_RELOC_SIZE32
BFD_RELOC_SIZE64
Size relocations.

BFD_RELOC_68K_GLOB_DAT
BFD_RELOC_68K_JMP_SLOT
BFD_RELOC_68K_RELATIVE
BFD_RELOC_68K_TLS_GD32
BFD_RELOC_68K_TLS_GD16
BFD_RELOC_68K_TLS_GD8
BFD_RELOC_68K_TLS_LDM32
BFD_RELOC_68K_TLS_LDM16
BFD_RELOC_68K_TLS_LDM8
BFD_RELOC_68K_TLS_LDO32
BFD_RELOC_68K_TLS_LDO16
BFD_RELOC_68K_TLS_LDO8
BFD_RELOC_68K_TLS_IE32
BFD_RELOC_68K_TLS_IE16
BFD_RELOC_68K_TLS_IE8
BFD_RELOC_68K_TLS_LE32
BFD_RELOC_68K_TLS_LE16
BFD_RELOC_68K_TLS_LE8
Relocations used by 68K ELF.

`BFD_RELOC_32_BASEREL`
`BFD_RELOC_16_BASEREL`
`BFD_RELOC_LO16_BASEREL`
`BFD_RELOC_HI16_BASEREL`
`BFD_RELOC_HI16_S_BASEREL`
`BFD_RELOC_8_BASEREL`
`BFD_RELOC_RVA`
Linkage-table relative.

`BFD_RELOC_8_FFnn`
Absolute 8-bit relocation, but used to form an address like 0xFFnn.

`BFD_RELOC_32_PCREL_S2`
`BFD_RELOC_16_PCREL_S2`
`BFD_RELOC_23_PCREL_S2`
These PC-relative relocations are stored as word displacements – i.e., byte displacements shifted right two bits. The 30-bit word displacement (<<32_PCREL_S2>> – 32 bits, shifted 2) is used on the SPARC. (SPARC tools generally refer to this as <<WDISP30>>.) The signed 16-bit displacement is used on the MIPS, and the 23-bit displacement is used on the Alpha.

`BFD_RELOC_HI22`
`BFD_RELOC_LO10`
High 22 bits and low 10 bits of 32-bit value, placed into lower bits of the target word. These are used on the SPARC.

`BFD_RELOC_GPREL16`
`BFD_RELOC_GPREL32`
For systems that allocate a Global Pointer register, these are displacements off that register. These relocation types are handled specially, because the value the register will have is decided relatively late.

`BFD_RELOC_NONE`
`BFD_RELOC_SPARC_WDISP22`
`BFD_RELOC_SPARC22`
`BFD_RELOC_SPARC13`
`BFD_RELOC_SPARC_GOT10`
`BFD_RELOC_SPARC_GOT13`
`BFD_RELOC_SPARC_GOT22`
`BFD_RELOC_SPARC_PC10`
`BFD_RELOC_SPARC_PC22`
`BFD_RELOC_SPARC_WPLT30`
`BFD_RELOC_SPARC_COPY`
`BFD_RELOC_SPARC_GLOB_DAT`
`BFD_RELOC_SPARC_JMP_SLOT`
`BFD_RELOC_SPARC_RELATIVE`
`BFD_RELOC_SPARC_UA16`
`BFD_RELOC_SPARC_UA32`
`BFD_RELOC_SPARC_UA64`

BFD_RELOC_SPARC_GOTDATA_HIX22
BFD_RELOC_SPARC_GOTDATA_LOX10
BFD_RELOC_SPARC_GOTDATA_OP_HIX22
BFD_RELOC_SPARC_GOTDATA_OP_LOX10
BFD_RELOC_SPARC_GOTDATA_OP
BFD_RELOC_SPARC_JMP_IREL
BFD_RELOC_SPARC_IRELATIVE

SPARC ELF relocations. There is probably some overlap with other relocation types already defined.

BFD_RELOC_SPARC_BASE13
BFD_RELOC_SPARC_BASE22

I think these are specific to SPARC a.out (e.g., Sun 4).

BFD_RELOC_SPARC_64
BFD_RELOC_SPARC_10
BFD_RELOC_SPARC_11
BFD_RELOC_SPARC_OLO10
BFD_RELOC_SPARC_HH22
BFD_RELOC_SPARC_HM10
BFD_RELOC_SPARC_LM22
BFD_RELOC_SPARC_PC_HH22
BFD_RELOC_SPARC_PC_HM10
BFD_RELOC_SPARC_PC_LM22
BFD_RELOC_SPARC_WDISP16
BFD_RELOC_SPARC_WDISP19
BFD_RELOC_SPARC_7
BFD_RELOC_SPARC_6
BFD_RELOC_SPARC_5
BFD_RELOC_SPARC_DISP64
BFD_RELOC_SPARC_PLT32
BFD_RELOC_SPARC_PLT64
BFD_RELOC_SPARC_HIX22
BFD_RELOC_SPARC_LOX10
BFD_RELOC_SPARC_H44
BFD_RELOC_SPARC_M44
BFD_RELOC_SPARC_L44
BFD_RELOC_SPARC_REGISTER
BFD_RELOC_SPARC_H34
BFD_RELOC_SPARC_SIZE32
BFD_RELOC_SPARC_SIZE64
BFD_RELOC_SPARC_WDISP10

SPARC64 relocations

BFD_RELOC_SPARC_REV32

SPARC little endian relocation

BFD_RELOC_SPARC_TLS_GD_HI22
BFD_RELOC_SPARC_TLS_GD_LO10

BFD_RELOC_SPARC_TLS_GD_ADD
BFD_RELOC_SPARC_TLS_GD_CALL
BFD_RELOC_SPARC_TLS_LDM_HI22
BFD_RELOC_SPARC_TLS_LDM_LO10
BFD_RELOC_SPARC_TLS_LDM_ADD
BFD_RELOC_SPARC_TLS_LDM_CALL
BFD_RELOC_SPARC_TLS_LDO_HIX22
BFD_RELOC_SPARC_TLS_LDO_LOX10
BFD_RELOC_SPARC_TLS_LDO_ADD
BFD_RELOC_SPARC_TLS_IE_HI22
BFD_RELOC_SPARC_TLS_IE_LO10
BFD_RELOC_SPARC_TLS_IE_LD
BFD_RELOC_SPARC_TLS_IE_LDX
BFD_RELOC_SPARC_TLS_IE_ADD
BFD_RELOC_SPARC_TLS_LE_HIX22
BFD_RELOC_SPARC_TLS_LE_LOX10
BFD_RELOC_SPARC_TLS_DTPMOD32
BFD_RELOC_SPARC_TLS_DTPMOD64
BFD_RELOC_SPARC_TLS_DTPOFF32
BFD_RELOC_SPARC_TLS_DTPOFF64
BFD_RELOC_SPARC_TLS_TPOFF32
BFD_RELOC_SPARC_TLS_TPOFF64
: SPARC TLS relocations

BFD_RELOC_SPU_IMM7
BFD_RELOC_SPU_IMM8
BFD_RELOC_SPU_IMM10
BFD_RELOC_SPU_IMM10W
BFD_RELOC_SPU_IMM16
BFD_RELOC_SPU_IMM16W
BFD_RELOC_SPU_IMM18
BFD_RELOC_SPU_PCREL9a
BFD_RELOC_SPU_PCREL9b
BFD_RELOC_SPU_PCREL16
BFD_RELOC_SPU_LO16
BFD_RELOC_SPU_HI16
BFD_RELOC_SPU_PPU32
BFD_RELOC_SPU_PPU64
BFD_RELOC_SPU_ADD_PIC
: SPU Relocations.

BFD_RELOC_ALPHA_GPDISP_HI16
: Alpha ECOFF and ELF relocations. Some of these treat the symbol or "addend" in some special way. For GPDISP_HI16 ("gpdisp") relocations, the symbol is ignored when writing; when reading, it will be the absolute section symbol. The addend is the displacement in bytes of the "lda" instruction from the "ldah" instruction (which is at the address of this reloc).

`BFD_RELOC_ALPHA_GPDISP_LO16`
For GPDISP_LO16 ("ignore") relocations, the symbol is handled as with GPDISP_HI16 relocs. The addend is ignored when writing the relocations out, and is filled in with the file's GP value on reading, for convenience.

`BFD_RELOC_ALPHA_GPDISP`
The ELF GPDISP relocation is exactly the same as the GPDISP_HI16 relocation except that there is no accompanying GPDISP_LO16 relocation.

`BFD_RELOC_ALPHA_LITERAL`
`BFD_RELOC_ALPHA_ELF_LITERAL`
`BFD_RELOC_ALPHA_LITUSE`
The Alpha LITERAL/LITUSE relocs are produced by a symbol reference; the assembler turns it into a LDQ instruction to load the address of the symbol, and then fills in a register in the real instruction.

The LITERAL reloc, at the LDQ instruction, refers to the .lita section symbol. The addend is ignored when writing, but is filled in with the file's GP value on reading, for convenience, as with the GPDISP_LO16 reloc.

The ELF_LITERAL reloc is somewhere between 16_GOTOFF and GPDISP_LO16. It should refer to the symbol to be referenced, as with 16_GOTOFF, but it generates output not based on the position within the .got section, but relative to the GP value chosen for the file during the final link stage.

The LITUSE reloc, on the instruction using the loaded address, gives information to the linker that it might be able to use to optimize away some literal section references. The symbol is ignored (read as the absolute section symbol), and the "addend" indicates the type of instruction using the register: 1 - "memory" fmt insn 2 - byte-manipulation (byte offset reg) 3 - jsr (target of branch)

`BFD_RELOC_ALPHA_HINT`
The HINT relocation indicates a value that should be filled into the "hint" field of a jmp/jsr/ret instruction, for possible branch- prediction logic which may be provided on some processors.

`BFD_RELOC_ALPHA_LINKAGE`
The LINKAGE relocation outputs a linkage pair in the object file, which is filled by the linker.

`BFD_RELOC_ALPHA_CODEADDR`
The CODEADDR relocation outputs a STO_CA in the object file, which is filled by the linker.

`BFD_RELOC_ALPHA_GPREL_HI16`
`BFD_RELOC_ALPHA_GPREL_LO16`
The GPREL_HI/LO relocations together form a 32-bit offset from the GP register.

`BFD_RELOC_ALPHA_BRSGP`
Like BFD_RELOC_23_PCREL_S2, except that the source and target must share a common GP, and the target address is adjusted for STO_ALPHA_STD_GPLOAD.

`BFD_RELOC_ALPHA_NOP`
The NOP relocation outputs a NOP if the longword displacement between two procedure entry points is < 2^21.

`BFD_RELOC_ALPHA_BSR`
The BSR relocation outputs a BSR if the longword displacement between two procedure entry points is < 2^21.

`BFD_RELOC_ALPHA_LDA`
The LDA relocation outputs a LDA if the longword displacement between two procedure entry points is < 2^16.

`BFD_RELOC_ALPHA_BOH`
The BOH relocation outputs a BSR if the longword displacement between two procedure entry points is < 2^21, or else a hint.

`BFD_RELOC_ALPHA_TLSGD`
`BFD_RELOC_ALPHA_TLSLDM`
`BFD_RELOC_ALPHA_DTPMOD64`
`BFD_RELOC_ALPHA_GOTDTPREL16`
`BFD_RELOC_ALPHA_DTPREL64`
`BFD_RELOC_ALPHA_DTPREL_HI16`
`BFD_RELOC_ALPHA_DTPREL_LO16`
`BFD_RELOC_ALPHA_DTPREL16`
`BFD_RELOC_ALPHA_GOTTPREL16`
`BFD_RELOC_ALPHA_TPREL64`
`BFD_RELOC_ALPHA_TPREL_HI16`
`BFD_RELOC_ALPHA_TPREL_LO16`
`BFD_RELOC_ALPHA_TPREL16`
Alpha thread-local storage relocations.

`BFD_RELOC_MIPS_JMP`
`BFD_RELOC_MICROMIPS_JMP`
The MIPS jump instruction.

`BFD_RELOC_MIPS16_JMP`
The MIPS16 jump instruction.

`BFD_RELOC_MIPS16_GPREL`
MIPS16 GP relative reloc.

`BFD_RELOC_HI16`
High 16 bits of 32-bit value; simple reloc.

`BFD_RELOC_HI16_S`
High 16 bits of 32-bit value but the low 16 bits will be sign extended and added to form the final result. If the low 16 bits form a negative number, we need to add one to the high value to compensate for the borrow when the low bits are added.

`BFD_RELOC_LO16`
Low 16 bits.

`BFD_RELOC_HI16_PCREL`
High 16 bits of 32-bit pc-relative value

`BFD_RELOC_HI16_S_PCREL`
High 16 bits of 32-bit pc-relative value, adjusted

`BFD_RELOC_LO16_PCREL`
Low 16 bits of pc-relative value

`BFD_RELOC_MIPS16_GOT16`
`BFD_RELOC_MIPS16_CALL16`
Equivalent of BFD_RELOC_MIPS_*, but with the MIPS16 layout of 16-bit immediate fields

`BFD_RELOC_MIPS16_HI16`
MIPS16 high 16 bits of 32-bit value.

`BFD_RELOC_MIPS16_HI16_S`
MIPS16 high 16 bits of 32-bit value but the low 16 bits will be sign extended and added to form the final result. If the low 16 bits form a negative number, we need to add one to the high value to compensate for the borrow when the low bits are added.

`BFD_RELOC_MIPS16_LO16`
MIPS16 low 16 bits.

`BFD_RELOC_MIPS16_TLS_GD`
`BFD_RELOC_MIPS16_TLS_LDM`
`BFD_RELOC_MIPS16_TLS_DTPREL_HI16`
`BFD_RELOC_MIPS16_TLS_DTPREL_LO16`
`BFD_RELOC_MIPS16_TLS_GOTTPREL`
`BFD_RELOC_MIPS16_TLS_TPREL_HI16`
`BFD_RELOC_MIPS16_TLS_TPREL_LO16`
MIPS16 TLS relocations

`BFD_RELOC_MIPS_LITERAL`
`BFD_RELOC_MICROMIPS_LITERAL`
Relocation against a MIPS literal section.

`BFD_RELOC_MICROMIPS_7_PCREL_S1`
`BFD_RELOC_MICROMIPS_10_PCREL_S1`
`BFD_RELOC_MICROMIPS_16_PCREL_S1`
microMIPS PC-relative relocations.

`BFD_RELOC_MIPS16_16_PCREL_S1`
MIPS16 PC-relative relocation.

`BFD_RELOC_MIPS_21_PCREL_S2`
`BFD_RELOC_MIPS_26_PCREL_S2`
`BFD_RELOC_MIPS_18_PCREL_S3`
`BFD_RELOC_MIPS_19_PCREL_S2`
MIPS PC-relative relocations.

BFD_RELOC_MICROMIPS_GPREL16
BFD_RELOC_MICROMIPS_HI16
BFD_RELOC_MICROMIPS_HI16_S
BFD_RELOC_MICROMIPS_LO16
microMIPS versions of generic BFD relocs.

BFD_RELOC_MIPS_GOT16
BFD_RELOC_MICROMIPS_GOT16
BFD_RELOC_MIPS_CALL16
BFD_RELOC_MICROMIPS_CALL16
BFD_RELOC_MIPS_GOT_HI16
BFD_RELOC_MICROMIPS_GOT_HI16
BFD_RELOC_MIPS_GOT_LO16
BFD_RELOC_MICROMIPS_GOT_LO16
BFD_RELOC_MIPS_CALL_HI16
BFD_RELOC_MICROMIPS_CALL_HI16
BFD_RELOC_MIPS_CALL_LO16
BFD_RELOC_MICROMIPS_CALL_LO16
BFD_RELOC_MIPS_SUB
BFD_RELOC_MICROMIPS_SUB
BFD_RELOC_MIPS_GOT_PAGE
BFD_RELOC_MICROMIPS_GOT_PAGE
BFD_RELOC_MIPS_GOT_OFST
BFD_RELOC_MICROMIPS_GOT_OFST
BFD_RELOC_MIPS_GOT_DISP
BFD_RELOC_MICROMIPS_GOT_DISP
BFD_RELOC_MIPS_SHIFT5
BFD_RELOC_MIPS_SHIFT6
BFD_RELOC_MIPS_INSERT_A
BFD_RELOC_MIPS_INSERT_B
BFD_RELOC_MIPS_DELETE
BFD_RELOC_MIPS_HIGHEST
BFD_RELOC_MICROMIPS_HIGHEST
BFD_RELOC_MIPS_HIGHER
BFD_RELOC_MICROMIPS_HIGHER
BFD_RELOC_MIPS_SCN_DISP
BFD_RELOC_MICROMIPS_SCN_DISP
BFD_RELOC_MIPS_REL16
BFD_RELOC_MIPS_RELGOT
BFD_RELOC_MIPS_JALR
BFD_RELOC_MICROMIPS_JALR
BFD_RELOC_MIPS_TLS_DTPMOD32
BFD_RELOC_MIPS_TLS_DTPREL32
BFD_RELOC_MIPS_TLS_DTPMOD64
BFD_RELOC_MIPS_TLS_DTPREL64
BFD_RELOC_MIPS_TLS_GD
BFD_RELOC_MICROMIPS_TLS_GD

BFD_RELOC_MIPS_TLS_LDM
BFD_RELOC_MICROMIPS_TLS_LDM
BFD_RELOC_MIPS_TLS_DTPREL_HI16
BFD_RELOC_MICROMIPS_TLS_DTPREL_HI16
BFD_RELOC_MIPS_TLS_DTPREL_LO16
BFD_RELOC_MICROMIPS_TLS_DTPREL_LO16
BFD_RELOC_MIPS_TLS_GOTTPREL
BFD_RELOC_MICROMIPS_TLS_GOTTPREL
BFD_RELOC_MIPS_TLS_TPREL32
BFD_RELOC_MIPS_TLS_TPREL64
BFD_RELOC_MIPS_TLS_TPREL_HI16
BFD_RELOC_MICROMIPS_TLS_TPREL_HI16
BFD_RELOC_MIPS_TLS_TPREL_LO16
BFD_RELOC_MICROMIPS_TLS_TPREL_LO16
BFD_RELOC_MIPS_EH
MIPS ELF relocations.

BFD_RELOC_MIPS_COPY
BFD_RELOC_MIPS_JUMP_SLOT
MIPS ELF relocations (VxWorks and PLT extensions).

BFD_RELOC_MOXIE_10_PCREL
Moxie ELF relocations.

BFD_RELOC_FT32_10
BFD_RELOC_FT32_20
BFD_RELOC_FT32_17
BFD_RELOC_FT32_18
BFD_RELOC_FT32_RELAX
BFD_RELOC_FT32_SC0
BFD_RELOC_FT32_SC1
BFD_RELOC_FT32_15
BFD_RELOC_FT32_DIFF32
FT32 ELF relocations.

BFD_RELOC_FRV_LABEL16
BFD_RELOC_FRV_LABEL24
BFD_RELOC_FRV_LO16
BFD_RELOC_FRV_HI16
BFD_RELOC_FRV_GPREL12
BFD_RELOC_FRV_GPRELU12
BFD_RELOC_FRV_GPREL32
BFD_RELOC_FRV_GPRELHI
BFD_RELOC_FRV_GPRELLO
BFD_RELOC_FRV_GOT12
BFD_RELOC_FRV_GOTHI
BFD_RELOC_FRV_GOTLO
BFD_RELOC_FRV_FUNCDESC

BFD_RELOC_FRV_FUNCDESC_GOT12
BFD_RELOC_FRV_FUNCDESC_GOTHI
BFD_RELOC_FRV_FUNCDESC_GOTLO
BFD_RELOC_FRV_FUNCDESC_VALUE
BFD_RELOC_FRV_FUNCDESC_GOTOFF12
BFD_RELOC_FRV_FUNCDESC_GOTOFFHI
BFD_RELOC_FRV_FUNCDESC_GOTOFFLO
BFD_RELOC_FRV_GOTOFF12
BFD_RELOC_FRV_GOTOFFHI
BFD_RELOC_FRV_GOTOFFLO
BFD_RELOC_FRV_GETTLSOFF
BFD_RELOC_FRV_TLSDESC_VALUE
BFD_RELOC_FRV_GOTTLSDESC12
BFD_RELOC_FRV_GOTTLSDESCHI
BFD_RELOC_FRV_GOTTLSDESCLO
BFD_RELOC_FRV_TLSMOFF12
BFD_RELOC_FRV_TLSMOFFHI
BFD_RELOC_FRV_TLSMOFFLO
BFD_RELOC_FRV_GOTTLSOFF12
BFD_RELOC_FRV_GOTTLSOFFHI
BFD_RELOC_FRV_GOTTLSOFFLO
BFD_RELOC_FRV_TLSOFF
BFD_RELOC_FRV_TLSDESC_RELAX
BFD_RELOC_FRV_GETTLSOFF_RELAX
BFD_RELOC_FRV_TLSOFF_RELAX
BFD_RELOC_FRV_TLSMOFF
: Fujitsu Frv Relocations.

BFD_RELOC_MN10300_GOTOFF24
: This is a 24bit GOT-relative reloc for the mn10300.

BFD_RELOC_MN10300_GOT32
: This is a 32bit GOT-relative reloc for the mn10300, offset by two bytes in the instruction.

BFD_RELOC_MN10300_GOT24
: This is a 24bit GOT-relative reloc for the mn10300, offset by two bytes in the instruction.

BFD_RELOC_MN10300_GOT16
: This is a 16bit GOT-relative reloc for the mn10300, offset by two bytes in the instruction.

BFD_RELOC_MN10300_COPY
: Copy symbol at runtime.

BFD_RELOC_MN10300_GLOB_DAT
: Create GOT entry.

`BFD_RELOC_MN10300_JMP_SLOT`
Create PLT entry.

`BFD_RELOC_MN10300_RELATIVE`
Adjust by program base.

`BFD_RELOC_MN10300_SYM_DIFF`
Together with another reloc targeted at the same location, allows for a value that is the difference of two symbols in the same section.

`BFD_RELOC_MN10300_ALIGN`
The addend of this reloc is an alignment power that must be honoured at the offset's location, regardless of linker relaxation.

`BFD_RELOC_MN10300_TLS_GD`
`BFD_RELOC_MN10300_TLS_LD`
`BFD_RELOC_MN10300_TLS_LDO`
`BFD_RELOC_MN10300_TLS_GOTIE`
`BFD_RELOC_MN10300_TLS_IE`
`BFD_RELOC_MN10300_TLS_LE`
`BFD_RELOC_MN10300_TLS_DTPMOD`
`BFD_RELOC_MN10300_TLS_DTPOFF`
`BFD_RELOC_MN10300_TLS_TPOFF`
Various TLS-related relocations.

`BFD_RELOC_MN10300_32_PCREL`
This is a 32bit pcrel reloc for the mn10300, offset by two bytes in the instruction.

`BFD_RELOC_MN10300_16_PCREL`
This is a 16bit pcrel reloc for the mn10300, offset by two bytes in the instruction.

`BFD_RELOC_386_GOT32`
`BFD_RELOC_386_PLT32`
`BFD_RELOC_386_COPY`
`BFD_RELOC_386_GLOB_DAT`
`BFD_RELOC_386_JUMP_SLOT`
`BFD_RELOC_386_RELATIVE`
`BFD_RELOC_386_GOTOFF`
`BFD_RELOC_386_GOTPC`
`BFD_RELOC_386_TLS_TPOFF`
`BFD_RELOC_386_TLS_IE`
`BFD_RELOC_386_TLS_GOTIE`
`BFD_RELOC_386_TLS_LE`
`BFD_RELOC_386_TLS_GD`
`BFD_RELOC_386_TLS_LDM`
`BFD_RELOC_386_TLS_LDO_32`
`BFD_RELOC_386_TLS_IE_32`
`BFD_RELOC_386_TLS_LE_32`
`BFD_RELOC_386_TLS_DTPMOD32`

BFD_RELOC_386_TLS_DTPOFF32
BFD_RELOC_386_TLS_TPOFF32
BFD_RELOC_386_TLS_GOTDESC
BFD_RELOC_386_TLS_DESC_CALL
BFD_RELOC_386_TLS_DESC
BFD_RELOC_386_IRELATIVE
BFD_RELOC_386_GOT32X
i386/elf relocations

BFD_RELOC_X86_64_GOT32
BFD_RELOC_X86_64_PLT32
BFD_RELOC_X86_64_COPY
BFD_RELOC_X86_64_GLOB_DAT
BFD_RELOC_X86_64_JUMP_SLOT
BFD_RELOC_X86_64_RELATIVE
BFD_RELOC_X86_64_GOTPCREL
BFD_RELOC_X86_64_32S
BFD_RELOC_X86_64_DTPMOD64
BFD_RELOC_X86_64_DTPOFF64
BFD_RELOC_X86_64_TPOFF64
BFD_RELOC_X86_64_TLSGD
BFD_RELOC_X86_64_TLSLD
BFD_RELOC_X86_64_DTPOFF32
BFD_RELOC_X86_64_GOTTPOFF
BFD_RELOC_X86_64_TPOFF32
BFD_RELOC_X86_64_GOTOFF64
BFD_RELOC_X86_64_GOTPC32
BFD_RELOC_X86_64_GOT64
BFD_RELOC_X86_64_GOTPCREL64
BFD_RELOC_X86_64_GOTPC64
BFD_RELOC_X86_64_GOTPLT64
BFD_RELOC_X86_64_PLTOFF64
BFD_RELOC_X86_64_GOTPC32_TLSDESC
BFD_RELOC_X86_64_TLSDESC_CALL
BFD_RELOC_X86_64_TLSDESC
BFD_RELOC_X86_64_IRELATIVE
BFD_RELOC_X86_64_PC32_BND
BFD_RELOC_X86_64_PLT32_BND
BFD_RELOC_X86_64_GOTPCRELX
BFD_RELOC_X86_64_REX_GOTPCRELX
x86-64/elf relocations

BFD_RELOC_NS32K_IMM_8
BFD_RELOC_NS32K_IMM_16
BFD_RELOC_NS32K_IMM_32
BFD_RELOC_NS32K_IMM_8_PCREL
BFD_RELOC_NS32K_IMM_16_PCREL
BFD_RELOC_NS32K_IMM_32_PCREL

BFD_RELOC_NS32K_DISP_8
BFD_RELOC_NS32K_DISP_16
BFD_RELOC_NS32K_DISP_32
BFD_RELOC_NS32K_DISP_8_PCREL
BFD_RELOC_NS32K_DISP_16_PCREL
BFD_RELOC_NS32K_DISP_32_PCREL
ns32k relocations

BFD_RELOC_PDP11_DISP_8_PCREL
BFD_RELOC_PDP11_DISP_6_PCREL
PDP11 relocations

BFD_RELOC_PJ_CODE_HI16
BFD_RELOC_PJ_CODE_LO16
BFD_RELOC_PJ_CODE_DIR16
BFD_RELOC_PJ_CODE_DIR32
BFD_RELOC_PJ_CODE_REL16
BFD_RELOC_PJ_CODE_REL32
Picojava relocs. Not all of these appear in object files.

BFD_RELOC_PPC_B26
BFD_RELOC_PPC_BA26
BFD_RELOC_PPC_TOC16
BFD_RELOC_PPC_TOC16_LO
BFD_RELOC_PPC_TOC16_HI
BFD_RELOC_PPC_B16
BFD_RELOC_PPC_B16_BRTAKEN
BFD_RELOC_PPC_B16_BRNTAKEN
BFD_RELOC_PPC_BA16
BFD_RELOC_PPC_BA16_BRTAKEN
BFD_RELOC_PPC_BA16_BRNTAKEN
BFD_RELOC_PPC_COPY
BFD_RELOC_PPC_GLOB_DAT
BFD_RELOC_PPC_JMP_SLOT
BFD_RELOC_PPC_RELATIVE
BFD_RELOC_PPC_LOCAL24PC
BFD_RELOC_PPC_EMB_NADDR32
BFD_RELOC_PPC_EMB_NADDR16
BFD_RELOC_PPC_EMB_NADDR16_LO
BFD_RELOC_PPC_EMB_NADDR16_HI
BFD_RELOC_PPC_EMB_NADDR16_HA
BFD_RELOC_PPC_EMB_SDAI16
BFD_RELOC_PPC_EMB_SDA2I16
BFD_RELOC_PPC_EMB_SDA2REL
BFD_RELOC_PPC_EMB_SDA21
BFD_RELOC_PPC_EMB_MRKREF
BFD_RELOC_PPC_EMB_RELSEC16
BFD_RELOC_PPC_EMB_RELST_LO

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BFD_RELOC_PPC_EMB_RELST_HI
BFD_RELOC_PPC_EMB_RELST_HA
BFD_RELOC_PPC_EMB_BIT_FLD
BFD_RELOC_PPC_EMB_RELSDA
BFD_RELOC_PPC_VLE_REL8
BFD_RELOC_PPC_VLE_REL15
BFD_RELOC_PPC_VLE_REL24
BFD_RELOC_PPC_VLE_LO16A
BFD_RELOC_PPC_VLE_LO16D
BFD_RELOC_PPC_VLE_HI16A
BFD_RELOC_PPC_VLE_HI16D
BFD_RELOC_PPC_VLE_HA16A
BFD_RELOC_PPC_VLE_HA16D
BFD_RELOC_PPC_VLE_SDA21
BFD_RELOC_PPC_VLE_SDA21_LO
BFD_RELOC_PPC_VLE_SDAREL_LO16A
BFD_RELOC_PPC_VLE_SDAREL_LO16D
BFD_RELOC_PPC_VLE_SDAREL_HI16A
BFD_RELOC_PPC_VLE_SDAREL_HI16D
BFD_RELOC_PPC_VLE_SDAREL_HA16A
BFD_RELOC_PPC_VLE_SDAREL_HA16D
BFD_RELOC_PPC_16DX_HA
BFD_RELOC_PPC_REL16DX_HA
BFD_RELOC_PPC_NEG
BFD_RELOC_PPC64_HIGHER
BFD_RELOC_PPC64_HIGHER_S
BFD_RELOC_PPC64_HIGHEST
BFD_RELOC_PPC64_HIGHEST_S
BFD_RELOC_PPC64_TOC16_LO
BFD_RELOC_PPC64_TOC16_HI
BFD_RELOC_PPC64_TOC16_HA
BFD_RELOC_PPC64_TOC
BFD_RELOC_PPC64_PLTGOT16
BFD_RELOC_PPC64_PLTGOT16_LO
BFD_RELOC_PPC64_PLTGOT16_HI
BFD_RELOC_PPC64_PLTGOT16_HA
BFD_RELOC_PPC64_ADDR16_DS
BFD_RELOC_PPC64_ADDR16_LO_DS
BFD_RELOC_PPC64_GOT16_DS
BFD_RELOC_PPC64_GOT16_LO_DS
BFD_RELOC_PPC64_PLT16_LO_DS
BFD_RELOC_PPC64_SECTOFF_DS
BFD_RELOC_PPC64_SECTOFF_LO_DS
BFD_RELOC_PPC64_TOC16_DS
BFD_RELOC_PPC64_TOC16_LO_DS
BFD_RELOC_PPC64_PLTGOT16_DS
BFD_RELOC_PPC64_PLTGOT16_LO_DS
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BFD_RELOC_PPC64_ADDR16_HIGH
BFD_RELOC_PPC64_ADDR16_HIGHA
BFD_RELOC_PPC64_REL16_HIGH
BFD_RELOC_PPC64_REL16_HIGHA
BFD_RELOC_PPC64_REL16_HIGHER
BFD_RELOC_PPC64_REL16_HIGHERA
BFD_RELOC_PPC64_REL16_HIGHEST
BFD_RELOC_PPC64_REL16_HIGHESTA
BFD_RELOC_PPC64_ADDR64_LOCAL
BFD_RELOC_PPC64_ENTRY
BFD_RELOC_PPC64_REL24_NOTOC
BFD_RELOC_PPC64_REL24_P9NOTOC
BFD_RELOC_PPC64_D34
BFD_RELOC_PPC64_D34_LO
BFD_RELOC_PPC64_D34_HI30
BFD_RELOC_PPC64_D34_HA30
BFD_RELOC_PPC64_PCREL34
BFD_RELOC_PPC64_GOT_PCREL34
BFD_RELOC_PPC64_PLT_PCREL34
BFD_RELOC_PPC64_ADDR16_HIGHER34
BFD_RELOC_PPC64_ADDR16_HIGHERA34
BFD_RELOC_PPC64_ADDR16_HIGHEST34
BFD_RELOC_PPC64_ADDR16_HIGHESTA34
BFD_RELOC_PPC64_REL16_HIGHER34
BFD_RELOC_PPC64_REL16_HIGHERA34
BFD_RELOC_PPC64_REL16_HIGHEST34
BFD_RELOC_PPC64_REL16_HIGHESTA34
BFD_RELOC_PPC64_D28
BFD_RELOC_PPC64_PCREL28

Power(rs6000) and PowerPC relocations.

BFD_RELOC_PPC_TLS
BFD_RELOC_PPC_TLSGD
BFD_RELOC_PPC_TLSLD
BFD_RELOC_PPC_TLSLE
BFD_RELOC_PPC_TLSIE
BFD_RELOC_PPC_TLSM
BFD_RELOC_PPC_TLSML
BFD_RELOC_PPC_DTPMOD
BFD_RELOC_PPC_TPREL16
BFD_RELOC_PPC_TPREL16_LO
BFD_RELOC_PPC_TPREL16_HI
BFD_RELOC_PPC_TPREL16_HA
BFD_RELOC_PPC_TPREL
BFD_RELOC_PPC_DTPREL16
BFD_RELOC_PPC_DTPREL16_LO
BFD_RELOC_PPC_DTPREL16_HI

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BFD_RELOC_PPC_DTPREL16_HA
BFD_RELOC_PPC_DTPREL
BFD_RELOC_PPC_GOT_TLSGD16
BFD_RELOC_PPC_GOT_TLSGD16_LO
BFD_RELOC_PPC_GOT_TLSGD16_HI
BFD_RELOC_PPC_GOT_TLSGD16_HA
BFD_RELOC_PPC_GOT_TLSLD16
BFD_RELOC_PPC_GOT_TLSLD16_LO
BFD_RELOC_PPC_GOT_TLSLD16_HI
BFD_RELOC_PPC_GOT_TLSLD16_HA
BFD_RELOC_PPC_GOT_TPREL16
BFD_RELOC_PPC_GOT_TPREL16_LO
BFD_RELOC_PPC_GOT_TPREL16_HI
BFD_RELOC_PPC_GOT_TPREL16_HA
BFD_RELOC_PPC_GOT_DTPREL16
BFD_RELOC_PPC_GOT_DTPREL16_LO
BFD_RELOC_PPC_GOT_DTPREL16_HI
BFD_RELOC_PPC_GOT_DTPREL16_HA
BFD_RELOC_PPC64_TLSGD
BFD_RELOC_PPC64_TLSLD
BFD_RELOC_PPC64_TLSLE
BFD_RELOC_PPC64_TLSIE
BFD_RELOC_PPC64_TLSM
BFD_RELOC_PPC64_TLSML
BFD_RELOC_PPC64_TPREL16_DS
BFD_RELOC_PPC64_TPREL16_LO_DS
BFD_RELOC_PPC64_TPREL16_HIGH
BFD_RELOC_PPC64_TPREL16_HIGHA
BFD_RELOC_PPC64_TPREL16_HIGHER
BFD_RELOC_PPC64_TPREL16_HIGHERA
BFD_RELOC_PPC64_TPREL16_HIGHEST
BFD_RELOC_PPC64_TPREL16_HIGHESTA
BFD_RELOC_PPC64_DTPREL16_DS
BFD_RELOC_PPC64_DTPREL16_LO_DS
BFD_RELOC_PPC64_DTPREL16_HIGH
BFD_RELOC_PPC64_DTPREL16_HIGHA
BFD_RELOC_PPC64_DTPREL16_HIGHER
BFD_RELOC_PPC64_DTPREL16_HIGHERA
BFD_RELOC_PPC64_DTPREL16_HIGHEST
BFD_RELOC_PPC64_DTPREL16_HIGHESTA
BFD_RELOC_PPC64_TPREL34
BFD_RELOC_PPC64_DTPREL34
BFD_RELOC_PPC64_GOT_TLSGD_PCREL34
BFD_RELOC_PPC64_GOT_TLSLD_PCREL34
BFD_RELOC_PPC64_GOT_TPREL_PCREL34
BFD_RELOC_PPC64_GOT_DTPREL_PCREL34
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`BFD_RELOC_PPC64_TLS_PCREL`
PowerPC and PowerPC64 thread-local storage relocations.

`BFD_RELOC_I370_D12`
IBM 370/390 relocations

`BFD_RELOC_CTOR`
The type of reloc used to build a constructor table - at the moment probably a 32 bit wide absolute relocation, but the target can choose. It generally does map to one of the other relocation types.

`BFD_RELOC_ARM_PCREL_BRANCH`
ARM 26 bit pc-relative branch. The lowest two bits must be zero and are not stored in the instruction.

`BFD_RELOC_ARM_PCREL_BLX`
ARM 26 bit pc-relative branch. The lowest bit must be zero and is not stored in the instruction. The 2nd lowest bit comes from a 1 bit field in the instruction.

`BFD_RELOC_THUMB_PCREL_BLX`
Thumb 22 bit pc-relative branch. The lowest bit must be zero and is not stored in the instruction. The 2nd lowest bit comes from a 1 bit field in the instruction.

`BFD_RELOC_ARM_PCREL_CALL`
ARM 26-bit pc-relative branch for an unconditional BL or BLX instruction.

`BFD_RELOC_ARM_PCREL_JUMP`
ARM 26-bit pc-relative branch for B or conditional BL instruction.

`BFD_RELOC_THUMB_PCREL_BRANCH5`
ARM 5-bit pc-relative branch for Branch Future instructions.

`BFD_RELOC_THUMB_PCREL_BFCSEL`
ARM 6-bit pc-relative branch for BFCSEL instruction.

`BFD_RELOC_ARM_THUMB_BF17`
ARM 17-bit pc-relative branch for Branch Future instructions.

`BFD_RELOC_ARM_THUMB_BF13`
ARM 13-bit pc-relative branch for BFCSEL instruction.

`BFD_RELOC_ARM_THUMB_BF19`
ARM 19-bit pc-relative branch for Branch Future Link instruction.

`BFD_RELOC_ARM_THUMB_LOOP12`
ARM 12-bit pc-relative branch for Low Overhead Loop instructions.

`BFD_RELOC_THUMB_PCREL_BRANCH7`
`BFD_RELOC_THUMB_PCREL_BRANCH9`
`BFD_RELOC_THUMB_PCREL_BRANCH12`
`BFD_RELOC_THUMB_PCREL_BRANCH20`
`BFD_RELOC_THUMB_PCREL_BRANCH23`

`BFD_RELOC_THUMB_PCREL_BRANCH25`
: Thumb 7-, 9-, 12-, 20-, 23-, and 25-bit pc-relative branches. The lowest bit must be zero and is not stored in the instruction. Note that the corresponding ELF R_ARM_THM_JUMPnn constant has an "nn" one smaller in all cases. Note further that BRANCH23 corresponds to R_ARM_THM_CALL.

`BFD_RELOC_ARM_OFFSET_IMM`
: 12-bit immediate offset, used in ARM-format ldr and str instructions.

`BFD_RELOC_ARM_THUMB_OFFSET`
: 5-bit immediate offset, used in Thumb-format ldr and str instructions.

`BFD_RELOC_ARM_TARGET1`
: Pc-relative or absolute relocation depending on target. Used for entries in .init_array sections.

`BFD_RELOC_ARM_ROSEGREL32`
: Read-only segment base relative address.

`BFD_RELOC_ARM_SBREL32`
: Data segment base relative address.

`BFD_RELOC_ARM_TARGET2`
: This reloc is used for references to RTTI data from exception handling tables. The actual definition depends on the target. It may be a pc-relative or some form of GOT-indirect relocation.

`BFD_RELOC_ARM_PREL31`
: 31-bit PC relative address.

`BFD_RELOC_ARM_MOVW`
`BFD_RELOC_ARM_MOVT`
`BFD_RELOC_ARM_MOVW_PCREL`
`BFD_RELOC_ARM_MOVT_PCREL`
`BFD_RELOC_ARM_THUMB_MOVW`
`BFD_RELOC_ARM_THUMB_MOVT`
`BFD_RELOC_ARM_THUMB_MOVW_PCREL`
`BFD_RELOC_ARM_THUMB_MOVT_PCREL`
: Low and High halfword relocations for MOVW and MOVT instructions.

`BFD_RELOC_ARM_GOTFUNCDESC`
`BFD_RELOC_ARM_GOTOFFFUNCDESC`
`BFD_RELOC_ARM_FUNCDESC`
`BFD_RELOC_ARM_FUNCDESC_VALUE`
`BFD_RELOC_ARM_TLS_GD32_FDPIC`
`BFD_RELOC_ARM_TLS_LDM32_FDPIC`
`BFD_RELOC_ARM_TLS_IE32_FDPIC`
: ARM FDPIC specific relocations.

BFD_RELOC_ARM_JUMP_SLOT
BFD_RELOC_ARM_GLOB_DAT
BFD_RELOC_ARM_GOT32
BFD_RELOC_ARM_PLT32
BFD_RELOC_ARM_RELATIVE
BFD_RELOC_ARM_GOTOFF
BFD_RELOC_ARM_GOTPC
BFD_RELOC_ARM_GOT_PREL

Relocations for setting up GOTs and PLTs for shared libraries.

BFD_RELOC_ARM_TLS_GD32
BFD_RELOC_ARM_TLS_LDO32
BFD_RELOC_ARM_TLS_LDM32
BFD_RELOC_ARM_TLS_DTPOFF32
BFD_RELOC_ARM_TLS_DTPMOD32
BFD_RELOC_ARM_TLS_TPOFF32
BFD_RELOC_ARM_TLS_IE32
BFD_RELOC_ARM_TLS_LE32
BFD_RELOC_ARM_TLS_GOTDESC
BFD_RELOC_ARM_TLS_CALL
BFD_RELOC_ARM_THM_TLS_CALL
BFD_RELOC_ARM_TLS_DESCSEQ
BFD_RELOC_ARM_THM_TLS_DESCSEQ
BFD_RELOC_ARM_TLS_DESC

ARM thread-local storage relocations.

BFD_RELOC_ARM_ALU_PC_G0_NC
BFD_RELOC_ARM_ALU_PC_G0
BFD_RELOC_ARM_ALU_PC_G1_NC
BFD_RELOC_ARM_ALU_PC_G1
BFD_RELOC_ARM_ALU_PC_G2
BFD_RELOC_ARM_LDR_PC_G0
BFD_RELOC_ARM_LDR_PC_G1
BFD_RELOC_ARM_LDR_PC_G2
BFD_RELOC_ARM_LDRS_PC_G0
BFD_RELOC_ARM_LDRS_PC_G1
BFD_RELOC_ARM_LDRS_PC_G2
BFD_RELOC_ARM_LDC_PC_G0
BFD_RELOC_ARM_LDC_PC_G1
BFD_RELOC_ARM_LDC_PC_G2
BFD_RELOC_ARM_ALU_SB_G0_NC
BFD_RELOC_ARM_ALU_SB_G0
BFD_RELOC_ARM_ALU_SB_G1_NC
BFD_RELOC_ARM_ALU_SB_G1
BFD_RELOC_ARM_ALU_SB_G2
BFD_RELOC_ARM_LDR_SB_G0
BFD_RELOC_ARM_LDR_SB_G1
BFD_RELOC_ARM_LDR_SB_G2

BFD_RELOC_ARM_LDRS_SB_G0
BFD_RELOC_ARM_LDRS_SB_G1
BFD_RELOC_ARM_LDRS_SB_G2
BFD_RELOC_ARM_LDC_SB_G0
BFD_RELOC_ARM_LDC_SB_G1
BFD_RELOC_ARM_LDC_SB_G2
: ARM group relocations.

BFD_RELOC_ARM_V4BX
: Annotation of BX instructions.

BFD_RELOC_ARM_IRELATIVE
: ARM support for STT_GNU_IFUNC.

BFD_RELOC_ARM_THUMB_ALU_ABS_G0_NC
BFD_RELOC_ARM_THUMB_ALU_ABS_G1_NC
BFD_RELOC_ARM_THUMB_ALU_ABS_G2_NC
BFD_RELOC_ARM_THUMB_ALU_ABS_G3_NC
: Thumb1 relocations to support execute-only code.

BFD_RELOC_ARM_IMMEDIATE
BFD_RELOC_ARM_ADRL_IMMEDIATE
BFD_RELOC_ARM_T32_IMMEDIATE
BFD_RELOC_ARM_T32_ADD_IMM
BFD_RELOC_ARM_T32_IMM12
BFD_RELOC_ARM_T32_ADD_PC12
BFD_RELOC_ARM_SHIFT_IMM
BFD_RELOC_ARM_SMC
BFD_RELOC_ARM_HVC
BFD_RELOC_ARM_SWI
BFD_RELOC_ARM_MULTI
BFD_RELOC_ARM_CP_OFF_IMM
BFD_RELOC_ARM_CP_OFF_IMM_S2
BFD_RELOC_ARM_T32_CP_OFF_IMM
BFD_RELOC_ARM_T32_CP_OFF_IMM_S2
BFD_RELOC_ARM_T32_VLDR_VSTR_OFF_IMM
BFD_RELOC_ARM_ADR_IMM
BFD_RELOC_ARM_LDR_IMM
BFD_RELOC_ARM_LITERAL
BFD_RELOC_ARM_IN_POOL
BFD_RELOC_ARM_OFFSET_IMM8
BFD_RELOC_ARM_T32_OFFSET_U8
BFD_RELOC_ARM_T32_OFFSET_IMM
BFD_RELOC_ARM_HWLITERAL
BFD_RELOC_ARM_THUMB_ADD
BFD_RELOC_ARM_THUMB_IMM
BFD_RELOC_ARM_THUMB_SHIFT
: These relocs are only used within the ARM assembler. They are not (at present) written to any object files.

```
BFD_RELOC_SH_PCDISP8BY2
BFD_RELOC_SH_PCDISP12BY2
BFD_RELOC_SH_IMM3
BFD_RELOC_SH_IMM3U
BFD_RELOC_SH_DISP12
BFD_RELOC_SH_DISP12BY2
BFD_RELOC_SH_DISP12BY4
BFD_RELOC_SH_DISP12BY8
BFD_RELOC_SH_DISP20
BFD_RELOC_SH_DISP20BY8
BFD_RELOC_SH_IMM4
BFD_RELOC_SH_IMM4BY2
BFD_RELOC_SH_IMM4BY4
BFD_RELOC_SH_IMM8
BFD_RELOC_SH_IMM8BY2
BFD_RELOC_SH_IMM8BY4
BFD_RELOC_SH_PCRELIMM8BY2
BFD_RELOC_SH_PCRELIMM8BY4
BFD_RELOC_SH_SWITCH16
BFD_RELOC_SH_SWITCH32
BFD_RELOC_SH_USES
BFD_RELOC_SH_COUNT
BFD_RELOC_SH_ALIGN
BFD_RELOC_SH_CODE
BFD_RELOC_SH_DATA
BFD_RELOC_SH_LABEL
BFD_RELOC_SH_LOOP_START
BFD_RELOC_SH_LOOP_END
BFD_RELOC_SH_COPY
BFD_RELOC_SH_GLOB_DAT
BFD_RELOC_SH_JMP_SLOT
BFD_RELOC_SH_RELATIVE
BFD_RELOC_SH_GOTPC
BFD_RELOC_SH_GOT_LOW16
BFD_RELOC_SH_GOT_MEDLOW16
BFD_RELOC_SH_GOT_MEDHI16
BFD_RELOC_SH_GOT_HI16
BFD_RELOC_SH_GOTPLT_LOW16
BFD_RELOC_SH_GOTPLT_MEDLOW16
BFD_RELOC_SH_GOTPLT_MEDHI16
BFD_RELOC_SH_GOTPLT_HI16
BFD_RELOC_SH_PLT_LOW16
BFD_RELOC_SH_PLT_MEDLOW16
BFD_RELOC_SH_PLT_MEDHI16
BFD_RELOC_SH_PLT_HI16
BFD_RELOC_SH_GOTOFF_LOW16
BFD_RELOC_SH_GOTOFF_MEDLOW16
```

```
BFD_RELOC_SH_GOTOFF_MEDHI16
BFD_RELOC_SH_GOTOFF_HI16
BFD_RELOC_SH_GOTPC_LOW16
BFD_RELOC_SH_GOTPC_MEDLOW16
BFD_RELOC_SH_GOTPC_MEDHI16
BFD_RELOC_SH_GOTPC_HI16
BFD_RELOC_SH_COPY64
BFD_RELOC_SH_GLOB_DAT64
BFD_RELOC_SH_JMP_SLOT64
BFD_RELOC_SH_RELATIVE64
BFD_RELOC_SH_GOT10BY4
BFD_RELOC_SH_GOT10BY8
BFD_RELOC_SH_GOTPLT10BY4
BFD_RELOC_SH_GOTPLT10BY8
BFD_RELOC_SH_GOTPLT32
BFD_RELOC_SH_SHMEDIA_CODE
BFD_RELOC_SH_IMMU5
BFD_RELOC_SH_IMMS6
BFD_RELOC_SH_IMMS6BY32
BFD_RELOC_SH_IMMU6
BFD_RELOC_SH_IMMS10
BFD_RELOC_SH_IMMS10BY2
BFD_RELOC_SH_IMMS10BY4
BFD_RELOC_SH_IMMS10BY8
BFD_RELOC_SH_IMMS16
BFD_RELOC_SH_IMMU16
BFD_RELOC_SH_IMM_LOW16
BFD_RELOC_SH_IMM_LOW16_PCREL
BFD_RELOC_SH_IMM_MEDLOW16
BFD_RELOC_SH_IMM_MEDLOW16_PCREL
BFD_RELOC_SH_IMM_MEDHI16
BFD_RELOC_SH_IMM_MEDHI16_PCREL
BFD_RELOC_SH_IMM_HI16
BFD_RELOC_SH_IMM_HI16_PCREL
BFD_RELOC_SH_PT_16
BFD_RELOC_SH_TLS_GD_32
BFD_RELOC_SH_TLS_LD_32
BFD_RELOC_SH_TLS_LDO_32
BFD_RELOC_SH_TLS_IE_32
BFD_RELOC_SH_TLS_LE_32
BFD_RELOC_SH_TLS_DTPMOD32
BFD_RELOC_SH_TLS_DTPOFF32
BFD_RELOC_SH_TLS_TPOFF32
BFD_RELOC_SH_GOT20
BFD_RELOC_SH_GOTOFF20
BFD_RELOC_SH_GOTFUNCDESC
BFD_RELOC_SH_GOTFUNCDESC20
```

BFD_RELOC_SH_GOTOFFFUNCDESC
BFD_RELOC_SH_GOTOFFFUNCDESC20
BFD_RELOC_SH_FUNCDESC
Renesas / SuperH SH relocs. Not all of these appear in object files.

BFD_RELOC_ARC_NONE
BFD_RELOC_ARC_8
BFD_RELOC_ARC_16
BFD_RELOC_ARC_24
BFD_RELOC_ARC_32
BFD_RELOC_ARC_N8
BFD_RELOC_ARC_N16
BFD_RELOC_ARC_N24
BFD_RELOC_ARC_N32
BFD_RELOC_ARC_SDA
BFD_RELOC_ARC_SECTOFF
BFD_RELOC_ARC_S21H_PCREL
BFD_RELOC_ARC_S21W_PCREL
BFD_RELOC_ARC_S25H_PCREL
BFD_RELOC_ARC_S25W_PCREL
BFD_RELOC_ARC_SDA32
BFD_RELOC_ARC_SDA_LDST
BFD_RELOC_ARC_SDA_LDST1
BFD_RELOC_ARC_SDA_LDST2
BFD_RELOC_ARC_SDA16_LD
BFD_RELOC_ARC_SDA16_LD1
BFD_RELOC_ARC_SDA16_LD2
BFD_RELOC_ARC_S13_PCREL
BFD_RELOC_ARC_W
BFD_RELOC_ARC_32_ME
BFD_RELOC_ARC_32_ME_S
BFD_RELOC_ARC_N32_ME
BFD_RELOC_ARC_SECTOFF_ME
BFD_RELOC_ARC_SDA32_ME
BFD_RELOC_ARC_W_ME
BFD_RELOC_AC_SECTOFF_U8
BFD_RELOC_AC_SECTOFF_U8_1
BFD_RELOC_AC_SECTOFF_U8_2
BFD_RELOC_AC_SECTOFF_S9
BFD_RELOC_AC_SECTOFF_S9_1
BFD_RELOC_AC_SECTOFF_S9_2
BFD_RELOC_ARC_SECTOFF_ME_1
BFD_RELOC_ARC_SECTOFF_ME_2
BFD_RELOC_ARC_SECTOFF_1
BFD_RELOC_ARC_SECTOFF_2
BFD_RELOC_ARC_SDA_12
BFD_RELOC_ARC_SDA16_ST2

BFD_RELOC_ARC_32_PCREL
BFD_RELOC_ARC_PC32
BFD_RELOC_ARC_GOT32
BFD_RELOC_ARC_GOTPC32
BFD_RELOC_ARC_PLT32
BFD_RELOC_ARC_COPY
BFD_RELOC_ARC_GLOB_DAT
BFD_RELOC_ARC_JMP_SLOT
BFD_RELOC_ARC_RELATIVE
BFD_RELOC_ARC_GOTOFF
BFD_RELOC_ARC_GOTPC
BFD_RELOC_ARC_S21W_PCREL_PLT
BFD_RELOC_ARC_S25H_PCREL_PLT
BFD_RELOC_ARC_TLS_DTPMOD
BFD_RELOC_ARC_TLS_TPOFF
BFD_RELOC_ARC_TLS_GD_GOT
BFD_RELOC_ARC_TLS_GD_LD
BFD_RELOC_ARC_TLS_GD_CALL
BFD_RELOC_ARC_TLS_IE_GOT
BFD_RELOC_ARC_TLS_DTPOFF
BFD_RELOC_ARC_TLS_DTPOFF_S9
BFD_RELOC_ARC_TLS_LE_S9
BFD_RELOC_ARC_TLS_LE_32
BFD_RELOC_ARC_S25W_PCREL_PLT
BFD_RELOC_ARC_S21H_PCREL_PLT
BFD_RELOC_ARC_NPS_CMEM16
BFD_RELOC_ARC_JLI_SECTOFF
ARC relocs.

BFD_RELOC_BFIN_16_IMM
ADI Blackfin 16 bit immediate absolute reloc.

BFD_RELOC_BFIN_16_HIGH
ADI Blackfin 16 bit immediate absolute reloc higher 16 bits.

BFD_RELOC_BFIN_4_PCREL
ADI Blackfin 'a' part of LSETUP.

BFD_RELOC_BFIN_5_PCREL
ADI Blackfin.

BFD_RELOC_BFIN_16_LOW
ADI Blackfin 16 bit immediate absolute reloc lower 16 bits.

BFD_RELOC_BFIN_10_PCREL
ADI Blackfin.

BFD_RELOC_BFIN_11_PCREL
ADI Blackfin 'b' part of LSETUP.

`BFD_RELOC_BFIN_12_PCREL_JUMP`
ADI Blackfin.

`BFD_RELOC_BFIN_12_PCREL_JUMP_S`
ADI Blackfin Short jump, pcrel.

`BFD_RELOC_BFIN_24_PCREL_CALL_X`
ADI Blackfin Call.x not implemented.

`BFD_RELOC_BFIN_24_PCREL_JUMP_L`
ADI Blackfin Long Jump pcrel.

`BFD_RELOC_BFIN_GOT17M4`
`BFD_RELOC_BFIN_GOTHI`
`BFD_RELOC_BFIN_GOTLO`
`BFD_RELOC_BFIN_FUNCDESC`
`BFD_RELOC_BFIN_FUNCDESC_GOT17M4`
`BFD_RELOC_BFIN_FUNCDESC_GOTHI`
`BFD_RELOC_BFIN_FUNCDESC_GOTLO`
`BFD_RELOC_BFIN_FUNCDESC_VALUE`
`BFD_RELOC_BFIN_FUNCDESC_GOTOFF17M4`
`BFD_RELOC_BFIN_FUNCDESC_GOTOFFHI`
`BFD_RELOC_BFIN_FUNCDESC_GOTOFFLO`
`BFD_RELOC_BFIN_GOTOFF17M4`
`BFD_RELOC_BFIN_GOTOFFHI`
`BFD_RELOC_BFIN_GOTOFFLO`
ADI Blackfin FD-PIC relocations.

`BFD_RELOC_BFIN_GOT`
ADI Blackfin GOT relocation.

`BFD_RELOC_BFIN_PLTPC`
ADI Blackfin PLTPC relocation.

`BFD_ARELOC_BFIN_PUSH`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_CONST`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_ADD`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_SUB`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_MULT`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_DIV`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_MOD`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_LSHIFT`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_RSHIFT`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_AND`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_OR`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_XOR`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_LAND`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_LOR`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_LEN`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_NEG`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_COMP`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_PAGE`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_HWPAGE`
ADI Blackfin arithmetic relocation.

`BFD_ARELOC_BFIN_ADDR`
ADI Blackfin arithmetic relocation.

`BFD_RELOC_D10V_10_PCREL_R`
Mitsubishi D10V relocs. This is a 10-bit reloc with the right 2 bits assumed to be 0.

`BFD_RELOC_D10V_10_PCREL_L`
Mitsubishi D10V relocs. This is a 10-bit reloc with the right 2 bits assumed to be 0. This is the same as the previous reloc except it is in the left container, i.e., shifted left 15 bits.

`BFD_RELOC_D10V_18`
This is an 18-bit reloc with the right 2 bits assumed to be 0.

`BFD_RELOC_D10V_18_PCREL`
This is an 18-bit reloc with the right 2 bits assumed to be 0.

`BFD_RELOC_D30V_6`
Mitsubishi D30V relocs. This is a 6-bit absolute reloc.

`BFD_RELOC_D30V_9_PCREL`
This is a 6-bit pc-relative reloc with the right 3 bits assumed to be 0.

`BFD_RELOC_D30V_9_PCREL_R`
This is a 6-bit pc-relative reloc with the right 3 bits assumed to be 0. Same as the previous reloc but on the right side of the container.

`BFD_RELOC_D30V_15`
This is a 12-bit absolute reloc with the right 3 bitsassumed to be 0.

`BFD_RELOC_D30V_15_PCREL`
This is a 12-bit pc-relative reloc with the right 3 bits assumed to be 0.

`BFD_RELOC_D30V_15_PCREL_R`
This is a 12-bit pc-relative reloc with the right 3 bits assumed to be 0. Same as the previous reloc but on the right side of the container.

`BFD_RELOC_D30V_21`
This is an 18-bit absolute reloc with the right 3 bits assumed to be 0.

`BFD_RELOC_D30V_21_PCREL`
This is an 18-bit pc-relative reloc with the right 3 bits assumed to be 0.

`BFD_RELOC_D30V_21_PCREL_R`
This is an 18-bit pc-relative reloc with the right 3 bits assumed to be 0. Same as the previous reloc but on the right side of the container.

`BFD_RELOC_D30V_32`
This is a 32-bit absolute reloc.

`BFD_RELOC_D30V_32_PCREL`
This is a 32-bit pc-relative reloc.

`BFD_RELOC_DLX_HI16_S`
DLX relocs

`BFD_RELOC_DLX_LO16`
DLX relocs

`BFD_RELOC_DLX_JMP26`
DLX relocs

`BFD_RELOC_M32C_HI8`
`BFD_RELOC_M32C_RL_JUMP`
`BFD_RELOC_M32C_RL_1ADDR`
`BFD_RELOC_M32C_RL_2ADDR`
Renesas M16C/M32C Relocations.

`BFD_RELOC_M32R_24`
Renesas M32R (formerly Mitsubishi M32R) relocs. This is a 24 bit absolute address.

`BFD_RELOC_M32R_10_PCREL`
This is a 10-bit pc-relative reloc with the right 2 bits assumed to be 0.

`BFD_RELOC_M32R_18_PCREL`
This is an 18-bit reloc with the right 2 bits assumed to be 0.

`BFD_RELOC_M32R_26_PCREL`
This is a 26-bit reloc with the right 2 bits assumed to be 0.

`BFD_RELOC_M32R_HI16_ULO`
This is a 16-bit reloc containing the high 16 bits of an address used when the lower 16 bits are treated as unsigned.

`BFD_RELOC_M32R_HI16_SLO`
This is a 16-bit reloc containing the high 16 bits of an address used when the lower 16 bits are treated as signed.

`BFD_RELOC_M32R_LO16`
This is a 16-bit reloc containing the lower 16 bits of an address.

`BFD_RELOC_M32R_SDA16`
This is a 16-bit reloc containing the small data area offset for use in add3, load, and store instructions.

`BFD_RELOC_M32R_GOT24`
`BFD_RELOC_M32R_26_PLTREL`
`BFD_RELOC_M32R_COPY`
`BFD_RELOC_M32R_GLOB_DAT`
`BFD_RELOC_M32R_JMP_SLOT`
`BFD_RELOC_M32R_RELATIVE`
`BFD_RELOC_M32R_GOTOFF`
`BFD_RELOC_M32R_GOTOFF_HI_ULO`
`BFD_RELOC_M32R_GOTOFF_HI_SLO`
`BFD_RELOC_M32R_GOTOFF_LO`
`BFD_RELOC_M32R_GOTPC24`
`BFD_RELOC_M32R_GOT16_HI_ULO`
`BFD_RELOC_M32R_GOT16_HI_SLO`
`BFD_RELOC_M32R_GOT16_LO`
`BFD_RELOC_M32R_GOTPC_HI_ULO`
`BFD_RELOC_M32R_GOTPC_HI_SLO`
`BFD_RELOC_M32R_GOTPC_LO`
For PIC.

`BFD_RELOC_NDS32_20`
NDS32 relocs. This is a 20 bit absolute address.

`BFD_RELOC_NDS32_9_PCREL`
This is a 9-bit pc-relative reloc with the right 1 bit assumed to be 0.

`BFD_RELOC_NDS32_WORD_9_PCREL`
This is a 9-bit pc-relative reloc with the right 1 bit assumed to be 0.

`BFD_RELOC_NDS32_15_PCREL`
This is an 15-bit reloc with the right 1 bit assumed to be 0.

`BFD_RELOC_NDS32_17_PCREL`
This is an 17-bit reloc with the right 1 bit assumed to be 0.

`BFD_RELOC_NDS32_25_PCREL`
This is a 25-bit reloc with the right 1 bit assumed to be 0.

`BFD_RELOC_NDS32_HI20`
This is a 20-bit reloc containing the high 20 bits of an address used with the lower 12 bits

`BFD_RELOC_NDS32_LO12S3`
This is a 12-bit reloc containing the lower 12 bits of an address then shift right by 3. This is used with ldi,sdi...

`BFD_RELOC_NDS32_LO12S2`
This is a 12-bit reloc containing the lower 12 bits of an address then shift left by 2. This is used with lwi,swi...

`BFD_RELOC_NDS32_LO12S1`
This is a 12-bit reloc containing the lower 12 bits of an address then shift left by 1. This is used with lhi,shi...

`BFD_RELOC_NDS32_LO12S0`
This is a 12-bit reloc containing the lower 12 bits of an address then shift left by 0. This is used with lbisbi...

`BFD_RELOC_NDS32_LO12S0_ORI`
This is a 12-bit reloc containing the lower 12 bits of an address then shift left by 0. This is only used with branch relaxations

`BFD_RELOC_NDS32_SDA15S3`
This is a 15-bit reloc containing the small data area 18-bit signed offset and shift left by 3 for use in ldi, sdi...

`BFD_RELOC_NDS32_SDA15S2`
This is a 15-bit reloc containing the small data area 17-bit signed offset and shift left by 2 for use in lwi, swi...

`BFD_RELOC_NDS32_SDA15S1`
This is a 15-bit reloc containing the small data area 16-bit signed offset and shift left by 1 for use in lhi, shi...

`BFD_RELOC_NDS32_SDA15S0`
This is a 15-bit reloc containing the small data area 15-bit signed offset and shift left by 0 for use in lbi, sbi...

`BFD_RELOC_NDS32_SDA16S3`
This is a 16-bit reloc containing the small data area 16-bit signed offset and shift left by 3

`BFD_RELOC_NDS32_SDA17S2`
This is a 17-bit reloc containing the small data area 17-bit signed offset and shift left by 2 for use in lwi.gp, swi.gp...

`BFD_RELOC_NDS32_SDA18S1`
This is a 18-bit reloc containing the small data area 18-bit signed offset and shift left by 1 for use in lhi.gp, shi.gp...

`BFD_RELOC_NDS32_SDA19S0`
This is a 19-bit reloc containing the small data area 19-bit signed offset and shift left by 0 for use in lbi.gp, sbi.gp...

`BFD_RELOC_NDS32_GOT20`
`BFD_RELOC_NDS32_9_PLTREL`
`BFD_RELOC_NDS32_25_PLTREL`
`BFD_RELOC_NDS32_COPY`
`BFD_RELOC_NDS32_GLOB_DAT`
`BFD_RELOC_NDS32_JMP_SLOT`
`BFD_RELOC_NDS32_RELATIVE`
`BFD_RELOC_NDS32_GOTOFF`
`BFD_RELOC_NDS32_GOTOFF_HI20`
`BFD_RELOC_NDS32_GOTOFF_LO12`
`BFD_RELOC_NDS32_GOTPC20`
`BFD_RELOC_NDS32_GOT_HI20`
`BFD_RELOC_NDS32_GOT_LO12`
`BFD_RELOC_NDS32_GOTPC_HI20`
`BFD_RELOC_NDS32_GOTPC_LO12`
for PIC

`BFD_RELOC_NDS32_INSN16`
`BFD_RELOC_NDS32_LABEL`
`BFD_RELOC_NDS32_LONGCALL1`
`BFD_RELOC_NDS32_LONGCALL2`
`BFD_RELOC_NDS32_LONGCALL3`
`BFD_RELOC_NDS32_LONGJUMP1`
`BFD_RELOC_NDS32_LONGJUMP2`
`BFD_RELOC_NDS32_LONGJUMP3`
`BFD_RELOC_NDS32_LOADSTORE`
`BFD_RELOC_NDS32_9_FIXED`
`BFD_RELOC_NDS32_15_FIXED`
`BFD_RELOC_NDS32_17_FIXED`
`BFD_RELOC_NDS32_25_FIXED`
`BFD_RELOC_NDS32_LONGCALL4`
`BFD_RELOC_NDS32_LONGCALL5`
`BFD_RELOC_NDS32_LONGCALL6`

BFD_RELOC_NDS32_LONGJUMP4
BFD_RELOC_NDS32_LONGJUMP5
BFD_RELOC_NDS32_LONGJUMP6
BFD_RELOC_NDS32_LONGJUMP7
for relax

BFD_RELOC_NDS32_PLTREL_HI20
BFD_RELOC_NDS32_PLTREL_LO12
BFD_RELOC_NDS32_PLT_GOTREL_HI20
BFD_RELOC_NDS32_PLT_GOTREL_LO12
for PIC

BFD_RELOC_NDS32_SDA12S2_DP
BFD_RELOC_NDS32_SDA12S2_SP
BFD_RELOC_NDS32_LO12S2_DP
BFD_RELOC_NDS32_LO12S2_SP
for floating point

BFD_RELOC_NDS32_DWARF2_OP1
BFD_RELOC_NDS32_DWARF2_OP2
BFD_RELOC_NDS32_DWARF2_LEB
for dwarf2 debug_line.

BFD_RELOC_NDS32_UPDATE_TA
for eliminate 16-bit instructions

BFD_RELOC_NDS32_PLT_GOTREL_LO20
BFD_RELOC_NDS32_PLT_GOTREL_LO15
BFD_RELOC_NDS32_PLT_GOTREL_LO19
BFD_RELOC_NDS32_GOT_LO15
BFD_RELOC_NDS32_GOT_LO19
BFD_RELOC_NDS32_GOTOFF_LO15
BFD_RELOC_NDS32_GOTOFF_LO19
BFD_RELOC_NDS32_GOT15S2
BFD_RELOC_NDS32_GOT17S2
for PIC object relaxation

BFD_RELOC_NDS32_5
NDS32 relocs. This is a 5 bit absolute address.

BFD_RELOC_NDS32_10_UPCREL
This is a 10-bit unsigned pc-relative reloc with the right 1 bit assumed to be 0.

BFD_RELOC_NDS32_SDA_FP7U2_RELA
If fp were omitted, fp can used as another gp.

BFD_RELOC_NDS32_RELAX_ENTRY
BFD_RELOC_NDS32_GOT_SUFF
BFD_RELOC_NDS32_GOTOFF_SUFF
BFD_RELOC_NDS32_PLT_GOT_SUFF

BFD_RELOC_NDS32_MULCALL_SUFF
BFD_RELOC_NDS32_PTR
BFD_RELOC_NDS32_PTR_COUNT
BFD_RELOC_NDS32_PTR_RESOLVED
BFD_RELOC_NDS32_PLTBLOCK
BFD_RELOC_NDS32_RELAX_REGION_BEGIN
BFD_RELOC_NDS32_RELAX_REGION_END
BFD_RELOC_NDS32_MINUEND
BFD_RELOC_NDS32_SUBTRAHEND
BFD_RELOC_NDS32_DIFF8
BFD_RELOC_NDS32_DIFF16
BFD_RELOC_NDS32_DIFF32
BFD_RELOC_NDS32_DIFF_ULEB128
BFD_RELOC_NDS32_EMPTY
relaxation relative relocation types

BFD_RELOC_NDS32_25_ABS
This is a 25 bit absolute address.

BFD_RELOC_NDS32_DATA
BFD_RELOC_NDS32_TRAN
BFD_RELOC_NDS32_17IFC_PCREL
BFD_RELOC_NDS32_10IFCU_PCREL
For ex9 and ifc using.

BFD_RELOC_NDS32_TPOFF
BFD_RELOC_NDS32_GOTTPOFF
BFD_RELOC_NDS32_TLS_LE_HI20
BFD_RELOC_NDS32_TLS_LE_LO12
BFD_RELOC_NDS32_TLS_LE_20
BFD_RELOC_NDS32_TLS_LE_15S0
BFD_RELOC_NDS32_TLS_LE_15S1
BFD_RELOC_NDS32_TLS_LE_15S2
BFD_RELOC_NDS32_TLS_LE_ADD
BFD_RELOC_NDS32_TLS_LE_LS
BFD_RELOC_NDS32_TLS_IE_HI20
BFD_RELOC_NDS32_TLS_IE_LO12
BFD_RELOC_NDS32_TLS_IE_LO12S2
BFD_RELOC_NDS32_TLS_IEGP_HI20
BFD_RELOC_NDS32_TLS_IEGP_LO12
BFD_RELOC_NDS32_TLS_IEGP_LO12S2
BFD_RELOC_NDS32_TLS_IEGP_LW
BFD_RELOC_NDS32_TLS_DESC
BFD_RELOC_NDS32_TLS_DESC_HI20
BFD_RELOC_NDS32_TLS_DESC_LO12
BFD_RELOC_NDS32_TLS_DESC_20
BFD_RELOC_NDS32_TLS_DESC_SDA17S2
BFD_RELOC_NDS32_TLS_DESC_ADD

BFD_RELOC_NDS32_TLS_DESC_FUNC
BFD_RELOC_NDS32_TLS_DESC_CALL
BFD_RELOC_NDS32_TLS_DESC_MEM
BFD_RELOC_NDS32_REMOVE
BFD_RELOC_NDS32_GROUP
For TLS.

BFD_RELOC_NDS32_LSI
For floating load store relaxation.

BFD_RELOC_V850_9_PCREL
This is a 9-bit reloc

BFD_RELOC_V850_22_PCREL
This is a 22-bit reloc

BFD_RELOC_V850_SDA_16_16_OFFSET
This is a 16 bit offset from the short data area pointer.

BFD_RELOC_V850_SDA_15_16_OFFSET
This is a 16 bit offset (of which only 15 bits are used) from the short data area pointer.

BFD_RELOC_V850_ZDA_16_16_OFFSET
This is a 16 bit offset from the zero data area pointer.

BFD_RELOC_V850_ZDA_15_16_OFFSET
This is a 16 bit offset (of which only 15 bits are used) from the zero data area pointer.

BFD_RELOC_V850_TDA_6_8_OFFSET
This is an 8 bit offset (of which only 6 bits are used) from the tiny data area pointer.

BFD_RELOC_V850_TDA_7_8_OFFSET
This is an 8bit offset (of which only 7 bits are used) from the tiny data area pointer.

BFD_RELOC_V850_TDA_7_7_OFFSET
This is a 7 bit offset from the tiny data area pointer.

BFD_RELOC_V850_TDA_16_16_OFFSET
This is a 16 bit offset from the tiny data area pointer.

BFD_RELOC_V850_TDA_4_5_OFFSET
This is a 5 bit offset (of which only 4 bits are used) from the tiny data area pointer.

BFD_RELOC_V850_TDA_4_4_OFFSET
This is a 4 bit offset from the tiny data area pointer.

BFD_RELOC_V850_SDA_16_16_SPLIT_OFFSET
This is a 16 bit offset from the short data area pointer, with the bits placed non-contiguously in the instruction.

BFD_RELOC_V850_ZDA_16_16_SPLIT_OFFSET
This is a 16 bit offset from the zero data area pointer, with the bits placed non-contiguously in the instruction.

`BFD_RELOC_V850_CALLT_6_7_OFFSET`
This is a 6 bit offset from the call table base pointer.

`BFD_RELOC_V850_CALLT_16_16_OFFSET`
This is a 16 bit offset from the call table base pointer.

`BFD_RELOC_V850_LONGCALL`
Used for relaxing indirect function calls.

`BFD_RELOC_V850_LONGJUMP`
Used for relaxing indirect jumps.

`BFD_RELOC_V850_ALIGN`
Used to maintain alignment whilst relaxing.

`BFD_RELOC_V850_LO16_SPLIT_OFFSET`
This is a variation of BFD_RELOC_LO16 that can be used in v850e ld.bu instructions.

`BFD_RELOC_V850_16_PCREL`
This is a 16-bit reloc.

`BFD_RELOC_V850_17_PCREL`
This is a 17-bit reloc.

`BFD_RELOC_V850_23`
This is a 23-bit reloc.

`BFD_RELOC_V850_32_PCREL`
This is a 32-bit reloc.

`BFD_RELOC_V850_32_ABS`
This is a 32-bit reloc.

`BFD_RELOC_V850_16_SPLIT_OFFSET`
This is a 16-bit reloc.

`BFD_RELOC_V850_16_S1`
This is a 16-bit reloc.

`BFD_RELOC_V850_LO16_S1`
Low 16 bits. 16 bit shifted by 1.

`BFD_RELOC_V850_CALLT_15_16_OFFSET`
This is a 16 bit offset from the call table base pointer.

`BFD_RELOC_V850_32_GOTPCREL`
DSO relocations.

`BFD_RELOC_V850_16_GOT`
DSO relocations.

`BFD_RELOC_V850_32_GOT`
DSO relocations.

`BFD_RELOC_V850_22_PLT_PCREL`
DSO relocations.

`BFD_RELOC_V850_32_PLT_PCREL`
DSO relocations.

`BFD_RELOC_V850_COPY`
DSO relocations.

`BFD_RELOC_V850_GLOB_DAT`
DSO relocations.

`BFD_RELOC_V850_JMP_SLOT`
DSO relocations.

`BFD_RELOC_V850_RELATIVE`
DSO relocations.

`BFD_RELOC_V850_16_GOTOFF`
DSO relocations.

`BFD_RELOC_V850_32_GOTOFF`
DSO relocations.

`BFD_RELOC_V850_CODE`
start code.

`BFD_RELOC_V850_DATA`
start data in text.

`BFD_RELOC_TIC30_LDP`
This is a 8bit DP reloc for the tms320c30, where the most significant 8 bits of a 24 bit word are placed into the least significant 8 bits of the opcode.

`BFD_RELOC_TIC54X_PARTLS7`
This is a 7bit reloc for the tms320c54x, where the least significant 7 bits of a 16 bit word are placed into the least significant 7 bits of the opcode.

`BFD_RELOC_TIC54X_PARTMS9`
This is a 9bit DP reloc for the tms320c54x, where the most significant 9 bits of a 16 bit word are placed into the least significant 9 bits of the opcode.

`BFD_RELOC_TIC54X_23`
This is an extended address 23-bit reloc for the tms320c54x.

`BFD_RELOC_TIC54X_16_OF_23`
This is a 16-bit reloc for the tms320c54x, where the least significant 16 bits of a 23-bit extended address are placed into the opcode.

`BFD_RELOC_TIC54X_MS7_OF_23`
This is a reloc for the tms320c54x, where the most significant 7 bits of a 23-bit extended address are placed into the opcode.

BFD_RELOC_C6000_PCR_S21
BFD_RELOC_C6000_PCR_S12
BFD_RELOC_C6000_PCR_S10
BFD_RELOC_C6000_PCR_S7
BFD_RELOC_C6000_ABS_S16
BFD_RELOC_C6000_ABS_L16
BFD_RELOC_C6000_ABS_H16
BFD_RELOC_C6000_SBR_U15_B
BFD_RELOC_C6000_SBR_U15_H
BFD_RELOC_C6000_SBR_U15_W
BFD_RELOC_C6000_SBR_S16
BFD_RELOC_C6000_SBR_L16_B
BFD_RELOC_C6000_SBR_L16_H
BFD_RELOC_C6000_SBR_L16_W
BFD_RELOC_C6000_SBR_H16_B
BFD_RELOC_C6000_SBR_H16_H
BFD_RELOC_C6000_SBR_H16_W
BFD_RELOC_C6000_SBR_GOT_U15_W
BFD_RELOC_C6000_SBR_GOT_L16_W
BFD_RELOC_C6000_SBR_GOT_H16_W
BFD_RELOC_C6000_DSBT_INDEX
BFD_RELOC_C6000_PREL31
BFD_RELOC_C6000_COPY
BFD_RELOC_C6000_JUMP_SLOT
BFD_RELOC_C6000_EHTYPE
BFD_RELOC_C6000_PCR_H16
BFD_RELOC_C6000_PCR_L16
BFD_RELOC_C6000_ALIGN
BFD_RELOC_C6000_FPHEAD
BFD_RELOC_C6000_NOCMP
TMS320C6000 relocations.

BFD_RELOC_FR30_48
This is a 48 bit reloc for the FR30 that stores 32 bits.

BFD_RELOC_FR30_20
This is a 32 bit reloc for the FR30 that stores 20 bits split up into two sections.

BFD_RELOC_FR30_6_IN_4
This is a 16 bit reloc for the FR30 that stores a 6 bit word offset in 4 bits.

BFD_RELOC_FR30_8_IN_8
This is a 16 bit reloc for the FR30 that stores an 8 bit byte offset into 8 bits.

BFD_RELOC_FR30_9_IN_8
This is a 16 bit reloc for the FR30 that stores a 9 bit short offset into 8 bits.

BFD_RELOC_FR30_10_IN_8
This is a 16 bit reloc for the FR30 that stores a 10 bit word offset into 8 bits.

BFD_RELOC_FR30_9_PCREL
This is a 16 bit reloc for the FR30 that stores a 9 bit pc relative short offset into 8 bits.

BFD_RELOC_FR30_12_PCREL
This is a 16 bit reloc for the FR30 that stores a 12 bit pc relative short offset into 11 bits.

BFD_RELOC_MCORE_PCREL_IMM8BY4
BFD_RELOC_MCORE_PCREL_IMM11BY2
BFD_RELOC_MCORE_PCREL_IMM4BY2
BFD_RELOC_MCORE_PCREL_32
BFD_RELOC_MCORE_PCREL_JSR_IMM11BY2
BFD_RELOC_MCORE_RVA
Motorola Mcore relocations.

BFD_RELOC_MEP_8
BFD_RELOC_MEP_16
BFD_RELOC_MEP_32
BFD_RELOC_MEP_PCREL8A2
BFD_RELOC_MEP_PCREL12A2
BFD_RELOC_MEP_PCREL17A2
BFD_RELOC_MEP_PCREL24A2
BFD_RELOC_MEP_PCABS24A2
BFD_RELOC_MEP_LOW16
BFD_RELOC_MEP_HI16U
BFD_RELOC_MEP_HI16S
BFD_RELOC_MEP_GPREL
BFD_RELOC_MEP_TPREL
BFD_RELOC_MEP_TPREL7
BFD_RELOC_MEP_TPREL7A2
BFD_RELOC_MEP_TPREL7A4
BFD_RELOC_MEP_UIMM24
BFD_RELOC_MEP_ADDR24A4
BFD_RELOC_MEP_GNU_VTINHERIT
BFD_RELOC_MEP_GNU_VTENTRY
Toshiba Media Processor Relocations.

BFD_RELOC_METAG_HIADDR16
BFD_RELOC_METAG_LOADDR16
BFD_RELOC_METAG_RELBRANCH
BFD_RELOC_METAG_GETSETOFF
BFD_RELOC_METAG_HIOG
BFD_RELOC_METAG_LOOG
BFD_RELOC_METAG_REL8
BFD_RELOC_METAG_REL16
BFD_RELOC_METAG_HI16_GOTOFF
BFD_RELOC_METAG_LO16_GOTOFF

BFD_RELOC_METAG_GETSET_GOTOFF
BFD_RELOC_METAG_GETSET_GOT
BFD_RELOC_METAG_HI16_GOTPC
BFD_RELOC_METAG_LO16_GOTPC
BFD_RELOC_METAG_HI16_PLT
BFD_RELOC_METAG_LO16_PLT
BFD_RELOC_METAG_RELBRANCH_PLT
BFD_RELOC_METAG_GOTOFF
BFD_RELOC_METAG_PLT
BFD_RELOC_METAG_COPY
BFD_RELOC_METAG_JMP_SLOT
BFD_RELOC_METAG_RELATIVE
BFD_RELOC_METAG_GLOB_DAT
BFD_RELOC_METAG_TLS_GD
BFD_RELOC_METAG_TLS_LDM
BFD_RELOC_METAG_TLS_LDO_HI16
BFD_RELOC_METAG_TLS_LDO_LO16
BFD_RELOC_METAG_TLS_LDO
BFD_RELOC_METAG_TLS_IE
BFD_RELOC_METAG_TLS_IENONPIC
BFD_RELOC_METAG_TLS_IENONPIC_HI16
BFD_RELOC_METAG_TLS_IENONPIC_LO16
BFD_RELOC_METAG_TLS_TPOFF
BFD_RELOC_METAG_TLS_DTPMOD
BFD_RELOC_METAG_TLS_DTPOFF
BFD_RELOC_METAG_TLS_LE
BFD_RELOC_METAG_TLS_LE_HI16
BFD_RELOC_METAG_TLS_LE_LO16

Imagination Technologies Meta relocations.

BFD_RELOC_MMIX_GETA
BFD_RELOC_MMIX_GETA_1
BFD_RELOC_MMIX_GETA_2
BFD_RELOC_MMIX_GETA_3

These are relocations for the GETA instruction.

BFD_RELOC_MMIX_CBRANCH
BFD_RELOC_MMIX_CBRANCH_J
BFD_RELOC_MMIX_CBRANCH_1
BFD_RELOC_MMIX_CBRANCH_2
BFD_RELOC_MMIX_CBRANCH_3

These are relocations for a conditional branch instruction.

BFD_RELOC_MMIX_PUSHJ
BFD_RELOC_MMIX_PUSHJ_1
BFD_RELOC_MMIX_PUSHJ_2
BFD_RELOC_MMIX_PUSHJ_3

`BFD_RELOC_MMIX_PUSHJ_STUBBABLE`
: These are relocations for the PUSHJ instruction.

`BFD_RELOC_MMIX_JMP`
`BFD_RELOC_MMIX_JMP_1`
`BFD_RELOC_MMIX_JMP_2`
`BFD_RELOC_MMIX_JMP_3`
: These are relocations for the JMP instruction.

`BFD_RELOC_MMIX_ADDR19`
: This is a relocation for a relative address as in a GETA instruction or a branch.

`BFD_RELOC_MMIX_ADDR27`
: This is a relocation for a relative address as in a JMP instruction.

`BFD_RELOC_MMIX_REG_OR_BYTE`
: This is a relocation for an instruction field that may be a general register or a value 0..255.

`BFD_RELOC_MMIX_REG`
: This is a relocation for an instruction field that may be a general register.

`BFD_RELOC_MMIX_BASE_PLUS_OFFSET`
: This is a relocation for two instruction fields holding a register and an offset, the equivalent of the relocation.

`BFD_RELOC_MMIX_LOCAL`
: This relocation is an assertion that the expression is not allocated as a global register. It does not modify contents.

`BFD_RELOC_AVR_7_PCREL`
: This is a 16 bit reloc for the AVR that stores 8 bit pc relative short offset into 7 bits.

`BFD_RELOC_AVR_13_PCREL`
: This is a 16 bit reloc for the AVR that stores 13 bit pc relative short offset into 12 bits.

`BFD_RELOC_AVR_16_PM`
: This is a 16 bit reloc for the AVR that stores 17 bit value (usually program memory address) into 16 bits.

`BFD_RELOC_AVR_LO8_LDI`
: This is a 16 bit reloc for the AVR that stores 8 bit value (usually data memory address) into 8 bit immediate value of LDI insn.

`BFD_RELOC_AVR_HI8_LDI`
: This is a 16 bit reloc for the AVR that stores 8 bit value (high 8 bit of data memory address) into 8 bit immediate value of LDI insn.

`BFD_RELOC_AVR_HH8_LDI`
: This is a 16 bit reloc for the AVR that stores 8 bit value (most high 8 bit of program memory address) into 8 bit immediate value of LDI insn.

`BFD_RELOC_AVR_MS8_LDI`
This is a 16 bit reloc for the AVR that stores 8 bit value (most high 8 bit of 32 bit value) into 8 bit immediate value of LDI insn.

`BFD_RELOC_AVR_LO8_LDI_NEG`
This is a 16 bit reloc for the AVR that stores negated 8 bit value (usually data memory address) into 8 bit immediate value of SUBI insn.

`BFD_RELOC_AVR_HI8_LDI_NEG`
This is a 16 bit reloc for the AVR that stores negated 8 bit value (high 8 bit of data memory address) into 8 bit immediate value of SUBI insn.

`BFD_RELOC_AVR_HH8_LDI_NEG`
This is a 16 bit reloc for the AVR that stores negated 8 bit value (most high 8 bit of program memory address) into 8 bit immediate value of LDI or SUBI insn.

`BFD_RELOC_AVR_MS8_LDI_NEG`
This is a 16 bit reloc for the AVR that stores negated 8 bit value (msb of 32 bit value) into 8 bit immediate value of LDI insn.

`BFD_RELOC_AVR_LO8_LDI_PM`
This is a 16 bit reloc for the AVR that stores 8 bit value (usually command address) into 8 bit immediate value of LDI insn.

`BFD_RELOC_AVR_LO8_LDI_GS`
This is a 16 bit reloc for the AVR that stores 8 bit value (command address) into 8 bit immediate value of LDI insn. If the address is beyond the 128k boundary, the linker inserts a jump stub for this reloc in the lower 128k.

`BFD_RELOC_AVR_HI8_LDI_PM`
This is a 16 bit reloc for the AVR that stores 8 bit value (high 8 bit of command address) into 8 bit immediate value of LDI insn.

`BFD_RELOC_AVR_HI8_LDI_GS`
This is a 16 bit reloc for the AVR that stores 8 bit value (high 8 bit of command address) into 8 bit immediate value of LDI insn. If the address is beyond the 128k boundary, the linker inserts a jump stub for this reloc below 128k.

`BFD_RELOC_AVR_HH8_LDI_PM`
This is a 16 bit reloc for the AVR that stores 8 bit value (most high 8 bit of command address) into 8 bit immediate value of LDI insn.

`BFD_RELOC_AVR_LO8_LDI_PM_NEG`
This is a 16 bit reloc for the AVR that stores negated 8 bit value (usually command address) into 8 bit immediate value of SUBI insn.

`BFD_RELOC_AVR_HI8_LDI_PM_NEG`
This is a 16 bit reloc for the AVR that stores negated 8 bit value (high 8 bit of 16 bit command address) into 8 bit immediate value of SUBI insn.

`BFD_RELOC_AVR_HH8_LDI_PM_NEG`
This is a 16 bit reloc for the AVR that stores negated 8 bit value (high 6 bit of 22 bit command address) into 8 bit immediate value of SUBI insn.

`BFD_RELOC_AVR_CALL`
This is a 32 bit reloc for the AVR that stores 23 bit value into 22 bits.

`BFD_RELOC_AVR_LDI`
This is a 16 bit reloc for the AVR that stores all needed bits for absolute addressing with ldi with overflow check to linktime

`BFD_RELOC_AVR_6`
This is a 6 bit reloc for the AVR that stores offset for ldd/std instructions

`BFD_RELOC_AVR_6_ADIW`
This is a 6 bit reloc for the AVR that stores offset for adiw/sbiw instructions

`BFD_RELOC_AVR_8_LO`
This is a 8 bit reloc for the AVR that stores bits 0..7 of a symbol in .byte lo8(symbol)

`BFD_RELOC_AVR_8_HI`
This is a 8 bit reloc for the AVR that stores bits 8..15 of a symbol in .byte hi8(symbol)

`BFD_RELOC_AVR_8_HLO`
This is a 8 bit reloc for the AVR that stores bits 16..23 of a symbol in .byte hlo8(symbol)

`BFD_RELOC_AVR_DIFF8`
`BFD_RELOC_AVR_DIFF16`
`BFD_RELOC_AVR_DIFF32`
AVR relocations to mark the difference of two local symbols. These are only needed to support linker relaxation and can be ignored when not relaxing. The field is set to the value of the difference assuming no relaxation. The relocation encodes the position of the second symbol so the linker can determine whether to adjust the field value.

`BFD_RELOC_AVR_LDS_STS_16`
This is a 7 bit reloc for the AVR that stores SRAM address for 16bit lds and sts instructions supported only tiny core.

`BFD_RELOC_AVR_PORT6`
This is a 6 bit reloc for the AVR that stores an I/O register number for the IN and OUT instructions

`BFD_RELOC_AVR_PORT5`
This is a 5 bit reloc for the AVR that stores an I/O register number for the SBIC, SBIS, SBI and CBI instructions

`BFD_RELOC_RISCV_HI20`
`BFD_RELOC_RISCV_PCREL_HI20`
`BFD_RELOC_RISCV_PCREL_LO12_I`

BFD_RELOC_RISCV_PCREL_LO12_S
BFD_RELOC_RISCV_LO12_I
BFD_RELOC_RISCV_LO12_S
BFD_RELOC_RISCV_GPREL12_I
BFD_RELOC_RISCV_GPREL12_S
BFD_RELOC_RISCV_TPREL_HI20
BFD_RELOC_RISCV_TPREL_LO12_I
BFD_RELOC_RISCV_TPREL_LO12_S
BFD_RELOC_RISCV_TPREL_ADD
BFD_RELOC_RISCV_CALL
BFD_RELOC_RISCV_CALL_PLT
BFD_RELOC_RISCV_ADD8
BFD_RELOC_RISCV_ADD16
BFD_RELOC_RISCV_ADD32
BFD_RELOC_RISCV_ADD64
BFD_RELOC_RISCV_SUB8
BFD_RELOC_RISCV_SUB16
BFD_RELOC_RISCV_SUB32
BFD_RELOC_RISCV_SUB64
BFD_RELOC_RISCV_GOT_HI20
BFD_RELOC_RISCV_TLS_GOT_HI20
BFD_RELOC_RISCV_TLS_GD_HI20
BFD_RELOC_RISCV_JMP
BFD_RELOC_RISCV_TLS_DTPMOD32
BFD_RELOC_RISCV_TLS_DTPREL32
BFD_RELOC_RISCV_TLS_DTPMOD64
BFD_RELOC_RISCV_TLS_DTPREL64
BFD_RELOC_RISCV_TLS_TPREL32
BFD_RELOC_RISCV_TLS_TPREL64
BFD_RELOC_RISCV_ALIGN
BFD_RELOC_RISCV_RVC_BRANCH
BFD_RELOC_RISCV_RVC_JUMP
BFD_RELOC_RISCV_RVC_LUI
BFD_RELOC_RISCV_GPREL_I
BFD_RELOC_RISCV_GPREL_S
BFD_RELOC_RISCV_TPREL_I
BFD_RELOC_RISCV_TPREL_S
BFD_RELOC_RISCV_RELAX
BFD_RELOC_RISCV_CFA
BFD_RELOC_RISCV_SUB6
BFD_RELOC_RISCV_SET6
BFD_RELOC_RISCV_SET8
BFD_RELOC_RISCV_SET16
BFD_RELOC_RISCV_SET32
BFD_RELOC_RISCV_32_PCREL

RISC-V relocations.

BFD_RELOC_RL78_NEG8
BFD_RELOC_RL78_NEG16
BFD_RELOC_RL78_NEG24
BFD_RELOC_RL78_NEG32
BFD_RELOC_RL78_16_OP
BFD_RELOC_RL78_24_OP
BFD_RELOC_RL78_32_OP
BFD_RELOC_RL78_8U
BFD_RELOC_RL78_16U
BFD_RELOC_RL78_24U
BFD_RELOC_RL78_DIR3U_PCREL
BFD_RELOC_RL78_DIFF
BFD_RELOC_RL78_GPRELB
BFD_RELOC_RL78_GPRELW
BFD_RELOC_RL78_GPRELL
BFD_RELOC_RL78_SYM
BFD_RELOC_RL78_OP_SUBTRACT
BFD_RELOC_RL78_OP_NEG
BFD_RELOC_RL78_OP_AND
BFD_RELOC_RL78_OP_SHRA
BFD_RELOC_RL78_ABS8
BFD_RELOC_RL78_ABS16
BFD_RELOC_RL78_ABS16_REV
BFD_RELOC_RL78_ABS32
BFD_RELOC_RL78_ABS32_REV
BFD_RELOC_RL78_ABS16U
BFD_RELOC_RL78_ABS16UW
BFD_RELOC_RL78_ABS16UL
BFD_RELOC_RL78_RELAX
BFD_RELOC_RL78_HI16
BFD_RELOC_RL78_HI8
BFD_RELOC_RL78_LO16
BFD_RELOC_RL78_CODE
BFD_RELOC_RL78_SADDR

Renesas RL78 Relocations.

BFD_RELOC_RX_NEG8
BFD_RELOC_RX_NEG16
BFD_RELOC_RX_NEG24
BFD_RELOC_RX_NEG32
BFD_RELOC_RX_16_OP
BFD_RELOC_RX_24_OP
BFD_RELOC_RX_32_OP
BFD_RELOC_RX_8U
BFD_RELOC_RX_16U
BFD_RELOC_RX_24U
BFD_RELOC_RX_DIR3U_PCREL

BFD_RELOC_RX_DIFF
BFD_RELOC_RX_GPRELB
BFD_RELOC_RX_GPRELW
BFD_RELOC_RX_GPRELL
BFD_RELOC_RX_SYM
BFD_RELOC_RX_OP_SUBTRACT
BFD_RELOC_RX_OP_NEG
BFD_RELOC_RX_ABS8
BFD_RELOC_RX_ABS16
BFD_RELOC_RX_ABS16_REV
BFD_RELOC_RX_ABS32
BFD_RELOC_RX_ABS32_REV
BFD_RELOC_RX_ABS16U
BFD_RELOC_RX_ABS16UW
BFD_RELOC_RX_ABS16UL
BFD_RELOC_RX_RELAX
: Renesas RX Relocations.

BFD_RELOC_390_12
: Direct 12 bit.

BFD_RELOC_390_GOT12
: 12 bit GOT offset.

BFD_RELOC_390_PLT32
: 32 bit PC relative PLT address.

BFD_RELOC_390_COPY
: Copy symbol at runtime.

BFD_RELOC_390_GLOB_DAT
: Create GOT entry.

BFD_RELOC_390_JMP_SLOT
: Create PLT entry.

BFD_RELOC_390_RELATIVE
: Adjust by program base.

BFD_RELOC_390_GOTPC
: 32 bit PC relative offset to GOT.

BFD_RELOC_390_GOT16
: 16 bit GOT offset.

BFD_RELOC_390_PC12DBL
: PC relative 12 bit shifted by 1.

BFD_RELOC_390_PLT12DBL
: 12 bit PC rel. PLT shifted by 1.

`BFD_RELOC_390_PC16DBL`
PC relative 16 bit shifted by 1.

`BFD_RELOC_390_PLT16DBL`
16 bit PC rel. PLT shifted by 1.

`BFD_RELOC_390_PC24DBL`
PC relative 24 bit shifted by 1.

`BFD_RELOC_390_PLT24DBL`
24 bit PC rel. PLT shifted by 1.

`BFD_RELOC_390_PC32DBL`
PC relative 32 bit shifted by 1.

`BFD_RELOC_390_PLT32DBL`
32 bit PC rel. PLT shifted by 1.

`BFD_RELOC_390_GOTPCDBL`
32 bit PC rel. GOT shifted by 1.

`BFD_RELOC_390_GOT64`
64 bit GOT offset.

`BFD_RELOC_390_PLT64`
64 bit PC relative PLT address.

`BFD_RELOC_390_GOTENT`
32 bit rel. offset to GOT entry.

`BFD_RELOC_390_GOTOFF64`
64 bit offset to GOT.

`BFD_RELOC_390_GOTPLT12`
12-bit offset to symbol-entry within GOT, with PLT handling.

`BFD_RELOC_390_GOTPLT16`
16-bit offset to symbol-entry within GOT, with PLT handling.

`BFD_RELOC_390_GOTPLT32`
32-bit offset to symbol-entry within GOT, with PLT handling.

`BFD_RELOC_390_GOTPLT64`
64-bit offset to symbol-entry within GOT, with PLT handling.

`BFD_RELOC_390_GOTPLTENT`
32-bit rel. offset to symbol-entry within GOT, with PLT handling.

`BFD_RELOC_390_PLTOFF16`
16-bit rel. offset from the GOT to a PLT entry.

`BFD_RELOC_390_PLTOFF32`
32-bit rel. offset from the GOT to a PLT entry.

`BFD_RELOC_390_PLTOFF64`
: 64-bit rel. offset from the GOT to a PLT entry.

`BFD_RELOC_390_TLS_LOAD`
`BFD_RELOC_390_TLS_GDCALL`
`BFD_RELOC_390_TLS_LDCALL`
`BFD_RELOC_390_TLS_GD32`
`BFD_RELOC_390_TLS_GD64`
`BFD_RELOC_390_TLS_GOTIE12`
`BFD_RELOC_390_TLS_GOTIE32`
`BFD_RELOC_390_TLS_GOTIE64`
`BFD_RELOC_390_TLS_LDM32`
`BFD_RELOC_390_TLS_LDM64`
`BFD_RELOC_390_TLS_IE32`
`BFD_RELOC_390_TLS_IE64`
`BFD_RELOC_390_TLS_IEENT`
`BFD_RELOC_390_TLS_LE32`
`BFD_RELOC_390_TLS_LE64`
`BFD_RELOC_390_TLS_LDO32`
`BFD_RELOC_390_TLS_LDO64`
`BFD_RELOC_390_TLS_DTPMOD`
`BFD_RELOC_390_TLS_DTPOFF`
`BFD_RELOC_390_TLS_TPOFF`
: s390 tls relocations.

`BFD_RELOC_390_20`
`BFD_RELOC_390_GOT20`
`BFD_RELOC_390_GOTPLT20`
`BFD_RELOC_390_TLS_GOTIE20`
: Long displacement extension.

`BFD_RELOC_390_IRELATIVE`
: STT_GNU_IFUNC relocation.

`BFD_RELOC_SCORE_GPREL15`
: Score relocations Low 16 bit for load/store

`BFD_RELOC_SCORE_DUMMY2`
`BFD_RELOC_SCORE_JMP`
: This is a 24-bit reloc with the right 1 bit assumed to be 0

`BFD_RELOC_SCORE_BRANCH`
: This is a 19-bit reloc with the right 1 bit assumed to be 0

`BFD_RELOC_SCORE_IMM30`
: This is a 32-bit reloc for 48-bit instructions.

`BFD_RELOC_SCORE_IMM32`
: This is a 32-bit reloc for 48-bit instructions.

`BFD_RELOC_SCORE16_JMP`
This is a 11-bit reloc with the right 1 bit assumed to be 0

`BFD_RELOC_SCORE16_BRANCH`
This is a 8-bit reloc with the right 1 bit assumed to be 0

`BFD_RELOC_SCORE_BCMP`
This is a 9-bit reloc with the right 1 bit assumed to be 0

`BFD_RELOC_SCORE_GOT15`
`BFD_RELOC_SCORE_GOT_LO16`
`BFD_RELOC_SCORE_CALL15`
`BFD_RELOC_SCORE_DUMMY_HI16`
Undocumented Score relocs

`BFD_RELOC_IP2K_FR9`
Scenix IP2K - 9-bit register number / data address

`BFD_RELOC_IP2K_BANK`
Scenix IP2K - 4-bit register/data bank number

`BFD_RELOC_IP2K_ADDR16CJP`
Scenix IP2K - low 13 bits of instruction word address

`BFD_RELOC_IP2K_PAGE3`
Scenix IP2K - high 3 bits of instruction word address

`BFD_RELOC_IP2K_LO8DATA`
`BFD_RELOC_IP2K_HI8DATA`
`BFD_RELOC_IP2K_EX8DATA`
Scenix IP2K - ext/low/high 8 bits of data address

`BFD_RELOC_IP2K_LO8INSN`
`BFD_RELOC_IP2K_HI8INSN`
Scenix IP2K - low/high 8 bits of instruction word address

`BFD_RELOC_IP2K_PC_SKIP`
Scenix IP2K - even/odd PC modifier to modify snb pcl.0

`BFD_RELOC_IP2K_TEXT`
Scenix IP2K - 16 bit word address in text section.

`BFD_RELOC_IP2K_FR_OFFSET`
Scenix IP2K - 7-bit sp or dp offset

`BFD_RELOC_VPE4KMATH_DATA`
`BFD_RELOC_VPE4KMATH_INSN`
Scenix VPE4K coprocessor - data/insn-space addressing

BFD_RELOC_VTABLE_INHERIT
BFD_RELOC_VTABLE_ENTRY

These two relocations are used by the linker to determine which of the entries in a C++ virtual function table are actually used. When the –gc-sections option is given, the linker will zero out the entries that are not used, so that the code for those functions need not be included in the output.

VTABLE_INHERIT is a zero-space relocation used to describe to the linker the inheritance tree of a C++ virtual function table. The relocation's symbol should be the parent class' vtable, and the relocation should be located at the child vtable.

VTABLE_ENTRY is a zero-space relocation that describes the use of a virtual function table entry. The reloc's symbol should refer to the table of the class mentioned in the code. Off of that base, an offset describes the entry that is being used. For Rela hosts, this offset is stored in the reloc's addend. For Rel hosts, we are forced to put this offset in the reloc's section offset.

BFD_RELOC_IA64_IMM14
BFD_RELOC_IA64_IMM22
BFD_RELOC_IA64_IMM64
BFD_RELOC_IA64_DIR32MSB
BFD_RELOC_IA64_DIR32LSB
BFD_RELOC_IA64_DIR64MSB
BFD_RELOC_IA64_DIR64LSB
BFD_RELOC_IA64_GPREL22
BFD_RELOC_IA64_GPREL64I
BFD_RELOC_IA64_GPREL32MSB
BFD_RELOC_IA64_GPREL32LSB
BFD_RELOC_IA64_GPREL64MSB
BFD_RELOC_IA64_GPREL64LSB
BFD_RELOC_IA64_LTOFF22
BFD_RELOC_IA64_LTOFF64I
BFD_RELOC_IA64_PLTOFF22
BFD_RELOC_IA64_PLTOFF64I
BFD_RELOC_IA64_PLTOFF64MSB
BFD_RELOC_IA64_PLTOFF64LSB
BFD_RELOC_IA64_FPTR64I
BFD_RELOC_IA64_FPTR32MSB
BFD_RELOC_IA64_FPTR32LSB
BFD_RELOC_IA64_FPTR64MSB
BFD_RELOC_IA64_FPTR64LSB
BFD_RELOC_IA64_PCREL21B
BFD_RELOC_IA64_PCREL21BI
BFD_RELOC_IA64_PCREL21M
BFD_RELOC_IA64_PCREL21F
BFD_RELOC_IA64_PCREL22
BFD_RELOC_IA64_PCREL60B
BFD_RELOC_IA64_PCREL64I
BFD_RELOC_IA64_PCREL32MSB

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BFD_RELOC_IA64_PCREL32LSB
BFD_RELOC_IA64_PCREL64MSB
BFD_RELOC_IA64_PCREL64LSB
BFD_RELOC_IA64_LTOFF_FPTR22
BFD_RELOC_IA64_LTOFF_FPTR64I
BFD_RELOC_IA64_LTOFF_FPTR32MSB
BFD_RELOC_IA64_LTOFF_FPTR32LSB
BFD_RELOC_IA64_LTOFF_FPTR64MSB
BFD_RELOC_IA64_LTOFF_FPTR64LSB
BFD_RELOC_IA64_SEGREL32MSB
BFD_RELOC_IA64_SEGREL32LSB
BFD_RELOC_IA64_SEGREL64MSB
BFD_RELOC_IA64_SEGREL64LSB
BFD_RELOC_IA64_SECREL32MSB
BFD_RELOC_IA64_SECREL32LSB
BFD_RELOC_IA64_SECREL64MSB
BFD_RELOC_IA64_SECREL64LSB
BFD_RELOC_IA64_REL32MSB
BFD_RELOC_IA64_REL32LSB
BFD_RELOC_IA64_REL64MSB
BFD_RELOC_IA64_REL64LSB
BFD_RELOC_IA64_LTV32MSB
BFD_RELOC_IA64_LTV32LSB
BFD_RELOC_IA64_LTV64MSB
BFD_RELOC_IA64_LTV64LSB
BFD_RELOC_IA64_IPLTMSB
BFD_RELOC_IA64_IPLTLSB
BFD_RELOC_IA64_COPY
BFD_RELOC_IA64_LTOFF22X
BFD_RELOC_IA64_LDXMOV
BFD_RELOC_IA64_TPREL14
BFD_RELOC_IA64_TPREL22
BFD_RELOC_IA64_TPREL64I
BFD_RELOC_IA64_TPREL64MSB
BFD_RELOC_IA64_TPREL64LSB
BFD_RELOC_IA64_LTOFF_TPREL22
BFD_RELOC_IA64_DTPMOD64MSB
BFD_RELOC_IA64_DTPMOD64LSB
BFD_RELOC_IA64_LTOFF_DTPMOD22
BFD_RELOC_IA64_DTPREL14
BFD_RELOC_IA64_DTPREL22
BFD_RELOC_IA64_DTPREL64I
BFD_RELOC_IA64_DTPREL32MSB
BFD_RELOC_IA64_DTPREL32LSB
BFD_RELOC_IA64_DTPREL64MSB
BFD_RELOC_IA64_DTPREL64LSB
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`BFD_RELOC_IA64_LTOFF_DTPREL22`
Intel IA64 Relocations.

`BFD_RELOC_M68HC11_HI8`
Motorola 68HC11 reloc. This is the 8 bit high part of an absolute address.

`BFD_RELOC_M68HC11_LO8`
Motorola 68HC11 reloc. This is the 8 bit low part of an absolute address.

`BFD_RELOC_M68HC11_3B`
Motorola 68HC11 reloc. This is the 3 bit of a value.

`BFD_RELOC_M68HC11_RL_JUMP`
Motorola 68HC11 reloc. This reloc marks the beginning of a jump/call instruction. It is used for linker relaxation to correctly identify beginning of instruction and change some branches to use PC-relative addressing mode.

`BFD_RELOC_M68HC11_RL_GROUP`
Motorola 68HC11 reloc. This reloc marks a group of several instructions that gcc generates and for which the linker relaxation pass can modify and/or remove some of them.

`BFD_RELOC_M68HC11_LO16`
Motorola 68HC11 reloc. This is the 16-bit lower part of an address. It is used for 'call' instruction to specify the symbol address without any special transformation (due to memory bank window).

`BFD_RELOC_M68HC11_PAGE`
Motorola 68HC11 reloc. This is a 8-bit reloc that specifies the page number of an address. It is used by 'call' instruction to specify the page number of the symbol.

`BFD_RELOC_M68HC11_24`
Motorola 68HC11 reloc. This is a 24-bit reloc that represents the address with a 16-bit value and a 8-bit page number. The symbol address is transformed to follow the 16K memory bank of 68HC12 (seen as mapped in the window).

`BFD_RELOC_M68HC12_5B`
Motorola 68HC12 reloc. This is the 5 bits of a value.

`BFD_RELOC_XGATE_RL_JUMP`
Freescale XGATE reloc. This reloc marks the beginning of a bra/jal instruction.

`BFD_RELOC_XGATE_RL_GROUP`
Freescale XGATE reloc. This reloc marks a group of several instructions that gcc generates and for which the linker relaxation pass can modify and/or remove some of them.

`BFD_RELOC_XGATE_LO16`
Freescale XGATE reloc. This is the 16-bit lower part of an address. It is used for the '16-bit' instructions.

`BFD_RELOC_XGATE_GPAGE`
Freescale XGATE reloc.

`BFD_RELOC_XGATE_24`
Freescale XGATE reloc.

`BFD_RELOC_XGATE_PCREL_9`
Freescale XGATE reloc. This is a 9-bit pc-relative reloc.

`BFD_RELOC_XGATE_PCREL_10`
Freescale XGATE reloc. This is a 10-bit pc-relative reloc.

`BFD_RELOC_XGATE_IMM8_LO`
Freescale XGATE reloc. This is the 16-bit lower part of an address. It is used for the '16-bit' instructions.

`BFD_RELOC_XGATE_IMM8_HI`
Freescale XGATE reloc. This is the 16-bit higher part of an address. It is used for the '16-bit' instructions.

`BFD_RELOC_XGATE_IMM3`
Freescale XGATE reloc. This is a 3-bit pc-relative reloc.

`BFD_RELOC_XGATE_IMM4`
Freescale XGATE reloc. This is a 4-bit pc-relative reloc.

`BFD_RELOC_XGATE_IMM5`
Freescale XGATE reloc. This is a 5-bit pc-relative reloc.

`BFD_RELOC_M68HC12_9B`
Motorola 68HC12 reloc. This is the 9 bits of a value.

`BFD_RELOC_M68HC12_16B`
Motorola 68HC12 reloc. This is the 16 bits of a value.

`BFD_RELOC_M68HC12_9_PCREL`
Motorola 68HC12/XGATE reloc. This is a PCREL9 branch.

`BFD_RELOC_M68HC12_10_PCREL`
Motorola 68HC12/XGATE reloc. This is a PCREL10 branch.

`BFD_RELOC_M68HC12_LO8XG`
Motorola 68HC12/XGATE reloc. This is the 8 bit low part of an absolute address and immediately precedes a matching HI8XG part.

`BFD_RELOC_M68HC12_HI8XG`
Motorola 68HC12/XGATE reloc. This is the 8 bit high part of an absolute address and immediately follows a matching LO8XG part.

`BFD_RELOC_S12Z_15_PCREL`
Freescale S12Z reloc. This is a 15 bit relative address. If the most significant bits are all zero then it may be truncated to 8 bits.

BFD_RELOC_CR16_NUM8
BFD_RELOC_CR16_NUM16
BFD_RELOC_CR16_NUM32
BFD_RELOC_CR16_NUM32a
BFD_RELOC_CR16_REGREL0
BFD_RELOC_CR16_REGREL4
BFD_RELOC_CR16_REGREL4a
BFD_RELOC_CR16_REGREL14
BFD_RELOC_CR16_REGREL14a
BFD_RELOC_CR16_REGREL16
BFD_RELOC_CR16_REGREL20
BFD_RELOC_CR16_REGREL20a
BFD_RELOC_CR16_ABS20
BFD_RELOC_CR16_ABS24
BFD_RELOC_CR16_IMM4
BFD_RELOC_CR16_IMM8
BFD_RELOC_CR16_IMM16
BFD_RELOC_CR16_IMM20
BFD_RELOC_CR16_IMM24
BFD_RELOC_CR16_IMM32
BFD_RELOC_CR16_IMM32a
BFD_RELOC_CR16_DISP4
BFD_RELOC_CR16_DISP8
BFD_RELOC_CR16_DISP16
BFD_RELOC_CR16_DISP20
BFD_RELOC_CR16_DISP24
BFD_RELOC_CR16_DISP24a
BFD_RELOC_CR16_SWITCH8
BFD_RELOC_CR16_SWITCH16
BFD_RELOC_CR16_SWITCH32
BFD_RELOC_CR16_GOT_REGREL20
BFD_RELOC_CR16_GOTC_REGREL20
BFD_RELOC_CR16_GLOB_DAT

NS CR16 Relocations.

BFD_RELOC_CRX_REL4
BFD_RELOC_CRX_REL8
BFD_RELOC_CRX_REL8_CMP
BFD_RELOC_CRX_REL16
BFD_RELOC_CRX_REL24
BFD_RELOC_CRX_REL32
BFD_RELOC_CRX_REGREL12
BFD_RELOC_CRX_REGREL22
BFD_RELOC_CRX_REGREL28
BFD_RELOC_CRX_REGREL32
BFD_RELOC_CRX_ABS16
BFD_RELOC_CRX_ABS32

BFD_RELOC_CRX_NUM8
BFD_RELOC_CRX_NUM16
BFD_RELOC_CRX_NUM32
BFD_RELOC_CRX_IMM16
BFD_RELOC_CRX_IMM32
BFD_RELOC_CRX_SWITCH8
BFD_RELOC_CRX_SWITCH16
BFD_RELOC_CRX_SWITCH32
NS CRX Relocations.

BFD_RELOC_CRIS_BDISP8
BFD_RELOC_CRIS_UNSIGNED_5
BFD_RELOC_CRIS_SIGNED_6
BFD_RELOC_CRIS_UNSIGNED_6
BFD_RELOC_CRIS_SIGNED_8
BFD_RELOC_CRIS_UNSIGNED_8
BFD_RELOC_CRIS_SIGNED_16
BFD_RELOC_CRIS_UNSIGNED_16
BFD_RELOC_CRIS_LAPCQ_OFFSET
BFD_RELOC_CRIS_UNSIGNED_4
These relocs are only used within the CRIS assembler. They are not (at present) written to any object files.

BFD_RELOC_CRIS_COPY
BFD_RELOC_CRIS_GLOB_DAT
BFD_RELOC_CRIS_JUMP_SLOT
BFD_RELOC_CRIS_RELATIVE
Relocs used in ELF shared libraries for CRIS.

BFD_RELOC_CRIS_32_GOT
32-bit offset to symbol-entry within GOT.

BFD_RELOC_CRIS_16_GOT
16-bit offset to symbol-entry within GOT.

BFD_RELOC_CRIS_32_GOTPLT
32-bit offset to symbol-entry within GOT, with PLT handling.

BFD_RELOC_CRIS_16_GOTPLT
16-bit offset to symbol-entry within GOT, with PLT handling.

BFD_RELOC_CRIS_32_GOTREL
32-bit offset to symbol, relative to GOT.

BFD_RELOC_CRIS_32_PLT_GOTREL
32-bit offset to symbol with PLT entry, relative to GOT.

BFD_RELOC_CRIS_32_PLT_PCREL
32-bit offset to symbol with PLT entry, relative to this relocation.

BFD_RELOC_CRIS_32_GOT_GD
BFD_RELOC_CRIS_16_GOT_GD
BFD_RELOC_CRIS_32_GD
BFD_RELOC_CRIS_DTP
BFD_RELOC_CRIS_32_DTPREL
BFD_RELOC_CRIS_16_DTPREL
BFD_RELOC_CRIS_32_GOT_TPREL
BFD_RELOC_CRIS_16_GOT_TPREL
BFD_RELOC_CRIS_32_TPREL
BFD_RELOC_CRIS_16_TPREL
BFD_RELOC_CRIS_DTPMOD
BFD_RELOC_CRIS_32_IE

Relocs used in TLS code for CRIS.

BFD_RELOC_OR1K_REL_26
BFD_RELOC_OR1K_SLO16
BFD_RELOC_OR1K_PCREL_PG21
BFD_RELOC_OR1K_LO13
BFD_RELOC_OR1K_SLO13
BFD_RELOC_OR1K_GOTPC_HI16
BFD_RELOC_OR1K_GOTPC_LO16
BFD_RELOC_OR1K_GOT_AHI16
BFD_RELOC_OR1K_GOT16
BFD_RELOC_OR1K_GOT_PG21
BFD_RELOC_OR1K_GOT_LO13
BFD_RELOC_OR1K_PLT26
BFD_RELOC_OR1K_PLTA26
BFD_RELOC_OR1K_GOTOFF_SLO16
BFD_RELOC_OR1K_COPY
BFD_RELOC_OR1K_GLOB_DAT
BFD_RELOC_OR1K_JMP_SLOT
BFD_RELOC_OR1K_RELATIVE
BFD_RELOC_OR1K_TLS_GD_HI16
BFD_RELOC_OR1K_TLS_GD_LO16
BFD_RELOC_OR1K_TLS_GD_PG21
BFD_RELOC_OR1K_TLS_GD_LO13
BFD_RELOC_OR1K_TLS_LDM_HI16
BFD_RELOC_OR1K_TLS_LDM_LO16
BFD_RELOC_OR1K_TLS_LDM_PG21
BFD_RELOC_OR1K_TLS_LDM_LO13
BFD_RELOC_OR1K_TLS_LDO_HI16
BFD_RELOC_OR1K_TLS_LDO_LO16
BFD_RELOC_OR1K_TLS_IE_HI16
BFD_RELOC_OR1K_TLS_IE_AHI16
BFD_RELOC_OR1K_TLS_IE_LO16
BFD_RELOC_OR1K_TLS_IE_PG21
BFD_RELOC_OR1K_TLS_IE_LO13

BFD_RELOC_OR1K_TLS_LE_HI16
BFD_RELOC_OR1K_TLS_LE_AHI16
BFD_RELOC_OR1K_TLS_LE_LO16
BFD_RELOC_OR1K_TLS_LE_SLO16
BFD_RELOC_OR1K_TLS_TPOFF
BFD_RELOC_OR1K_TLS_DTPOFF
BFD_RELOC_OR1K_TLS_DTPMOD
OpenRISC 1000 Relocations.

BFD_RELOC_H8_DIR16A8
BFD_RELOC_H8_DIR16R8
BFD_RELOC_H8_DIR24A8
BFD_RELOC_H8_DIR24R8
BFD_RELOC_H8_DIR32A16
BFD_RELOC_H8_DISP32A16
H8 elf Relocations.

BFD_RELOC_XSTORMY16_REL_12
BFD_RELOC_XSTORMY16_12
BFD_RELOC_XSTORMY16_24
BFD_RELOC_XSTORMY16_FPTR16
Sony Xstormy16 Relocations.

BFD_RELOC_RELC
Self-describing complex relocations.

BFD_RELOC_XC16X_PAG
BFD_RELOC_XC16X_POF
BFD_RELOC_XC16X_SEG
BFD_RELOC_XC16X_SOF
Infineon Relocations.

BFD_RELOC_VAX_GLOB_DAT
BFD_RELOC_VAX_JMP_SLOT
BFD_RELOC_VAX_RELATIVE
Relocations used by VAX ELF.

BFD_RELOC_MT_PC16
Morpho MT - 16 bit immediate relocation.

BFD_RELOC_MT_HI16
Morpho MT - Hi 16 bits of an address.

BFD_RELOC_MT_LO16
Morpho MT - Low 16 bits of an address.

BFD_RELOC_MT_GNU_VTINHERIT
Morpho MT - Used to tell the linker which vtable entries are used.

BFD_RELOC_MT_GNU_VTENTRY
Morpho MT - Used to tell the linker which vtable entries are used.

BFD_RELOC_MT_PCINSN8
Morpho MT - 8 bit immediate relocation.

BFD_RELOC_MSP430_10_PCREL
BFD_RELOC_MSP430_16_PCREL
BFD_RELOC_MSP430_16
BFD_RELOC_MSP430_16_PCREL_BYTE
BFD_RELOC_MSP430_16_BYTE
BFD_RELOC_MSP430_2X_PCREL
BFD_RELOC_MSP430_RL_PCREL
BFD_RELOC_MSP430_ABS8
BFD_RELOC_MSP430X_PCR20_EXT_SRC
BFD_RELOC_MSP430X_PCR20_EXT_DST
BFD_RELOC_MSP430X_PCR20_EXT_ODST
BFD_RELOC_MSP430X_ABS20_EXT_SRC
BFD_RELOC_MSP430X_ABS20_EXT_DST
BFD_RELOC_MSP430X_ABS20_EXT_ODST
BFD_RELOC_MSP430X_ABS20_ADR_SRC
BFD_RELOC_MSP430X_ABS20_ADR_DST
BFD_RELOC_MSP430X_PCR16
BFD_RELOC_MSP430X_PCR20_CALL
BFD_RELOC_MSP430X_ABS16
BFD_RELOC_MSP430_ABS_HI16
BFD_RELOC_MSP430_PREL31
BFD_RELOC_MSP430_SYM_DIFF
BFD_RELOC_MSP430_SET_ULEB128
BFD_RELOC_MSP430_SUB_ULEB128
msp430 specific relocation codes

BFD_RELOC_NIOS2_S16
BFD_RELOC_NIOS2_U16
BFD_RELOC_NIOS2_CALL26
BFD_RELOC_NIOS2_IMM5
BFD_RELOC_NIOS2_CACHE_OPX
BFD_RELOC_NIOS2_IMM6
BFD_RELOC_NIOS2_IMM8
BFD_RELOC_NIOS2_HI16
BFD_RELOC_NIOS2_LO16
BFD_RELOC_NIOS2_HIADJ16
BFD_RELOC_NIOS2_GPREL
BFD_RELOC_NIOS2_UJMP
BFD_RELOC_NIOS2_CJMP
BFD_RELOC_NIOS2_CALLR
BFD_RELOC_NIOS2_ALIGN
BFD_RELOC_NIOS2_GOT16
BFD_RELOC_NIOS2_CALL16
BFD_RELOC_NIOS2_GOTOFF_LO
BFD_RELOC_NIOS2_GOTOFF_HA

BFD_RELOC_NIOS2_PCREL_LO
BFD_RELOC_NIOS2_PCREL_HA
BFD_RELOC_NIOS2_TLS_GD16
BFD_RELOC_NIOS2_TLS_LDM16
BFD_RELOC_NIOS2_TLS_LDO16
BFD_RELOC_NIOS2_TLS_IE16
BFD_RELOC_NIOS2_TLS_LE16
BFD_RELOC_NIOS2_TLS_DTPMOD
BFD_RELOC_NIOS2_TLS_DTPREL
BFD_RELOC_NIOS2_TLS_TPREL
BFD_RELOC_NIOS2_COPY
BFD_RELOC_NIOS2_GLOB_DAT
BFD_RELOC_NIOS2_JUMP_SLOT
BFD_RELOC_NIOS2_RELATIVE
BFD_RELOC_NIOS2_GOTOFF
BFD_RELOC_NIOS2_CALL26_NOAT
BFD_RELOC_NIOS2_GOT_LO
BFD_RELOC_NIOS2_GOT_HA
BFD_RELOC_NIOS2_CALL_LO
BFD_RELOC_NIOS2_CALL_HA
BFD_RELOC_NIOS2_R2_S12
BFD_RELOC_NIOS2_R2_I10_1_PCREL
BFD_RELOC_NIOS2_R2_T1I7_1_PCREL
BFD_RELOC_NIOS2_R2_T1I7_2
BFD_RELOC_NIOS2_R2_T2I4
BFD_RELOC_NIOS2_R2_T2I4_1
BFD_RELOC_NIOS2_R2_T2I4_2
BFD_RELOC_NIOS2_R2_X1I7_2
BFD_RELOC_NIOS2_R2_X2L5
BFD_RELOC_NIOS2_R2_F1I5_2
BFD_RELOC_NIOS2_R2_L5I4X1
BFD_RELOC_NIOS2_R2_T1X1I6
BFD_RELOC_NIOS2_R2_T1X1I6_2

Relocations used by the Altera Nios II core.

BFD_RELOC_PRU_U16

PRU LDI 16-bit unsigned data-memory relocation.

BFD_RELOC_PRU_U16_PMEMIMM

PRU LDI 16-bit unsigned instruction-memory relocation.

BFD_RELOC_PRU_LDI32

PRU relocation for two consecutive LDI load instructions that load a 32 bit value into a register. If the higher bits are all zero, then the second instruction may be relaxed.

BFD_RELOC_PRU_S10_PCREL

PRU QBBx 10-bit signed PC-relative relocation.

`BFD_RELOC_PRU_U8_PCREL`
: PRU 8-bit unsigned relocation used for the LOOP instruction.

`BFD_RELOC_PRU_32_PMEM`
`BFD_RELOC_PRU_16_PMEM`
: PRU Program Memory relocations. Used to convert from byte addressing to 32-bit word addressing.

`BFD_RELOC_PRU_GNU_DIFF8`
`BFD_RELOC_PRU_GNU_DIFF16`
`BFD_RELOC_PRU_GNU_DIFF32`
`BFD_RELOC_PRU_GNU_DIFF16_PMEM`
`BFD_RELOC_PRU_GNU_DIFF32_PMEM`
: PRU relocations to mark the difference of two local symbols. These are only needed to support linker relaxation and can be ignored when not relaxing. The field is set to the value of the difference assuming no relaxation. The relocation encodes the position of the second symbol so the linker can determine whether to adjust the field value. The PMEM variants encode the word difference, instead of byte difference between symbols.

`BFD_RELOC_IQ2000_OFFSET_16`
`BFD_RELOC_IQ2000_OFFSET_21`
`BFD_RELOC_IQ2000_UHI16`
: IQ2000 Relocations.

`BFD_RELOC_XTENSA_RTLD`
: Special Xtensa relocation used only by PLT entries in ELF shared objects to indicate that the runtime linker should set the value to one of its own internal functions or data structures.

`BFD_RELOC_XTENSA_GLOB_DAT`
`BFD_RELOC_XTENSA_JMP_SLOT`
`BFD_RELOC_XTENSA_RELATIVE`
: Xtensa relocations for ELF shared objects.

`BFD_RELOC_XTENSA_PLT`
: Xtensa relocation used in ELF object files for symbols that may require PLT entries. Otherwise, this is just a generic 32-bit relocation.

`BFD_RELOC_XTENSA_DIFF8`
`BFD_RELOC_XTENSA_DIFF16`
`BFD_RELOC_XTENSA_DIFF32`
: Xtensa relocations for backward compatibility. These have been replaced by BFD_RELOC_XTENSA_PDIFF and BFD_RELOC_XTENSA_NDIFF. Xtensa relocations to mark the difference of two local symbols. These are only needed to support linker relaxation and can be ignored when not relaxing. The field is set to the value of the difference assuming no relaxation. The relocation encodes the position of the first symbol so the linker can determine whether to adjust the field value.

BFD_RELOC_XTENSA_SLOT0_OP
BFD_RELOC_XTENSA_SLOT1_OP
BFD_RELOC_XTENSA_SLOT2_OP
BFD_RELOC_XTENSA_SLOT3_OP
BFD_RELOC_XTENSA_SLOT4_OP
BFD_RELOC_XTENSA_SLOT5_OP
BFD_RELOC_XTENSA_SLOT6_OP
BFD_RELOC_XTENSA_SLOT7_OP
BFD_RELOC_XTENSA_SLOT8_OP
BFD_RELOC_XTENSA_SLOT9_OP
BFD_RELOC_XTENSA_SLOT10_OP
BFD_RELOC_XTENSA_SLOT11_OP
BFD_RELOC_XTENSA_SLOT12_OP
BFD_RELOC_XTENSA_SLOT13_OP
BFD_RELOC_XTENSA_SLOT14_OP

Generic Xtensa relocations for instruction operands. Only the slot number is encoded in the relocation. The relocation applies to the last PC-relative immediate operand, or if there are no PC-relative immediates, to the last immediate operand.

BFD_RELOC_XTENSA_SLOT0_ALT
BFD_RELOC_XTENSA_SLOT1_ALT
BFD_RELOC_XTENSA_SLOT2_ALT
BFD_RELOC_XTENSA_SLOT3_ALT
BFD_RELOC_XTENSA_SLOT4_ALT
BFD_RELOC_XTENSA_SLOT5_ALT
BFD_RELOC_XTENSA_SLOT6_ALT
BFD_RELOC_XTENSA_SLOT7_ALT
BFD_RELOC_XTENSA_SLOT8_ALT
BFD_RELOC_XTENSA_SLOT9_ALT
BFD_RELOC_XTENSA_SLOT10_ALT
BFD_RELOC_XTENSA_SLOT11_ALT
BFD_RELOC_XTENSA_SLOT12_ALT
BFD_RELOC_XTENSA_SLOT13_ALT
BFD_RELOC_XTENSA_SLOT14_ALT

Alternate Xtensa relocations. Only the slot is encoded in the relocation. The meaning of these relocations is opcode-specific.

BFD_RELOC_XTENSA_OP0
BFD_RELOC_XTENSA_OP1
BFD_RELOC_XTENSA_OP2

Xtensa relocations for backward compatibility. These have all been replaced by BFD_RELOC_XTENSA_SLOT0_OP.

BFD_RELOC_XTENSA_ASM_EXPAND

Xtensa relocation to mark that the assembler expanded the instructions from an original target. The expansion size is encoded in the reloc size.

`BFD_RELOC_XTENSA_ASM_SIMPLIFY`
: Xtensa relocation to mark that the linker should simplify assembler-expanded instructions. This is commonly used internally by the linker after analysis of a BFD_RELOC_XTENSA_ASM_EXPAND.

`BFD_RELOC_XTENSA_TLSDESC_FN`
`BFD_RELOC_XTENSA_TLSDESC_ARG`
`BFD_RELOC_XTENSA_TLS_DTPOFF`
`BFD_RELOC_XTENSA_TLS_TPOFF`
`BFD_RELOC_XTENSA_TLS_FUNC`
`BFD_RELOC_XTENSA_TLS_ARG`
`BFD_RELOC_XTENSA_TLS_CALL`
: Xtensa TLS relocations.

`BFD_RELOC_XTENSA_PDIFF8`
`BFD_RELOC_XTENSA_PDIFF16`
`BFD_RELOC_XTENSA_PDIFF32`
`BFD_RELOC_XTENSA_NDIFF8`
`BFD_RELOC_XTENSA_NDIFF16`
`BFD_RELOC_XTENSA_NDIFF32`
: Xtensa relocations to mark the difference of two local symbols. These are only needed to support linker relaxation and can be ignored when not relaxing. The field is set to the value of the difference assuming no relaxation. The relocation encodes the position of the subtracted symbol so the linker can determine whether to adjust the field value. PDIFF relocations are used for positive differences, NDIFF relocations are used for negative differences. The difference value is treated as unsigned with these relocation types, giving full 8/16 value ranges.

`BFD_RELOC_Z80_DISP8`
: 8 bit signed offset in (ix+d) or (iy+d).

`BFD_RELOC_Z80_BYTE0`
: First 8 bits of multibyte (32, 24 or 16 bit) value.

`BFD_RELOC_Z80_BYTE1`
: Second 8 bits of multibyte (32, 24 or 16 bit) value.

`BFD_RELOC_Z80_BYTE2`
: Third 8 bits of multibyte (32 or 24 bit) value.

`BFD_RELOC_Z80_BYTE3`
: Fourth 8 bits of multibyte (32 bit) value.

`BFD_RELOC_Z80_WORD0`
: Lowest 16 bits of multibyte (32 or 24 bit) value.

`BFD_RELOC_Z80_WORD1`
: Highest 16 bits of multibyte (32 or 24 bit) value.

`BFD_RELOC_Z80_16_BE`
: Like BFD_RELOC_16 but big-endian.

`BFD_RELOC_Z8K_DISP7`
DJNZ offset.

`BFD_RELOC_Z8K_CALLR`
CALR offset.

`BFD_RELOC_Z8K_IMM4L`
4 bit value.

`BFD_RELOC_LM32_CALL`
`BFD_RELOC_LM32_BRANCH`
`BFD_RELOC_LM32_16_GOT`
`BFD_RELOC_LM32_GOTOFF_HI16`
`BFD_RELOC_LM32_GOTOFF_LO16`
`BFD_RELOC_LM32_COPY`
`BFD_RELOC_LM32_GLOB_DAT`
`BFD_RELOC_LM32_JMP_SLOT`
`BFD_RELOC_LM32_RELATIVE`
Lattice Mico32 relocations.

`BFD_RELOC_MACH_O_SECTDIFF`
Difference between two section addreses. Must be followed by a BFD_RELOC_MACH_O_PAIR.

`BFD_RELOC_MACH_O_LOCAL_SECTDIFF`
Like BFD_RELOC_MACH_O_SECTDIFF but with a local symbol.

`BFD_RELOC_MACH_O_PAIR`
Pair of relocation. Contains the first symbol.

`BFD_RELOC_MACH_O_SUBTRACTOR32`
Symbol will be substracted. Must be followed by a BFD_RELOC_32.

`BFD_RELOC_MACH_O_SUBTRACTOR64`
Symbol will be substracted. Must be followed by a BFD_RELOC_64.

`BFD_RELOC_MACH_O_X86_64_BRANCH32`
`BFD_RELOC_MACH_O_X86_64_BRANCH8`
PCREL relocations. They are marked as branch to create PLT entry if required.

`BFD_RELOC_MACH_O_X86_64_GOT`
Used when referencing a GOT entry.

`BFD_RELOC_MACH_O_X86_64_GOT_LOAD`
Used when loading a GOT entry with movq. It is specially marked so that the linker could optimize the movq to a leaq if possible.

`BFD_RELOC_MACH_O_X86_64_PCREL32_1`
Same as BFD_RELOC_32_PCREL but with an implicit -1 addend.

`BFD_RELOC_MACH_O_X86_64_PCREL32_2`
Same as BFD_RELOC_32_PCREL but with an implicit -2 addend.

`BFD_RELOC_MACH_O_X86_64_PCREL32_4`
Same as BFD_RELOC_32_PCREL but with an implicit -4 addend.

`BFD_RELOC_MACH_O_X86_64_TLV`
Used when referencing a TLV entry.

`BFD_RELOC_MACH_O_ARM64_ADDEND`
Addend for PAGE or PAGEOFF.

`BFD_RELOC_MACH_O_ARM64_GOT_LOAD_PAGE21`
Relative offset to page of GOT slot.

`BFD_RELOC_MACH_O_ARM64_GOT_LOAD_PAGEOFF12`
Relative offset within page of GOT slot.

`BFD_RELOC_MACH_O_ARM64_POINTER_TO_GOT`
Address of a GOT entry.

`BFD_RELOC_MICROBLAZE_32_LO`
This is a 32 bit reloc for the microblaze that stores the low 16 bits of a value

`BFD_RELOC_MICROBLAZE_32_LO_PCREL`
This is a 32 bit pc-relative reloc for the microblaze that stores the low 16 bits of a value

`BFD_RELOC_MICROBLAZE_32_ROSDA`
This is a 32 bit reloc for the microblaze that stores a value relative to the read-only small data area anchor

`BFD_RELOC_MICROBLAZE_32_RWSDA`
This is a 32 bit reloc for the microblaze that stores a value relative to the read-write small data area anchor

`BFD_RELOC_MICROBLAZE_32_SYM_OP_SYM`
This is a 32 bit reloc for the microblaze to handle expressions of the form "Symbol Op Symbol"

`BFD_RELOC_MICROBLAZE_64_NONE`
This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). No relocation is done here - only used for relaxing

`BFD_RELOC_MICROBLAZE_64_GOTPC`
This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). The relocation is PC-relative GOT offset

`BFD_RELOC_MICROBLAZE_64_GOT`
This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). The relocation is GOT offset

`BFD_RELOC_MICROBLAZE_64_PLT`
This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). The relocation is PC-relative offset into PLT

`BFD_RELOC_MICROBLAZE_64_GOTOFF`
This is a 64 bit reloc that stores the 32 bit GOT relative value in two words (with an imm instruction). The relocation is relative offset from _GLOBAL_OFFSET_TABLE_

`BFD_RELOC_MICROBLAZE_32_GOTOFF`
This is a 32 bit reloc that stores the 32 bit GOT relative value in a word. The relocation is relative offset from

`BFD_RELOC_MICROBLAZE_COPY`
This is used to tell the dynamic linker to copy the value out of the dynamic object into the runtime process image.

`BFD_RELOC_MICROBLAZE_64_TLS`
Unused Reloc

`BFD_RELOC_MICROBLAZE_64_TLSGD`
This is a 64 bit reloc that stores the 32 bit GOT relative value of the GOT TLS GD info entry in two words (with an imm instruction). The relocation is GOT offset.

`BFD_RELOC_MICROBLAZE_64_TLSLD`
This is a 64 bit reloc that stores the 32 bit GOT relative value of the GOT TLS LD info entry in two words (with an imm instruction). The relocation is GOT offset.

`BFD_RELOC_MICROBLAZE_32_TLSDTPMOD`
This is a 32 bit reloc that stores the Module ID to GOT(n).

`BFD_RELOC_MICROBLAZE_32_TLSDTPREL`
This is a 32 bit reloc that stores TLS offset to GOT(n+1).

`BFD_RELOC_MICROBLAZE_64_TLSDTPREL`
This is a 32 bit reloc for storing TLS offset to two words (uses imm instruction)

`BFD_RELOC_MICROBLAZE_64_TLSGOTTPREL`
This is a 64 bit reloc that stores 32-bit thread pointer relative offset to two words (uses imm instruction).

`BFD_RELOC_MICROBLAZE_64_TLSTPREL`
This is a 64 bit reloc that stores 32-bit thread pointer relative offset to two words (uses imm instruction).

`BFD_RELOC_MICROBLAZE_64_TEXTPCREL`
This is a 64 bit reloc that stores the 32 bit pc relative value in two words (with an imm instruction). The relocation is PC-relative offset from start of TEXT.

`BFD_RELOC_MICROBLAZE_64_TEXTREL`
This is a 64 bit reloc that stores the 32 bit offset value in two words (with an imm instruction). The relocation is relative offset from start of TEXT.

`BFD_RELOC_AARCH64_RELOC_START`
AArch64 pseudo relocation code to mark the start of the AArch64 relocation enumerators. N.B. the order of the enumerators is important as several tables in the AArch64 bfd backend are indexed by these enumerators; make sure they are all synced.

`BFD_RELOC_AARCH64_NULL`
Deprecated AArch64 null relocation code.

`BFD_RELOC_AARCH64_NONE`
AArch64 null relocation code.

`BFD_RELOC_AARCH64_64`
`BFD_RELOC_AARCH64_32`
`BFD_RELOC_AARCH64_16`
Basic absolute relocations of N bits. These are equivalent to BFD_RELOC_N and they were added to assist the indexing of the howto table.

`BFD_RELOC_AARCH64_64_PCREL`
`BFD_RELOC_AARCH64_32_PCREL`
`BFD_RELOC_AARCH64_16_PCREL`
PC-relative relocations. These are equivalent to BFD_RELOC_N_PCREL and they were added to assist the indexing of the howto table.

`BFD_RELOC_AARCH64_MOVW_G0`
AArch64 MOV[NZK] instruction with most significant bits 0 to 15 of an unsigned address/value.

`BFD_RELOC_AARCH64_MOVW_G0_NC`
AArch64 MOV[NZK] instruction with less significant bits 0 to 15 of an address/value. No overflow checking.

`BFD_RELOC_AARCH64_MOVW_G1`
AArch64 MOV[NZK] instruction with most significant bits 16 to 31 of an unsigned address/value.

`BFD_RELOC_AARCH64_MOVW_G1_NC`
AArch64 MOV[NZK] instruction with less significant bits 16 to 31 of an address/value. No overflow checking.

`BFD_RELOC_AARCH64_MOVW_G2`
AArch64 MOV[NZK] instruction with most significant bits 32 to 47 of an unsigned address/value.

`BFD_RELOC_AARCH64_MOVW_G2_NC`
AArch64 MOV[NZK] instruction with less significant bits 32 to 47 of an address/value. No overflow checking.

`BFD_RELOC_AARCH64_MOVW_G3`
AArch64 MOV[NZK] instruction with most signficant bits 48 to 64 of a signed or unsigned address/value.

`BFD_RELOC_AARCH64_MOVW_G0_S`
AArch64 MOV[NZ] instruction with most significant bits 0 to 15 of a signed value. Changes instruction to MOVZ or MOVN depending on the value's sign.

`BFD_RELOC_AARCH64_MOVW_G1_S`
AArch64 MOV[NZ] instruction with most significant bits 16 to 31 of a signed value. Changes instruction to MOVZ or MOVN depending on the value's sign.

`BFD_RELOC_AARCH64_MOVW_G2_S`
AArch64 MOV[NZ] instruction with most significant bits 32 to 47 of a signed value. Changes instruction to MOVZ or MOVN depending on the value's sign.

`BFD_RELOC_AARCH64_MOVW_PREL_G0`
AArch64 MOV[NZ] instruction with most significant bits 0 to 15 of a signed value. Changes instruction to MOVZ or MOVN depending on the value's sign.

`BFD_RELOC_AARCH64_MOVW_PREL_G0_NC`
AArch64 MOV[NZ] instruction with most significant bits 0 to 15 of a signed value. Changes instruction to MOVZ or MOVN depending on the value's sign.

`BFD_RELOC_AARCH64_MOVW_PREL_G1`
AArch64 MOVK instruction with most significant bits 16 to 31 of a signed value.

`BFD_RELOC_AARCH64_MOVW_PREL_G1_NC`
AArch64 MOVK instruction with most significant bits 16 to 31 of a signed value.

`BFD_RELOC_AARCH64_MOVW_PREL_G2`
AArch64 MOVK instruction with most significant bits 32 to 47 of a signed value.

`BFD_RELOC_AARCH64_MOVW_PREL_G2_NC`
AArch64 MOVK instruction with most significant bits 32 to 47 of a signed value.

`BFD_RELOC_AARCH64_MOVW_PREL_G3`
AArch64 MOVK instruction with most significant bits 47 to 63 of a signed value.

`BFD_RELOC_AARCH64_LD_LO19_PCREL`
AArch64 Load Literal instruction, holding a 19 bit pc-relative word offset. The lowest two bits must be zero and are not stored in the instruction, giving a 21 bit signed byte offset.

`BFD_RELOC_AARCH64_ADR_LO21_PCREL`
AArch64 ADR instruction, holding a simple 21 bit pc-relative byte offset.

`BFD_RELOC_AARCH64_ADR_HI21_PCREL`
AArch64 ADRP instruction, with bits 12 to 32 of a pc-relative page offset, giving a 4KB aligned page base address.

`BFD_RELOC_AARCH64_ADR_HI21_NC_PCREL`
AArch64 ADRP instruction, with bits 12 to 32 of a pc-relative page offset, giving a 4KB aligned page base address, but with no overflow checking.

`BFD_RELOC_AARCH64_ADD_LO12`
AArch64 ADD immediate instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD_RELOC_AARCH64_ADR_HI21_PCREL.

`BFD_RELOC_AARCH64_LDST8_LO12`
AArch64 8-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD_RELOC_AARCH64_ADR_HI21_PCREL.

`BFD_RELOC_AARCH64_TSTBR14`
AArch64 14 bit pc-relative test bit and branch. The lowest two bits must be zero and are not stored in the instruction, giving a 16 bit signed byte offset.

`BFD_RELOC_AARCH64_BRANCH19`
AArch64 19 bit pc-relative conditional branch and compare & branch. The lowest two bits must be zero and are not stored in the instruction, giving a 21 bit signed byte offset.

`BFD_RELOC_AARCH64_JUMP26`
AArch64 26 bit pc-relative unconditional branch. The lowest two bits must be zero and are not stored in the instruction, giving a 28 bit signed byte offset.

`BFD_RELOC_AARCH64_CALL26`
AArch64 26 bit pc-relative unconditional branch and link. The lowest two bits must be zero and are not stored in the instruction, giving a 28 bit signed byte offset.

`BFD_RELOC_AARCH64_LDST16_LO12`
AArch64 16-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD_RELOC_AARCH64_ADR_HI21_PCREL.

`BFD_RELOC_AARCH64_LDST32_LO12`
AArch64 32-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD_RELOC_AARCH64_ADR_HI21_PCREL.

`BFD_RELOC_AARCH64_LDST64_LO12`
AArch64 64-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD_RELOC_AARCH64_ADR_HI21_PCREL.

`BFD_RELOC_AARCH64_LDST128_LO12`
AArch64 128-bit load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD_RELOC_AARCH64_ADR_HI21_PCREL.

`BFD_RELOC_AARCH64_GOT_LD_PREL19`
AArch64 Load Literal instruction, holding a 19 bit PC relative word offset of the global offset table entry for a symbol. The lowest two bits must be zero and are not stored in the instruction, giving a 21 bit signed byte offset. This relocation type requires signed overflow checking.

`BFD_RELOC_AARCH64_ADR_GOT_PAGE`
Get to the page base of the global offset table entry for a symbol as part of an ADRP instruction using a 21 bit PC relative value.Used in conjunction with BFD_RELOC_AARCH64_LD64_GOT_LO12_NC.

`BFD_RELOC_AARCH64_LD64_GOT_LO12_NC`
Unsigned 12 bit byte offset for 64 bit load/store from the page of the GOT entry for this symbol. Used in conjunction with BFD_RELOC_AARCH64_ADR_GOT_PAGE. Valid in LP64 ABI only.

`BFD_RELOC_AARCH64_LD32_GOT_LO12_NC`
Unsigned 12 bit byte offset for 32 bit load/store from the page of the GOT entry for this symbol. Used in conjunction with BFD_RELOC_AARCH64_ADR_GOT_PAGE. Valid in ILP32 ABI only.

`BFD_RELOC_AARCH64_MOVW_GOTOFF_G0_NC`
Unsigned 16 bit byte offset for 64 bit load/store from the GOT entry for this symbol. Valid in LP64 ABI only.

`BFD_RELOC_AARCH64_MOVW_GOTOFF_G1`
Unsigned 16 bit byte higher offset for 64 bit load/store from the GOT entry for this symbol. Valid in LP64 ABI only.

`BFD_RELOC_AARCH64_LD64_GOTOFF_LO15`
Unsigned 15 bit byte offset for 64 bit load/store from the page of the GOT entry for this symbol. Valid in LP64 ABI only.

`BFD_RELOC_AARCH64_LD32_GOTPAGE_LO14`
Scaled 14 bit byte offset to the page base of the global offset table.

`BFD_RELOC_AARCH64_LD64_GOTPAGE_LO15`
Scaled 15 bit byte offset to the page base of the global offset table.

`BFD_RELOC_AARCH64_TLSGD_ADR_PAGE21`
Get to the page base of the global offset table entry for a symbols tls_index structure as part of an adrp instruction using a 21 bit PC relative value. Used in conjunction with BFD_RELOC_AARCH64_TLSGD_ADD_LO12_NC.

`BFD_RELOC_AARCH64_TLSGD_ADR_PREL21`
AArch64 TLS General Dynamic

`BFD_RELOC_AARCH64_TLSGD_ADD_LO12_NC`
Unsigned 12 bit byte offset to global offset table entry for a symbols tls_index structure. Used in conjunction with BFD_RELOC_AARCH64_TLSGD_ADR_PAGE21.

`BFD_RELOC_AARCH64_TLSGD_MOVW_G0_NC`
AArch64 TLS General Dynamic relocation.

`BFD_RELOC_AARCH64_TLSGD_MOVW_G1`
AArch64 TLS General Dynamic relocation.

`BFD_RELOC_AARCH64_TLSIE_ADR_GOTTPREL_PAGE21`
AArch64 TLS INITIAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSIE_LD64_GOTTPREL_LO12_NC`
AArch64 TLS INITIAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSIE_LD32_GOTTPREL_LO12_NC`
AArch64 TLS INITIAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSIE_LD_GOTTPREL_PREL19`
AArch64 TLS INITIAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSIE_MOVW_GOTTPREL_G0_NC`
AArch64 TLS INITIAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSIE_MOVW_GOTTPREL_G1`
AArch64 TLS INITIAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSLD_ADD_DTPREL_HI12`
bit[23:12] of byte offset to module TLS base address.

`BFD_RELOC_AARCH64_TLSLD_ADD_DTPREL_LO12`
Unsigned 12 bit byte offset to module TLS base address.

`BFD_RELOC_AARCH64_TLSLD_ADD_DTPREL_LO12_NC`
No overflow check version of BFD_RELOC_AARCH64_TLSLD_ADD_DTPREL_LO12.

`BFD_RELOC_AARCH64_TLSLD_ADD_LO12_NC`
Unsigned 12 bit byte offset to global offset table entry for a symbols tls_index structure. Used in conjunction with BFD_RELOC_AARCH64_TLSLD_ADR_PAGE21.

`BFD_RELOC_AARCH64_TLSLD_ADR_PAGE21`
GOT entry page address for AArch64 TLS Local Dynamic, used with ADRP instruction.

`BFD_RELOC_AARCH64_TLSLD_ADR_PREL21`
GOT entry address for AArch64 TLS Local Dynamic, used with ADR instruction.

`BFD_RELOC_AARCH64_TLSLD_LDST16_DTPREL_LO12`
bit[11:1] of byte offset to module TLS base address, encoded in ldst instructions.

`BFD_RELOC_AARCH64_TLSLD_LDST16_DTPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLD_LDST16_DTPREL_LO12, but no overflow check.

`BFD_RELOC_AARCH64_TLSLD_LDST32_DTPREL_LO12`
bit[11:2] of byte offset to module TLS base address, encoded in ldst instructions.

`BFD_RELOC_AARCH64_TLSLD_LDST32_DTPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLD_LDST32_DTPREL_LO12, but no overflow check.

`BFD_RELOC_AARCH64_TLSLD_LDST64_DTPREL_LO12`
bit[11:3] of byte offset to module TLS base address, encoded in ldst instructions.

`BFD_RELOC_AARCH64_TLSLD_LDST64_DTPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLD_LDST64_DTPREL_LO12, but no overflow check.

`BFD_RELOC_AARCH64_TLSLD_LDST8_DTPREL_LO12`
bit[11:0] of byte offset to module TLS base address, encoded in ldst instructions.

`BFD_RELOC_AARCH64_TLSLD_LDST8_DTPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLD_LDST8_DTPREL_LO12, but no overflow check.

`BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G0`
bit[15:0] of byte offset to module TLS base address.

`BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G0_NC`
No overflow check version of BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G0

`BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G1`
bit[31:16] of byte offset to module TLS base address.

`BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G1_NC`
No overflow check version of BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G1

`BFD_RELOC_AARCH64_TLSLD_MOVW_DTPREL_G2`
bit[47:32] of byte offset to module TLS base address.

`BFD_RELOC_AARCH64_TLSLE_MOVW_TPREL_G2`
AArch64 TLS LOCAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSLE_MOVW_TPREL_G1`
AArch64 TLS LOCAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSLE_MOVW_TPREL_G1_NC`
AArch64 TLS LOCAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSLE_MOVW_TPREL_G0`
AArch64 TLS LOCAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSLE_MOVW_TPREL_G0_NC`
AArch64 TLS LOCAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSLE_ADD_TPREL_HI12`
AArch64 TLS LOCAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSLE_ADD_TPREL_LO12`
AArch64 TLS LOCAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSLE_ADD_TPREL_LO12_NC`
AArch64 TLS LOCAL EXEC relocation.

`BFD_RELOC_AARCH64_TLSLE_LDST16_TPREL_LO12`
bit[11:1] of byte offset to module TLS base address, encoded in ldst instructions.

`BFD_RELOC_AARCH64_TLSLE_LDST16_TPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLE_LDST16_TPREL_LO12, but no overflow check.

`BFD_RELOC_AARCH64_TLSLE_LDST32_TPREL_LO12`
bit[11:2] of byte offset to module TLS base address, encoded in ldst instructions.

`BFD_RELOC_AARCH64_TLSLE_LDST32_TPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLE_LDST32_TPREL_LO12, but no overflow check.

`BFD_RELOC_AARCH64_TLSLE_LDST64_TPREL_LO12`
bit[11:3] of byte offset to module TLS base address, encoded in ldst instructions.

`BFD_RELOC_AARCH64_TLSLE_LDST64_TPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLE_LDST64_TPREL_LO12, but no overflow check.

`BFD_RELOC_AARCH64_TLSLE_LDST8_TPREL_LO12`
bit[11:0] of byte offset to module TLS base address, encoded in ldst instructions.

`BFD_RELOC_AARCH64_TLSLE_LDST8_TPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLE_LDST8_TPREL_LO12, but no overflow check.

`BFD_RELOC_AARCH64_TLSDESC_LD_PREL19`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_ADR_PREL21`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_ADR_PAGE21`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_LD64_LO12`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_LD32_LO12_NC`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_ADD_LO12`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_OFF_G1`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_OFF_G0_NC`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_LDR`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_ADD`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_TLSDESC_CALL`
AArch64 TLS DESC relocation.

`BFD_RELOC_AARCH64_COPY`
AArch64 TLS relocation.

`BFD_RELOC_AARCH64_GLOB_DAT`
AArch64 TLS relocation.

`BFD_RELOC_AARCH64_JUMP_SLOT`
AArch64 TLS relocation.

`BFD_RELOC_AARCH64_RELATIVE`
AArch64 TLS relocation.

`BFD_RELOC_AARCH64_TLS_DTPMOD`
AArch64 TLS relocation.

`BFD_RELOC_AARCH64_TLS_DTPREL`
AArch64 TLS relocation.

`BFD_RELOC_AARCH64_TLS_TPREL`
AArch64 TLS relocation.

`BFD_RELOC_AARCH64_TLSDESC`
AArch64 TLS relocation.

`BFD_RELOC_AARCH64_IRELATIVE`
AArch64 support for STT_GNU_IFUNC.

`BFD_RELOC_AARCH64_RELOC_END`
AArch64 pseudo relocation code to mark the end of the AArch64 relocation enumerators that have direct mapping to ELF reloc codes. There are a few more enumerators after this one; those are mainly used by the AArch64 assembler for the internal fixup or to select one of the above enumerators.

`BFD_RELOC_AARCH64_GAS_INTERNAL_FIXUP`
AArch64 pseudo relocation code to be used internally by the AArch64 assembler and not (currently) written to any object files.

`BFD_RELOC_AARCH64_LDST_LO12`
AArch64 unspecified load/store instruction, holding bits 0 to 11 of the address. Used in conjunction with BFD_RELOC_AARCH64_ADR_HI21_PCREL.

`BFD_RELOC_AARCH64_TLSLD_LDST_DTPREL_LO12`
AArch64 pseudo relocation code for TLS local dynamic mode. It's to be used internally by the AArch64 assembler and not (currently) written to any object files.

`BFD_RELOC_AARCH64_TLSLD_LDST_DTPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLD_LDST_DTPREL_LO12, but no overflow check.

`BFD_RELOC_AARCH64_TLSLE_LDST_TPREL_LO12`
AArch64 pseudo relocation code for TLS local exec mode. It's to be used internally by the AArch64 assembler and not (currently) written to any object files.

`BFD_RELOC_AARCH64_TLSLE_LDST_TPREL_LO12_NC`
Similar as BFD_RELOC_AARCH64_TLSLE_LDST_TPREL_LO12, but no overflow check.

BFD_RELOC_AARCH64_LD_GOT_LO12_NC
: AArch64 pseudo relocation code to be used internally by the AArch64 assembler and not (currently) written to any object files.

BFD_RELOC_AARCH64_TLSIE_LD_GOTTPREL_LO12_NC
: AArch64 pseudo relocation code to be used internally by the AArch64 assembler and not (currently) written to any object files.

BFD_RELOC_AARCH64_TLSDESC_LD_LO12_NC
: AArch64 pseudo relocation code to be used internally by the AArch64 assembler and not (currently) written to any object files.

BFD_RELOC_TILEPRO_COPY
BFD_RELOC_TILEPRO_GLOB_DAT
BFD_RELOC_TILEPRO_JMP_SLOT
BFD_RELOC_TILEPRO_RELATIVE
BFD_RELOC_TILEPRO_BROFF_X1
BFD_RELOC_TILEPRO_JOFFLONG_X1
BFD_RELOC_TILEPRO_JOFFLONG_X1_PLT
BFD_RELOC_TILEPRO_IMM8_X0
BFD_RELOC_TILEPRO_IMM8_Y0
BFD_RELOC_TILEPRO_IMM8_X1
BFD_RELOC_TILEPRO_IMM8_Y1
BFD_RELOC_TILEPRO_DEST_IMM8_X1
BFD_RELOC_TILEPRO_MT_IMM15_X1
BFD_RELOC_TILEPRO_MF_IMM15_X1
BFD_RELOC_TILEPRO_IMM16_X0
BFD_RELOC_TILEPRO_IMM16_X1
BFD_RELOC_TILEPRO_IMM16_X0_LO
BFD_RELOC_TILEPRO_IMM16_X1_LO
BFD_RELOC_TILEPRO_IMM16_X0_HI
BFD_RELOC_TILEPRO_IMM16_X1_HI
BFD_RELOC_TILEPRO_IMM16_X0_HA
BFD_RELOC_TILEPRO_IMM16_X1_HA
BFD_RELOC_TILEPRO_IMM16_X0_PCREL
BFD_RELOC_TILEPRO_IMM16_X1_PCREL
BFD_RELOC_TILEPRO_IMM16_X0_LO_PCREL
BFD_RELOC_TILEPRO_IMM16_X1_LO_PCREL
BFD_RELOC_TILEPRO_IMM16_X0_HI_PCREL
BFD_RELOC_TILEPRO_IMM16_X1_HI_PCREL
BFD_RELOC_TILEPRO_IMM16_X0_HA_PCREL
BFD_RELOC_TILEPRO_IMM16_X1_HA_PCREL
BFD_RELOC_TILEPRO_IMM16_X0_GOT
BFD_RELOC_TILEPRO_IMM16_X1_GOT
BFD_RELOC_TILEPRO_IMM16_X0_GOT_LO
BFD_RELOC_TILEPRO_IMM16_X1_GOT_LO
BFD_RELOC_TILEPRO_IMM16_X0_GOT_HI
BFD_RELOC_TILEPRO_IMM16_X1_GOT_HI

BFD_RELOC_TILEPRO_IMM16_X0_GOT_HA
BFD_RELOC_TILEPRO_IMM16_X1_GOT_HA
BFD_RELOC_TILEPRO_MMSTART_X0
BFD_RELOC_TILEPRO_MMEND_X0
BFD_RELOC_TILEPRO_MMSTART_X1
BFD_RELOC_TILEPRO_MMEND_X1
BFD_RELOC_TILEPRO_SHAMT_X0
BFD_RELOC_TILEPRO_SHAMT_X1
BFD_RELOC_TILEPRO_SHAMT_Y0
BFD_RELOC_TILEPRO_SHAMT_Y1
BFD_RELOC_TILEPRO_TLS_GD_CALL
BFD_RELOC_TILEPRO_IMM8_X0_TLS_GD_ADD
BFD_RELOC_TILEPRO_IMM8_X1_TLS_GD_ADD
BFD_RELOC_TILEPRO_IMM8_Y0_TLS_GD_ADD
BFD_RELOC_TILEPRO_IMM8_Y1_TLS_GD_ADD
BFD_RELOC_TILEPRO_TLS_IE_LOAD
BFD_RELOC_TILEPRO_IMM16_X0_TLS_GD
BFD_RELOC_TILEPRO_IMM16_X1_TLS_GD
BFD_RELOC_TILEPRO_IMM16_X0_TLS_GD_LO
BFD_RELOC_TILEPRO_IMM16_X1_TLS_GD_LO
BFD_RELOC_TILEPRO_IMM16_X0_TLS_GD_HI
BFD_RELOC_TILEPRO_IMM16_X1_TLS_GD_HI
BFD_RELOC_TILEPRO_IMM16_X0_TLS_GD_HA
BFD_RELOC_TILEPRO_IMM16_X1_TLS_GD_HA
BFD_RELOC_TILEPRO_IMM16_X0_TLS_IE
BFD_RELOC_TILEPRO_IMM16_X1_TLS_IE
BFD_RELOC_TILEPRO_IMM16_X0_TLS_IE_LO
BFD_RELOC_TILEPRO_IMM16_X1_TLS_IE_LO
BFD_RELOC_TILEPRO_IMM16_X0_TLS_IE_HI
BFD_RELOC_TILEPRO_IMM16_X1_TLS_IE_HI
BFD_RELOC_TILEPRO_IMM16_X0_TLS_IE_HA
BFD_RELOC_TILEPRO_IMM16_X1_TLS_IE_HA
BFD_RELOC_TILEPRO_TLS_DTPMOD32
BFD_RELOC_TILEPRO_TLS_DTPOFF32
BFD_RELOC_TILEPRO_TLS_TPOFF32
BFD_RELOC_TILEPRO_IMM16_X0_TLS_LE
BFD_RELOC_TILEPRO_IMM16_X1_TLS_LE
BFD_RELOC_TILEPRO_IMM16_X0_TLS_LE_LO
BFD_RELOC_TILEPRO_IMM16_X1_TLS_LE_LO
BFD_RELOC_TILEPRO_IMM16_X0_TLS_LE_HI
BFD_RELOC_TILEPRO_IMM16_X1_TLS_LE_HI
BFD_RELOC_TILEPRO_IMM16_X0_TLS_LE_HA
BFD_RELOC_TILEPRO_IMM16_X1_TLS_LE_HA

Tilera TILEPro Relocations.

BFD_RELOC_TILEGX_HW0
BFD_RELOC_TILEGX_HW1

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BFD_RELOC_TILEGX_HW2
BFD_RELOC_TILEGX_HW3
BFD_RELOC_TILEGX_HW0_LAST
BFD_RELOC_TILEGX_HW1_LAST
BFD_RELOC_TILEGX_HW2_LAST
BFD_RELOC_TILEGX_COPY
BFD_RELOC_TILEGX_GLOB_DAT
BFD_RELOC_TILEGX_JMP_SLOT
BFD_RELOC_TILEGX_RELATIVE
BFD_RELOC_TILEGX_BROFF_X1
BFD_RELOC_TILEGX_JUMPOFF_X1
BFD_RELOC_TILEGX_JUMPOFF_X1_PLT
BFD_RELOC_TILEGX_IMM8_X0
BFD_RELOC_TILEGX_IMM8_Y0
BFD_RELOC_TILEGX_IMM8_X1
BFD_RELOC_TILEGX_IMM8_Y1
BFD_RELOC_TILEGX_DEST_IMM8_X1
BFD_RELOC_TILEGX_MT_IMM14_X1
BFD_RELOC_TILEGX_MF_IMM14_X1
BFD_RELOC_TILEGX_MMSTART_X0
BFD_RELOC_TILEGX_MMEND_X0
BFD_RELOC_TILEGX_SHAMT_X0
BFD_RELOC_TILEGX_SHAMT_X1
BFD_RELOC_TILEGX_SHAMT_Y0
BFD_RELOC_TILEGX_SHAMT_Y1
BFD_RELOC_TILEGX_IMM16_X0_HW0
BFD_RELOC_TILEGX_IMM16_X1_HW0
BFD_RELOC_TILEGX_IMM16_X0_HW1
BFD_RELOC_TILEGX_IMM16_X1_HW1
BFD_RELOC_TILEGX_IMM16_X0_HW2
BFD_RELOC_TILEGX_IMM16_X1_HW2
BFD_RELOC_TILEGX_IMM16_X0_HW3
BFD_RELOC_TILEGX_IMM16_X1_HW3
BFD_RELOC_TILEGX_IMM16_X0_HW0_LAST
BFD_RELOC_TILEGX_IMM16_X1_HW0_LAST
BFD_RELOC_TILEGX_IMM16_X0_HW1_LAST
BFD_RELOC_TILEGX_IMM16_X1_HW1_LAST
BFD_RELOC_TILEGX_IMM16_X0_HW2_LAST
BFD_RELOC_TILEGX_IMM16_X1_HW2_LAST
BFD_RELOC_TILEGX_IMM16_X0_HW0_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW0_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW1_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW1_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW2_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW2_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW3_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW3_PCREL
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BFD_RELOC_TILEGX_IMM16_X0_HW0_LAST_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW0_LAST_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW1_LAST_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW1_LAST_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW2_LAST_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW2_LAST_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW0_GOT
BFD_RELOC_TILEGX_IMM16_X1_HW0_GOT
BFD_RELOC_TILEGX_IMM16_X0_HW0_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW0_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW1_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW1_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW2_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW2_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW0_LAST_GOT
BFD_RELOC_TILEGX_IMM16_X1_HW0_LAST_GOT
BFD_RELOC_TILEGX_IMM16_X0_HW1_LAST_GOT
BFD_RELOC_TILEGX_IMM16_X1_HW1_LAST_GOT
BFD_RELOC_TILEGX_IMM16_X0_HW3_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW3_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW0_TLS_GD
BFD_RELOC_TILEGX_IMM16_X1_HW0_TLS_GD
BFD_RELOC_TILEGX_IMM16_X0_HW0_TLS_LE
BFD_RELOC_TILEGX_IMM16_X1_HW0_TLS_LE
BFD_RELOC_TILEGX_IMM16_X0_HW0_LAST_TLS_LE
BFD_RELOC_TILEGX_IMM16_X1_HW0_LAST_TLS_LE
BFD_RELOC_TILEGX_IMM16_X0_HW1_LAST_TLS_LE
BFD_RELOC_TILEGX_IMM16_X1_HW1_LAST_TLS_LE
BFD_RELOC_TILEGX_IMM16_X0_HW0_LAST_TLS_GD
BFD_RELOC_TILEGX_IMM16_X1_HW0_LAST_TLS_GD
BFD_RELOC_TILEGX_IMM16_X0_HW1_LAST_TLS_GD
BFD_RELOC_TILEGX_IMM16_X1_HW1_LAST_TLS_GD
BFD_RELOC_TILEGX_IMM16_X0_HW0_TLS_IE
BFD_RELOC_TILEGX_IMM16_X1_HW0_TLS_IE
BFD_RELOC_TILEGX_IMM16_X0_HW0_LAST_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW0_LAST_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW1_LAST_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW1_LAST_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW2_LAST_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X1_HW2_LAST_PLT_PCREL
BFD_RELOC_TILEGX_IMM16_X0_HW0_LAST_TLS_IE
BFD_RELOC_TILEGX_IMM16_X1_HW0_LAST_TLS_IE
BFD_RELOC_TILEGX_IMM16_X0_HW1_LAST_TLS_IE
BFD_RELOC_TILEGX_IMM16_X1_HW1_LAST_TLS_IE
BFD_RELOC_TILEGX_TLS_DTPMOD64
BFD_RELOC_TILEGX_TLS_DTPOFF64
BFD_RELOC_TILEGX_TLS_TPOFF64
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BFD_RELOC_TILEGX_TLS_DTPMOD32
BFD_RELOC_TILEGX_TLS_DTPOFF32
BFD_RELOC_TILEGX_TLS_TPOFF32
BFD_RELOC_TILEGX_TLS_GD_CALL
BFD_RELOC_TILEGX_IMM8_X0_TLS_GD_ADD
BFD_RELOC_TILEGX_IMM8_X1_TLS_GD_ADD
BFD_RELOC_TILEGX_IMM8_Y0_TLS_GD_ADD
BFD_RELOC_TILEGX_IMM8_Y1_TLS_GD_ADD
BFD_RELOC_TILEGX_TLS_IE_LOAD
BFD_RELOC_TILEGX_IMM8_X0_TLS_ADD
BFD_RELOC_TILEGX_IMM8_X1_TLS_ADD
BFD_RELOC_TILEGX_IMM8_Y0_TLS_ADD
BFD_RELOC_TILEGX_IMM8_Y1_TLS_ADD
Tilera TILE-Gx Relocations.

BFD_RELOC_BPF_64
BFD_RELOC_BPF_32
BFD_RELOC_BPF_16
BFD_RELOC_BPF_DISP16
BFD_RELOC_BPF_DISP32
Linux eBPF relocations.

BFD_RELOC_EPIPHANY_SIMM8
Adapteva EPIPHANY - 8 bit signed pc-relative displacement

BFD_RELOC_EPIPHANY_SIMM24
Adapteva EPIPHANY - 24 bit signed pc-relative displacement

BFD_RELOC_EPIPHANY_HIGH
Adapteva EPIPHANY - 16 most-significant bits of absolute address

BFD_RELOC_EPIPHANY_LOW
Adapteva EPIPHANY - 16 least-significant bits of absolute address

BFD_RELOC_EPIPHANY_SIMM11
Adapteva EPIPHANY - 11 bit signed number - add/sub immediate

BFD_RELOC_EPIPHANY_IMM11
Adapteva EPIPHANY - 11 bit sign-magnitude number (ld/st displacement)

BFD_RELOC_EPIPHANY_IMM8
Adapteva EPIPHANY - 8 bit immediate for 16 bit mov instruction.

BFD_RELOC_VISIUM_HI16
BFD_RELOC_VISIUM_LO16
BFD_RELOC_VISIUM_IM16
BFD_RELOC_VISIUM_REL16
BFD_RELOC_VISIUM_HI16_PCREL
BFD_RELOC_VISIUM_LO16_PCREL
BFD_RELOC_VISIUM_IM16_PCREL
Visium Relocations.

BFD_RELOC_WASM32_LEB128
BFD_RELOC_WASM32_LEB128_GOT
BFD_RELOC_WASM32_LEB128_GOT_CODE
BFD_RELOC_WASM32_LEB128_PLT
BFD_RELOC_WASM32_PLT_INDEX
BFD_RELOC_WASM32_ABS32_CODE
BFD_RELOC_WASM32_COPY
BFD_RELOC_WASM32_CODE_POINTER
BFD_RELOC_WASM32_INDEX
BFD_RELOC_WASM32_PLT_SIG

WebAssembly relocations.

BFD_RELOC_CKCORE_NONE
BFD_RELOC_CKCORE_ADDR32
BFD_RELOC_CKCORE_PCREL_IMM8BY4
BFD_RELOC_CKCORE_PCREL_IMM11BY2
BFD_RELOC_CKCORE_PCREL_IMM4BY2
BFD_RELOC_CKCORE_PCREL32
BFD_RELOC_CKCORE_PCREL_JSR_IMM11BY2
BFD_RELOC_CKCORE_GNU_VTINHERIT
BFD_RELOC_CKCORE_GNU_VTENTRY
BFD_RELOC_CKCORE_RELATIVE
BFD_RELOC_CKCORE_COPY
BFD_RELOC_CKCORE_GLOB_DAT
BFD_RELOC_CKCORE_JUMP_SLOT
BFD_RELOC_CKCORE_GOTOFF
BFD_RELOC_CKCORE_GOTPC
BFD_RELOC_CKCORE_GOT32
BFD_RELOC_CKCORE_PLT32
BFD_RELOC_CKCORE_ADDRGOT
BFD_RELOC_CKCORE_ADDRPLT
BFD_RELOC_CKCORE_PCREL_IMM26BY2
BFD_RELOC_CKCORE_PCREL_IMM16BY2
BFD_RELOC_CKCORE_PCREL_IMM16BY4
BFD_RELOC_CKCORE_PCREL_IMM10BY2
BFD_RELOC_CKCORE_PCREL_IMM10BY4
BFD_RELOC_CKCORE_ADDR_HI16
BFD_RELOC_CKCORE_ADDR_LO16
BFD_RELOC_CKCORE_GOTPC_HI16
BFD_RELOC_CKCORE_GOTPC_LO16
BFD_RELOC_CKCORE_GOTOFF_HI16
BFD_RELOC_CKCORE_GOTOFF_LO16
BFD_RELOC_CKCORE_GOT12
BFD_RELOC_CKCORE_GOT_HI16
BFD_RELOC_CKCORE_GOT_LO16
BFD_RELOC_CKCORE_PLT12
BFD_RELOC_CKCORE_PLT_HI16

BFD_RELOC_CKCORE_PLT_LO16
BFD_RELOC_CKCORE_ADDRGOT_HI16
BFD_RELOC_CKCORE_ADDRGOT_LO16
BFD_RELOC_CKCORE_ADDRPLT_HI16
BFD_RELOC_CKCORE_ADDRPLT_LO16
BFD_RELOC_CKCORE_PCREL_JSR_IMM26BY2
BFD_RELOC_CKCORE_TOFFSET_LO16
BFD_RELOC_CKCORE_DOFFSET_LO16
BFD_RELOC_CKCORE_PCREL_IMM18BY2
BFD_RELOC_CKCORE_DOFFSET_IMM18
BFD_RELOC_CKCORE_DOFFSET_IMM18BY2
BFD_RELOC_CKCORE_DOFFSET_IMM18BY4
BFD_RELOC_CKCORE_GOTOFF_IMM18
BFD_RELOC_CKCORE_GOT_IMM18BY4
BFD_RELOC_CKCORE_PLT_IMM18BY4
BFD_RELOC_CKCORE_PCREL_IMM7BY4
BFD_RELOC_CKCORE_TLS_LE32
BFD_RELOC_CKCORE_TLS_IE32
BFD_RELOC_CKCORE_TLS_GD32
BFD_RELOC_CKCORE_TLS_LDM32
BFD_RELOC_CKCORE_TLS_LDO32
BFD_RELOC_CKCORE_TLS_DTPMOD32
BFD_RELOC_CKCORE_TLS_DTPOFF32
BFD_RELOC_CKCORE_TLS_TPOFF32
BFD_RELOC_CKCORE_PCREL_FLRW_IMM8BY4
BFD_RELOC_CKCORE_NOJSRI
BFD_RELOC_CKCORE_CALLGRAPH
BFD_RELOC_CKCORE_IRELATIVE
BFD_RELOC_CKCORE_PCREL_BLOOP_IMM4BY4
BFD_RELOC_CKCORE_PCREL_BLOOP_IMM12BY4
C-SKY relocations.

BFD_RELOC_S12Z_OPR
S12Z relocations.

BFD_RELOC_LARCH_TLS_DTPMOD32
BFD_RELOC_LARCH_TLS_DTPREL32
BFD_RELOC_LARCH_TLS_DTPMOD64
BFD_RELOC_LARCH_TLS_DTPREL64
BFD_RELOC_LARCH_TLS_TPREL32
BFD_RELOC_LARCH_TLS_TPREL64
BFD_RELOC_LARCH_MARK_LA
BFD_RELOC_LARCH_MARK_PCREL
BFD_RELOC_LARCH_SOP_PUSH_PCREL
BFD_RELOC_LARCH_SOP_PUSH_ABSOLUTE
BFD_RELOC_LARCH_SOP_PUSH_DUP
BFD_RELOC_LARCH_SOP_PUSH_GPREL
BFD_RELOC_LARCH_SOP_PUSH_TLS_TPREL

```
BFD_RELOC_LARCH_SOP_PUSH_TLS_GOT
BFD_RELOC_LARCH_SOP_PUSH_TLS_GD
BFD_RELOC_LARCH_SOP_PUSH_PLT_PCREL
BFD_RELOC_LARCH_SOP_ASSERT
BFD_RELOC_LARCH_SOP_NOT
BFD_RELOC_LARCH_SOP_SUB
BFD_RELOC_LARCH_SOP_SL
BFD_RELOC_LARCH_SOP_SR
BFD_RELOC_LARCH_SOP_ADD
BFD_RELOC_LARCH_SOP_AND
BFD_RELOC_LARCH_SOP_IF_ELSE
BFD_RELOC_LARCH_SOP_POP_32_S_10_5
BFD_RELOC_LARCH_SOP_POP_32_U_10_12
BFD_RELOC_LARCH_SOP_POP_32_S_10_12
BFD_RELOC_LARCH_SOP_POP_32_S_10_16
BFD_RELOC_LARCH_SOP_POP_32_S_10_16_S2
BFD_RELOC_LARCH_SOP_POP_32_S_5_20
BFD_RELOC_LARCH_SOP_POP_32_S_0_5_10_16_S2
BFD_RELOC_LARCH_SOP_POP_32_S_0_10_10_16_S2
BFD_RELOC_LARCH_SOP_POP_32_U
BFD_RELOC_LARCH_ADD8
BFD_RELOC_LARCH_ADD16
BFD_RELOC_LARCH_ADD24
BFD_RELOC_LARCH_ADD32
BFD_RELOC_LARCH_ADD64
BFD_RELOC_LARCH_SUB8
BFD_RELOC_LARCH_SUB16
BFD_RELOC_LARCH_SUB24
BFD_RELOC_LARCH_SUB32
BFD_RELOC_LARCH_SUB64
```

LARCH relocations.

```
typedef enum bfd_reloc_code_real bfd_reloc_code_real_type;
```

2.10.2.2 bfd_reloc_type_lookup

Synopsis

```
reloc_howto_type *bfd_reloc_type_lookup
   (bfd *abfd, bfd_reloc_code_real_type code);
reloc_howto_type *bfd_reloc_name_lookup
   (bfd *abfd, const char *reloc_name);
```

Description

Return a pointer to a howto structure which, when invoked, will perform the relocation *code* on data from the architecture noted.

2.10.2.3 bfd_default_reloc_type_lookup

Synopsis

```
reloc_howto_type *bfd_default_reloc_type_lookup
   (bfd *abfd, bfd_reloc_code_real_type  code);
```

Description
Provides a default relocation lookup routine for any architecture.

2.10.2.4 bfd_get_reloc_code_name

Synopsis

```
const char *bfd_get_reloc_code_name (bfd_reloc_code_real_type code);
```

Description
Provides a printable name for the supplied relocation code. Useful mainly for printing error messages.

2.10.2.5 bfd_generic_relax_section

Synopsis

```
bool bfd_generic_relax_section
   (bfd *abfd,
    asection *section,
    struct bfd_link_info *,
    bool *);
```

Description
Provides default handling for relaxing for back ends which don't do relaxing.

2.10.2.6 bfd_generic_gc_sections

Synopsis

```
bool bfd_generic_gc_sections
   (bfd *, struct bfd_link_info *);
```

Description
Provides default handling for relaxing for back ends which don't do section gc – i.e., does nothing.

2.10.2.7 bfd_generic_lookup_section_flags

Synopsis

```
bool bfd_generic_lookup_section_flags
   (struct bfd_link_info *, struct flag_info *, asection *);
```

Description
Provides default handling for section flags lookup – i.e., does nothing. Returns FALSE if the section should be omitted, otherwise TRUE.

2.10.2.8 bfd_generic_merge_sections

Synopsis

```
bool bfd_generic_merge_sections
   (bfd *, struct bfd_link_info *);
```

Description
Provides default handling for SEC_MERGE section merging for back ends which don't have SEC_MERGE support – i.e., does nothing.

2.10.2.9 bfd_generic_get_relocated_section_contents

Synopsis

```
bfd_byte *bfd_generic_get_relocated_section_contents
   (bfd *abfd,
    struct bfd_link_info *link_info,
    struct bfd_link_order *link_order,
    bfd_byte *data,
    bool relocatable,
    asymbol **symbols);
```

Description

Provides default handling of relocation effort for back ends which can't be bothered to do it efficiently.

2.10.2.10 _bfd_generic_set_reloc

Synopsis

```
void _bfd_generic_set_reloc
   (bfd *abfd,
    sec_ptr section,
    arelent **relptr,
    unsigned int count);
```

Description

Installs a new set of internal relocations in SECTION.

2.10.2.11 _bfd_unrecognized_reloc

Synopsis

```
bool _bfd_unrecognized_reloc
   (bfd * abfd,
    sec_ptr section,
    unsigned int r_type);
```

Description

Reports an unrecognized reloc. Written as a function in order to reduce code duplication. Returns FALSE so that it can be called from a return statement.

2.11 Core files

2.11.1 Core file functions

Description

These are functions pertaining to core files.

2.11.1.1 bfd_core_file_failing_command

Synopsis

```
const char *bfd_core_file_failing_command (bfd *abfd);
```

Description

Return a read-only string explaining which program was running when it failed and produced the core file *abfd*.

2.11.1.2 bfd_core_file_failing_signal

Synopsis

```
int bfd_core_file_failing_signal (bfd *abfd);
```

Description

Returns the signal number which caused the core dump which generated the file the BFD *abfd* is attached to.

2.11.1.3 bfd_core_file_pid

Synopsis

```
int bfd_core_file_pid (bfd *abfd);
```

Description

Returns the PID of the process the core dump the BFD *abfd* is attached to was generated from.

2.11.1.4 core_file_matches_executable_p

Synopsis

```
bool core_file_matches_executable_p
   (bfd *core_bfd, bfd *exec_bfd);
```

Description

Return `TRUE` if the core file attached to *core_bfd* was generated by a run of the executable file attached to *exec_bfd*, `FALSE` otherwise.

2.11.1.5 generic_core_file_matches_executable_p

Synopsis

```
bool generic_core_file_matches_executable_p
   (bfd *core_bfd, bfd *exec_bfd);
```

Description

Return TRUE if the core file attached to *core_bfd* was generated by a run of the executable file attached to *exec_bfd*. The match is based on executable basenames only.

Note: When not able to determine the core file failing command or the executable name, we still return TRUE even though we're not sure that core file and executable match. This is to avoid generating a false warning in situations where we really don't know whether they match or not.

2.12 Targets

Description

Each port of BFD to a different machine requires the creation of a target back end. All the back end provides to the root part of BFD is a structure containing pointers to functions which perform certain low level operations on files. BFD translates the applications's requests through a pointer into calls to the back end routines.

When a file is opened with `bfd_openr`, its format and target are unknown. BFD uses various mechanisms to determine how to interpret the file. The operations performed are:

- Create a BFD by calling the internal routine `_bfd_new_bfd`, then call `bfd_find_target` with the target string supplied to `bfd_openr` and the new BFD pointer.

- If a null target string was provided to `bfd_find_target`, look up the environment variable `GNUTARGET` and use that as the target string.
- If the target string is still `NULL`, or the target string is `default`, then use the first item in the target vector as the target type, and set `target_defaulted` in the BFD to cause `bfd_check_format` to loop through all the targets. See Section 2.12.1 [bfd_target], page 151. See Section 2.9 [Formats], page 59.
- Otherwise, inspect the elements in the target vector one by one, until a match on target name is found. When found, use it.
- Otherwise return the error `bfd_error_invalid_target` to `bfd_openr`.
- `bfd_openr` attempts to open the file using `bfd_open_file`, and returns the BFD.

Once the BFD has been opened and the target selected, the file format may be determined. This is done by calling `bfd_check_format` on the BFD with a suggested format. If `target_defaulted` has been set, each possible target type is tried to see if it recognizes the specified format. `bfd_check_format` returns `TRUE` when the caller guesses right.

2.12.1 bfd_target

Description

This structure contains everything that BFD knows about a target. It includes things like its byte order, name, and which routines to call to do various operations.

Every BFD points to a target structure with its `xvec` member.

The macros below are used to dispatch to functions through the `bfd_target` vector. They are used in a number of macros further down in `bfd.h`, and are also used when calling various routines by hand inside the BFD implementation. The *arglist* argument must be parenthesized; it contains all the arguments to the called function.

They make the documentation (more) unpleasant to read, so if someone wants to fix this and not break the above, please do.

```
#define BFD_SEND(bfd, message, arglist) \
  ((*((bfd)->xvec->message)) arglist)

#ifdef DEBUG_BFD_SEND
#undef BFD_SEND
#define BFD_SEND(bfd, message, arglist) \
  (((bfd) && (bfd)->xvec && (bfd)->xvec->message) ? \
    ((*((bfd)->xvec->message)) arglist) : \
    (bfd_assert (__FILE__,__LINE__), NULL))
#endif
```

For operations which index on the BFD format:

```
#define BFD_SEND_FMT(bfd, message, arglist) \
  (((bfd)->xvec->message[(int) ((bfd)->format)]) arglist)

#ifdef DEBUG_BFD_SEND
#undef BFD_SEND_FMT
#define BFD_SEND_FMT(bfd, message, arglist) \
  (((bfd) && (bfd)->xvec && (bfd)->xvec->message) ? \
```

```
   (((bfd)->xvec->message[(int) ((bfd)->format)]) arglist) : \
   (bfd_assert (__FILE__,__LINE__), NULL))
#endif

/* Defined to TRUE if unused section symbol should be kept.  */
#ifndef TARGET_KEEP_UNUSED_SECTION_SYMBOLS
#define TARGET_KEEP_UNUSED_SECTION_SYMBOLS true
#endif
```

This is the structure which defines the type of BFD this is. The `xvec` member of the struct `bfd` itself points here. Each module that implements access to a different target under BFD, defines one of these.

FIXME, these names should be rationalised with the names of the entry points which call them. Too bad we can't have one macro to define them both!

```
enum bfd_flavour
{
  /* N.B. Update bfd_flavour_name if you change this.  */
  bfd_target_unknown_flavour,
  bfd_target_aout_flavour,
  bfd_target_coff_flavour,
  bfd_target_ecoff_flavour,
  bfd_target_xcoff_flavour,
  bfd_target_elf_flavour,
  bfd_target_tekhex_flavour,
  bfd_target_srec_flavour,
  bfd_target_verilog_flavour,
  bfd_target_ihex_flavour,
  bfd_target_som_flavour,
  bfd_target_os9k_flavour,
  bfd_target_versados_flavour,
  bfd_target_msdos_flavour,
  bfd_target_ovax_flavour,
  bfd_target_evax_flavour,
  bfd_target_mmo_flavour,
  bfd_target_mach_o_flavour,
  bfd_target_pef_flavour,
  bfd_target_pef_xlib_flavour,
  bfd_target_sym_flavour
};

enum bfd_endian { BFD_ENDIAN_BIG, BFD_ENDIAN_LITTLE, BFD_ENDIAN_UNKNOWN };

/* Forward declaration.  */
typedef struct bfd_link_info _bfd_link_info;

/* Forward declaration.  */
```

```
typedef struct flag_info flag_info;

typedef void (*bfd_cleanup) (bfd *);

typedef struct bfd_target
{
  /* Identifies the kind of target, e.g., SunOS4, Ultrix, etc.  */
  const char *name;

 /* The "flavour" of a back end is a general indication about
    the contents of a file.  */
  enum bfd_flavour flavour;

  /* The order of bytes within the data area of a file.  */
  enum bfd_endian byteorder;

 /* The order of bytes within the header parts of a file.  */
  enum bfd_endian header_byteorder;

  /* A mask of all the flags which an executable may have set -
     from the set BFD_NO_FLAGS, HAS_RELOC, ...D_PAGED.  */
  flagword object_flags;

 /* A mask of all the flags which a section may have set - from
    the set SEC_NO_FLAGS, SEC_ALLOC, ...SET_NEVER_LOAD.  */
  flagword section_flags;

 /* The character normally found at the front of a symbol.
    (if any), perhaps '_'.  */
  char symbol_leading_char;

 /* The pad character for file names within an archive header.  */
  char ar_pad_char;

  /* The maximum number of characters in an archive header.  */
  unsigned char ar_max_namelen;

  /* How well this target matches, used to select between various
     possible targets when more than one target matches.  */
  unsigned char match_priority;

 /* TRUE if unused section symbols should be kept.  */
  bool keep_unused_section_symbols;

  /* Entries for byte swapping for data. These are different from the
     other entry points, since they don't take a BFD as the first argument.
     Certain other handlers could do the same.  */
```

```
  bfd_uint64_t   (*bfd_getx64) (const void *);
  bfd_int64_t    (*bfd_getx_signed_64) (const void *);
  void           (*bfd_putx64) (bfd_uint64_t, void *);
  bfd_vma        (*bfd_getx32) (const void *);
  bfd_signed_vma (*bfd_getx_signed_32) (const void *);
  void           (*bfd_putx32) (bfd_vma, void *);
  bfd_vma        (*bfd_getx16) (const void *);
  bfd_signed_vma (*bfd_getx_signed_16) (const void *);
  void           (*bfd_putx16) (bfd_vma, void *);

  /* Byte swapping for the headers.  */
  bfd_uint64_t   (*bfd_h_getx64) (const void *);
  bfd_int64_t    (*bfd_h_getx_signed_64) (const void *);
  void           (*bfd_h_putx64) (bfd_uint64_t, void *);
  bfd_vma        (*bfd_h_getx32) (const void *);
  bfd_signed_vma (*bfd_h_getx_signed_32) (const void *);
  void           (*bfd_h_putx32) (bfd_vma, void *);
  bfd_vma        (*bfd_h_getx16) (const void *);
  bfd_signed_vma (*bfd_h_getx_signed_16) (const void *);
  void           (*bfd_h_putx16) (bfd_vma, void *);

  /* Format dependent routines: these are vectors of entry points
     within the target vector structure, one for each format to check.  */

  /* Check the format of a file being read.  Return a bfd_cleanup on
     success or zero on failure.  */
  bfd_cleanup (*_bfd_check_format[bfd_type_end]) (bfd *);

  /* Set the format of a file being written.  */
  bool (*_bfd_set_format[bfd_type_end]) (bfd *);

  /* Write cached information into a file being written, at bfd_close.  */
  bool (*_bfd_write_contents[bfd_type_end]) (bfd *);
```

The general target vector. These vectors are initialized using the BFD_JUMP_TABLE macros.

```
  /* Generic entry points.  */
#define BFD_JUMP_TABLE_GENERIC(NAME) \
  NAME##_close_and_cleanup, \
  NAME##_bfd_free_cached_info, \
  NAME##_new_section_hook, \
  NAME##_get_section_contents, \
  NAME##_get_section_contents_in_window

  /* Called when the BFD is being closed to do any necessary cleanup.  */
```

```
  bool (*_close_and_cleanup) (bfd *);
  /* Ask the BFD to free all cached information.  */
  bool (*_bfd_free_cached_info) (bfd *);
  /* Called when a new section is created.  */
  bool (*_new_section_hook) (bfd *, sec_ptr);
  /* Read the contents of a section.  */
  bool (*_bfd_get_section_contents) (bfd *, sec_ptr, void *, file_ptr,
                                     bfd_size_type);
  bool (*_bfd_get_section_contents_in_window) (bfd *, sec_ptr, bfd_window *,
                                               file_ptr, bfd_size_type);

  /* Entry points to copy private data.  */
#define BFD_JUMP_TABLE_COPY(NAME) \
  NAME##_bfd_copy_private_bfd_data, \
  NAME##_bfd_merge_private_bfd_data, \
  _bfd_generic_init_private_section_data, \
  NAME##_bfd_copy_private_section_data, \
  NAME##_bfd_copy_private_symbol_data, \
  NAME##_bfd_copy_private_header_data, \
  NAME##_bfd_set_private_flags, \
  NAME##_bfd_print_private_bfd_data

  /* Called to copy BFD general private data from one object file
     to another.  */
  bool (*_bfd_copy_private_bfd_data) (bfd *, bfd *);
  /* Called to merge BFD general private data from one object file
     to a common output file when linking.  */
  bool (*_bfd_merge_private_bfd_data) (bfd *, struct bfd_link_info *);
  /* Called to initialize BFD private section data from one object file
     to another.  */
#define bfd_init_private_section_data(ibfd, isec, obfd, osec, link_info) \
       BFD_SEND (obfd, _bfd_init_private_section_data, \
                 (ibfd, isec, obfd, osec, link_info))
  bool (*_bfd_init_private_section_data) (bfd *, sec_ptr, bfd *, sec_ptr,
                                          struct bfd_link_info *);
  /* Called to copy BFD private section data from one object file
     to another.  */
  bool (*_bfd_copy_private_section_data) (bfd *, sec_ptr, bfd *, sec_ptr);
  /* Called to copy BFD private symbol data from one symbol
     to another.  */
  bool (*_bfd_copy_private_symbol_data) (bfd *, asymbol *,
                                         bfd *, asymbol *);
  /* Called to copy BFD private header data from one object file
     to another.  */
  bool (*_bfd_copy_private_header_data) (bfd *, bfd *);
  /* Called to set private backend flags.  */
  bool (*_bfd_set_private_flags) (bfd *, flagword);
```

```
  /* Called to print private BFD data.  */
  bool (*_bfd_print_private_bfd_data) (bfd *, void *);

  /* Core file entry points.  */
#define BFD_JUMP_TABLE_CORE(NAME) \
  NAME##_core_file_failing_command, \
  NAME##_core_file_failing_signal, \
  NAME##_core_file_matches_executable_p, \
  NAME##_core_file_pid

  char *(*_core_file_failing_command) (bfd *);
  int   (*_core_file_failing_signal) (bfd *);
  bool  (*_core_file_matches_executable_p) (bfd *, bfd *);
  int   (*_core_file_pid) (bfd *);

  /* Archive entry points.  */
#define BFD_JUMP_TABLE_ARCHIVE(NAME) \
  NAME##_slurp_armap, \
  NAME##_slurp_extended_name_table, \
  NAME##_construct_extended_name_table, \
  NAME##_truncate_arname, \
  NAME##_write_armap, \
  NAME##_read_ar_hdr, \
  NAME##_write_ar_hdr, \
  NAME##_openr_next_archived_file, \
  NAME##_get_elt_at_index, \
  NAME##_generic_stat_arch_elt, \
  NAME##_update_armap_timestamp

  bool (*_bfd_slurp_armap) (bfd *);
  bool (*_bfd_slurp_extended_name_table) (bfd *);
  bool (*_bfd_construct_extended_name_table) (bfd *, char **,
                                               bfd_size_type *,
                                               const char **);
  void (*_bfd_truncate_arname) (bfd *, const char *, char *);
  bool (*write_armap) (bfd *, unsigned, struct orl *, unsigned, int);
  void *(*_bfd_read_ar_hdr_fn) (bfd *);
  bool (*_bfd_write_ar_hdr_fn) (bfd *, bfd *);
  bfd *(*openr_next_archived_file) (bfd *, bfd *);
#define bfd_get_elt_at_index(b,i) \
       BFD_SEND (b, _bfd_get_elt_at_index, (b,i))
  bfd *(*_bfd_get_elt_at_index) (bfd *, symindex);
  int  (*_bfd_stat_arch_elt) (bfd *, struct stat *);
  bool (*_bfd_update_armap_timestamp) (bfd *);

  /* Entry points used for symbols.  */
```

```
#define BFD_JUMP_TABLE_SYMBOLS(NAME) \
  NAME##_get_symtab_upper_bound, \
  NAME##_canonicalize_symtab, \
  NAME##_make_empty_symbol, \
  NAME##_print_symbol, \
  NAME##_get_symbol_info, \
  NAME##_get_symbol_version_string, \
  NAME##_bfd_is_local_label_name, \
  NAME##_bfd_is_target_special_symbol, \
  NAME##_get_lineno, \
  NAME##_find_nearest_line, \
  NAME##_find_line, \
  NAME##_find_inliner_info, \
  NAME##_bfd_make_debug_symbol, \
  NAME##_read_minisymbols, \
  NAME##_minisymbol_to_symbol

  long (*_bfd_get_symtab_upper_bound) (bfd *);
  long (*_bfd_canonicalize_symtab) (bfd *, struct bfd_symbol **);
  struct bfd_symbol *
       (*_bfd_make_empty_symbol) (bfd *);
  void (*_bfd_print_symbol) (bfd *, void *, struct bfd_symbol *,
                             bfd_print_symbol_type);
#define bfd_print_symbol(b,p,s,e) \
       BFD_SEND (b, _bfd_print_symbol, (b,p,s,e))
  void  (*_bfd_get_symbol_info) (bfd *, struct bfd_symbol *, symbol_info *);
#define bfd_get_symbol_info(b,p,e) \
       BFD_SEND (b, _bfd_get_symbol_info, (b,p,e))
  const char *
       (*_bfd_get_symbol_version_string) (bfd *, struct bfd_symbol *,
                                          bool, bool *);
#define bfd_get_symbol_version_string(b,s,p,h) \
       BFD_SEND (b, _bfd_get_symbol_version_string, (b,s,p,h))
  bool (*_bfd_is_local_label_name) (bfd *, const char *);
  bool (*_bfd_is_target_special_symbol) (bfd *, asymbol *);
  alent *
       (*_get_lineno) (bfd *, struct bfd_symbol *);
  bool (*_bfd_find_nearest_line) (bfd *, struct bfd_symbol **,
                                  struct bfd_section *, bfd_vma,
                                  const char **, const char **,
                                  unsigned int *, unsigned int *);
  bool (*_bfd_find_line) (bfd *, struct bfd_symbol **,
                          struct bfd_symbol *, const char **,
                          unsigned int *);
  bool (*_bfd_find_inliner_info)
    (bfd *, const char **, const char **, unsigned int *);
 /* Back-door to allow format-aware applications to create debug symbols
```

```
    while using BFD for everything else.  Currently used by the assembler
    when creating COFF files.  */
  asymbol *
       (*_bfd_make_debug_symbol) (bfd *, void *, unsigned long size);
#define bfd_read_minisymbols(b, d, m, s) \
       BFD_SEND (b, _read_minisymbols, (b, d, m, s))
  long (*_read_minisymbols) (bfd *, bool, void **, unsigned int *);
#define bfd_minisymbol_to_symbol(b, d, m, f) \
       BFD_SEND (b, _minisymbol_to_symbol, (b, d, m, f))
  asymbol *
       (*_minisymbol_to_symbol) (bfd *, bool, const void *, asymbol *);

  /* Routines for relocs.  */
#define BFD_JUMP_TABLE_RELOCS(NAME) \
  NAME##_get_reloc_upper_bound, \
  NAME##_canonicalize_reloc, \
  NAME##_set_reloc, \
  NAME##_bfd_reloc_type_lookup, \
  NAME##_bfd_reloc_name_lookup

  long (*_get_reloc_upper_bound) (bfd *, sec_ptr);
  long (*_bfd_canonicalize_reloc) (bfd *, sec_ptr, arelent **,
                                   struct bfd_symbol **);
  void (*_bfd_set_reloc) (bfd *, sec_ptr, arelent **, unsigned int);
  /* See documentation on reloc types.  */
  reloc_howto_type *
       (*reloc_type_lookup) (bfd *, bfd_reloc_code_real_type);
  reloc_howto_type *
       (*reloc_name_lookup) (bfd *, const char *);

  /* Routines used when writing an object file.  */
#define BFD_JUMP_TABLE_WRITE(NAME) \
  NAME##_set_arch_mach, \
  NAME##_set_section_contents

  bool (*_bfd_set_arch_mach) (bfd *, enum bfd_architecture,
                                     unsigned long);
  bool (*_bfd_set_section_contents) (bfd *, sec_ptr, const void *,
                                     file_ptr, bfd_size_type);

  /* Routines used by the linker.  */
#define BFD_JUMP_TABLE_LINK(NAME) \
  NAME##_sizeof_headers, \
  NAME##_bfd_get_relocated_section_contents, \
  NAME##_bfd_relax_section, \
  NAME##_bfd_link_hash_table_create, \
  NAME##_bfd_link_add_symbols, \
```

```
  NAME##_bfd_link_just_syms, \
  NAME##_bfd_copy_link_hash_symbol_type, \
  NAME##_bfd_final_link, \
  NAME##_bfd_link_split_section, \
  NAME##_bfd_link_check_relocs, \
  NAME##_bfd_gc_sections, \
  NAME##_bfd_lookup_section_flags, \
  NAME##_bfd_merge_sections, \
  NAME##_bfd_is_group_section, \
  NAME##_bfd_group_name, \
  NAME##_bfd_discard_group, \
  NAME##_section_already_linked, \
  NAME##_bfd_define_common_symbol, \
  NAME##_bfd_link_hide_symbol, \
  NAME##_bfd_define_start_stop

  int  (*_bfd_sizeof_headers) (bfd *, struct bfd_link_info *);
  bfd_byte *
       (*_bfd_get_relocated_section_contents) (bfd *,
                                               struct bfd_link_info *,
                                               struct bfd_link_order *,
                                               bfd_byte *, bool,
                                               struct bfd_symbol **);

  bool (*_bfd_relax_section) (bfd *, struct bfd_section *,
                              struct bfd_link_info *, bool *);

  /* Create a hash table for the linker.  Different backends store
     different information in this table.  */
  struct bfd_link_hash_table *
       (*_bfd_link_hash_table_create) (bfd *);

  /* Add symbols from this object file into the hash table.  */
  bool (*_bfd_link_add_symbols) (bfd *, struct bfd_link_info *);

  /* Indicate that we are only retrieving symbol values from this section.  */
  void (*_bfd_link_just_syms) (asection *, struct bfd_link_info *);

  /* Copy the symbol type and other attributes for a linker script
     assignment of one symbol to another.  */
#define bfd_copy_link_hash_symbol_type(b, t, f) \
       BFD_SEND (b, _bfd_copy_link_hash_symbol_type, (b, t, f))
  void (*_bfd_copy_link_hash_symbol_type) (bfd *,
                                           struct bfd_link_hash_entry *,
                                           struct bfd_link_hash_entry *);

  /* Do a link based on the link_order structures attached to each
```

```
   section of the BFD.  */
bool (*_bfd_final_link) (bfd *, struct bfd_link_info *);

/* Should this section be split up into smaller pieces during linking.  */
bool (*_bfd_link_split_section) (bfd *, struct bfd_section *);

/* Check the relocations in the bfd for validity.  */
bool (* _bfd_link_check_relocs)(bfd *, struct bfd_link_info *);

/* Remove sections that are not referenced from the output.  */
bool (*_bfd_gc_sections) (bfd *, struct bfd_link_info *);

/* Sets the bitmask of allowed and disallowed section flags.  */
bool (*_bfd_lookup_section_flags) (struct bfd_link_info *,
                                   struct flag_info *, asection *);

/* Attempt to merge SEC_MERGE sections.  */
bool (*_bfd_merge_sections) (bfd *, struct bfd_link_info *);

/* Is this section a member of a group?  */
bool (*_bfd_is_group_section) (bfd *, const struct bfd_section *);

/* The group name, if section is a member of a group.  */
const char *(*_bfd_group_name) (bfd *, const struct bfd_section *);

/* Discard members of a group.  */
bool (*_bfd_discard_group) (bfd *, struct bfd_section *);

/* Check if SEC has been already linked during a reloceatable or
   final link.  */
bool (*_section_already_linked) (bfd *, asection *,
                                 struct bfd_link_info *);

/* Define a common symbol.  */
bool (*_bfd_define_common_symbol) (bfd *, struct bfd_link_info *,
                                   struct bfd_link_hash_entry *);

/* Hide a symbol.  */
void (*_bfd_link_hide_symbol) (bfd *, struct bfd_link_info *,
                               struct bfd_link_hash_entry *);

/* Define a __start, __stop, .startof. or .sizeof. symbol.  */
struct bfd_link_hash_entry *
     (*_bfd_define_start_stop) (struct bfd_link_info *, const char *,
                                asection *);

/* Routines to handle dynamic symbols and relocs.  */
```

```
#define BFD_JUMP_TABLE_DYNAMIC(NAME) \
  NAME##_get_dynamic_symtab_upper_bound, \
  NAME##_canonicalize_dynamic_symtab, \
  NAME##_get_synthetic_symtab, \
  NAME##_get_dynamic_reloc_upper_bound, \
  NAME##_canonicalize_dynamic_reloc

  /* Get the amount of memory required to hold the dynamic symbols.  */
  long (*_bfd_get_dynamic_symtab_upper_bound) (bfd *);
  /* Read in the dynamic symbols.  */
  long (*_bfd_canonicalize_dynamic_symtab) (bfd *, struct bfd_symbol **);
  /* Create synthetized symbols.  */
  long (*_bfd_get_synthetic_symtab) (bfd *, long, struct bfd_symbol **,
                                     long, struct bfd_symbol **,
                                     struct bfd_symbol **);
  /* Get the amount of memory required to hold the dynamic relocs.  */
  long (*_bfd_get_dynamic_reloc_upper_bound) (bfd *);
  /* Read in the dynamic relocs.  */
  long (*_bfd_canonicalize_dynamic_reloc) (bfd *, arelent **,
                                           struct bfd_symbol **);
```

A pointer to an alternative bfd_target in case the current one is not satisfactory. This can happen when the target cpu supports both big and little endian code, and target chosen by the linker has the wrong endianness. The function open_output() in ld/ldlang.c uses this field to find an alternative output format that is suitable.

```
  /* Opposite endian version of this target.  */
  const struct bfd_target *alternative_target;

  /* Data for use by back-end routines, which isn't
     generic enough to belong in this structure.  */
  const void *backend_data;

} bfd_target;

static inline const char *
bfd_get_target (const bfd *abfd)
{
  return abfd->xvec->name;
}

static inline enum bfd_flavour
bfd_get_flavour (const bfd *abfd)
{
  return abfd->xvec->flavour;
}
```

```
static inline flagword
bfd_applicable_file_flags (const bfd *abfd)
{
  return abfd->xvec->object_flags;
}

static inline bool
bfd_family_coff (const bfd *abfd)
{
  return (bfd_get_flavour (abfd) == bfd_target_coff_flavour
          || bfd_get_flavour (abfd) == bfd_target_xcoff_flavour);
}

static inline bool
bfd_big_endian (const bfd *abfd)
{
  return abfd->xvec->byteorder == BFD_ENDIAN_BIG;
}
static inline bool
bfd_little_endian (const bfd *abfd)
{
  return abfd->xvec->byteorder == BFD_ENDIAN_LITTLE;
}

static inline bool
bfd_header_big_endian (const bfd *abfd)
{
  return abfd->xvec->header_byteorder == BFD_ENDIAN_BIG;
}

static inline bool
bfd_header_little_endian (const bfd *abfd)
{
  return abfd->xvec->header_byteorder == BFD_ENDIAN_LITTLE;
}

static inline flagword
bfd_applicable_section_flags (const bfd *abfd)
{
  return abfd->xvec->section_flags;
}

static inline char
bfd_get_symbol_leading_char (const bfd *abfd)
{
  return abfd->xvec->symbol_leading_char;
}
```

```
static inline enum bfd_flavour
bfd_asymbol_flavour (const asymbol *sy)
{
  if ((sy->flags & BSF_SYNTHETIC) != 0)
    return bfd_target_unknown_flavour;
  return sy->the_bfd->xvec->flavour;
}

static inline bool
bfd_keep_unused_section_symbols (const bfd *abfd)
{
  return abfd->xvec->keep_unused_section_symbols;
}
```

2.12.1.1 bfd_set_default_target

Synopsis

```
bool bfd_set_default_target (const char *name);
```

Description

Set the default target vector to use when recognizing a BFD. This takes the name of the target, which may be a BFD target name or a configuration triplet.

2.12.1.2 bfd_find_target

Synopsis

```
const bfd_target *bfd_find_target (const char *target_name, bfd *abfd);
```

Description

Return a pointer to the transfer vector for the object target named *target_name*. If *target_name* is `NULL`, choose the one in the environment variable `GNUTARGET`; if that is null or not defined, then choose the first entry in the target list. Passing in the string "default" or setting the environment variable to "default" will cause the first entry in the target list to be returned, and "target_defaulted" will be set in the BFD if *abfd* isn't `NULL`. This causes `bfd_check_format` to loop over all the targets to find the one that matches the file being read.

2.12.1.3 bfd_get_target_info

Synopsis

```
const bfd_target *bfd_get_target_info (const char *target_name,
    bfd *abfd,
    bool *is_bigendian,
    int *underscoring,
    const char **def_target_arch);
```

Description

Return a pointer to the transfer vector for the object target named *target_name*. If *target_name* is `NULL`, choose the one in the environment variable `GNUTARGET`; if that is null or

not defined, then choose the first entry in the target list. Passing in the string "default" or setting the environment variable to "default" will cause the first entry in the target list to be returned, and "target_defaulted" will be set in the BFD if *abfd* isn't NULL. This causes bfd_check_format to loop over all the targets to find the one that matches the file being read. If *is_bigendian* is not NULL, then set this value to target's endian mode. True for big-endian, FALSE for little-endian or for invalid target. If *underscoring* is not NULL, then set this value to target's underscoring mode. Zero for none-underscoring, -1 for invalid target, else the value of target vector's symbol underscoring. If *def_target_arch* is not NULL, then set it to the architecture string specified by the target_name.

2.12.1.4 bfd_target_list

Synopsis

```
const char ** bfd_target_list (void);
```

Description

Return a freshly malloced NULL-terminated vector of the names of all the valid BFD targets. Do not modify the names.

2.12.1.5 bfd_iterate_over_targets

Synopsis

```
const bfd_target *bfd_iterate_over_targets
   (int (*func) (const bfd_target *, void *),
    void *data);
```

Description

Call *func* for each target in the list of BFD target vectors, passing *data* to *func*. Stop iterating if *func* returns a non-zero result, and return that target vector. Return NULL if *func* always returns zero.

2.12.1.6 bfd_flavour_name

Synopsis

```
const char *bfd_flavour_name (enum bfd_flavour flavour);
```

Description

Return the string form of *flavour*.

2.13 Architectures

BFD keeps one atom in a BFD describing the architecture of the data attached to the BFD: a pointer to a bfd_arch_info_type.

Pointers to structures can be requested independently of a BFD so that an architecture's information can be interrogated without access to an open BFD.

The architecture information is provided by each architecture package. The set of default architectures is selected by the macro SELECT_ARCHITECTURES. This is normally set up in the config/*target*.mt file of your choice. If the name is not defined, then all the architectures supported are included.

When BFD starts up, all the architectures are called with an initialize method. It is up to the architecture back end to insert as many items into the list of architectures as it wants

to; generally this would be one for each machine and one for the default case (an item with a machine field of 0).

BFD's idea of an architecture is implemented in `archures.c`.

2.13.1 bfd_architecture

Description

This enum gives the object file's CPU architecture, in a global sense—i.e., what processor family does it belong to? Another field indicates which processor within the family is in use. The machine gives a number which distinguishes different versions of the architecture, containing, for example, 68020 for Motorola 68020.

```
enum bfd_architecture
{
  bfd_arch_unknown,   /* File arch not known.  */
  bfd_arch_obscure,   /* Arch known, not one of these.  */
  bfd_arch_m68k,      /* Motorola 68xxx.  */
#define bfd_mach_m68000                1
#define bfd_mach_m68008                2
#define bfd_mach_m68010                3
#define bfd_mach_m68020                4
#define bfd_mach_m68030                5
#define bfd_mach_m68040                6
#define bfd_mach_m68060                7
#define bfd_mach_cpu32                 8
#define bfd_mach_fido                  9
#define bfd_mach_mcf_isa_a_nodiv       10
#define bfd_mach_mcf_isa_a             11
#define bfd_mach_mcf_isa_a_mac         12
#define bfd_mach_mcf_isa_a_emac        13
#define bfd_mach_mcf_isa_aplus         14
#define bfd_mach_mcf_isa_aplus_mac     15
#define bfd_mach_mcf_isa_aplus_emac    16
#define bfd_mach_mcf_isa_b_nousp       17
#define bfd_mach_mcf_isa_b_nousp_mac   18
#define bfd_mach_mcf_isa_b_nousp_emac  19
#define bfd_mach_mcf_isa_b             20
#define bfd_mach_mcf_isa_b_mac         21
#define bfd_mach_mcf_isa_b_emac        22
#define bfd_mach_mcf_isa_b_float       23
#define bfd_mach_mcf_isa_b_float_mac   24
#define bfd_mach_mcf_isa_b_float_emac  25
#define bfd_mach_mcf_isa_c             26
#define bfd_mach_mcf_isa_c_mac         27
#define bfd_mach_mcf_isa_c_emac        28
#define bfd_mach_mcf_isa_c_nodiv       29
#define bfd_mach_mcf_isa_c_nodiv_mac   30
#define bfd_mach_mcf_isa_c_nodiv_emac  31
```

```
  bfd_arch_vax,       /* DEC Vax.  */

  bfd_arch_or1k,      /* OpenRISC 1000.  */
#define bfd_mach_or1k          1
#define bfd_mach_or1knd        2

  bfd_arch_sparc,     /* SPARC.  */
#define bfd_mach_sparc                 1
/* The difference between v8plus and v9 is that v9 is a true 64 bit env.  */
#define bfd_mach_sparc_sparclet        2
#define bfd_mach_sparc_sparclite       3
#define bfd_mach_sparc_v8plus          4
#define bfd_mach_sparc_v8plusa         5 /* with ultrasparc add'ns.  */
#define bfd_mach_sparc_sparclite_le    6
#define bfd_mach_sparc_v9              7
#define bfd_mach_sparc_v9a             8 /* with ultrasparc add'ns.  */
#define bfd_mach_sparc_v8plusb         9 /* with cheetah add'ns.  */
#define bfd_mach_sparc_v9b             10 /* with cheetah add'ns.  */
#define bfd_mach_sparc_v8plusc         11 /* with UA2005 and T1 add'ns.  */
#define bfd_mach_sparc_v9c             12 /* with UA2005 and T1 add'ns.  */
#define bfd_mach_sparc_v8plusd         13 /* with UA2007 and T3 add'ns.  */
#define bfd_mach_sparc_v9d             14 /* with UA2007 and T3 add'ns.  */
#define bfd_mach_sparc_v8pluse         15 /* with OSA2001 and T4 add'ns (no IMA).  */
#define bfd_mach_sparc_v9e             16 /* with OSA2001 and T4 add'ns (no IMA).  */
#define bfd_mach_sparc_v8plusv         17 /* with OSA2011 and T4 and IMA and FJMAU add
#define bfd_mach_sparc_v9v             18 /* with OSA2011 and T4 and IMA and FJMAU add
#define bfd_mach_sparc_v8plusm         19 /* with OSA2015 and M7 add'ns.  */
#define bfd_mach_sparc_v9m             20 /* with OSA2015 and M7 add'ns.  */
#define bfd_mach_sparc_v8plusm8        21 /* with OSA2017 and M8 add'ns.  */
#define bfd_mach_sparc_v9m8            22 /* with OSA2017 and M8 add'ns.  */
/* Nonzero if MACH has the v9 instruction set.  */
#define bfd_mach_sparc_v9_p(mach) \
  ((mach) >= bfd_mach_sparc_v8plus && (mach) <= bfd_mach_sparc_v9m8 \
   && (mach) != bfd_mach_sparc_sparclite_le)
/* Nonzero if MACH is a 64 bit sparc architecture.  */
#define bfd_mach_sparc_64bit_p(mach) \
  ((mach) >= bfd_mach_sparc_v9 \
   && (mach) != bfd_mach_sparc_v8plusb \
   && (mach) != bfd_mach_sparc_v8plusc \
   && (mach) != bfd_mach_sparc_v8plusd \
   && (mach) != bfd_mach_sparc_v8pluse \
   && (mach) != bfd_mach_sparc_v8plusv \
   && (mach) != bfd_mach_sparc_v8plusm \
   && (mach) != bfd_mach_sparc_v8plusm8)
  bfd_arch_spu,       /* PowerPC SPU.  */
#define bfd_mach_spu           256
  bfd_arch_mips,      /* MIPS Rxxxx.  */
```

```
#define bfd_mach_mips3000              3000
#define bfd_mach_mips3900              3900
#define bfd_mach_mips4000              4000
#define bfd_mach_mips4010              4010
#define bfd_mach_mips4100              4100
#define bfd_mach_mips4111              4111
#define bfd_mach_mips4120              4120
#define bfd_mach_mips4300              4300
#define bfd_mach_mips4400              4400
#define bfd_mach_mips4600              4600
#define bfd_mach_mips4650              4650
#define bfd_mach_mips5000              5000
#define bfd_mach_mips5400              5400
#define bfd_mach_mips5500              5500
#define bfd_mach_mips5900              5900
#define bfd_mach_mips6000              6000
#define bfd_mach_mips7000              7000
#define bfd_mach_mips8000              8000
#define bfd_mach_mips9000              9000
#define bfd_mach_mips10000             10000
#define bfd_mach_mips12000             12000
#define bfd_mach_mips14000             14000
#define bfd_mach_mips16000             16000
#define bfd_mach_mips16                16
#define bfd_mach_mips5                 5
#define bfd_mach_mips_loongson_2e      3001
#define bfd_mach_mips_loongson_2f      3002
#define bfd_mach_mips_gs464            3003
#define bfd_mach_mips_gs464e           3004
#define bfd_mach_mips_gs264e           3005
#define bfd_mach_mips_sb1              12310201 /* octal 'SB', 01.  */
#define bfd_mach_mips_octeon           6501
#define bfd_mach_mips_octeonp          6601
#define bfd_mach_mips_octeon2          6502
#define bfd_mach_mips_octeon3          6503
#define bfd_mach_mips_xlr              887682   /* decimal 'XLR'.  */
#define bfd_mach_mips_interaptiv_mr2   736550   /* decimal 'IA2'.  */
#define bfd_mach_mipsisa32             32
#define bfd_mach_mipsisa32r2           33
#define bfd_mach_mipsisa32r3           34
#define bfd_mach_mipsisa32r5           36
#define bfd_mach_mipsisa32r6           37
#define bfd_mach_mipsisa64             64
#define bfd_mach_mipsisa64r2           65
#define bfd_mach_mipsisa64r3           66
#define bfd_mach_mipsisa64r5           68
#define bfd_mach_mipsisa64r6           69
```

```
#define bfd_mach_mips_micromips        96
  bfd_arch_i386,      /* Intel 386.  */
#define bfd_mach_i386_intel_syntax     (1 << 0)
#define bfd_mach_i386_i8086            (1 << 1)
#define bfd_mach_i386_i386             (1 << 2)
#define bfd_mach_x86_64                (1 << 3)
#define bfd_mach_x64_32                (1 << 4)
#define bfd_mach_i386_i386_intel_syntax (bfd_mach_i386_i386 | bfd_mach_i386_intel_synt
#define bfd_mach_x86_64_intel_syntax   (bfd_mach_x86_64 | bfd_mach_i386_intel_syntax)
#define bfd_mach_x64_32_intel_syntax   (bfd_mach_x64_32 | bfd_mach_i386_intel_syntax)
  bfd_arch_l1om,      /* Intel L1OM.  */
#define bfd_mach_l1om                  (1 << 5)
#define bfd_mach_l1om_intel_syntax     (bfd_mach_l1om | bfd_mach_i386_intel_syntax)
  bfd_arch_k1om,      /* Intel K1OM.  */
#define bfd_mach_k1om                  (1 << 6)
#define bfd_mach_k1om_intel_syntax     (bfd_mach_k1om | bfd_mach_i386_intel_syntax)
  bfd_arch_iamcu,     /* Intel MCU.  */
#define bfd_mach_iamcu                 (1 << 8)
#define bfd_mach_i386_iamcu            (bfd_mach_i386_i386 | bfd_mach_iamcu)
#define bfd_mach_i386_iamcu_intel_syntax (bfd_mach_i386_iamcu | bfd_mach_i386_intel_sy
  bfd_arch_romp,      /* IBM ROMP PC/RT.  */
  bfd_arch_convex,    /* Convex.  */
  bfd_arch_m98k,      /* Motorola 98xxx.  */
  bfd_arch_pyramid,   /* Pyramid Technology.  */
  bfd_arch_h8300,     /* Renesas H8/300 (formerly Hitachi H8/300).  */
#define bfd_mach_h8300         1
#define bfd_mach_h8300h        2
#define bfd_mach_h8300s        3
#define bfd_mach_h8300hn       4
#define bfd_mach_h8300sn       5
#define bfd_mach_h8300sx       6
#define bfd_mach_h8300sxn      7
  bfd_arch_pdp11,     /* DEC PDP-11.  */
  bfd_arch_powerpc,   /* PowerPC.  */
#define bfd_mach_ppc           32
#define bfd_mach_ppc64         64
#define bfd_mach_ppc_403       403
#define bfd_mach_ppc_403gc     4030
#define bfd_mach_ppc_405       405
#define bfd_mach_ppc_505       505
#define bfd_mach_ppc_601       601
#define bfd_mach_ppc_602       602
#define bfd_mach_ppc_603       603
#define bfd_mach_ppc_ec603e    6031
#define bfd_mach_ppc_604       604
#define bfd_mach_ppc_620       620
#define bfd_mach_ppc_630       630
```

```
#define bfd_mach_ppc_750       750
#define bfd_mach_ppc_860       860
#define bfd_mach_ppc_a35       35
#define bfd_mach_ppc_rs64ii    642
#define bfd_mach_ppc_rs64iii   643
#define bfd_mach_ppc_7400      7400
#define bfd_mach_ppc_e500      500
#define bfd_mach_ppc_e500mc    5001
#define bfd_mach_ppc_e500mc64  5005
#define bfd_mach_ppc_e5500     5006
#define bfd_mach_ppc_e6500     5007
#define bfd_mach_ppc_titan     83
#define bfd_mach_ppc_vle       84
  bfd_arch_rs6000,    /* IBM RS/6000.  */
#define bfd_mach_rs6k          6000
#define bfd_mach_rs6k_rs1      6001
#define bfd_mach_rs6k_rsc      6003
#define bfd_mach_rs6k_rs2      6002
  bfd_arch_hppa,      /* HP PA RISC.  */
#define bfd_mach_hppa10        10
#define bfd_mach_hppa11        11
#define bfd_mach_hppa20        20
#define bfd_mach_hppa20w       25
  bfd_arch_d10v,      /* Mitsubishi D10V.  */
#define bfd_mach_d10v          1
#define bfd_mach_d10v_ts2      2
#define bfd_mach_d10v_ts3      3
  bfd_arch_d30v,      /* Mitsubishi D30V.  */
  bfd_arch_dlx,       /* DLX.  */
  bfd_arch_m68hc11,   /* Motorola 68HC11.  */
  bfd_arch_m68hc12,   /* Motorola 68HC12.  */
#define bfd_mach_m6812_default 0
#define bfd_mach_m6812         1
#define bfd_mach_m6812s        2
  bfd_arch_m9s12x,    /* Freescale S12X.  */
  bfd_arch_m9s12xg,   /* Freescale XGATE.  */
  bfd_arch_s12z,    /* Freescale S12Z.  */
#define bfd_mach_s12z_default 0
  bfd_arch_z8k,       /* Zilog Z8000.  */
#define bfd_mach_z8001         1
#define bfd_mach_z8002         2
  bfd_arch_sh,        /* Renesas / SuperH SH (formerly Hitachi SH).  */
#define bfd_mach_sh                            1
#define bfd_mach_sh2                           0x20
#define bfd_mach_sh_dsp                        0x2d
#define bfd_mach_sh2a                          0x2a
#define bfd_mach_sh2a_nofpu                    0x2b
```

```
#define bfd_mach_sh2a_nofpu_or_sh4_nommu_nofpu 0x2a1
#define bfd_mach_sh2a_nofpu_or_sh3_nommu       0x2a2
#define bfd_mach_sh2a_or_sh4                   0x2a3
#define bfd_mach_sh2a_or_sh3e                  0x2a4
#define bfd_mach_sh2e                          0x2e
#define bfd_mach_sh3                           0x30
#define bfd_mach_sh3_nommu                     0x31
#define bfd_mach_sh3_dsp                       0x3d
#define bfd_mach_sh3e                          0x3e
#define bfd_mach_sh4                           0x40
#define bfd_mach_sh4_nofpu                     0x41
#define bfd_mach_sh4_nommu_nofpu               0x42
#define bfd_mach_sh4a                          0x4a
#define bfd_mach_sh4a_nofpu                    0x4b
#define bfd_mach_sh4al_dsp                     0x4d
  bfd_arch_alpha,     /* Dec Alpha.  */
#define bfd_mach_alpha_ev4     0x10
#define bfd_mach_alpha_ev5     0x20
#define bfd_mach_alpha_ev6     0x30
  bfd_arch_arm,       /* Advanced Risc Machines ARM.  */
#define bfd_mach_arm_unknown   0
#define bfd_mach_arm_2         1
#define bfd_mach_arm_2a        2
#define bfd_mach_arm_3         3
#define bfd_mach_arm_3M        4
#define bfd_mach_arm_4         5
#define bfd_mach_arm_4T        6
#define bfd_mach_arm_5         7
#define bfd_mach_arm_5T        8
#define bfd_mach_arm_5TE       9
#define bfd_mach_arm_XScale    10
#define bfd_mach_arm_ep9312    11
#define bfd_mach_arm_iWMMXt    12
#define bfd_mach_arm_iWMMXt2   13
#define bfd_mach_arm_5TEJ      14
#define bfd_mach_arm_6         15
#define bfd_mach_arm_6KZ       16
#define bfd_mach_arm_6T2       17
#define bfd_mach_arm_6K        18
#define bfd_mach_arm_7         19
#define bfd_mach_arm_6M        20
#define bfd_mach_arm_6SM       21
#define bfd_mach_arm_7EM       22
#define bfd_mach_arm_8         23
#define bfd_mach_arm_8R        24
#define bfd_mach_arm_8M_BASE   25
#define bfd_mach_arm_8M_MAIN   26
```

```
#define bfd_mach_arm_8_1M_MAIN 27
#define bfd_mach_arm_9         28
  bfd_arch_nds32,     /* Andes NDS32.  */
#define bfd_mach_n1            1
#define bfd_mach_n1h           2
#define bfd_mach_n1h_v2        3
#define bfd_mach_n1h_v3        4
#define bfd_mach_n1h_v3m       5
  bfd_arch_ns32k,     /* National Semiconductors ns32000.  */
  bfd_arch_tic30,     /* Texas Instruments TMS320C30.  */
  bfd_arch_tic4x,     /* Texas Instruments TMS320C3X/4X.  */
#define bfd_mach_tic3x         30
#define bfd_mach_tic4x         40
  bfd_arch_tic54x,    /* Texas Instruments TMS320C54X.  */
  bfd_arch_tic6x,     /* Texas Instruments TMS320C6X.  */
  bfd_arch_v850,      /* NEC V850.  */
  bfd_arch_v850_rh850,/* NEC V850 (using RH850 ABI).  */
#define bfd_mach_v850          1
#define bfd_mach_v850e         'E'
#define bfd_mach_v850e1        '1'
#define bfd_mach_v850e2        0x4532
#define bfd_mach_v850e2v3      0x45325633
#define bfd_mach_v850e3v5      0x45335635 /* ('E'|'3'|'V'|'5').  */
  bfd_arch_arc,       /* ARC Cores.  */
#define bfd_mach_arc_a4        0
#define bfd_mach_arc_a5        1
#define bfd_mach_arc_arc600    2
#define bfd_mach_arc_arc601    4
#define bfd_mach_arc_arc700    3
#define bfd_mach_arc_arcv2     5
 bfd_arch_m32c,       /* Renesas M16C/M32C.  */
#define bfd_mach_m16c          0x75
#define bfd_mach_m32c          0x78
  bfd_arch_m32r,      /* Renesas M32R (formerly Mitsubishi M32R/D).  */
#define bfd_mach_m32r          1 /* For backwards compatibility.  */
#define bfd_mach_m32rx         'x'
#define bfd_mach_m32r2         '2'
  bfd_arch_mn10200,   /* Matsushita MN10200.  */
  bfd_arch_mn10300,   /* Matsushita MN10300.  */
#define bfd_mach_mn10300       300
#define bfd_mach_am33          330
#define bfd_mach_am33_2        332
  bfd_arch_fr30,
#define bfd_mach_fr30          0x46523330
  bfd_arch_frv,
#define bfd_mach_frv           1
#define bfd_mach_frvsimple     2
```

```
#define bfd_mach_fr300         300
#define bfd_mach_fr400         400
#define bfd_mach_fr450         450
#define bfd_mach_frvtomcat     499     /* fr500 prototype.  */
#define bfd_mach_fr500         500
#define bfd_mach_fr550         550
  bfd_arch_moxie,     /* The moxie processor.  */
#define bfd_mach_moxie         1
  bfd_arch_ft32,      /* The ft32 processor.  */
#define bfd_mach_ft32          1
#define bfd_mach_ft32b         2
  bfd_arch_mcore,
  bfd_arch_mep,
#define bfd_mach_mep           1
#define bfd_mach_mep_h1        0x6831
#define bfd_mach_mep_c5        0x6335
  bfd_arch_metag,
#define bfd_mach_metag         1
  bfd_arch_ia64,      /* HP/Intel ia64.  */
#define bfd_mach_ia64_elf64    64
#define bfd_mach_ia64_elf32    32
  bfd_arch_ip2k,      /* Ubicom IP2K microcontrollers. */
#define bfd_mach_ip2022        1
#define bfd_mach_ip2022ext     2
 bfd_arch_iq2000,     /* Vitesse IQ2000.  */
#define bfd_mach_iq2000        1
#define bfd_mach_iq10          2
  bfd_arch_bpf,       /* Linux eBPF.  */
#define bfd_mach_bpf           1
#define bfd_mach_xbpf          2
  bfd_arch_epiphany,  /* Adapteva EPIPHANY.  */
#define bfd_mach_epiphany16    1
#define bfd_mach_epiphany32    2
  bfd_arch_mt,
#define bfd_mach_ms1           1
#define bfd_mach_mrisc2        2
#define bfd_mach_ms2           3
  bfd_arch_pj,
  bfd_arch_avr,       /* Atmel AVR microcontrollers.  */
#define bfd_mach_avr1          1
#define bfd_mach_avr2          2
#define bfd_mach_avr25         25
#define bfd_mach_avr3          3
#define bfd_mach_avr31         31
#define bfd_mach_avr35         35
#define bfd_mach_avr4          4
#define bfd_mach_avr5          5
```

```
#define bfd_mach_avr51         51
#define bfd_mach_avr6          6
#define bfd_mach_avrtiny       100
#define bfd_mach_avrxmega1     101
#define bfd_mach_avrxmega2     102
#define bfd_mach_avrxmega3     103
#define bfd_mach_avrxmega4     104
#define bfd_mach_avrxmega5     105
#define bfd_mach_avrxmega6     106
#define bfd_mach_avrxmega7     107
  bfd_arch_bfin,      /* ADI Blackfin.  */
#define bfd_mach_bfin          1
  bfd_arch_cr16,      /* National Semiconductor CompactRISC (ie CR16).  */
#define bfd_mach_cr16          1
  bfd_arch_crx,       /*  National Semiconductor CRX.  */
#define bfd_mach_crx           1
  bfd_arch_cris,      /* Axis CRIS.  */
#define bfd_mach_cris_v0_v10   255
#define bfd_mach_cris_v32      32
#define bfd_mach_cris_v10_v32  1032
  bfd_arch_riscv,
#define bfd_mach_riscv32       132
#define bfd_mach_riscv64       164
  bfd_arch_rl78,
#define bfd_mach_rl78          0x75
  bfd_arch_rx,        /* Renesas RX.  */
#define bfd_mach_rx            0x75
#define bfd_mach_rx_v2         0x76
#define bfd_mach_rx_v3         0x77
  bfd_arch_s390,      /* IBM s390.  */
#define bfd_mach_s390_31       31
#define bfd_mach_s390_64       64
  bfd_arch_score,     /* Sunplus score.  */
#define bfd_mach_score3        3
#define bfd_mach_score7        7
  bfd_arch_mmix,      /* Donald Knuth's educational processor.  */
  bfd_arch_xstormy16,
#define bfd_mach_xstormy16     1
  bfd_arch_msp430,    /* Texas Instruments MSP430 architecture.  */
#define bfd_mach_msp11         11
#define bfd_mach_msp110        110
#define bfd_mach_msp12         12
#define bfd_mach_msp13         13
#define bfd_mach_msp14         14
#define bfd_mach_msp15         15
#define bfd_mach_msp16         16
#define bfd_mach_msp20         20
```

```
#define bfd_mach_msp21         21
#define bfd_mach_msp22         22
#define bfd_mach_msp23         23
#define bfd_mach_msp24         24
#define bfd_mach_msp26         26
#define bfd_mach_msp31         31
#define bfd_mach_msp32         32
#define bfd_mach_msp33         33
#define bfd_mach_msp41         41
#define bfd_mach_msp42         42
#define bfd_mach_msp43         43
#define bfd_mach_msp44         44
#define bfd_mach_msp430x       45
#define bfd_mach_msp46         46
#define bfd_mach_msp47         47
#define bfd_mach_msp54         54
  bfd_arch_xc16x,     /* Infineon's XC16X Series.  */
#define bfd_mach_xc16x         1
#define bfd_mach_xc16xl        2
#define bfd_mach_xc16xs        3
  bfd_arch_xgate,     /* Freescale XGATE.  */
#define bfd_mach_xgate         1
  bfd_arch_xtensa,    /* Tensilica's Xtensa cores.  */
#define bfd_mach_xtensa        1
  bfd_arch_z80,
/* Zilog Z80 without undocumented opcodes.  */
#define bfd_mach_z80strict     1
/* Zilog Z180: successor with additional instructions, but without
 halves of ix and iy.  */
#define bfd_mach_z180          2
/* Zilog Z80 with ixl, ixh, iyl, and iyh.  */
#define bfd_mach_z80           3
/* Zilog eZ80 (successor of Z80 & Z180) in Z80 (16-bit address) mode.  */
#define bfd_mach_ez80_z80      4
/* Zilog eZ80 (successor of Z80 & Z180) in ADL (24-bit address) mode.  */
#define bfd_mach_ez80_adl      5
/* Z80N */
#define bfd_mach_z80n          6
/* Zilog Z80 with all undocumented instructions.  */
#define bfd_mach_z80full       7
/* GameBoy Z80 (reduced instruction set).  */
#define bfd_mach_gbz80         8
/* ASCII R800: successor with multiplication.  */
#define bfd_mach_r800          11
  bfd_arch_lm32,      /* Lattice Mico32.  */
#define bfd_mach_lm32          1
  bfd_arch_microblaze,/* Xilinx MicroBlaze.  */
```

```
  bfd_arch_tilepro,   /* Tilera TILEPro.  */
  bfd_arch_tilegx,    /* Tilera TILE-Gx.  */
#define bfd_mach_tilepro       1
#define bfd_mach_tilegx        1
#define bfd_mach_tilegx32      2
  bfd_arch_aarch64,   /* AArch64.  */
#define bfd_mach_aarch64 0
#define bfd_mach_aarch64_8R    1
#define bfd_mach_aarch64_ilp32 32
  bfd_arch_nios2,     /* Nios II.  */
#define bfd_mach_nios2         0
#define bfd_mach_nios2r1       1
#define bfd_mach_nios2r2       2
  bfd_arch_visium,    /* Visium.  */
#define bfd_mach_visium        1
  bfd_arch_wasm32,    /* WebAssembly.  */
#define bfd_mach_wasm32        1
  bfd_arch_pru,       /* PRU.  */
#define bfd_mach_pru           0
  bfd_arch_nfp,       /* Netronome Flow Processor */
#define bfd_mach_nfp3200       0x3200
#define bfd_mach_nfp6000       0x6000
  bfd_arch_csky,      /* C-SKY.  */
#define bfd_mach_ck_unknown    0
#define bfd_mach_ck510         1
#define bfd_mach_ck610         2
#define bfd_mach_ck801         3
#define bfd_mach_ck802         4
#define bfd_mach_ck803         5
#define bfd_mach_ck807         6
#define bfd_mach_ck810         7
#define bfd_mach_ck860         8
  bfd_arch_loongarch,       /* LoongArch */
#define bfd_mach_loongarch32   1
#define bfd_mach_loongarch64   2
  bfd_arch_last
  };
```

2.13.2 bfd_arch_info

Description

This structure contains information on architectures for use within BFD.

```
typedef struct bfd_arch_info
{
  int bits_per_word;
  int bits_per_address;
```

```
  int bits_per_byte;
  enum bfd_architecture arch;
  unsigned long mach;
  const char *arch_name;
  const char *printable_name;
  unsigned int section_align_power;
  /* TRUE if this is the default machine for the architecture.
     The default arch should be the first entry for an arch so that
     all the entries for that arch can be accessed via next.  */
  bool the_default;
  const struct bfd_arch_info * (*compatible) (const struct bfd_arch_info *,
                                              const struct bfd_arch_info *);

  bool (*scan) (const struct bfd_arch_info *, const char *);

  /* Allocate via bfd_malloc and return a fill buffer of size COUNT.  If
     IS_BIGENDIAN is TRUE, the order of bytes is big endian.  If CODE is
     TRUE, the buffer contains code.  */
  void *(*fill) (bfd_size_type count, bool is_bigendian, bool code);

  const struct bfd_arch_info *next;

  /* On some architectures the offset for a relocation can point into
     the middle of an instruction.  This field specifies the maximum
     offset such a relocation can have (in octets).  This affects the
     behaviour of the disassembler, since a value greater than zero
     means that it may need to disassemble an instruction twice, once
     to get its length and then a second time to display it.  If the
     value is negative then this has to be done for every single
     instruction, regardless of the offset of the reloc.  */
  signed int max_reloc_offset_into_insn;
}
bfd_arch_info_type;
```

2.13.2.1 bfd_printable_name

Synopsis

```
const char *bfd_printable_name (bfd *abfd);
```

Description

Return a printable string representing the architecture and machine from the pointer to the architecture info structure.

2.13.2.2 bfd_scan_arch

Synopsis

```
const bfd_arch_info_type *bfd_scan_arch (const char *string);
```

Description
Figure out if BFD supports any cpu which could be described with the name *string*. Return a pointer to an `arch_info` structure if a machine is found, otherwise NULL.

2.13.2.3 bfd_arch_list

Synopsis

```
const char **bfd_arch_list (void);
```

Description
Return a freshly malloced NULL-terminated vector of the names of all the valid BFD architectures. Do not modify the names.

2.13.2.4 bfd_arch_get_compatible

Synopsis

```
const bfd_arch_info_type *bfd_arch_get_compatible
   (const bfd *abfd, const bfd *bbfd, bool accept_unknowns);
```

Description
Determine whether two BFDs' architectures and machine types are compatible. Calculates the lowest common denominator between the two architectures and machine types implied by the BFDs and returns a pointer to an `arch_info` structure describing the compatible machine.

2.13.2.5 bfd_default_arch_struct

Description
The `bfd_default_arch_struct` is an item of `bfd_arch_info_type` which has been initialized to a fairly generic state. A BFD starts life by pointing to this structure, until the correct back end has determined the real architecture of the file.

```
extern const bfd_arch_info_type bfd_default_arch_struct;
```

2.13.2.6 bfd_set_arch_info

Synopsis

```
void bfd_set_arch_info (bfd *abfd, const bfd_arch_info_type *arg);
```

Description
Set the architecture info of *abfd* to *arg*.

2.13.2.7 bfd_default_set_arch_mach

Synopsis

```
bool bfd_default_set_arch_mach
   (bfd *abfd, enum bfd_architecture arch, unsigned long mach);
```

Description
Set the architecture and machine type in BFD *abfd* to *arch* and *mach*. Find the correct pointer to a structure and insert it into the `arch_info` pointer.

2.13.2.8 bfd_get_arch

Synopsis

```
enum bfd_architecture bfd_get_arch (const bfd *abfd);
```

Description

Return the enumerated type which describes the BFD *abfd*'s architecture.

2.13.2.9 bfd_get_mach

Synopsis

```
unsigned long bfd_get_mach (const bfd *abfd);
```

Description

Return the long type which describes the BFD *abfd*'s machine.

2.13.2.10 bfd_arch_bits_per_byte

Synopsis

```
unsigned int bfd_arch_bits_per_byte (const bfd *abfd);
```

Description

Return the number of bits in one of the BFD *abfd*'s architecture's bytes.

2.13.2.11 bfd_arch_bits_per_address

Synopsis

```
unsigned int bfd_arch_bits_per_address (const bfd *abfd);
```

Description

Return the number of bits in one of the BFD *abfd*'s architecture's addresses.

2.13.2.12 bfd_default_compatible

Synopsis

```
const bfd_arch_info_type *bfd_default_compatible
   (const bfd_arch_info_type *a, const bfd_arch_info_type *b);
```

Description

The default function for testing for compatibility.

2.13.2.13 bfd_default_scan

Synopsis

```
bool bfd_default_scan
   (const struct bfd_arch_info *info, const char *string);
```

Description

The default function for working out whether this is an architecture hit and a machine hit.

2.13.2.14 bfd_get_arch_info

Synopsis

```
const bfd_arch_info_type *bfd_get_arch_info (bfd *abfd);
```

Description

Return the architecture info struct in *abfd*.

2.13.2.15 bfd_lookup_arch

Synopsis

```
const bfd_arch_info_type *bfd_lookup_arch
   (enum bfd_architecture arch, unsigned long machine);
```

Description
Look for the architecture info structure which matches the arguments *arch* and *machine*. A machine of 0 matches the machine/architecture structure which marks itself as the default.

2.13.2.16 bfd_printable_arch_mach

Synopsis

```
const char *bfd_printable_arch_mach
   (enum bfd_architecture arch, unsigned long machine);
```

Description
Return a printable string representing the architecture and machine type.

This routine is depreciated.

2.13.2.17 bfd_octets_per_byte

Synopsis

```
unsigned int bfd_octets_per_byte (const bfd *abfd,
    const asection *sec);
```

Description
Return the number of octets (8-bit quantities) per target byte (minimum addressable unit). In most cases, this will be one, but some DSP targets have 16, 32, or even 48 bits per byte.

2.13.2.18 bfd_arch_mach_octets_per_byte

Synopsis

```
unsigned int bfd_arch_mach_octets_per_byte
   (enum bfd_architecture arch, unsigned long machine);
```

Description
See bfd_octets_per_byte.

This routine is provided for those cases where a bfd * is not available

2.13.2.19 bfd_arch_default_fill

Synopsis

```
void *bfd_arch_default_fill (bfd_size_type count,
    bool is_bigendian,
    bool code);
```

Description
Allocate via bfd_malloc and return a fill buffer of size COUNT. If IS_BIGENDIAN is TRUE, the order of bytes is big endian. If CODE is TRUE, the buffer contains code.

```
/* Set to N to open the next N BFDs using an alternate id space.  */
extern unsigned int bfd_use_reserved_id;
```

2.14 Opening and closing BFDs

2.14.1 Functions for opening and closing

2.14.1.1 bfd_fopen

Synopsis

```
bfd *bfd_fopen (const char *filename, const char *target,
    const char *mode, int fd);
```

Description

Open the file *filename* with the target *target*. Return a pointer to the created BFD. If *fd* is not -1, then `fdopen` is used to open the file; otherwise, `fopen` is used. *mode* is passed directly to `fopen` or `fdopen`.

Calls `bfd_find_target`, so *target* is interpreted as by that function.

The new BFD is marked as cacheable iff *fd* is -1.

If `NULL` is returned then an error has occured. Possible errors are `bfd_error_no_memory`, `bfd_error_invalid_target` or `system_call` error.

On error, *fd* is always closed.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the bfd_get_filename() macro.

2.14.1.2 bfd_openr

Synopsis

```
bfd *bfd_openr (const char *filename, const char *target);
```

Description

Open the file *filename* (using `fopen`) with the target *target*. Return a pointer to the created BFD.

Calls `bfd_find_target`, so *target* is interpreted as by that function.

If `NULL` is returned then an error has occured. Possible errors are `bfd_error_no_memory`, `bfd_error_invalid_target` or `system_call` error.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the bfd_get_filename() macro.

2.14.1.3 bfd_fdopenr

Synopsis

```
bfd *bfd_fdopenr (const char *filename, const char *target, int fd);
```

Description

`bfd_fdopenr` is to `bfd_fopenr` much like `fdopen` is to `fopen`. It opens a BFD on a file already described by the *fd* supplied.

When the file is later `bfd_close`d, the file descriptor will be closed. If the caller desires that this file descriptor be cached by BFD (opened as needed, closed as needed to free descriptors for other opens), with the supplied *fd* used as an initial file descriptor (but subject to closure at any time), call bfd_set_cacheable(bfd, 1) on the returned BFD. The default is to assume

no caching; the file descriptor will remain open until `bfd_close`, and will not be affected by BFD operations on other files.

Possible errors are `bfd_error_no_memory`, `bfd_error_invalid_target` and `bfd_error_system_call`.

On error, *fd* is closed.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the bfd_get_filename() macro.

2.14.1.4 bfd_fdopenw

Synopsis

```
bfd *bfd_fdopenw (const char *filename, const char *target, int fd);
```

Description

`bfd_fdopenw` is exactly like `bfd_fdopenr` with the exception that the resulting BFD is suitable for output.

2.14.1.5 bfd_openstreamr

Synopsis

```
bfd *bfd_openstreamr (const char * filename, const char * target,
    void * stream);
```

Description

Open a BFD for read access on an existing stdio stream. When the BFD is passed to `bfd_close`, the stream will be closed.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the bfd_get_filename() macro.

2.14.1.6 bfd_openr_iovec

Synopsis

```
bfd *bfd_openr_iovec (const char *filename, const char *target,
    void *(*open_func) (struct bfd *nbfd,
    void *open_closure),
    void *open_closure,
    file_ptr (*pread_func) (struct bfd *nbfd,
    void *stream,
    void *buf,
    file_ptr nbytes,
    file_ptr offset),
    int (*close_func) (struct bfd *nbfd,
    void *stream),
    int (*stat_func) (struct bfd *abfd,
    void *stream,
    struct stat *sb));
```

Description

Create and return a BFD backed by a read-only *stream*. The *stream* is created using *open_func*, accessed using *pread_func* and destroyed using *close_func*.

Calls `bfd_find_target`, so *target* is interpreted as by that function.

Calls *open_func* (which can call `bfd_zalloc` and `bfd_get_filename`) to obtain the read-only stream backing the BFD. *open_func* either succeeds returning the non-`NULL` *stream*, or fails returning `NULL` (setting `bfd_error`).

Calls *pread_func* to request *nbytes* of data from *stream* starting at *offset* (e.g., via a call to `bfd_read`). *pread_func* either succeeds returning the number of bytes read (which can be less than *nbytes* when end-of-file), or fails returning -1 (setting `bfd_error`).

Calls *close_func* when the BFD is later closed using `bfd_close`. *close_func* either succeeds returning 0, or fails returning -1 (setting `bfd_error`).

Calls *stat_func* to fill in a stat structure for bfd_stat, bfd_get_size, and bfd_get_mtime calls. *stat_func* returns 0 on success, or returns -1 on failure (setting `bfd_error`).

If `bfd_openr_iovec` returns `NULL` then an error has occurred. Possible errors are `bfd_error_no_memory`, `bfd_error_invalid_target` and `bfd_error_system_call`.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the bfd_get_filename() macro.

2.14.1.7 `bfd_openw`

Synopsis

```
bfd *bfd_openw (const char *filename, const char *target);
```

Description
Create a BFD, associated with file *filename*, using the file format *target*, and return a pointer to it.

Possible errors are `bfd_error_system_call`, `bfd_error_no_memory`, `bfd_error_invalid_target`.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the bfd_get_filename() macro.

2.14.1.8 `bfd_close`

Synopsis

```
bool bfd_close (bfd *abfd);
```

Description
Close a BFD. If the BFD was open for writing, then pending operations are completed and the file written out and closed. If the created file is executable, then `chmod` is called to mark it as such.

All memory attached to the BFD is released.

The file descriptor associated with the BFD is closed (even if it was passed in to BFD by `bfd_fdopenr`).

Returns
`TRUE` is returned if all is ok, otherwise `FALSE`.

2.14.1.9 bfd_close_all_done

Synopsis

```
bool bfd_close_all_done (bfd *);
```

Description
Close a BFD. Differs from `bfd_close` since it does not complete any pending operations. This routine would be used if the application had just used BFD for swapping and didn't want to use any of the writing code.

If the created file is executable, then `chmod` is called to mark it as such.

All memory attached to the BFD is released.

Returns
`TRUE` is returned if all is ok, otherwise `FALSE`.

2.14.1.10 bfd_create

Synopsis

```
bfd *bfd_create (const char *filename, bfd *templ);
```

Description
Create a new BFD in the manner of `bfd_openw`, but without opening a file. The new BFD takes the target from the target used by *templ*. The format is always set to `bfd_object`.

A copy of the *filename* argument is stored in the newly created BFD. It can be accessed via the bfd_get_filename() macro.

2.14.1.11 bfd_make_writable

Synopsis

```
bool bfd_make_writable (bfd *abfd);
```

Description
Takes a BFD as created by `bfd_create` and converts it into one like as returned by `bfd_openw`. It does this by converting the BFD to BFD_IN_MEMORY. It's assumed that you will call `bfd_make_readable` on this bfd later.

Returns
`TRUE` is returned if all is ok, otherwise `FALSE`.

2.14.1.12 bfd_make_readable

Synopsis

```
bool bfd_make_readable (bfd *abfd);
```

Description
Takes a BFD as created by `bfd_create` and `bfd_make_writable` and converts it into one like as returned by `bfd_openr`. It does this by writing the contents out to the memory buffer, then reversing the direction.

Returns
`TRUE` is returned if all is ok, otherwise `FALSE`.

2.14.1.13 bfd_alloc

Synopsis

```
void *bfd_alloc (bfd *abfd, bfd_size_type wanted);
```

Description

Allocate a block of *wanted* bytes of memory attached to `abfd` and return a pointer to it.

2.14.1.14 bfd_zalloc

Synopsis

```
void *bfd_zalloc (bfd *abfd, bfd_size_type wanted);
```

Description

Allocate a block of *wanted* bytes of zeroed memory attached to `abfd` and return a pointer to it.

2.14.1.15 bfd_calc_gnu_debuglink_crc32

Synopsis

```
unsigned long bfd_calc_gnu_debuglink_crc32
   (unsigned long crc, const unsigned char *buf, bfd_size_type len);
```

Description

Computes a CRC value as used in the .gnu_debuglink section. Advances the previously computed *crc* value by computing and adding in the crc32 for *len* bytes of *buf*.

Returns

Return the updated CRC32 value.

2.14.1.16 bfd_get_debug_link_info_1

Synopsis

```
char *bfd_get_debug_link_info_1 (bfd *abfd, void *crc32_out);
```

Description

Extracts the filename and CRC32 value for any separate debug information file associated with *abfd*.

The *crc32_out* parameter is an untyped pointer because this routine is used as a `get_func_type` function, but it is expected to be an unsigned long pointer.

Returns

The filename of the associated debug information file, or NULL if there is no such file. If the filename was found then the contents of *crc32_out* are updated to hold the corresponding CRC32 value for the file.

The returned filename is allocated with `malloc`; freeing it is the responsibility of the caller.

2.14.1.17 bfd_get_debug_link_info

Synopsis

```
char *bfd_get_debug_link_info (bfd *abfd, unsigned long *crc32_out);
```

Description

Extracts the filename and CRC32 value for any separate debug information file associated with *abfd*.

Returns
The filename of the associated debug information file, or NULL if there is no such file. If the filename was found then the contents of *crc32_out* are updated to hold the corresponding CRC32 value for the file.

The returned filename is allocated with `malloc`; freeing it is the responsibility of the caller.

2.14.1.18 bfd_get_alt_debug_link_info

Synopsis

```
char *bfd_get_alt_debug_link_info (bfd * abfd,
    bfd_size_type *buildid_len,
    bfd_byte **buildid_out);
```

Description
Fetch the filename and BuildID value for any alternate debuginfo associated with *abfd*. Return NULL if no such info found, otherwise return filename and update *buildid_len* and *buildid_out*. The returned filename and build_id are allocated with `malloc`; freeing them is the responsibility of the caller.

2.14.1.19 separate_debug_file_exists

Synopsis

```
bool separate_debug_file_exists
   (char *name, void *crc32_p);
```

Description
Checks to see if *name* is a file and if its contents match *crc32*, which is a pointer to an `unsigned long` containing a CRC32.

The *crc32_p* parameter is an untyped pointer because this routine is used as a `check_func_type` function.

2.14.1.20 separate_alt_debug_file_exists

Synopsis

```
bool separate_alt_debug_file_exists
   (char *name, void *unused);
```

Description
Checks to see if *name* is a file.

2.14.1.21 find_separate_debug_file

Synopsis

```
char *find_separate_debug_file
   (bfd *abfd, const char *dir, bool include_dirs,
    get_func_type get, check_func_type check, void *data);
```

Description
Searches for a debug information file corresponding to *abfd*.

The name of the separate debug info file is returned by the *get* function. This function scans various fixed locations in the filesystem, including the file tree rooted at *dir*. If the

include_dirs parameter is true then the directory components of *abfd*'s filename will be included in the searched locations.

data is passed unmodified to the *get* and *check* functions. It is generally used to implement build-id-like matching in the callback functions.

Returns
Returns the filename of the first file to be found which receives a TRUE result from the *check* function. Returns NULL if no valid file could be found.

2.14.1.22 bfd_follow_gnu_debuglink

Synopsis

```
char *bfd_follow_gnu_debuglink (bfd *abfd, const char *dir);
```

Description
Takes a BFD and searches it for a .gnu_debuglink section. If this section is found, it examines the section for the name and checksum of a '.debug' file containing auxiliary debugging information. It then searches the filesystem for this .debug file in some standard locations, including the directory tree rooted at *dir*, and if found returns the full filename.

If *dir* is NULL, the search will take place starting at the current directory.

Returns
`NULL` on any errors or failure to locate the .debug file, otherwise a pointer to a heap-allocated string containing the filename. The caller is responsible for freeing this string.

2.14.1.23 bfd_follow_gnu_debugaltlink

Synopsis

```
char *bfd_follow_gnu_debugaltlink (bfd *abfd, const char *dir);
```

Description
Takes a BFD and searches it for a .gnu_debugaltlink section. If this section is found, it examines the section for the name of a file containing auxiliary debugging information. It then searches the filesystem for this file in a set of standard locations, including the directory tree rooted at *dir*, and if found returns the full filename.

If *dir* is NULL, the search will take place starting at the current directory.

Returns
`NULL` on any errors or failure to locate the debug file, otherwise a pointer to a heap-allocated string containing the filename. The caller is responsible for freeing this string.

2.14.1.24 bfd_create_gnu_debuglink_section

Synopsis

```
struct bfd_section *bfd_create_gnu_debuglink_section
   (bfd *abfd, const char *filename);
```

Description
Takes a *BFD* and adds a .gnu_debuglink section to it. The section is sized to be big enough to contain a link to the specified *filename*.

Returns
A pointer to the new section is returned if all is ok. Otherwise `NULL` is returned and bfd_error is set.

2.14.1.25 bfd_fill_in_gnu_debuglink_section

Synopsis

```
bool bfd_fill_in_gnu_debuglink_section
   (bfd *abfd, struct bfd_section *sect, const char *filename);
```

Description

Takes a *BFD* and containing a .gnu_debuglink section *SECT* and fills in the contents of the section to contain a link to the specified *filename*. The filename should be relative to the current directory.

Returns

`TRUE` is returned if all is ok. Otherwise `FALSE` is returned and bfd_error is set.

2.14.1.26 get_build_id

Synopsis

```
struct bfd_build_id * get_build_id (bfd *abfd);
```

Description

Finds the build-id associated with *abfd*. If the build-id is extracted from the note section then a build-id structure is built for it, using memory allocated to *abfd*, and this is then attached to the *abfd*.

Returns

Returns a pointer to the build-id structure if a build-id could be found. If no build-id is found NULL is returned and error code is set.

2.14.1.27 get_build_id_name

Synopsis

```
char * get_build_id_name (bfd *abfd, void *build_id_out_p)
```

Description

Searches *abfd* for a build-id, and then constructs a pathname from it. The path is computed as .build-id/NN/NN+NN.debug where NNNN+NN is the build-id value as a hexadecimal string.

Returns

Returns the constructed filename or NULL upon error. It is the caller's responsibility to free the memory used to hold the filename. If a filename is returned then the *build_id_out_p* parameter (which points to a `struct bfd_build_id` pointer) is set to a pointer to the build_id structure.

2.14.1.28 check_build_id_file

Synopsis

```
bool check_build_id_file (char *name, void *buildid_p);
```

Description

Checks to see if *name* is a readable file and if its build-id matches *buildid*.

Returns

Returns TRUE if the file exists, is readable, and contains a build-id which matches the build-id pointed at by *build_id_p* (which is really a `struct bfd_build_id **`).

2.14.1.29 bfd_follow_build_id_debuglink

Synopsis

```
char *bfd_follow_build_id_debuglink (bfd *abfd, const char *dir);
```

Description

Takes *abfd* and searches it for a .note.gnu.build-id section. If this section is found, it extracts the value of the NT_GNU_BUILD_ID note, which should be a hexadecimal value *NNNN+NN* (for 32+ hex digits). It then searches the filesystem for a file named *.build-id/NN/NN+NN.debug* in a set of standard locations, including the directory tree rooted at *dir*. The filename of the first matching file to be found is returned. A matching file should contain a .note.gnu.build-id section with the same *NNNN+NN* note as *abfd*, although this check is currently not implemented.

If *dir* is NULL, the search will take place starting at the current directory.

Returns

`NULL` on any errors or failure to locate the debug file, otherwise a pointer to a heap-allocated string containing the filename. The caller is responsible for freeing this string.

2.14.1.30 bfd_set_filename

Synopsis

```
const char *bfd_set_filename (bfd *abfd, const char *filename);
```

Description

Set the filename of *abfd*, copying the FILENAME parameter to bfd_alloc'd memory owned by *abfd*. Returns a pointer the newly allocated name, or NULL if the allocation failed.

2.15 Implementation details

2.15.1 Internal functions

Description

These routines are used within BFD. They are not intended for export, but are documented here for completeness.

2.15.1.1 bfd_malloc

Synopsis

```
extern void * bfd_malloc (bfd_size_type SIZE) ATTRIBUTE_HIDDEN;
```

Description

Returns a pointer to an allocated block of memory that is at least SIZE bytes long. If SIZE is 0 then it will be treated as if it were 1. If SIZE is too big then NULL will be returned. Returns NULL upon error and sets bfd_error.

2.15.1.2 bfd_realloc

Synopsis

```
extern void * bfd_realloc (void * MEM, bfd_size_type SIZE) ATTRIBUTE_HIDDEN;
```

Description

Returns a pointer to an allocated block of memory that is at least SIZE bytes long. If SIZE

is 0 then it will be treated as if it were 1. If SIZE is too big then NULL will be returned. If MEM is not NULL then it must point to an allocated block of memory. If this block is large enough then MEM may be used as the return value for this function, but this is not guaranteed.

If MEM is not returned then the first N bytes in the returned block will be identical to the first N bytes in region pointed to by MEM, where N is the lessor of SIZE and the length of the region of memory currently addressed by MEM.

Returns NULL upon error and sets bfd_error.

2.15.1.3 bfd_realloc_or_free

Synopsis

```
extern void * bfd_realloc_or_free (void * MEM, bfd_size_type SIZE) ATTRIBUTE_HIDDEN;
```

Description

Returns a pointer to an allocated block of memory that is at least SIZE bytes long. If SIZE is 0 then no memory will be allocated, MEM will be freed, and NULL will be returned. This will not cause bfd_error to be set.

If SIZE is too big then NULL will be returned and bfd_error will be set. If MEM is not NULL then it must point to an allocated block of memory. If this block is large enough then MEM may be used as the return value for this function, but this is not guaranteed.

If MEM is not returned then the first N bytes in the returned block will be identical to the first N bytes in region pointed to by MEM, where N is the lessor of SIZE and the length of the region of memory currently addressed by MEM.

2.15.1.4 bfd_zmalloc

Synopsis

```
extern void * bfd_zmalloc (bfd_size_type SIZE) ATTRIBUTE_HIDDEN;
```

Description

Returns a pointer to an allocated block of memory that is at least SIZE bytes long. If SIZE is 0 then it will be treated as if it were 1. If SIZE is too big then NULL will be returned. Returns NULL upon error and sets bfd_error.

If NULL is not returned then the allocated block of memory will have been cleared.

2.15.1.5 bfd_write_bigendian_4byte_int

Synopsis

```
bool bfd_write_bigendian_4byte_int (bfd *, unsigned int);
```

Description

Write a 4 byte integer *i* to the output BFD *abfd*, in big endian order regardless of what else is going on. This is useful in archives.

2.15.1.6 bfd_put_size

2.15.1.7 bfd_get_size

Description

These macros as used for reading and writing raw data in sections; each access (except for

bytes) is vectored through the target format of the BFD and mangled accordingly. The mangling performs any necessary endian translations and removes alignment restrictions. Note that types accepted and returned by these macros are identical so they can be swapped around in macros—for example, `libaout.h` defines `GET_WORD` to either `bfd_get_32` or `bfd_get_64`.

In the put routines, *val* must be a `bfd_vma`. If we are on a system without prototypes, the caller is responsible for making sure that is true, with a cast if necessary. We don't cast them in the macro definitions because that would prevent `lint` or `gcc -Wall` from detecting sins such as passing a pointer. To detect calling these with less than a `bfd_vma`, use `gcc -Wconversion` on a host with 64 bit `bfd_vma`'s.

```
/* Byte swapping macros for user section data.  */

#define bfd_put_8(abfd, val, ptr) \
  ((void) (*((unsigned char *) (ptr)) = (val) & 0xff))
#define bfd_put_signed_8 \
  bfd_put_8
#define bfd_get_8(abfd, ptr) \
  ((bfd_vma) *(const unsigned char *) (ptr) & 0xff)
#define bfd_get_signed_8(abfd, ptr) \
  ((((bfd_signed_vma) *(const unsigned char *) (ptr) & 0xff) ^ 0x80) - 0x80)

#define bfd_put_16(abfd, val, ptr) \
  BFD_SEND (abfd, bfd_putx16, ((val),(ptr)))
#define bfd_put_signed_16 \
  bfd_put_16
#define bfd_get_16(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx16, (ptr))
#define bfd_get_signed_16(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx_signed_16, (ptr))

#define bfd_put_24(abfd, val, ptr) \
  do                                  \
    if (bfd_big_endian (abfd))        \
      bfd_putb24 ((val), (ptr));      \
    else                              \
      bfd_putl24 ((val), (ptr));      \
  while (0)

bfd_vma bfd_getb24 (const void *p);
bfd_vma bfd_getl24 (const void *p);

#define bfd_get_24(abfd, ptr) \
  (bfd_big_endian (abfd) ? bfd_getb24 (ptr) : bfd_getl24 (ptr))

#define bfd_put_32(abfd, val, ptr) \
```

```
  BFD_SEND (abfd, bfd_putx32, ((val),(ptr)))
#define bfd_put_signed_32 \
  bfd_put_32
#define bfd_get_32(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx32, (ptr))
#define bfd_get_signed_32(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx_signed_32, (ptr))

#define bfd_put_64(abfd, val, ptr) \
  BFD_SEND (abfd, bfd_putx64, ((val), (ptr)))
#define bfd_put_signed_64 \
  bfd_put_64
#define bfd_get_64(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx64, (ptr))
#define bfd_get_signed_64(abfd, ptr) \
  BFD_SEND (abfd, bfd_getx_signed_64, (ptr))

#define bfd_get(bits, abfd, ptr)                       \
  ((bits) == 8 ? bfd_get_8 (abfd, ptr)                 \
   : (bits) == 16 ? bfd_get_16 (abfd, ptr)             \
   : (bits) == 32 ? bfd_get_32 (abfd, ptr)             \
   : (bits) == 64 ? bfd_get_64 (abfd, ptr)             \
   : (abort (), (bfd_vma) - 1))

#define bfd_put(bits, abfd, val, ptr)                  \
  ((bits) == 8 ? bfd_put_8  (abfd, val, ptr)           \
   : (bits) == 16 ? bfd_put_16 (abfd, val, ptr)        \
   : (bits) == 32 ? bfd_put_32 (abfd, val, ptr)        \
   : (bits) == 64 ? bfd_put_64 (abfd, val, ptr)        \
   : (abort (), (void) 0))
```

2.15.1.8 bfd_h_put_size

Description

These macros have the same function as their `bfd_get_x` brethren, except that they are used for removing information for the header records of object files. Believe it or not, some object files keep their header records in big endian order and their data in little endian order.

```
/* Byte swapping macros for file header data.  */

#define bfd_h_put_8(abfd, val, ptr) \
  bfd_put_8 (abfd, val, ptr)
#define bfd_h_put_signed_8(abfd, val, ptr) \
  bfd_put_8 (abfd, val, ptr)
#define bfd_h_get_8(abfd, ptr) \
```

```
  bfd_get_8 (abfd, ptr)
#define bfd_h_get_signed_8(abfd, ptr) \
  bfd_get_signed_8 (abfd, ptr)

#define bfd_h_put_16(abfd, val, ptr) \
  BFD_SEND (abfd, bfd_h_putx16, (val, ptr))
#define bfd_h_put_signed_16 \
  bfd_h_put_16
#define bfd_h_get_16(abfd, ptr) \
  BFD_SEND (abfd, bfd_h_getx16, (ptr))
#define bfd_h_get_signed_16(abfd, ptr) \
  BFD_SEND (abfd, bfd_h_getx_signed_16, (ptr))

#define bfd_h_put_32(abfd, val, ptr) \
  BFD_SEND (abfd, bfd_h_putx32, (val, ptr))
#define bfd_h_put_signed_32 \
  bfd_h_put_32
#define bfd_h_get_32(abfd, ptr) \
  BFD_SEND (abfd, bfd_h_getx32, (ptr))
#define bfd_h_get_signed_32(abfd, ptr) \
  BFD_SEND (abfd, bfd_h_getx_signed_32, (ptr))

#define bfd_h_put_64(abfd, val, ptr) \
  BFD_SEND (abfd, bfd_h_putx64, (val, ptr))
#define bfd_h_put_signed_64 \
  bfd_h_put_64
#define bfd_h_get_64(abfd, ptr) \
  BFD_SEND (abfd, bfd_h_getx64, (ptr))
#define bfd_h_get_signed_64(abfd, ptr) \
  BFD_SEND (abfd, bfd_h_getx_signed_64, (ptr))

/* Aliases for the above, which should eventually go away.  */

#define H_PUT_64  bfd_h_put_64
#define H_PUT_32  bfd_h_put_32
#define H_PUT_16  bfd_h_put_16
#define H_PUT_8   bfd_h_put_8
#define H_PUT_S64 bfd_h_put_signed_64
#define H_PUT_S32 bfd_h_put_signed_32
#define H_PUT_S16 bfd_h_put_signed_16
#define H_PUT_S8  bfd_h_put_signed_8
#define H_GET_64  bfd_h_get_64
#define H_GET_32  bfd_h_get_32
#define H_GET_16  bfd_h_get_16
#define H_GET_8   bfd_h_get_8
#define H_GET_S64 bfd_h_get_signed_64
#define H_GET_S32 bfd_h_get_signed_32
```

```
#define H_GET_S16 bfd_h_get_signed_16
#define H_GET_S8  bfd_h_get_signed_8
```

2.15.1.9 bfd_log2

Synopsis

```
unsigned int bfd_log2 (bfd_vma x);
```

Description

Return the log base 2 of the value supplied, rounded up. E.g., an *x* of 1025 returns 11. A *x* of 0 returns 0.

2.16 File caching

The file caching mechanism is embedded within BFD and allows the application to open as many BFDs as it wants without regard to the underlying operating system's file descriptor limit (often as low as 20 open files). The module in `cache.c` maintains a least recently used list of `bfd_cache_max_open` files, and exports the name `bfd_cache_lookup`, which runs around and makes sure that the required BFD is open. If not, then it chooses a file to close, closes it and opens the one wanted, returning its file handle.

2.16.1 Caching functions

2.16.1.1 bfd_cache_init

Synopsis

```
bool bfd_cache_init (bfd *abfd);
```

Description

Add a newly opened BFD to the cache.

2.16.1.2 bfd_cache_close

Synopsis

```
bool bfd_cache_close (bfd *abfd);
```

Description

Remove the BFD *abfd* from the cache. If the attached file is open, then close it too.

Returns

`FALSE` is returned if closing the file fails, `TRUE` is returned if all is well.

2.16.1.3 bfd_cache_close_all

Synopsis

```
bool bfd_cache_close_all (void);
```

Description

Remove all BFDs from the cache. If the attached file is open, then close it too.

Returns

`FALSE` is returned if closing one of the file fails, `TRUE` is returned if all is well.

2.16.1.4 bfd_open_file

Synopsis

```
FILE* bfd_open_file (bfd *abfd);
```

Description

Call the OS to open a file for *abfd*. Return the `FILE *` (possibly `NULL`) that results from this operation. Set up the BFD so that future accesses know the file is open. If the `FILE *` returned is `NULL`, then it won't have been put in the cache, so it won't have to be removed from it.

2.17 Linker Functions

The linker uses three special entry points in the BFD target vector. It is not necessary to write special routines for these entry points when creating a new BFD back end, since generic versions are provided. However, writing them can speed up linking and make it use significantly less runtime memory.

The first routine creates a hash table used by the other routines. The second routine adds the symbols from an object file to the hash table. The third routine takes all the object files and links them together to create the output file. These routines are designed so that the linker proper does not need to know anything about the symbols in the object files that it is linking. The linker merely arranges the sections as directed by the linker script and lets BFD handle the details of symbols and relocs.

The second routine and third routines are passed a pointer to a `struct bfd_link_info` structure (defined in `bfdlink.h`) which holds information relevant to the link, including the linker hash table (which was created by the first routine) and a set of callback functions to the linker proper.

The generic linker routines are in `linker.c`, and use the header file `genlink.h`. As of this writing, the only back ends which have implemented versions of these routines are a.out (in `aoutx.h`) and ECOFF (in `ecoff.c`). The a.out routines are used as examples throughout this section.

2.17.1 Creating a linker hash table

The linker routines must create a hash table, which must be derived from `struct bfd_link_hash_table` described in `bfdlink.c`. See Section 2.18 [Hash Tables], page 201, for information on how to create a derived hash table. This entry point is called using the target vector of the linker output file.

The `_bfd_link_hash_table_create` entry point must allocate and initialize an instance of the desired hash table. If the back end does not require any additional information to be stored with the entries in the hash table, the entry point may simply create a `struct bfd_link_hash_table`. Most likely, however, some additional information will be needed.

For example, with each entry in the hash table the a.out linker keeps the index the symbol has in the final output file (this index number is used so that when doing a relocatable link the symbol index used in the output file can be quickly filled in when copying over a reloc). The a.out linker code defines the required structures and functions for a hash table derived from `struct bfd_link_hash_table`. The a.out linker hash table is created

by the function `NAME(aout,link_hash_table_create)`; it simply allocates space for the hash table, initializes it, and returns a pointer to it.

When writing the linker routines for a new back end, you will generally not know exactly which fields will be required until you have finished. You should simply create a new hash table which defines no additional fields, and then simply add fields as they become necessary.

2.17.2 Adding symbols to the hash table

The linker proper will call the `_bfd_link_add_symbols` entry point for each object file or archive which is to be linked (typically these are the files named on the command line, but some may also come from the linker script). The entry point is responsible for examining the file. For an object file, BFD must add any relevant symbol information to the hash table. For an archive, BFD must determine which elements of the archive should be used and adding them to the link.

The a.out version of this entry point is `NAME(aout,link_add_symbols)`.

2.17.2.1 Differing file formats

Normally all the files involved in a link will be of the same format, but it is also possible to link together different format object files, and the back end must support that. The `_bfd_link_add_symbols` entry point is called via the target vector of the file to be added. This has an important consequence: the function may not assume that the hash table is the type created by the corresponding `_bfd_link_hash_table_create` vector. All the `_bfd_link_add_symbols` function can assume about the hash table is that it is derived from `struct bfd_link_hash_table`.

Sometimes the `_bfd_link_add_symbols` function must store some information in the hash table entry to be used by the `_bfd_final_link` function. In such a case the output bfd xvec must be checked to make sure that the hash table was created by an object file of the same format.

The `_bfd_final_link` routine must be prepared to handle a hash entry without any extra information added by the `_bfd_link_add_symbols` function. A hash entry without extra information will also occur when the linker script directs the linker to create a symbol. Note that, regardless of how a hash table entry is added, all the fields will be initialized to some sort of null value by the hash table entry initialization function.

See `ecoff_link_add_externals` for an example of how to check the output bfd before saving information (in this case, the ECOFF external symbol debugging information) in a hash table entry.

2.17.2.2 Adding symbols from an object file

When the `_bfd_link_add_symbols` routine is passed an object file, it must add all externally visible symbols in that object file to the hash table. The actual work of adding the symbol to the hash table is normally handled by the function `_bfd_generic_link_add_one_symbol`. The `_bfd_link_add_symbols` routine is responsible for reading all the symbols from the object file and passing the correct information to `_bfd_generic_link_add_one_symbol`.

The `_bfd_link_add_symbols` routine should not use `bfd_canonicalize_symtab` to read the symbols. The point of providing this routine is to avoid the overhead of converting the symbols into generic `asymbol` structures.

_bfd_generic_link_add_one_symbol handles the details of combining common symbols, warning about multiple definitions, and so forth. It takes arguments which describe the symbol to add, notably symbol flags, a section, and an offset. The symbol flags include such things as BSF_WEAK or BSF_INDIRECT. The section is a section in the object file, or something like bfd_und_section_ptr for an undefined symbol or bfd_com_section_ptr for a common symbol.

If the _bfd_final_link routine is also going to need to read the symbol information, the _bfd_link_add_symbols routine should save it somewhere attached to the object file BFD. However, the information should only be saved if the keep_memory field of the info argument is TRUE, so that the -no-keep-memory linker switch is effective.

The a.out function which adds symbols from an object file is aout_link_add_object_symbols, and most of the interesting work is in aout_link_add_symbols. The latter saves pointers to the hash tables entries created by _bfd_generic_link_add_one_symbol indexed by symbol number, so that the _bfd_final_link routine does not have to call the hash table lookup routine to locate the entry.

2.17.2.3 Adding symbols from an archive

When the _bfd_link_add_symbols routine is passed an archive, it must look through the symbols defined by the archive and decide which elements of the archive should be included in the link. For each such element it must call the add_archive_element linker callback, and it must add the symbols from the object file to the linker hash table. (The callback may in fact indicate that a replacement BFD should be used, in which case the symbols from that BFD should be added to the linker hash table instead.)

In most cases the work of looking through the symbols in the archive should be done by the _bfd_generic_link_add_archive_symbols function. _bfd_generic_link_add_archive_symbols is passed a function to call to make the final decision about adding an archive element to the link and to do the actual work of adding the symbols to the linker hash table. If the element is to be included, the add_archive_element linker callback routine must be called with the element as an argument, and the element's symbols must be added to the linker hash table just as though the element had itself been passed to the _bfd_link_add_symbols function.

When the a.out _bfd_link_add_symbols function receives an archive, it calls _bfd_generic_link_add_archive_symbols passing aout_link_check_archive_element as the function argument. aout_link_check_archive_element calls aout_link_check_ar_symbols. If the latter decides to add the element (an element is only added if it provides a real, non-common, definition for a previously undefined or common symbol) it calls the add_archive_element callback and then aout_link_check_archive_element calls aout_link_add_symbols to actually add the symbols to the linker hash table - possibly those of a substitute BFD, if the add_archive_element callback avails itself of that option.

The ECOFF back end is unusual in that it does not normally call _bfd_generic_link_add_archive_symbols, because ECOFF archives already contain a hash table of symbols. The ECOFF back end searches the archive itself to avoid the overhead of creating a new hash table.

2.17.3 Performing the final link

When all the input files have been processed, the linker calls the `_bfd_final_link` entry point of the output BFD. This routine is responsible for producing the final output file, which has several aspects. It must relocate the contents of the input sections and copy the data into the output sections. It must build an output symbol table including any local symbols from the input files and the global symbols from the hash table. When producing relocatable output, it must modify the input relocs and write them into the output file. There may also be object format dependent work to be done.

The linker will also call the `write_object_contents` entry point when the BFD is closed. The two entry points must work together in order to produce the correct output file.

The details of how this works are inevitably dependent upon the specific object file format. The a.out `_bfd_final_link` routine is `NAME(aout,final_link)`.

2.17.3.1 Information provided by the linker

Before the linker calls the `_bfd_final_link` entry point, it sets up some data structures for the function to use.

The `input_bfds` field of the `bfd_link_info` structure will point to a list of all the input files included in the link. These files are linked through the `link.next` field of the `bfd` structure.

Each section in the output file will have a list of `link_order` structures attached to the `map_head.link_order` field (the `link_order` structure is defined in `bfdlink.h`). These structures describe how to create the contents of the output section in terms of the contents of various input sections, fill constants, and, eventually, other types of information. They also describe relocs that must be created by the BFD backend, but do not correspond to any input file; this is used to support -Ur, which builds constructors while generating a relocatable object file.

2.17.3.2 Relocating the section contents

The `_bfd_final_link` function should look through the `link_order` structures attached to each section of the output file. Each `link_order` structure should either be handled specially, or it should be passed to the function `_bfd_default_link_order` which will do the right thing (`_bfd_default_link_order` is defined in `linker.c`).

For efficiency, a `link_order` of type `bfd_indirect_link_order` whose associated section belongs to a BFD of the same format as the output BFD must be handled specially. This type of `link_order` describes part of an output section in terms of a section belonging to one of the input files. The `_bfd_final_link` function should read the contents of the section and any associated relocs, apply the relocs to the section contents, and write out the modified section contents. If performing a relocatable link, the relocs themselves must also be modified and written out.

The functions `_bfd_relocate_contents` and `_bfd_final_link_relocate` provide some general support for performing the actual relocations, notably overflow checking. Their arguments include information about the symbol the relocation is against and a `reloc_howto_type` argument which describes the relocation to perform. These functions are defined in `reloc.c`.

The a.out function which handles reading, relocating, and writing section contents is `aout_link_input_section`. The actual relocation is done in `aout_link_input_section_std` and `aout_link_input_section_ext`.

2.17.3.3 Writing the symbol table

The `_bfd_final_link` function must gather all the symbols in the input files and write them out. It must also write out all the symbols in the global hash table. This must be controlled by the `strip` and `discard` fields of the `bfd_link_info` structure.

The local symbols of the input files will not have been entered into the linker hash table. The `_bfd_final_link` routine must consider each input file and include the symbols in the output file. It may be convenient to do this when looking through the `link_order` structures, or it may be done by stepping through the `input_bfds` list.

The `_bfd_final_link` routine must also traverse the global hash table to gather all the externally visible symbols. It is possible that most of the externally visible symbols may be written out when considering the symbols of each input file, but it is still necessary to traverse the hash table since the linker script may have defined some symbols that are not in any of the input files.

The `strip` field of the `bfd_link_info` structure controls which symbols are written out. The possible values are listed in `bfdlink.h`. If the value is `strip_some`, then the `keep_hash` field of the `bfd_link_info` structure is a hash table of symbols to keep; each symbol should be looked up in this hash table, and only symbols which are present should be included in the output file.

If the `strip` field of the `bfd_link_info` structure permits local symbols to be written out, the `discard` field is used to further controls which local symbols are included in the output file. If the value is `discard_l`, then all local symbols which begin with a certain prefix are discarded; this is controlled by the `bfd_is_local_label_name` entry point.

The a.out backend handles symbols by calling `aout_link_write_symbols` on each input BFD and then traversing the global hash table with the function `aout_link_write_other_symbol`. It builds a string table while writing out the symbols, which is written to the output file at the end of `NAME(aout,final_link)`.

2.17.3.4 `bfd_link_split_section`

Synopsis

```
bool bfd_link_split_section (bfd *abfd, asection *sec);
```

Description

Return nonzero if *sec* should be split during a reloceatable or final link.

```
#define bfd_link_split_section(abfd, sec) \
       BFD_SEND (abfd, _bfd_link_split_section, (abfd, sec))
```

2.17.3.5 `bfd_section_already_linked`

Synopsis

```
bool bfd_section_already_linked (bfd *abfd,
    asection *sec,
```

```
    struct bfd_link_info *info);
```

Description
Check if *data* has been already linked during a reloceatable or final link. Return TRUE if it has.

```
#define bfd_section_already_linked(abfd, sec, info) \
       BFD_SEND (abfd, _section_already_linked, (abfd, sec, info))
```

2.17.3.6 bfd_generic_define_common_symbol

Synopsis

```
bool bfd_generic_define_common_symbol
   (bfd *output_bfd, struct bfd_link_info *info,
    struct bfd_link_hash_entry *h);
```

Description
Convert common symbol *h* into a defined symbol. Return TRUE on success and FALSE on failure.

```
#define bfd_define_common_symbol(output_bfd, info, h) \
       BFD_SEND (output_bfd, _bfd_define_common_symbol, (output_bfd, info, h))
```

2.17.3.7 _bfd_generic_link_hide_symbol

Synopsis

```
void _bfd_generic_link_hide_symbol
   (bfd *output_bfd, struct bfd_link_info *info,
    struct bfd_link_hash_entry *h);
```

Description
Hide symbol *h*. This is an internal function. It should not be called from outside the BFD library.

```
#define bfd_link_hide_symbol(output_bfd, info, h) \
       BFD_SEND (output_bfd, _bfd_link_hide_symbol, (output_bfd, info, h))
```

2.17.3.8 bfd_generic_define_start_stop

Synopsis

```
struct bfd_link_hash_entry *bfd_generic_define_start_stop
   (struct bfd_link_info *info,
    const char *symbol, asection *sec);
```

Description
Define a __start, __stop, .startof. or .sizeof. symbol. Return the symbol or NULL if no such undefined symbol exists.

```
#define bfd_define_start_stop(output_bfd, info, symbol, sec) \
       BFD_SEND (output_bfd, _bfd_define_start_stop, (info, symbol, sec))
```

2.17.3.9 bfd_find_version_for_sym

Synopsis

```
struct bfd_elf_version_tree * bfd_find_version_for_sym
   (struct bfd_elf_version_tree *verdefs,
    const char *sym_name, bool *hide);
```

Description
Search an elf version script tree for symbol versioning info and export / don't-export status for a given symbol. Return non-NULL on success and NULL on failure; also sets the output '`hide`' boolean parameter.

2.17.3.10 bfd_hide_sym_by_version

Synopsis

```
bool bfd_hide_sym_by_version
   (struct bfd_elf_version_tree *verdefs, const char *sym_name);
```

Description
Search an elf version script tree for symbol versioning info for a given symbol. Return TRUE if the symbol is hidden.

2.17.3.11 bfd_link_check_relocs

Synopsis

```
bool bfd_link_check_relocs
   (bfd *abfd, struct bfd_link_info *info);
```

Description
Checks the relocs in ABFD for validity. Does not execute the relocs. Return TRUE if everything is OK, FALSE otherwise. This is the external entry point to this code.

2.17.3.12 _bfd_generic_link_check_relocs

Synopsis

```
bool _bfd_generic_link_check_relocs
   (bfd *abfd, struct bfd_link_info *info);
```

Description
Stub function for targets that do not implement reloc checking. Return TRUE. This is an internal function. It should not be called from outside the BFD library.

2.17.3.13 bfd_merge_private_bfd_data

Synopsis

```
bool bfd_merge_private_bfd_data
   (bfd *ibfd, struct bfd_link_info *info);
```

Description
Merge private BFD information from the BFD *ibfd* to the the output file BFD when linking. Return `TRUE` on success, `FALSE` on error. Possible error returns are:

- `bfd_error_no_memory` - Not enough memory exists to create private data for *obfd*.

```
#define bfd_merge_private_bfd_data(ibfd, info) \
       BFD_SEND ((info)->output_bfd, _bfd_merge_private_bfd_data, \
                 (ibfd, info))
```

2.17.3.14 _bfd_generic_verify_endian_match

Synopsis

```
bool _bfd_generic_verify_endian_match
   (bfd *ibfd, struct bfd_link_info *info);
```

Description

Can be used from / for bfd_merge_private_bfd_data to check that endianness matches between input and output file. Returns TRUE for a match, otherwise returns FALSE and emits an error.

2.18 Hash Tables

BFD provides a simple set of hash table functions. Routines are provided to initialize a hash table, to free a hash table, to look up a string in a hash table and optionally create an entry for it, and to traverse a hash table. There is currently no routine to delete an string from a hash table.

The basic hash table does not permit any data to be stored with a string. However, a hash table is designed to present a base class from which other types of hash tables may be derived. These derived types may store additional information with the string. Hash tables were implemented in this way, rather than simply providing a data pointer in a hash table entry, because they were designed for use by the linker back ends. The linker may create thousands of hash table entries, and the overhead of allocating private data and storing and following pointers becomes noticeable.

The basic hash table code is in `hash.c`.

2.18.1 Creating and freeing a hash table

To create a hash table, create an instance of a `struct bfd_hash_table` (defined in `bfd.h`) and call `bfd_hash_table_init` (if you know approximately how many entries you will need, the function `bfd_hash_table_init_n`, which takes a *size* argument, may be used). `bfd_hash_table_init` returns `FALSE` if some sort of error occurs.

The function `bfd_hash_table_init` take as an argument a function to use to create new entries. For a basic hash table, use the function `bfd_hash_newfunc`. See Section 2.18.4 [Deriving a New Hash Table Type], page 202, for why you would want to use a different value for this argument.

`bfd_hash_table_init` will create an objalloc which will be used to allocate new entries. You may allocate memory on this objalloc using `bfd_hash_allocate`.

Use `bfd_hash_table_free` to free up all the memory that has been allocated for a hash table. This will not free up the `struct bfd_hash_table` itself, which you must provide.

Use `bfd_hash_set_default_size` to set the default size of hash table to use.

2.18.2 Looking up or entering a string

The function `bfd_hash_lookup` is used both to look up a string in the hash table and to create a new entry.

If the *create* argument is `FALSE`, `bfd_hash_lookup` will look up a string. If the string is found, it will returns a pointer to a `struct bfd_hash_entry`. If the string is not found in the table `bfd_hash_lookup` will return `NULL`. You should not modify any of the fields in the returns `struct bfd_hash_entry`.

If the *create* argument is `TRUE`, the string will be entered into the hash table if it is not already there. Either way a pointer to a `struct bfd_hash_entry` will be returned, either to the existing structure or to a newly created one. In this case, a `NULL` return means that an error occurred.

If the *create* argument is `TRUE`, and a new entry is created, the *copy* argument is used to decide whether to copy the string onto the hash table objalloc or not. If *copy* is passed as `FALSE`, you must be careful not to deallocate or modify the string as long as the hash table exists.

2.18.3 Traversing a hash table

The function `bfd_hash_traverse` may be used to traverse a hash table, calling a function on each element. The traversal is done in a random order.

`bfd_hash_traverse` takes as arguments a function and a generic `void *` pointer. The function is called with a hash table entry (a `struct bfd_hash_entry *`) and the generic pointer passed to `bfd_hash_traverse`. The function must return a `boolean` value, which indicates whether to continue traversing the hash table. If the function returns `FALSE`, `bfd_hash_traverse` will stop the traversal and return immediately.

2.18.4 Deriving a new hash table type

Many uses of hash tables want to store additional information which each entry in the hash table. Some also find it convenient to store additional information with the hash table itself. This may be done using a derived hash table.

Since C is not an object oriented language, creating a derived hash table requires sticking together some boilerplate routines with a few differences specific to the type of hash table you want to create.

An example of a derived hash table is the linker hash table. The structures for this are defined in `bfdlink.h`. The functions are in `linker.c`.

You may also derive a hash table from an already derived hash table. For example, the a.out linker backend code uses a hash table derived from the linker hash table.

2.18.4.1 Define the derived structures

You must define a structure for an entry in the hash table, and a structure for the hash table itself.

The first field in the structure for an entry in the hash table must be of the type used for an entry in the hash table you are deriving from. If you are deriving from a basic hash table this is `struct bfd_hash_entry`, which is defined in `bfd.h`. The first field in the structure for the hash table itself must be of the type of the hash table you are deriving from itself. If you are deriving from a basic hash table, this is `struct bfd_hash_table`.

For example, the linker hash table defines `struct bfd_link_hash_entry` (in `bfdlink.h`). The first field, `root`, is of type `struct bfd_hash_entry`. Similarly, the first field in `struct bfd_link_hash_table`, `table`, is of type `struct bfd_hash_table`.

2.18.4.2 Write the derived creation routine

You must write a routine which will create and initialize an entry in the hash table. This routine is passed as the function argument to `bfd_hash_table_init`.

In order to permit other hash tables to be derived from the hash table you are creating, this routine must be written in a standard way.

The first argument to the creation routine is a pointer to a hash table entry. This may be `NULL`, in which case the routine should allocate the right amount of space. Otherwise the space has already been allocated by a hash table type derived from this one.

After allocating space, the creation routine must call the creation routine of the hash table type it is derived from, passing in a pointer to the space it just allocated. This will initialize any fields used by the base hash table.

Finally the creation routine must initialize any local fields for the new hash table type.

Here is a boilerplate example of a creation routine. *function_name* is the name of the routine. *entry_type* is the type of an entry in the hash table you are creating. *base_newfunc* is the name of the creation routine of the hash table type your hash table is derived from.

```
struct bfd_hash_entry *
function_name (struct bfd_hash_entry *entry,
                     struct bfd_hash_table *table,
                     const char *string)
{
  struct entry_type *ret = (entry_type *) entry;

 /* Allocate the structure if it has not already been allocated by a
    derived class.  */
  if (ret == NULL)
    {
      ret = bfd_hash_allocate (table, sizeof (* ret));
      if (ret == NULL)
        return NULL;
    }

 /* Call the allocation method of the base class.  */
  ret = ((entry_type *)
         base_newfunc ((struct bfd_hash_entry *) ret, table, string));

 /* Initialize the local fields here.  */

  return (struct bfd_hash_entry *) ret;
}
```

Description

The creation routine for the linker hash table, which is in `linker.c`, looks just like this example. *function_name* is `_bfd_link_hash_newfunc`. *entry_type* is `struct bfd_link_hash_entry`. *base_newfunc* is `bfd_hash_newfunc`, the creation routine for a basic hash table.

`_bfd_link_hash_newfunc` also initializes the local fields in a linker hash table entry: `type`, `written` and `next`.

2.18.4.3 Write other derived routines

You will want to write other routines for your new hash table, as well.

You will want an initialization routine which calls the initialization routine of the hash table you are deriving from and initializes any other local fields. For the linker hash table, this is `_bfd_link_hash_table_init` in `linker.c`.

You will want a lookup routine which calls the lookup routine of the hash table you are deriving from and casts the result. The linker hash table uses `bfd_link_hash_lookup` in `linker.c` (this actually takes an additional argument which it uses to decide how to return the looked up value).

You may want a traversal routine. This should just call the traversal routine of the hash table you are deriving from with appropriate casts. The linker hash table uses `bfd_link_hash_traverse` in `linker.c`.

These routines may simply be defined as macros. For example, the a.out backend linker hash table, which is derived from the linker hash table, uses macros for the lookup and traversal routines. These are `aout_link_hash_lookup` and `aout_link_hash_traverse` in aoutx.h.

3 BFD back ends

3.1 What to Put Where

All of BFD lives in one directory.

3.2 a.out backends

Description

BFD supports a number of different flavours of a.out format, though the major differences are only the sizes of the structures on disk, and the shape of the relocation information.

The support is split into a basic support file `aoutx.h` and other files which derive functions from the base. One derivation file is `aoutf1.h` (for a.out flavour 1), and adds to the basic a.out functions support for sun3, sun4, and 386 a.out files, to create a target jump vector for a specific target.

This information is further split out into more specific files for each machine, including `sunos.c` for sun3 and sun4, and `demo64.c` for a demonstration of a 64 bit a.out format.

The base file `aoutx.h` defines general mechanisms for reading and writing records to and from disk and various other methods which BFD requires. It is included by `aout32.c` and `aout64.c` to form the names `aout_32_swap_exec_header_in`, `aout_64_swap_exec_header_in`, etc.

As an example, this is what goes on to make the back end for a sun4, from `aout32.c`:

```
#define ARCH_SIZE 32
#include "aoutx.h"
```

Which exports names:

```
...
aout_32_canonicalize_reloc
aout_32_find_nearest_line
aout_32_get_lineno
aout_32_get_reloc_upper_bound
...
```

from `sunos.c`:

```
#define TARGET_NAME "a.out-sunos-big"
#define VECNAME    sparc_aout_sunos_be_vec
#include "aoutf1.h"
```

requires all the names from `aout32.c`, and produces the jump vector

```
sparc_aout_sunos_be_vec
```

The file `host-aout.c` is a special case. It is for a large set of hosts that use "more or less standard" a.out files, and for which cross-debugging is not interesting. It uses the standard 32-bit a.out support routines, but determines the file offsets and addresses of the text, data, and BSS sections, the machine architecture and machine type, and the entry point address, in a host-dependent manner. Once these values have been determined, generic code is used to handle the object file.

When porting it to run on a new system, you must supply:

```
HOST_PAGE_SIZE
HOST_SEGMENT_SIZE
HOST_MACHINE_ARCH       (optional)
HOST_MACHINE_MACHINE    (optional)
HOST_TEXT_START_ADDR
HOST_STACK_END_ADDR
```

in the file `../include/sys/h-XXX.h` (for your host). These values, plus the structures and macros defined in `a.out.h` on your host system, will produce a BFD target that will access ordinary a.out files on your host. To configure a new machine to use `host-aout.c`, specify:

```
TDEFAULTS = -DDEFAULT_VECTOR=host_aout_big_vec
TDEPFILES= host-aout.o trad-core.o
```

in the `config/XXX.mt` file, and modify `configure.ac` to use the `XXX.mt` file (by setting `"bfd_target=XXX"`) when your configuration is selected.

3.2.1 Relocations

Description

The file `aoutx.h` provides for both the *standard* and *extended* forms of a.out relocation records.

The standard records contain only an address, a symbol index, and a type field. The extended records also have a full integer for an addend.

3.2.2 Internal entry points

Description

`aoutx.h` exports several routines for accessing the contents of an a.out file, which are gathered and exported in turn by various format specific files (eg sunos.c).

3.2.2.1 aout_*size*_swap_exec_header_in

Synopsis

```
void aout_size_swap_exec_header_in,
   (bfd *abfd,
    struct external_exec *bytes,
    struct internal_exec *execp);
```

Description

Swap the information in an executable header *raw_bytes* taken from a raw byte stream memory image into the internal exec header structure *execp*.

3.2.2.2 aout_*size*_swap_exec_header_out

Synopsis

```
void aout_size_swap_exec_header_out
   (bfd *abfd,
    struct internal_exec *execp,
    struct external_exec *raw_bytes);
```

Description
Swap the information in an internal exec header structure *execp* into the buffer *raw_bytes* ready for writing to disk.

3.2.2.3 aout_*size*_some_aout_object_p

Synopsis

```
const bfd_target *aout_size_some_aout_object_p
   (bfd *abfd,
    struct internal_exec *execp,
    const bfd_target *(*callback_to_real_object_p) (bfd *));
```

Description
Some a.out variant thinks that the file open in *abfd* checking is an a.out file. Do some more checking, and set up for access if it really is. Call back to the calling environment's "finish up" function just before returning, to handle any last-minute setup.

3.2.2.4 aout_*size*_mkobject

Synopsis

```
bool aout_size_mkobject, (bfd *abfd);
```

Description
Initialize BFD *abfd* for use with a.out files.

3.2.2.5 aout_*size*_machine_type

Synopsis

```
enum machine_type  aout_size_machine_type
   (enum bfd_architecture arch,
    unsigned long machine,
    bool *unknown);
```

Description
Keep track of machine architecture and machine type for a.out's. Return the `machine_type` for a particular architecture and machine, or `M_UNKNOWN` if that exact architecture and machine can't be represented in a.out format.

If the architecture is understood, machine type 0 (default) is always understood.

3.2.2.6 aout_*size*_set_arch_mach

Synopsis

```
bool aout_size_set_arch_mach,
   (bfd *,
    enum bfd_architecture arch,
    unsigned long machine);
```

Description
Set the architecture and the machine of the BFD *abfd* to the values *arch* and *machine*. Verify that *abfd*'s format can support the architecture required.

3.2.2.7 aout_*size*_new_section_hook

Synopsis

```
bool aout_size_new_section_hook,
   (bfd *abfd,
    asection *newsect);
```

Description
Called by the BFD in response to a `bfd_make_section` request.

3.3 coff backends

BFD supports a number of different flavours of coff format. The major differences between formats are the sizes and alignments of fields in structures on disk, and the occasional extra field.

Coff in all its varieties is implemented with a few common files and a number of implementation specific files. For example, the i386 coff format is implemented in the file `coff-i386.c`. This file `#include`s `coff/i386.h` which defines the external structure of the coff format for the i386, and `coff/internal.h` which defines the internal structure. `coff-i386.c` also defines the relocations used by the i386 coff format See Section 2.10 [Relocations], page 60.

3.3.1 Porting to a new version of coff

The recommended method is to select from the existing implementations the version of coff which is most like the one you want to use. For example, we'll say that i386 coff is the one you select, and that your coff flavour is called foo. Copy `i386coff.c` to `foocoff.c`, copy `../include/coff/i386.h` to `../include/coff/foo.h`, and add the lines to `targets.c` and `Makefile.in` so that your new back end is used. Alter the shapes of the structures in `../include/coff/foo.h` so that they match what you need. You will probably also have to add `#ifdef`s to the code in `coff/internal.h` and `coffcode.h` if your version of coff is too wild.

You can verify that your new BFD backend works quite simply by building `objdump` from the `binutils` directory, and making sure that its version of what's going on and your host system's idea (assuming it has the pretty standard coff dump utility, usually called `att-dump` or just `dump`) are the same. Then clean up your code, and send what you've done to Cygnus. Then your stuff will be in the next release, and you won't have to keep integrating it.

3.3.2 How the coff backend works

3.3.2.1 File layout

The Coff backend is split into generic routines that are applicable to any Coff target and routines that are specific to a particular target. The target-specific routines are further split into ones which are basically the same for all Coff targets except that they use the external symbol format or use different values for certain constants.

The generic routines are in `coffgen.c`. These routines work for any Coff target. They use some hooks into the target specific code; the hooks are in a `bfd_coff_backend_data` structure, one of which exists for each target.

The essentially similar target-specific routines are in `coffcode.h`. This header file includes executable C code. The various Coff targets first include the appropriate Coff header file, make any special defines that are needed, and then include `coffcode.h`.

Some of the Coff targets then also have additional routines in the target source file itself.

3.3.2.2 Coff long section names

In the standard Coff object format, section names are limited to the eight bytes available in the `s_name` field of the `SCNHDR` section header structure. The format requires the field to be NUL-padded, but not necessarily NUL-terminated, so the longest section names permitted are a full eight characters.

The Microsoft PE variants of the Coff object file format add an extension to support the use of long section names. This extension is defined in section 4 of the Microsoft PE/COFF specification (rev 8.1). If a section name is too long to fit into the section header's `s_name` field, it is instead placed into the string table, and the `s_name` field is filled with a slash (`"/"`) followed by the ASCII decimal representation of the offset of the full name relative to the string table base.

Note that this implies that the extension can only be used in object files, as executables do not contain a string table. The standard specifies that long section names from objects emitted into executable images are to be truncated.

However, as a GNU extension, BFD can generate executable images that contain a string table and long section names. This would appear to be technically valid, as the standard only says that Coff debugging information is deprecated, not forbidden, and in practice it works, although some tools that parse PE files expecting the MS standard format may become confused; `PEview` is one known example.

The functionality is supported in BFD by code implemented under the control of the macro `COFF_LONG_SECTION_NAMES`. If not defined, the format does not support long section names in any way. If defined, it is used to initialise a flag, `_bfd_coff_long_section_names`, and a hook function pointer, `_bfd_coff_set_long_section_names`, in the Coff backend data structure. The flag controls the generation of long section names in output BFDs at runtime; if it is false, as it will be by default when generating an executable image, long section names are truncated; if true, the long section names extension is employed. The hook points to a function that allows the value of the flag to be altered at runtime, on formats that support long section names at all; on other formats it points to a stub that returns an error indication.

With input BFDs, the flag is set according to whether any long section names are detected while reading the section headers. For a completely new BFD, the flag is set to the default for the target format. This information can be used by a client of the BFD library when deciding what output format to generate, and means that a BFD that is opened for read and subsequently converted to a writeable BFD and modified in-place will retain whatever format it had on input.

If `COFF_LONG_SECTION_NAMES` is simply defined (blank), or is defined to the value `"1"`, then long section names are enabled by default; if it is defined to the value zero, they are disabled by default (but still accepted in input BFDs). The header `coffcode.h` defines a macro, `COFF_DEFAULT_LONG_SECTION_NAMES`, which is used in the backends to initialise the backend data structure fields appropriately; see the comments for further detail.

3.3.2.3 Bit twiddling

Each flavour of coff supported in BFD has its own header file describing the external layout of the structures. There is also an internal description of the coff layout, in `coff/internal.h`. A major function of the coff backend is swapping the bytes and twiddling the bits to translate the external form of the structures into the normal internal form. This is all performed in the `bfd_swap_`*thing*`_`*direction* routines. Some elements are different sizes between different versions of coff; it is the duty of the coff version specific include file to override the definitions of various packing routines in `coffcode.h`. E.g., the size of line number entry in coff is sometimes 16 bits, and sometimes 32 bits. `#define`ing `PUT_LNSZ_LNNO` and `GET_LNSZ_LNNO` will select the correct one. No doubt, some day someone will find a version of coff which has a varying field size not catered to at the moment. To port BFD, that person will have to add more `#defines`. Three of the bit twiddling routines are exported to `gdb`; `coff_swap_aux_in`, `coff_swap_sym_in` and `coff_swap_lineno_in`. `GDB` reads the symbol table on its own, but uses BFD to fix things up. More of the bit twiddlers are exported for `gas`; `coff_swap_aux_out`, `coff_swap_sym_out`, `coff_swap_lineno_out`, `coff_swap_reloc_out`, `coff_swap_filehdr_out`, `coff_swap_aouthdr_out`, `coff_swap_scnhdr_out`. `Gas` currently keeps track of all the symbol table and reloc drudgery itself, thereby saving the internal BFD overhead, but uses BFD to swap things on the way out, making cross ports much safer. Doing so also allows BFD (and thus the linker) to use the same header files as `gas`, which makes one avenue to disaster disappear.

3.3.2.4 Symbol reading

The simple canonical form for symbols used by BFD is not rich enough to keep all the information available in a coff symbol table. The back end gets around this problem by keeping the original symbol table around, "behind the scenes".

When a symbol table is requested (through a call to `bfd_canonicalize_symtab`), a request gets through to `coff_get_normalized_symtab`. This reads the symbol table from the coff file and swaps all the structures inside into the internal form. It also fixes up all the pointers in the table (represented in the file by offsets from the first symbol in the table) into physical pointers to elements in the new internal table. This involves some work since the meanings of fields change depending upon context: a field that is a pointer to another structure in the symbol table at one moment may be the size in bytes of a structure at the next. Another pass is made over the table. All symbols which mark file names (`C_FILE` symbols) are modified so that the internal string points to the value in the auxent (the real filename) rather than the normal text associated with the symbol (`".file"`).

At this time the symbol names are moved around. Coff stores all symbols less than nine characters long physically within the symbol table; longer strings are kept at the end of the file in the string table. This pass moves all strings into memory and replaces them with pointers to the strings.

The symbol table is massaged once again, this time to create the canonical table used by the BFD application. Each symbol is inspected in turn, and a decision made (using the `sclass` field) about the various flags to set in the `asymbol`. See Section 2.7 [Symbols], page 49. The generated canonical table shares strings with the hidden internal symbol table.

Any linenumbers are read from the coff file too, and attached to the symbols which own the functions the linenumbers belong to.

3.3.2.5 Symbol writing

Writing a symbol to a coff file which didn't come from a coff file will lose any debugging information. The `asymbol` structure remembers the BFD from which the symbol was taken, and on output the back end makes sure that the same destination target as source target is present.

When the symbols have come from a coff file then all the debugging information is preserved.

Symbol tables are provided for writing to the back end in a vector of pointers to pointers. This allows applications like the linker to accumulate and output large symbol tables without having to do too much byte copying.

This function runs through the provided symbol table and patches each symbol marked as a file place holder (`C_FILE`) to point to the next file place holder in the list. It also marks each `offset` field in the list with the offset from the first symbol of the current symbol.

Another function of this procedure is to turn the canonical value form of BFD into the form used by coff. Internally, BFD expects symbol values to be offsets from a section base; so a symbol physically at 0x120, but in a section starting at 0x100, would have the value 0x20. Coff expects symbols to contain their final value, so symbols have their values changed at this point to reflect their sum with their owning section. This transformation uses the `output_section` field of the `asymbol`'s `asection` See Section 2.6 [Sections], page 29.

- `coff_mangle_symbols`

This routine runs though the provided symbol table and uses the offsets generated by the previous pass and the pointers generated when the symbol table was read in to create the structured hierarchy required by coff. It changes each pointer to a symbol into the index into the symbol table of the asymbol.

- `coff_write_symbols`

This routine runs through the symbol table and patches up the symbols from their internal form into the coff way, calls the bit twiddlers, and writes out the table to the file.

3.3.2.6 `coff_symbol_type`

Description

The hidden information for an `asymbol` is described in a `combined_entry_type`:

```
typedef struct coff_ptr_struct
{
  /* Remembers the offset from the first symbol in the file for
     this symbol. Generated by coff_renumber_symbols.  */
  unsigned int offset;

  /* Should the value of this symbol be renumbered.  Used for
     XCOFF C_BSTAT symbols.  Set by coff_slurp_symbol_table.  */
  unsigned int fix_value : 1;

  /* Should the tag field of this symbol be renumbered.
     Created by coff_pointerize_aux.  */
  unsigned int fix_tag : 1;
```

```
  /* Should the endidx field of this symbol be renumbered.
     Created by coff_pointerize_aux.  */
  unsigned int fix_end : 1;

  /* Should the x_csect.x_scnlen field be renumbered.
     Created by coff_pointerize_aux.  */
  unsigned int fix_scnlen : 1;

  /* Fix up an XCOFF C_BINCL/C_EINCL symbol.  The value is the
     index into the line number entries.  Set by coff_slurp_symbol_table.  */
  unsigned int fix_line : 1;

  /* The container for the symbol structure as read and translated
     from the file.  */
  union
  {
    union internal_auxent auxent;
    struct internal_syment syment;
  } u;

 /* Selector for the union above.  */
 bool is_sym;

 /* An extra pointer which can used by format based on COFF (like XCOFF)
    to provide extra information to their backend.  */
 void *extrap;
} combined_entry_type;


/* Each canonical asymbol really looks like this: */

typedef struct coff_symbol_struct
{
  /* The actual symbol which the rest of BFD works with */
  asymbol symbol;

  /* A pointer to the hidden information for this symbol */
  combined_entry_type *native;

  /* A pointer to the linenumber information for this symbol */
  struct lineno_cache_entry *lineno;

  /* Have the line numbers been relocated yet ? */
  bool done_lineno;
} coff_symbol_type;
```

3.3.2.7 bfd_coff_backend_data

```
/* COFF symbol classifications.  */

enum coff_symbol_classification
{
  /* Global symbol.  */
  COFF_SYMBOL_GLOBAL,
  /* Common symbol.  */
  COFF_SYMBOL_COMMON,
  /* Undefined symbol.  */
  COFF_SYMBOL_UNDEFINED,
  /* Local symbol.  */
  COFF_SYMBOL_LOCAL,
  /* PE section symbol.  */
  COFF_SYMBOL_PE_SECTION
};

typedef asection * (*coff_gc_mark_hook_fn)
  (asection *, struct bfd_link_info *, struct internal_reloc *,
   struct coff_link_hash_entry *, struct internal_syment *);
```

Special entry points for gdb to swap in coff symbol table parts:

```
typedef struct
{
  void (*_bfd_coff_swap_aux_in)
    (bfd *, void *, int, int, int, int, void *);

  void (*_bfd_coff_swap_sym_in)
    (bfd *, void *, void *);

  void (*_bfd_coff_swap_lineno_in)
    (bfd *, void *, void *);

  unsigned int (*_bfd_coff_swap_aux_out)
    (bfd *, void *, int, int, int, int, void *);

  unsigned int (*_bfd_coff_swap_sym_out)
    (bfd *, void *, void *);

  unsigned int (*_bfd_coff_swap_lineno_out)
    (bfd *, void *, void *);

  unsigned int (*_bfd_coff_swap_reloc_out)
    (bfd *, void *, void *);

  unsigned int (*_bfd_coff_swap_filehdr_out)
```

```
  (bfd *, void *, void *);

unsigned int (*_bfd_coff_swap_aouthdr_out)
  (bfd *, void *, void *);

unsigned int (*_bfd_coff_swap_scnhdr_out)
  (bfd *, void *, void *);

unsigned int _bfd_filhsz;
unsigned int _bfd_aoutsz;
unsigned int _bfd_scnhsz;
unsigned int _bfd_symesz;
unsigned int _bfd_auxesz;
unsigned int _bfd_relsz;
unsigned int _bfd_linesz;
unsigned int _bfd_filnmlen;
bool _bfd_coff_long_filenames;

bool _bfd_coff_long_section_names;
bool (*_bfd_coff_set_long_section_names)
  (bfd *, int);

unsigned int _bfd_coff_default_section_alignment_power;
bool _bfd_coff_force_symnames_in_strings;
unsigned int _bfd_coff_debug_string_prefix_length;
unsigned int _bfd_coff_max_nscns;

void (*_bfd_coff_swap_filehdr_in)
  (bfd *, void *, void *);

void (*_bfd_coff_swap_aouthdr_in)
  (bfd *, void *, void *);

void (*_bfd_coff_swap_scnhdr_in)
  (bfd *, void *, void *);

void (*_bfd_coff_swap_reloc_in)
  (bfd *abfd, void *, void *);

bool (*_bfd_coff_bad_format_hook)
  (bfd *, void *);

bool (*_bfd_coff_set_arch_mach_hook)
  (bfd *, void *);

void * (*_bfd_coff_mkobject_hook)
  (bfd *, void *, void *);
```

```
bool (*_bfd_styp_to_sec_flags_hook)
  (bfd *, void *, const char *, asection *, flagword *);

void (*_bfd_set_alignment_hook)
  (bfd *, asection *, void *);

bool (*_bfd_coff_slurp_symbol_table)
  (bfd *);

bool (*_bfd_coff_symname_in_debug)
  (bfd *, struct internal_syment *);

bool (*_bfd_coff_pointerize_aux_hook)
  (bfd *, combined_entry_type *, combined_entry_type *,
   unsigned int, combined_entry_type *);

bool (*_bfd_coff_print_aux)
  (bfd *, FILE *, combined_entry_type *, combined_entry_type *,
   combined_entry_type *, unsigned int);

void (*_bfd_coff_reloc16_extra_cases)
  (bfd *, struct bfd_link_info *, struct bfd_link_order *, arelent *,
   bfd_byte *, unsigned int *, unsigned int *);

int (*_bfd_coff_reloc16_estimate)
  (bfd *, asection *, arelent *, unsigned int,
   struct bfd_link_info *);

enum coff_symbol_classification (*_bfd_coff_classify_symbol)
  (bfd *, struct internal_syment *);

bool (*_bfd_coff_compute_section_file_positions)
  (bfd *);

bool (*_bfd_coff_start_final_link)
  (bfd *, struct bfd_link_info *);

bool (*_bfd_coff_relocate_section)
  (bfd *, struct bfd_link_info *, bfd *, asection *, bfd_byte *,
   struct internal_reloc *, struct internal_syment *, asection **);

reloc_howto_type *(*_bfd_coff_rtype_to_howto)
  (bfd *, asection *, struct internal_reloc *,
   struct coff_link_hash_entry *, struct internal_syment *, bfd_vma *);

bool (*_bfd_coff_adjust_symndx)
```

```
    (bfd *, struct bfd_link_info *, bfd *, asection *,
     struct internal_reloc *, bool *);

  bool (*_bfd_coff_link_add_one_symbol)
    (struct bfd_link_info *, bfd *, const char *, flagword,
     asection *, bfd_vma, const char *, bool, bool,
     struct bfd_link_hash_entry **);

  bool (*_bfd_coff_link_output_has_begun)
    (bfd *, struct coff_final_link_info *);

  bool (*_bfd_coff_final_link_postscript)
    (bfd *, struct coff_final_link_info *);

  bool (*_bfd_coff_print_pdata)
    (bfd *, void *);

} bfd_coff_backend_data;

#define coff_backend_info(abfd) \
  ((bfd_coff_backend_data *) (abfd)->xvec->backend_data)

#define bfd_coff_swap_aux_in(a,e,t,c,ind,num,i) \
  ((coff_backend_info (a)->_bfd_coff_swap_aux_in) (a,e,t,c,ind,num,i))

#define bfd_coff_swap_sym_in(a,e,i) \
  ((coff_backend_info (a)->_bfd_coff_swap_sym_in) (a,e,i))

#define bfd_coff_swap_lineno_in(a,e,i) \
  ((coff_backend_info ( a)->_bfd_coff_swap_lineno_in) (a,e,i))

#define bfd_coff_swap_reloc_out(abfd, i, o) \
  ((coff_backend_info (abfd)->_bfd_coff_swap_reloc_out) (abfd, i, o))

#define bfd_coff_swap_lineno_out(abfd, i, o) \
  ((coff_backend_info (abfd)->_bfd_coff_swap_lineno_out) (abfd, i, o))

#define bfd_coff_swap_aux_out(a,i,t,c,ind,num,o) \
  ((coff_backend_info (a)->_bfd_coff_swap_aux_out) (a,i,t,c,ind,num,o))

#define bfd_coff_swap_sym_out(abfd, i,o) \
  ((coff_backend_info (abfd)->_bfd_coff_swap_sym_out) (abfd, i, o))

#define bfd_coff_swap_scnhdr_out(abfd, i,o) \
  ((coff_backend_info (abfd)->_bfd_coff_swap_scnhdr_out) (abfd, i, o))

#define bfd_coff_swap_filehdr_out(abfd, i,o) \
```

```
  ((coff_backend_info (abfd)->_bfd_coff_swap_filehdr_out) (abfd, i, o))

#define bfd_coff_swap_aouthdr_out(abfd, i,o) \
  ((coff_backend_info (abfd)->_bfd_coff_swap_aouthdr_out) (abfd, i, o))

#define bfd_coff_filhsz(abfd) (coff_backend_info (abfd)->_bfd_filhsz)
#define bfd_coff_aoutsz(abfd) (coff_backend_info (abfd)->_bfd_aoutsz)
#define bfd_coff_scnhsz(abfd) (coff_backend_info (abfd)->_bfd_scnhsz)
#define bfd_coff_symesz(abfd) (coff_backend_info (abfd)->_bfd_symesz)
#define bfd_coff_auxesz(abfd) (coff_backend_info (abfd)->_bfd_auxesz)
#define bfd_coff_relsz(abfd)  (coff_backend_info (abfd)->_bfd_relsz)
#define bfd_coff_linesz(abfd) (coff_backend_info (abfd)->_bfd_linesz)
#define bfd_coff_filnmlen(abfd) (coff_backend_info (abfd)->_bfd_filnmlen)
#define bfd_coff_long_filenames(abfd) \
  (coff_backend_info (abfd)->_bfd_coff_long_filenames)
#define bfd_coff_long_section_names(abfd) \
  (coff_backend_info (abfd)->_bfd_coff_long_section_names)
#define bfd_coff_set_long_section_names(abfd, enable) \
  ((coff_backend_info (abfd)->_bfd_coff_set_long_section_names) (abfd, enable))
#define bfd_coff_default_section_alignment_power(abfd) \
  (coff_backend_info (abfd)->_bfd_coff_default_section_alignment_power)
#define bfd_coff_max_nscns(abfd) \
  (coff_backend_info (abfd)->_bfd_coff_max_nscns)

#define bfd_coff_swap_filehdr_in(abfd, i,o) \
  ((coff_backend_info (abfd)->_bfd_coff_swap_filehdr_in) (abfd, i, o))

#define bfd_coff_swap_aouthdr_in(abfd, i,o) \
  ((coff_backend_info (abfd)->_bfd_coff_swap_aouthdr_in) (abfd, i, o))

#define bfd_coff_swap_scnhdr_in(abfd, i,o) \
  ((coff_backend_info (abfd)->_bfd_coff_swap_scnhdr_in) (abfd, i, o))

#define bfd_coff_swap_reloc_in(abfd, i, o) \
  ((coff_backend_info (abfd)->_bfd_coff_swap_reloc_in) (abfd, i, o))

#define bfd_coff_bad_format_hook(abfd, filehdr) \
  ((coff_backend_info (abfd)->_bfd_coff_bad_format_hook) (abfd, filehdr))

#define bfd_coff_set_arch_mach_hook(abfd, filehdr)\
  ((coff_backend_info (abfd)->_bfd_coff_set_arch_mach_hook) (abfd, filehdr))
#define bfd_coff_mkobject_hook(abfd, filehdr, aouthdr)\
  ((coff_backend_info (abfd)->_bfd_coff_mkobject_hook)\
   (abfd, filehdr, aouthdr))

#define bfd_coff_styp_to_sec_flags_hook(abfd, scnhdr, name, section, flags_ptr)\
  ((coff_backend_info (abfd)->_bfd_styp_to_sec_flags_hook)\
```

```
   (abfd, scnhdr, name, section, flags_ptr))

#define bfd_coff_set_alignment_hook(abfd, sec, scnhdr)\
  ((coff_backend_info (abfd)->_bfd_set_alignment_hook) (abfd, sec, scnhdr))

#define bfd_coff_slurp_symbol_table(abfd)\
  ((coff_backend_info (abfd)->_bfd_coff_slurp_symbol_table) (abfd))

#define bfd_coff_symname_in_debug(abfd, sym)\
  ((coff_backend_info (abfd)->_bfd_coff_symname_in_debug) (abfd, sym))

#define bfd_coff_force_symnames_in_strings(abfd)\
  (coff_backend_info (abfd)->_bfd_coff_force_symnames_in_strings)

#define bfd_coff_debug_string_prefix_length(abfd)\
  (coff_backend_info (abfd)->_bfd_coff_debug_string_prefix_length)

#define bfd_coff_print_aux(abfd, file, base, symbol, aux, indaux)\
  ((coff_backend_info (abfd)->_bfd_coff_print_aux)\
   (abfd, file, base, symbol, aux, indaux))

#define bfd_coff_reloc16_extra_cases(abfd, link_info, link_order,\
                                     reloc, data, src_ptr, dst_ptr)\
  ((coff_backend_info (abfd)->_bfd_coff_reloc16_extra_cases)\
   (abfd, link_info, link_order, reloc, data, src_ptr, dst_ptr))

#define bfd_coff_reloc16_estimate(abfd, section, reloc, shrink, link_info)\
  ((coff_backend_info (abfd)->_bfd_coff_reloc16_estimate)\
   (abfd, section, reloc, shrink, link_info))

#define bfd_coff_classify_symbol(abfd, sym)\
  ((coff_backend_info (abfd)->_bfd_coff_classify_symbol)\
   (abfd, sym))

#define bfd_coff_compute_section_file_positions(abfd)\
  ((coff_backend_info (abfd)->_bfd_coff_compute_section_file_positions)\
   (abfd))

#define bfd_coff_start_final_link(obfd, info)\
  ((coff_backend_info (obfd)->_bfd_coff_start_final_link)\
   (obfd, info))
#define bfd_coff_relocate_section(obfd,info,ibfd,o,con,rel,isyms,secs)\
  ((coff_backend_info (ibfd)->_bfd_coff_relocate_section)\
   (obfd, info, ibfd, o, con, rel, isyms, secs))
#define bfd_coff_rtype_to_howto(abfd, sec, rel, h, sym, addendp)\
  ((coff_backend_info (abfd)->_bfd_coff_rtype_to_howto)\
   (abfd, sec, rel, h, sym, addendp))
```

```
#define bfd_coff_adjust_symndx(obfd, info, ibfd, sec, rel, adjustedp)\
  ((coff_backend_info (abfd)->_bfd_coff_adjust_symndx)\
   (obfd, info, ibfd, sec, rel, adjustedp))
#define bfd_coff_link_add_one_symbol(info, abfd, name, flags, section,\
                                     value, string, cp, coll, hashp)\
  ((coff_backend_info (abfd)->_bfd_coff_link_add_one_symbol)\
   (info, abfd, name, flags, section, value, string, cp, coll, hashp))

#define bfd_coff_link_output_has_begun(a,p) \
  ((coff_backend_info (a)->_bfd_coff_link_output_has_begun) (a, p))
#define bfd_coff_final_link_postscript(a,p) \
  ((coff_backend_info (a)->_bfd_coff_final_link_postscript) (a, p))

#define bfd_coff_have_print_pdata(a) \
  (coff_backend_info (a)->_bfd_coff_print_pdata)
#define bfd_coff_print_pdata(a,p) \
  ((coff_backend_info (a)->_bfd_coff_print_pdata) (a, p))

/* Macro: Returns true if the bfd is a PE executable as opposed to a
   PE object file.  */
#define bfd_pei_p(abfd) \
  (startswith ((abfd)->xvec->name, "pei-"))
```

3.3.2.8 Writing relocations

To write relocations, the back end steps though the canonical relocation table and create an `internal_reloc`. The symbol index to use is removed from the `offset` field in the symbol table supplied. The address comes directly from the sum of the section base address and the relocation offset; the type is dug directly from the howto field. Then the `internal_reloc` is swapped into the shape of an `external_reloc` and written out to disk.

3.3.2.9 Reading linenumbers

Creating the linenumber table is done by reading in the entire coff linenumber table, and creating another table for internal use.

A coff linenumber table is structured so that each function is marked as having a line number of 0. Each line within the function is an offset from the first line in the function. The base of the line number information for the table is stored in the symbol associated with the function.

Note: The PE format uses line number 0 for a flag indicating a new source file.

The information is copied from the external to the internal table, and each symbol which marks a function is marked by pointing its...

How does this work ?

3.3.2.10 Reading relocations

Coff relocations are easily transformed into the internal BFD form (`arelent`).

Reading a coff relocation table is done in the following stages:

- Read the entire coff relocation table into memory.

- Process each relocation in turn; first swap it from the external to the internal form.
- Turn the symbol referenced in the relocation's symbol index into a pointer into the canonical symbol table. This table is the same as the one returned by a call to `bfd_canonicalize_symtab`. The back end will call that routine and save the result if a canonicalization hasn't been done.
- The reloc index is turned into a pointer to a howto structure, in a back end specific way. For instance, the 386 uses the `r_type` to directly produce an index into a howto table vector.

3.4 ELF backends

BFD support for ELF formats is being worked on. Currently, the best supported back ends are for sparc and i386 (running svr4 or Solaris 2).

Documentation of the internals of the support code still needs to be written. The code is changing quickly enough that we haven't bothered yet.

3.5 mmo backend

The mmo object format is used exclusively together with Professor Donald E. Knuth's educational 64-bit processor MMIX. The simulator `mmix` which is available at `http://mmix.cs.hm.edu/src/index.html` understands this format. That package also includes a combined assembler and linker called `mmixal`. The mmo format has no advantages feature-wise compared to e.g. ELF. It is a simple non-relocatable object format with no support for archives or debugging information, except for symbol value information and line numbers (which is not yet implemented in BFD). See `http://mmix.cs.hm.edu/` for more information about MMIX. The ELF format is used for intermediate object files in the BFD implementation.

3.5.1 File layout

The mmo file contents is not partitioned into named sections as with e.g. ELF. Memory areas is formed by specifying the location of the data that follows. Only the memory area '`0x0000...00`' to '`0x01ff...ff`' is executable, so it is used for code (and constants) and the area '`0x2000...00`' to '`0x20ff...ff`' is used for writable data. See Section 3.5.3 [mmo section mapping], page 224.

There is provision for specifying "special data" of 65536 different types. We use type 80 (decimal), arbitrarily chosen the same as the ELF `e_machine` number for MMIX, filling it with section information normally found in ELF objects. See Section 3.5.3 [mmo section mapping], page 224.

Contents is entered as 32-bit words, xor:ed over previous contents, always zero-initialized. A word that starts with the byte '`0x98`' forms a command called a '`lopcode`', where the next byte distinguished between the thirteen lopcodes. The two remaining bytes, called the '`Y`' and '`Z`' fields, or the '`YZ`' field (a 16-bit big-endian number), are used for various purposes different for each lopcode. As documented in `http://mmix.cs.hm.edu/doc/mmixal.pdf`, the lopcodes are:

`lop_quote`
: 0x98000001. The next word is contents, regardless of whether it starts with 0x98 or not.

`lop_loc` 0x9801YYZZ, where '`Z`' is 1 or 2. This is a location directive, setting the location for the next data to the next 32-bit word (for $Z = 1$) or 64-bit word (for $Z = 2$), plus $Y * 2^5 6$. Normally '`Y`' is 0 for the text segment and 2 for the data segment. Beware that the low bits of non- tetrabyte-aligned values are silently discarded when being automatically incremented and when storing contents (in contrast to e.g. its use as current location when followed by lop_fixo et al before the next possibly-quoted tetrabyte contents).

`lop_skip` 0x9802YYZZ. Increase the current location by '`YZ`' bytes.

`lop_fixo` 0x9803YYZZ, where '`Z`' is 1 or 2. Store the current location as 64 bits into the location pointed to by the next 32-bit ($Z = 1$) or 64-bit ($Z = 2$) word, plus $Y * 2^5 6$.

`lop_fixr` 0x9804YYZZ. '`YZ`' is stored into the current location plus $2 - 4 * YZ$.

`lop_fixrx`

0x980500ZZ. '`Z`' is 16 or 24. A value '`L`' derived from the following 32-bit word are used in a manner similar to '`YZ`' in lop_fixr: it is xor:ed into the current location minus $4 * L$. The first byte of the word is 0 or 1. If it is 1, then $L = (\mathit{lowest24bitsofword}) - 2^Z$, if 0, then $L = (\mathit{lowest24bitsofword})$.

`lop_file` 0x9806YYZZ. '`Y`' is the file number, '`Z`' is count of 32-bit words. Set the file number to '`Y`' and the line counter to 0. The next $Z * 4$ bytes contain the file name, padded with zeros if the count is not a multiple of four. The same '`Y`' may occur multiple times, but '`Z`' must be 0 for all but the first occurrence.

`lop_line` 0x9807YYZZ. '`YZ`' is the line number. Together with lop_file, it forms the source location for the next 32-bit word. Note that for each non-lopcode 32-bit word, line numbers are assumed incremented by one.

`lop_spec` 0x9808YYZZ. '`YZ`' is the type number. Data until the next lopcode other than lop_quote forms special data of type '`YZ`'. See Section 3.5.3 [mmo section mapping], page 224.

Other types than 80, (or type 80 with a content that does not parse) is stored in sections named `.MMIX.spec_data.n` where *n* is the '`YZ`'-type. The flags for such a sections say not to allocate or load the data. The vma is 0. Contents of multiple occurrences of special data *n* is concatenated to the data of the previous lop_spec *n*s. The location in data or code at which the lop_spec occurred is lost.

`lop_pre` 0x980901ZZ. The first lopcode in a file. The '`Z`' field forms the length of header information in 32-bit words, where the first word tells the time in seconds since '`00:00:00 GMT Jan 1 1970`'.

`lop_post` 0x980a00ZZ. $Z > 32$. This lopcode follows after all content-generating lopcodes in a program. The '`Z`' field denotes the value of '`rG`' at the beginning of the program. The following $256 - Z$ big-endian 64-bit words are loaded into global registers '`$G`' . . . '`$255`'.

`lop_stab` 0x980b0000. The next-to-last lopcode in a program. Must follow immediately after the lop_post lopcode and its data. After this lopcode follows all symbols in a compressed format (see Section 3.5.2 [Symbol-table], page 222).

`lop_end` 0x980cYYZZ. The last lopcode in a program. It must follow the lop_stab lopcode and its data. The '`YZ`' field contains the number of 32-bit words of symbol table information after the preceding lop_stab lopcode.

Note that the lopcode "fixups"; `lop_fixr`, `lop_fixrx` and `lop_fixo` are not generated by BFD, but are handled. They are generated by `mmixal`.

This trivial one-label, one-instruction file:

```
:Main TRAP 1,2,3
```

can be represented this way in mmo:

```
0x98090101 - lop_pre, one 32-bit word with timestamp.
<timestamp>
0x98010002 - lop_loc, text segment, using a 64-bit address.
             Note that mmixal does not emit this for the file above.
0x00000000 - Address, high 32 bits.
0x00000000 - Address, low 32 bits.
0x98060002 - lop_file, 2 32-bit words for file-name.
0x74657374 - "test"
0x2e730000 - ".s\0\0"
0x98070001 - lop_line, line 1.
0x00010203 - TRAP 1,2,3
0x980a00ff - lop_post, setting $255 to 0.
0x00000000
0x00000000
0x980b0000 - lop_stab for ":Main" = 0, serial 1.
0x203a4040   See Section 3.5.2 [Symbol-table], page 222.
0x10404020
0x4d206120
0x69016e00
0x81000000
0x980c0005 - lop_end; symbol table contained five 32-bit words.
```

3.5.2 Symbol table format

From mmixal.w (or really, the generated mmixal.tex) in the MMIXware package which also contains the `mmix` simulator: "Symbols are stored and retrieved by means of a '`ternary search trie`', following ideas of Bentley and Sedgewick. (See ACM–SIAM Symp. on Discrete Algorithms '`8`' (1997), 360–369; R.Sedgewick, '`Algorithms in C`' (Reading, Mass. Addison–Wesley, 1998), '`15.4`'.) Each trie node stores a character, and there are branches to subtries for the cases where a given character is less than, equal to, or greater than the character in the trie. There also is a pointer to a symbol table entry if a symbol ends at the current node."

So it's a tree encoded as a stream of bytes. The stream of bytes acts on a single virtual global symbol, adding and removing characters and signalling complete symbol points. Here, we read the stream and create symbols at the completion points.

First, there's a control byte `m`. If any of the listed bits in `m` is nonzero, we execute what stands at the right, in the listed order:

```
(MMO3_LEFT)
```

```
0x40 - Traverse left trie.
       (Read a new command byte and recurse.)

(MMO3_SYMBITS)
0x2f - Read the next byte as a character and store it in the
       current character position; increment character position.
       Test the bits of m:

       (MMO3_WCHAR)
       0x80 - The character is 16-bit (so read another byte,
              merge into current character.

       (MMO3_TYPEBITS)
       0xf  - We have a complete symbol; parse the type, value
              and serial number and do what should be done
              with a symbol.  The type and length information
              is in j = (m & 0xf).

              (MMO3_REGQUAL_BITS)
              j == 0xf: A register variable.  The following
                        byte tells which register.
              j <= 8:   An absolute symbol.  Read j bytes as the
                        big-endian number the symbol equals.
                        A j = 2 with two zero bytes denotes an
                        unknown symbol.
              j > 8:    As with j <= 8, but add (0x20 << 56)
                        to the value in the following j - 8
                        bytes.

              Then comes the serial number, as a variant of
              uleb128, but better named ubeb128:
              Read bytes and shift the previous value left 7
              (multiply by 128).  Add in the new byte, repeat
              until a byte has bit 7 set.  The serial number
              is the computed value minus 128.

       (MMO3_MIDDLE)
       0x20 - Traverse middle trie.  (Read a new command byte
              and recurse.)  Decrement character position.

(MMO3_RIGHT)
0x10 - Traverse right trie.  (Read a new command byte and
       recurse.)
```

Let's look again at the `lop_stab` for the trivial file (see Section 3.5.1 [File layout], page 220).

```
0x980b0000 - lop_stab for ":Main" = 0, serial 1.
0x203a4040
```

```
0x10404020
0x4d206120
0x69016e00
0x81000000
```

This forms the trivial trie (note that the path between ":" and "M" is redundant):

```
203a     ":"
40        /
40       /
10       \
40       /
40      /
204d  "M"
2061  "a"
2069  "i"
016e  "n" is the last character in a full symbol, and
      with a value represented in one byte.
00    The value is 0.
81    The serial number is 1.
```

3.5.3 mmo section mapping

The implementation in BFD uses special data type 80 (decimal) to encapsulate and describe named sections, containing e.g. debug information. If needed, any datum in the encapsulation will be quoted using lop_quote. First comes a 32-bit word holding the number of 32-bit words containing the zero-terminated zero-padded segment name. After the name there's a 32-bit word holding flags describing the section type. Then comes a 64-bit big-endian word with the section length (in bytes), then another with the section start address. Depending on the type of section, the contents might follow, zero-padded to 32-bit boundary. For a loadable section (such as data or code), the contents might follow at some later point, not necessarily immediately, as a lop_loc with the same start address as in the section description, followed by the contents. This in effect forms a descriptor that must be emitted before the actual contents. Sections described this way must not overlap.

For areas that don't have such descriptors, synthetic sections are formed by BFD. Consecutive contents in the two memory areas '`0x0000...00`' to '`0x01ff...ff`' and '`0x2000...00`' to '`0x20ff...ff`' are entered in sections named `.text` and `.data` respectively. If an area is not otherwise described, but would together with a neighboring lower area be less than '`0x40000000`' bytes long, it is joined with the lower area and the gap is zero-filled. For other cases, a new section is formed, named `.MMIX.sec.`*n*. Here, *n* is a number, a running count through the mmo file, starting at 0.

A loadable section specified as:

```
.section secname,"ax"
TETRA 1,2,3,4,-1,-2009
BYTE 80
```

and linked to address '`0x4`', is represented by the sequence:

```
0x98080050 - lop_spec 80
0x00000002 - two 32-bit words for the section name
```

```
0x7365636e - "secn"
0x616d6500 - "ame\0"
0x00000033 - flags CODE, READONLY, LOAD, ALLOC
0x00000000 - high 32 bits of section length
0x0000001c - section length is 28 bytes; 6 * 4 + 1 + alignment to 32 bits
0x00000000 - high 32 bits of section address
0x00000004 - section address is 4
0x98010002 - 64 bits with address of following data
0x00000000 - high 32 bits of address
0x00000004 - low 32 bits: data starts at address 4
0x00000001 - 1
0x00000002 - 2
0x00000003 - 3
0x00000004 - 4
0xffffffff - -1
0xfffff827 - -2009
0x50000000 - 80 as a byte, padded with zeros.
```

Note that the lop_spec wrapping does not include the section contents. Compare this to a non-loaded section specified as:

```
.section thirdsec
TETRA 200001,100002
BYTE 38,40
```

This, when linked to address '`0x200000000000001c`', is represented by:

```
0x98080050 - lop_spec 80
0x00000002 - two 32-bit words for the section name
0x7365636e - "thir"
0x616d6500 - "dsec"
0x00000010 - flag READONLY
0x00000000 - high 32 bits of section length
0x0000000c - section length is 12 bytes; 2 * 4 + 2 + alignment to 32 bits
0x20000000 - high 32 bits of address
0x0000001c - low 32 bits of address 0x200000000000001c
0x00030d41 - 200001
0x000186a2 - 100002
0x26280000 - 38, 40 as bytes, padded with zeros
```

For the latter example, the section contents must not be loaded in memory, and is therefore specified as part of the special data. The address is usually unimportant but might provide information for e.g. the DWARF 2 debugging format.

Version 1.3, 3 November 2008

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what is it? 1

The body of this manual is set in
cmr10,
with headings in **cmbx10**
and examples in `cmtt10`.
cmti10 and
cmsl10
are used for emphasis.